BIRATIONAL SUPERRIGIDITY DOES NOT IMPLY K-STABILITY

JIHAO LIU

Abstract. We disprove the conjecture of Kim, Okada, and Won that birationally rigid Fano varieties are K-stable. Our counterexample is a birationally superrigid Fano weighted complete intersection of dimension 25 which is K-semistable but not K-stable.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

This paper considers the relation between birational rigidity and K-stability of Fano varieties. Odaka and Okada proved that a birationally superrigid Fano manifold X of Fano index one is slope stable if $|-K_X|$ is base point free [OO13, Theorem 1.1]. The central conjecture relating birational rigidity and K-stability is the following.

Conjecture 1.1 ([KOW23, Conjecture 7.8])**.** *Let X be a $\mathbb{Q}$-factorial Fano variety of Picard number one with terminal singularities. If X is birationally rigid, then X is K-stable.*

Kim, Okada, and Won conjectured that birationally rigid Fano varieties are K-stable [KOW18, Conjecture 1.9]. The conjecture in [KOW18] also predicts that X admits a Kähler–Einstein metric if X has only quotient singularities, and this paper only considers the K-stability part. See also [Oka26, Conjecture 1.1].

Cheltsov, Sarikyan, and Zhuang formulated the conjecture for Fano varieties with terminal singularities whose class group has rank one, and attributed it to Kim, Okada, and Won.

Conjecture 1.2 ([CSZ23, Conjecture 1.1])**.** *Let X be a Fano variety with terminal singularities such that* $\operatorname{rk}\operatorname{Cl}(X) = 1$*. If X is birationally rigid, then X is K-stable.*

Date: October 29, 2026.
2020 *Mathematics Subject Classification.* 14J45, 14E05, 14E07, 32Q20, 14M10.
Key words and phrases. Fano varieties, birational rigidity, K-stability, weighted complete intersections.

By [CSZ23, Section 1], a Fano variety X with terminal singularities such that $\operatorname{rk}\operatorname{Cl}(X) = 1$ is a Mori fiber space, that is, X is $\mathbb{Q}$-factorial of Picard number one. For such X, birational rigidity in the sense of [CSZ23, Section 1], namely that X is not birational to any Mori fiber space other than X itself, agrees with birational rigidity in the sense of Definition 2.11. Thus Conjecture 1.1 implies Conjecture 1.2. The following version of the conjecture considers birationally superrigid Fano varieties.

Conjecture 1.3 ([KOW23, Conjecture 1.1])**.** *Let X be a $\mathbb{Q}$-factorial Fano variety of Picard number one with terminal singularities. If X is birationally superrigid, then X is K-stable.*

Since a birationally superrigid Fano variety is birationally rigid, Conjecture 1.1 also implies Conjecture 1.3. We refer the reader to Definitions 2.9, 2.11, and 2.12 for the definitions of weighted complete intersections, birational rigidity and superrigidity, and K-semistability and K-stability.

The Fano varieties in the following three lists are birationally rigid and K-stable. Thus Conjecture 1.1 holds for them. The first list consists of smooth Fano varieties.

(1) Smooth hypersurfaces $X \subset \mathbb{P}^{n+1}$ of degree $n+1$ with $n \geq 3$. Here X is birationally superrigid [dF13, Theorem A] and K-stable [Fuj19b, Corollary 1.5]. See [KOW23, Section 1.2].
(2) Smooth double covers X of $\mathbb{P}^3$ branched along a smooth sextic surface. Here X is birationally superrigid, see [KOW23, Section 1.2], and X is K-stable by [Fuj19b, Theorem 1.3], since its α-invariant (recalled below) is at least $5/6$ by [CPW14, Proposition 3.3] (see the paragraph before [CPW14, Corollary 3.4]).
(3) Smooth complete intersections of index one, codimension r, and dimension $n \geq 10r$ in $\mathbb{P}^{n+r}$ [Zhu20, Theorems 1.2 and 1.3].

The second list consists of hypersurfaces and complete intersections in projective spaces which may have singularities.

(4) Hypersurfaces $X \subset \mathbb{P}^{n+1}$ of degree $n+1$ with $n \geq 250$ whose singularities are ordinary of multiplicity at most $n-5$ [LZ21, Theorem 1.2], and, for integers $\delta \geq -1$ and $r \geq 1$, Fano complete intersections $X \subset \mathbb{P}^{n+r}$ of index one, codimension r, and dimension $n \geq N$ for a constant N depending only on δ and r, such that $\dim \operatorname{Sing}(X) \leq \delta$ and any projective tangent cone of X is a Fano complete intersection of index at least $4r + 2\delta + 2$ which is smooth in dimension $r + \delta$ [LZ21, Theorem 1.3].
(5) Hypersurfaces $X \subset \mathbb{P}^{n+1}$ of degree $n+1$ with $n \geq 5$ whose singularities are isolated ordinary double points [dF23, Theorems 1.2 and 1.3].

The third list consists of Fano threefolds in weighted projective spaces.

(6) Quasismooth Fano threefold weighted hypersurfaces of index one with terminal singularities in well-formed weighted projective spaces (see [KOW23, Section 1.2]). They form 95 families, and any member is birationally rigid [CPR00, CP17] (see [KOW23, Theorem 1.2]). A general member of each family is K-stable [Che09b, Che09a] (see [KOW23, Theorem 1.3]), a birationally superrigid member is K-stable [KOW23, Corollary 1.9], and any member is K-stable [CO24, Theorem 1.5].
(7) General birationally rigid quasismooth Fano threefolds of index one embedded in weighted projective spaces in codimension two (as weighted complete intersections) or three (as Pfaffian subvarieties), other than complete intersections of a quadric and a cubic in $\mathbb{P}^5$ [KOW18, Theorem 1.5 and Corollary 1.6].
(8) Birationally rigid well-formed quasismooth Fano threefold weighted complete intersections. Okada removed the generality assumption of [KOW18] in codimension two [Oka26, Theorem 1.5], and proved Conjecture 1.1 for well-formed quasismooth Fano threefold weighted complete intersections [Oka26, Corollary 1.6].

In general, Conjecture 1.2 has remained open [CSZ23, Section 1]. We refer the reader to [Che05] and [Kol19] for surveys on birational rigidity, and to [DGO25] for Fano threefold weighted complete intersections.

We now explain the approach through the α-invariant. For a Fano variety X with klt singularities, the α-invariant $\alpha(X)$ of X is the supremum of all $t > 0$ such that (X, tD) is log canonical for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$ [Tia87], [SZ19, Definition 1.1]. If $n = \dim X$, then X is K-semistable if $\alpha(X) \geq \frac{n}{n+1}$, and K-stable if $\alpha(X) > \frac{n}{n+1}$ [OS12, Theorem 1.4]. This is an algebraic counterpart of Tian's criterion for the existence of Kähler–Einstein metrics on Fano manifolds [Tia87, Theorem 2.1]. Stibitz and Zhuang proved that a birationally superrigid Fano variety X is K-semistable if $\alpha(X) \geq 1/2$, and K-stable if $\alpha(X) > 1/2$ [SZ19, Theorem 1.2], and they strengthened the second conclusion: X is K-stable already if $\alpha(X) \geq 1/2$ [SZ19, Corollary 3.2].

The bound $\alpha(X) \geq 1/2$ is a hypothesis of [SZ19, Theorem 1.2] in addition to birational superrigidity. By [SZ19, Theorem 2.3], a Fano variety X is birationally superrigid if and only if X is $\mathbb{Q}$-factorial of Picard number one with terminal singularities, and for any rational number $a > 0$ and any linear system $\mathcal{M}$ without fixed components such that $a\mathcal{M} \sim_{\mathbb{Q}} -K_X$, the pair $(X, a\mathcal{M})$ is canonical. Thus birational superrigidity controls only linear systems without fixed components, while the definition of $\alpha(X)$ involves all $\mathbb{Q}$-divisors $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$, including divisors that do not move in any linear system.

For example, the α-invariant of a smooth hypersurface $X \subset \mathbb{P}^{n+1}$ of degree $n+1$ with $n \geq 3$ is at least $n/(n+1)$ [Che01, Theorem 1.3] (see also [SZ19, Section 1]), and [SZ19, Corollary 1.3] recovers the K-stability of X from [SZ19, Theorem 1.2]. Similarly, [KOW23, Theorem 1.8] proved $\alpha(X) \geq 1/2$ for any quasismooth Fano threefold weighted hypersurface X of index one with terminal singularities in a well-formed weighted projective space. If X is a birationally superrigid Fano variety of dimension $n \geq 3$ such that $-K_X$ generates $\mathrm{Cl}(X)$ and $|-K_X|$ is base point free, then $\alpha(X) \geq 1/(n+1)$ [SZ19, Theorem 1.6]. Stibitz and Zhuang remark after [SZ19, Theorem 1.6] that most known birationally superrigid Fano varieties X satisfy the conditions that $-K_X$ generates $\mathrm{Cl}(X)$ and $|-K_X|$ is base point free.

Whether birational superrigidity implies $\alpha(X) > 1/2$, or $\alpha(X) \geq 1/2$, is asked in [SZ19, Question 1.5] and [CSZ23, Question 1.3], respectively. By [CSZ23, Theorem 1.4], for any $\epsilon > 0$ there exists a birationally superrigid Fano variety X with terminal singularities such that $\operatorname{rk} \mathrm{Cl}(X) = 1$ and

$$\frac{1}{2} \leq \alpha(X) \leq \frac{1}{2} + \epsilon.$$

Thus a lower bound on $\alpha(X)$ which holds for any birationally superrigid Fano variety X is at most $1/2$.

We have the following theorem.

Theorem 1.4. *Let $X \subset \mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ be the weighted complete intersection of four hypersurfaces of degree 1806 defined by*

$$\boldsymbol{v}_0 x^{1806} + \sum_{i=1}^{11} \boldsymbol{v}_i u_i^{e_i} + \sum_{j=1}^{18} \boldsymbol{v}_{11+j} y_j^{43} = 0,$$

where $\boldsymbol{v}_a := (1, a, a^2, a^3)^\top \in \mathbb{C}^4$ for $0 \leq a \leq 29$, $(e_1, \ldots, e_{11}) := (2, 2, 2, 3, 3, 3, 3, 3, 7, 7, 7)$, the coordinate x has weight 1, the coordinates u_1, u_2, u_3 have weight 903, the coordinates $u_4, \ldots, u_8$ have weight 602, the coordinates u_9, u_{10}, u_{11} have weight 258, and the coordinates $y_1, \ldots, y_{18}$ have weight 42. Let H be a Weil divisor on X with $\mathcal{O}_X(H) \cong \mathcal{O}_X(1)$. Then:

1. *X is a $\mathbb{Q}$-factorial Fano variety of dimension 25 with terminal singularities, $\mathrm{Cl}(X) = \mathbb{Z}[H]$, and $-K_X \sim 26H$,*
2. *X is birationally superrigid, and*
3. *X is K-semistable but not K-stable.*

In Theorem 1.4, any four of the vectors $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$ are linearly independent by the Vandermonde determinant.

Theorem 1.4 implies the following corollary.

Corollary 1.5. *The variety X in Theorem 1.4 satisfies the hypotheses of Conjectures 1.1, 1.2, and 1.3, but X is not K-stable. In particular, none of Conjectures 1.1, 1.2, and 1.3 holds.*

Remark 1.6. Let $P := (x = 0) \subset X$, which is a prime divisor with $26P \sim -K_X$ by Proposition 3.4(3)(5). The pair (X, tP) is not lc for any $t > 1$, so $\alpha(X) \leq \frac{1}{26}$. Hence Theorem 1.4 does not contradict [SZ19, Corollary 3.2], and X answers [SZ19, Question 1.5] and [CSZ23, Question 1.3] in the negative. Moreover, $H^0(X, \mathcal{O}_X(-K_X)) = \mathbb{C}x^{26}$ by Proposition 3.4(3)(4), so the linear system $|-K_X|$ consists of the single divisor $26P$ and is not base point free. Thus [OO13, Theorem 1.1] does not apply to X.

Sketch of the proof. We prove Theorem 1.4 in five steps.

(1) Part (1) is Proposition 3.4. Part (3) is Theorem 4.1: by [Fuj17, Corollary 1.6(1)] and [Zhu21, Corollary 4.13(1)], applied to the Galois cover $g : X \to \mathbb{P}^{25}$ of Construction-Proposition 4.2, X is K-semistable, and the prime divisor $P = (x = 0)$ satisfies $\beta_X(P) = 0$.

(2) By the Noether–Fano criterion [SZ19, Theorem 2.3], part (2) follows from Theorem 10.1:

$$\frac{26}{\ell} \operatorname{mult}_E \mathcal{M} \leq a(E, X) - 1$$

for movable $\mathcal{M} \subset |\ell H|$ and exceptional prime divisors E over X.

(3) If $\operatorname{center}_X E \not\subset P$, the inequality is Theorem 5.1, proved by [Suz17, Proposition 2.1] (Proposition 5.5) and Lemma 5.4 if $\dim \operatorname{center}_X E \geq 15$, and by Lemma 5.6 and Proposition 5.3 otherwise.

(4) If $\operatorname{center}_X E \subset P$, inversion of adjunction (Theorem 2.5) and Lemma 10.2 reduce the inequality to Theorem 9.1: $(P, \frac{42}{\ell} D_f)$ is lc for any non-zero $f \in (R_P)_\ell$, where R_P is the homogeneous coordinate ring of P and D_f is the divisor of zeros of f on P (Set-up 3.3).

(5) On the chart $y_j \neq 0$, P is a quotient of a family over $\mathbb{A}^{17}$ of fibers of $h : \mathbb{A}^{11} \to \mathbb{A}^4$ (Set-up 6.1). Proposition 9.4 combines Theorem 7.1 on the compactified fibers with Lemma 9.2 on the base, and Proposition 9.5 treats the locus $y_1 = \cdots = y_{18} = 0$.

The key point is the comparison at the end of (c) below. Off P, for general members $D_1, D_2 \in \mathcal{M}$, the bound $\ell^2/294$ of Proposition 5.3 on the multiplicity of $D_1 \cdot D_2$ at points outside P is smaller than the lower bound $4/(26/\ell)^2 = \ell^2/169$ which Lemma 5.6 gives at a non-canonical center. Most of the work is Theorem 7.1 at the singular points of the fibers of h.

We now explain the choice of X and the main difficulties in steps 4 and 5.

(a) The choice of X. By [SZ19, Corollary 3.2], a counterexample X to Conjecture 1.3 has $\alpha(X) < \frac{1}{2}$. We look for X with a prime divisor P such that $\operatorname{Cl}(X) = \mathbb{Z}[P]$ and $-K_X \sim (n+1)P$, where $n = \dim X$. Then $\alpha(X) \leq \frac{1}{n+1}$ (see Remark 1.6) and $\beta_X(P) = 0$, since $\operatorname{vol}(-K_X - tP) = (1 - \frac{t}{n+1})^n \operatorname{vol}(-K_X)$ for $0 \leq t \leq n+1$ (Definition 2.12). Birational superrigidity forces $|kP| = \{kP\}$ for $1 \leq k \leq n$ by [SZ19, Theorem 2.3] (Remark 3.5).

In X, the coordinate x of weight 1 defines P, and any coordinate other than x has weight at least $42 > 26$, so $|kP| = \{kP\}$ for $1 \leq k \leq 41$ (Proposition 3.4(4)(5)). For the hyperplane arrangement of Construction-Proposition 4.2, the relation $-K_X \sim 26P$ corresponds to equality in the criterion of Theorem 2.13 for the single hyperplane $g(P)$, and the other inequalities of the criterion are strict. The identity

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{43} + \frac{1}{1806} = 1, \quad \text{that is,} \quad 1 + 903 + 602 + 258 + 42 = 1806,$$

with at least two coordinates of each weight other than 1, makes the age of any non-trivial element of a local stabilizer of X larger than 1, so X is terminal by Reid's criterion [Ess25, Theorem 2.1] (Proposition 3.4(2)). With at least three coordinates of each weight other than 1,

as needed in (b), $-K_X \sim (n+1)P$ forces at least four equations of degree 1806, and for four equations it forces the numbers $3, 5, 3, 18$ of coordinates of weights $903, 602, 258, 42$ (Remark 3.6).

(b) Centers in P (step 4). Along P, X has non-Gorenstein quotient singularities, for instance at the points where only the coordinates y_j are non-zero, while the $4n^2$-inequality, which bounds from below the multiplicity of the intersection of two general members of a movable linear system at a non-canonical center (Lemma 5.6 at smooth points), is proved at smooth points and at complete intersection singularities of large dimension which are generic in the sense of [Puk17, Definition 1] (see [Puk17, Section 1]). Instead, a member $D_s \in \mathcal{M}$ not containing P satisfies $D_s|_P = D_f$ for a non-zero $f \in (R_P)_\ell$, and since $(K_X + P)|_P = K_P$ (Lemma 3.7), Theorem 9.1 and inversion of adjunction (Theorem 2.5) show that $(X, P + \frac{42}{\ell} D_s)$ is lc near P. Thus, for E with $\operatorname{center}_X E \subset P$,

$$a(E,X) - \frac{26}{\ell} \operatorname{mult}_E \mathcal{M} \geq \frac{16a(E,X) + 26 \operatorname{mult}_E P}{42}.$$

It suffices to show that the right-hand side is larger than 1. Terminality of X gives only $a(E,X) > 1$, and since $\operatorname{mult}_E P$ can be smaller than 1 at a quotient singularity, this does not imply $16a(E,X) + 26 \operatorname{mult}_E P > 42$. With the coefficient $\frac{26}{\ell}$ in place of $\frac{42}{\ell}$ in Theorem 9.1, inversion of adjunction would give only $a(E,X) - \frac{26}{\ell} \operatorname{mult}_E \mathcal{M} \geq \operatorname{mult}_E P$, which can be smaller than 1 at a quotient singularity. Lemma 10.2 yields one congruence for the multiplicities along E of the coordinates on a quotient chart of X, and the identity $1 + 903 + 602 + 258 + 42 = 1806$, with at least three coordinates of each weight other than 1, implies $16a(E,X) + 26 \operatorname{mult}_E P > 42$.

(c) The threshold on P (step 5). On the chart $y_j \neq 0$, a term-by-term estimate of the multiplicities of f fails, since the terms of f may cancel on the fibers of h. Definition-Lemma 9.3 expresses f uniquely as $\sum_\alpha p_\alpha m_\alpha$ with distinct monomials m_α in the u_i, in which the exponents of u_1, u_2, u_3 are at most one and the exponent of u_4 is at most two, and non-zero homogeneous polynomials p_α in the y_l, and $43j_\alpha + 42k_\alpha = \ell$, where j_α is the degree of m_α for the weights d_i of the u_i and $k_\alpha := \deg p_\alpha$. With $J := \max_\alpha j_\alpha$ and $k := \min_\alpha k_\alpha$, we have $\ell = 43J + 42k \geq 42 \max(J,k)$, and on the charts $y_j \neq 0$ the m_α give sections of degree at most J on the compactified fibers $\overline{V}_c$ of h (Set-up 6.1, $\deg U_i = d_i$), so that the coefficient $\frac{42}{\ell}$ follows from the threshold $\frac{1}{J}$ on the fibers and the degree k on the base. Lemma 8.1 controls the cancellation by the threshold $\frac{1}{J}$ of Theorem 7.1, through inversion of adjunction on the fibers.

Lemma 8.1 needs klt fibers, but over the codimension-four locus Y_0 of Set-up 8.2, the fiber is the cone $h^{-1}(0)$, which is not lc since $\sum_{i=1}^{11} \frac{42}{e_i} = 151 < 4 \cdot 42$. Thus we blow up Y_0, take a 42-nd root of a local equation of the exceptional divisor, and rescale the u_i (Construction 8.4), at the cost $\frac{17}{42} = \frac{4 \cdot 42 - 151}{42}$ in Proposition 8.6. Lemma 9.2(2), applied to $C = Y_0$, compensates the loss: for a prime divisor F over the base $Y = \mathbb{A}^{17}$ and a non-zero polynomial g of degree at most k on Y, $\operatorname{mult}_F g \leq k(a(F,Y) - 3 \operatorname{mult}_F \mathcal{I}_{Y_0})$, where $3 = 4 - 1$ and 4 is the codimension of Y_0, and $3 > \frac{17}{42}$.

(d) The threshold on the fibers of h (step 5). Proposition 7.3 bounds the multiplicities of sections at smooth points of the fibers by cutting cycles, like Proposition 5.3, and Proposition 7.4 deduces the threshold near the divisor at infinity, whose punctured affine cone $h^{-1}(0) \backslash \{0\}$ is smooth. At a singular point p of a fiber, a bound on the multiplicity does not imply log canonicity, and by Lemma 7.5, it suffices to bound the threshold of the lowest-degree part at p of a section on the exceptional divisor E_p of a weighted blow-up of the residual cone Z_p at p, an affine weighted cone onto which the fiber maps by the coordinates u_j vanishing at p (Construction 6.5). If only one coordinate is non-zero at p, Lemma 7.9 deduces the threshold on E_p from Proposition 7.3 at nearby smooth points, by homogeneity.

Otherwise the lowest degree M at p of a section of degree J can be up to $7J$ (Lemma 7.8). Since E_p is a Galois cover of a projective space branched along a hyperplane arrangement, Theorems 2.13 and 2.14, together with [BJ20, (3.1)], yield the threshold $\frac{\kappa_p}{M}$ (Proposition 7.11), where κ_p is a number computed from the weights d_j of the coordinates u_j vanishing at p

(Construction 6.5(5)). The threshold suffices unless exactly two of u_9, u_{10}, u_{11} are non-zero at p, where M can be $7J$ while $\kappa_p = 6$, so that $\frac{\kappa_p}{M}$ can be $\frac{6}{7J} < \frac{1}{J}$. There we restrict to the Cartier divisor $(u_i = u_i(p))$ for one such u_i, a fiber of a diagonal morphism $\mathbb{A}^{10} \to \mathbb{A}^4$ on which only one coordinate of exponent 7 is non-zero at p, so that the threshold suffices there (Proposition 7.13).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce the image and the ramification index of a vertical divisor over the base of a morphism, compute differents on quotient charts, compare log canonicity on a weighted cone and on the associated weighted projective variety, and recall the criterion of Fujita for hyperplane arrangements and the result of Zhuang on finite Galois morphisms.

In Section 3 we construct X and show that X is a $\mathbb{Q}$-factorial Fano variety with terminal singularities such that $\mathrm{Cl}(X) = \mathbb{Z}[H]$ and $-K_X \sim 26H$.

In Section 4 we show that X is K-semistable but not K-stable.

In Section 5 we show that $(X, \frac{26}{\ell}\mathcal{M})$ is canonical at the centers meeting $X \backslash P$ for any positive integer ℓ and any linear system $\mathcal{M} \subset |\ell H|$ without fixed components.

In Section 6 we study the fibers of a morphism $\mathbb{A}^{11} \to \mathbb{A}^4$ whose four components are linear combinations of $u_1^{e_1}, \dots, u_{11}^{e_{11}}$, together with the singularities of the fibers.

In Section 7 we bound from below the log canonical thresholds of the divisors of arbitrary sections on natural compactifications of the fibers of the morphism of Section 6 over the points of $\mathbb{A}^4$ other than the origin.

In Section 8 we prove two estimates on families of the fibers studied in Section 6.

In Section 9 we bound from below the log canonical thresholds of the divisors of arbitrary sections on the prime divisor P.

In Section 10 we show that $(X, \frac{26}{\ell}\mathcal{M})$ is canonical for any positive integer ℓ and any linear system $\mathcal{M} \subset |\ell H|$ without fixed components.

In Section 11 we prove Theorem 1.4 and Corollary 1.5.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and finite morphisms.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). If K_X is $\mathbb{Q}$-Cartier, then we write $a(E, X) := a(E, X, 0)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let X be a normal variety, E a prime divisor over X, and $h : Y \to X$ a birational morphism from a normal variety Y on which E is a divisor.

(1) For a $\mathbb{Q}$-Cartier Weil divisor D on X, we define $\operatorname{mult}_E D := \frac{1}{m}\operatorname{mult}_E h^*(mD)$, where m is a positive integer such that mD is Cartier. We extend mult_E $\mathbb{R}$-linearly to $\mathbb{R}$-linear combinations of $\mathbb{Q}$-Cartier Weil divisors on X.
(2) For a regular function φ on a neighborhood of the generic point of $\operatorname{center}_X E$, we define $\operatorname{mult}_E \varphi$ as the multiplicity of $\operatorname{div}(\varphi \circ h)$ along E. For an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$, we define $\operatorname{mult}_E \mathcal{I}$ as the minimum of $\operatorname{mult}_E \varphi$, where φ runs over the local sections of $\mathcal{I}$ near the generic point of $\operatorname{center}_X E$.
(3) For a linear system $\mathcal{M}$ of $\mathbb{Q}$-Cartier Weil divisors on X, we define

$$\operatorname{mult}_E \mathcal{M} := \min\{\operatorname{mult}_E D \mid D \in \mathcal{M}\}.$$

We say that $\mathcal{M}$ is *movable* if $\mathcal{M}$ has no fixed components.

Lemma 2.3. *Let $g : X' \to X$ be a finite surjective morphism between normal varieties and B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. Let B' be the $\mathbb{R}$-divisor on X' such that $K_{X'} + B' = g^*(K_X + B)$. Let $Y \to X$ be a birational morphism from a normal variety, Y' the normalization of the component of $Y \times_X X'$ dominating X', E' a prime divisor on Y', $E \subset Y$ the image of E', and r the ramification index of $Y' \to Y$ along E'. Then $a(E', X', B') = ra(E, X, B)$, and $\operatorname{mult}_{E'} g^*D = r \operatorname{mult}_E D$ for any $\mathbb{Q}$-Cartier Weil divisor D on X. In particular, (X, B) is sub-lc (resp. sub-klt) if and only if (X', B') is sub-lc (resp. sub-klt). If moreover $B \geq 0$ and $B' \geq 0$, then (X, B) is lc (resp. klt) if and only if (X', B') is lc (resp. klt).*

Proof. Let $f : Y \to X$, $f' : Y' \to X'$, and $g_Y : Y' \to Y$ be the induced morphisms, so that $f \circ g_Y = g \circ f'$. Let B_Y and $B_{Y'}$ be the $\mathbb{R}$-divisors such that $K_Y + B_Y = f^*(K_X + B)$ and $K_{Y'} + B_{Y'} = f'^*(K_{X'} + B')$. Then $a(E, X, B) = 1 - \operatorname{mult}_E B_Y$ and $a(E', X', B') = 1 - \operatorname{mult}_{E'} B_{Y'}$ [Kol13, Definition 2.4] (for $\mathbb{R}$-divisors, [Kol13, 2.20]). Since

$$K_{Y'} + B_{Y'} = f'^* g^*(K_X + B) = g_Y^* f^*(K_X + B) = g_Y^*(K_Y + B_Y),$$

the Hurwitz formula $K_{Y'} = g_Y^* K_Y + R(g_Y)$, where the coefficient of E' in the ramification divisor $R(g_Y)$ is $r - 1$, implies $B_{Y'} = g_Y^* B_Y - R(g_Y)$. Thus

$$\operatorname{mult}_{E'} B_{Y'} = r \operatorname{mult}_E B_Y - (r - 1),$$

that is, $a(E', X', B') = ra(E, X, B)$. Let D be a $\mathbb{Q}$-Cartier Weil divisor on X and m a positive integer such that mD is Cartier. Then

$$\operatorname{mult}_{E'} g^*D = \frac{1}{m}\operatorname{mult}_{E'} f'^* g^*(mD) = \frac{1}{m}\operatorname{mult}_{E'} g_Y^* f^*(mD) = \frac{r}{m}\operatorname{mult}_E f^*(mD) = r \operatorname{mult}_E D.$$

We show that any prime divisor over X' is of the form E' for some birational morphism $Y \to X$ from a normal variety. Let F' be a prime divisor on a normal variety Z' with a proper birational morphism $Z' \to X'$. By [Kol13, Lemma 2.22], applied to the composite $Z' \to X$ and the generic point of F', which does not dominate X, there is a proper birational morphism $Y \to X$ from a normal variety on which the image of the generic point of F' is a codimension one point (Y is the normalization of the last of the blow-ups in [Kol13, Lemma 2.22], and the normalization morphism is an isomorphism near the image of the generic point of F', which is a regular point by the lemma). Since $Y' \to Y$ is finite, the center of F' on Y' is a prime divisor. Conversely, any prime divisor E over X is a divisor on some Y, and E is the image of a prime divisor on Y', since $Y' \to Y$ is finite and surjective. Since $r \geq 1$, the equality $a(E', X', B') = ra(E, X, B)$ gives the equivalences for sub-lc and sub-klt sub-pairs (cf. [KM98, Proposition 5.20]). If $B \geq 0$ and $B' \geq 0$, then (X, B) and (X', B') are pairs, and the equivalences for lc and klt pairs follow. □

Log canonicity is passed between a variety and a divisor on it by inversion of adjunction (Theorem 2.5): along the prime divisor at infinity in Section 10, along the fibers of the family over prime divisors of the base in Section 8, and along the divisors at infinity and the exceptional divisors of weighted blow-ups in Section 7. Each time the different has to be computed first, and Lemma 2.4 computes it on quotient charts: it is zero on the divisors at infinity, and in general it is non-zero on the exceptional divisors of weighted blow-ups.

Lemma 2.4. *Let V be a normal variety, $S \subset V$ a normal prime divisor, and $v \in S$ a closed point. Assume that there exist an open neighborhood V° of v, a smooth variety $\widetilde{V}$ with a faithful action of a finite group G, and an isomorphism $\widetilde{V}/G \cong V^\circ$ such that the quotient morphism $q : \widetilde{V} \to V^\circ$ is étale in codimension one and $\widetilde{S} := q^{-1}(S \cap V^\circ)$ is smooth. For a prime divisor T on $S \cap V^\circ$, let s_T be the ramification index of $q|_{\widetilde{S}} : \widetilde{S} \to S \cap V^\circ$ along a prime divisor of $\widetilde{S}$ over T. Then $K_V + S$ is $\mathbb{Q}$-Cartier near v, and near v,*

$$(K_V + S)|_S = K_S + \sum_T \Big(1 - \frac{1}{s_T}\Big)T,$$

where $(K_V + S)|_S$ denotes K_S plus the different of (V, S) on S [FST11, 11.1]. In particular, if $q|_{\widetilde{S}}$ is étale in codimension one, then $(K_V + S)|_S = K_S$ near v.

Proof. Both statements of the lemma are local near v, so possibly replacing V with V°, we may assume that $V = V^\circ$. Since $\widetilde{V}$ is smooth, $\widetilde{V}$ is $\mathbb{Q}$-factorial, and so is V [KM98, Lemma 5.16]. In particular, $K_V + S$ is $\mathbb{Q}$-Cartier.

Write $q_S := q|_{\widetilde{S}} : \widetilde{S} \to S$. Since taking G-invariants is exact in characteristic zero, $\widetilde{S}/G = S$. Thus G acts transitively on the prime divisors of $\widetilde{S}$ over a prime divisor T of S, and s_T does not depend on the choice of such a prime divisor. Let $\widetilde{T}$ be the reduced preimage of T in $\widetilde{S}$. Then $q_S^* T = s_T \widetilde{T}$, and the ramification divisor of q_S is

$$R(q_S) = \sum_T (s_T - 1)\widetilde{T} = \sum_T \Big(1 - \frac{1}{s_T}\Big) q_S^* T.$$

The different of (V, S) on S in [FST11, 11.1] coincides with the different $\operatorname{Diff}_S(0)$ of [Kol13, Definition 4.2] by [Kol13, 4.7, (4.7.1)]. Let m be a positive integer such that $m(K_V + S)$ is Cartier. Possibly shrinking V around v, let ω be a generator of $\omega_V^{[m]}(mS)$. By [Kol13, (4.2.7) and (4.2.8)], the residue $\operatorname{Res}_S \omega$ is a rational section of $\omega_S^{[m]}$ with

$$\operatorname{div}(\operatorname{Res}_S \omega) = -m \operatorname{Diff}_S(0).$$

Since q is étale in codimension one and $\widetilde{V}$ is smooth, $q^*\omega$ generates $\omega_{\widetilde{V}}^{\otimes m}(m\widetilde{S})$, so by adjunction on $\widetilde{V}$, $\operatorname{Res}_{\widetilde{S}}(q^*\omega)$ generates $\omega_{\widetilde{S}}^{\otimes m}$. Since q is étale at the generic points of $\widetilde{S}$,

$$\operatorname{Res}_{\widetilde{S}}(q^*\omega) = q_S^*(\operatorname{Res}_S \omega).$$

Thus the Hurwitz formula for m-fold forms implies

$$0 = \operatorname{div}\big(q_S^*(\operatorname{Res}_S \omega)\big) = q_S^* \operatorname{div}(\operatorname{Res}_S \omega) + mR(q_S) = -mq_S^*\Big(\operatorname{Diff}_S(0) - \sum_T \Big(1 - \frac{1}{s_T}\Big)T\Big).$$

Since pullback under the finite surjective morphism q_S is injective on $\mathbb{R}$-divisors, $\operatorname{Diff}_S(0) = \sum_T (1 - \frac{1}{s_T})T$, which is the formula in Lemma 2.4. If q_S is étale in codimension one, then $s_T = 1$ for any T, and the last statement follows. □

Theorem 2.5 ([FST11, Theorem 11.3])**.** *Let V be a normal variety, S a reduced divisor on V, and $B \geq 0$ an $\mathbb{R}$-divisor on V with no common components with S. Assume that $K_V + S + B$ is $\mathbb{R}$-Cartier. Let $\nu : S^\nu \to S$ be the normalization and B_{S^ν} the different on S^ν, so that $K_{S^\nu} + B_{S^\nu} = \nu^*(K_V + S + B)$. Then $(V, S + B)$ is lc near S if and only if (S^ν, B_{S^ν}) is lc.*

This is the inversion of adjunction on log canonicity.

2.2. **Divisors over the base of a morphism.**

Definition-Lemma 2.6. Let $\pi : V \to Y$ be a dominant morphism between normal varieties, E a prime divisor over V, and $h : V' \to V$ a birational morphism from a normal variety V' on which E is a divisor. We say that E is *horizontal over Y* if $\pi \circ h|_E : E \to Y$ is dominant. Otherwise, we say that E is *vertical over Y*. If E is vertical over Y, then there exist a birational

morphism $h'' : V'' \to V$ from a normal variety V'' on which E is a divisor, a birational morphism $g : Y' \to Y$ from a smooth variety Y', a morphism $\pi'' : V'' \to Y'$ with $g \circ \pi'' = \pi \circ h''$, and a prime divisor F on Y', such that $\pi''(E)$ is dense in F. The prime divisor F over Y and the positive integer $r := \operatorname{mult}_E \pi''^* F$ do not depend on the choice of h'', g, and π''. We call F the *image* of E over Y, and we call r the *ramification index* of E over Y.

Proof. We first construct h'', g, π'', and F. Let η_E be the generic point of E. Only a neighborhood of η_E matters: let Y_0 be an affine open neighborhood of $\pi(h(\eta_E))$ in Y, and V_0' an affine open neighborhood of η_E in $h^{-1}(\pi^{-1}(Y_0))$. We construct h'', g, π'', and F below for $\pi|_{\pi^{-1}(Y_0)}$ and $h|_{V_0'}$, and then compose g and h'' with the open immersions $Y_0 \subset Y$ and $\pi^{-1}(Y_0) \subset V$. Then g and h'' are still birational morphisms. Thus we may assume that Y and V' are affine. Let $\overline{Y}$ be a normal projective variety containing Y as an open subset, $\overline{V}_0$ a projective variety containing V' as an open subset, and $\overline{V}'$ the closure of the graph of $\pi \circ h : V' \to \overline{Y}$ in $\overline{V}_0 \times \overline{Y}$. Then V' is an open subset of $\overline{V}'$, and the projection $\overline{V}' \to \overline{Y}$ is a projective surjective morphism extending $\pi \circ h$. By [AK00, Theorem 2.1] (see [AK00, Remark 2.2]) and [AK00, Proposition 4.4], there exist projective birational morphisms $\overline{V}_1 \to \overline{V}'$ and $g_1 : \overline{Y}_1 \to \overline{Y}$ and an equidimensional morphism $\pi_1 : \overline{V}_1 \to \overline{Y}_1$ such that $\overline{V}_1$ contains an open subset U_1 for which $U_1 \subset \overline{V}_1$ is a toroidal embedding, in particular $\overline{V}_1$ is normal [AK00, Definition 1.2], $\overline{Y}_1$ is smooth, and the composition $\overline{V}_1 \to \overline{V}' \to \overline{Y}$ is $g_1 \circ \pi_1$. Let V'' and Y' be the preimages of V' and Y in $\overline{V}_1$ and $\overline{Y}_1$ respectively, $h'' : V'' \to V$ the composition of $V'' \to V'$ with h, and $g := g_1|_{Y'}$. Since $g_1(\pi_1(V'')) \subset Y$, we have $\pi_1(V'') \subset Y'$, and we let $\pi'' := \pi_1|_{V''} : V'' \to Y'$. Then $g \circ \pi'' = \pi \circ h''$. Since V' is normal, $V'' \to V'$ is an isomorphism over a neighborhood of the generic point of E, so E is a divisor on V''. Let $\overline{E}$ be the closure of E in $\overline{V}_1$. Since π_1 is equidimensional,

$$\dim \pi_1(\overline{E}) \geq \dim E - (\dim V - \dim Y) = \dim Y - 1.$$

Since E is vertical over Y and g_1 is surjective, $\pi_1(\overline{E}) \neq \overline{Y}_1$. Thus $\pi_1(\overline{E})$ is a prime divisor on $\overline{Y}_1$, and we let $F := \pi_1(\overline{E}) \cap Y'$. Then $\pi''(E)$ is dense in F. Since Y' is smooth, F is Cartier, and $r = \operatorname{mult}_E \pi''^* F$ is a positive integer.

Let $h_j'' : V_j'' \to V$, $g_j : Y_j' \to Y$, $\pi_j'' : V_j'' \to Y_j'$, and F_j, $j = 1, 2$, be two such choices. Since E is a divisor on V_1'' and on V_2'', there exists an open subset $V_0 \subset V_1''$ containing the generic point of E such that the birational map $V_1'' \dashrightarrow V_2''$ over V restricts to an open immersion $\iota : V_0 \to V_2''$. Let Γ be the normalization of the closure of the image of V_0 under $(\pi_1''|_{V_0}, \pi_2'' \circ \iota)$ in $Y_1' \times_Y Y_2'$, $\pi_\Gamma : V_0 \to \Gamma$ the induced morphism, $f_j : \Gamma \to Y_j'$ the projections, and F_Γ the closure of $\pi_\Gamma(E \cap V_0)$. Since g_1 and g_2 are birational, f_1 and f_2 are birational. Since $f_j(F_\Gamma)$ is dense in F_j, F_Γ is a prime divisor on Γ. Let η be the generic point of F_Γ. Then $f_j(\eta)$ is the generic point of F_j, and since $\dim F_\Gamma = \dim F_j$, the residue field of η is a finite extension of that of $f_j(\eta)$, so η is a closed point of the fiber of f_j over $f_j(\eta)$. Any other point of Γ specializing to η is the generic point of Γ, since the only irreducible closed subset of Γ strictly containing F_Γ is Γ, and it maps to the generic point of Y_j', so η is an isolated point of $f_j^{-1}(f_j(\eta))$, that is, f_j is quasi-finite at η. By Zariski's Main Theorem, f_j is an open immersion near the generic point of F_Γ. Thus F_1 and F_2 coincide with F_Γ as prime divisors over Y, and

$$\operatorname{mult}_E \pi_1''^* F_1 = \operatorname{mult}_E \pi_\Gamma^* F_\Gamma = \operatorname{mult}_E \pi_2''^* F_2.$$

□

In Sections 8 and 9, for a vertical prime divisor E over the total space of a family, the estimates compare the log discrepancy and the multiplicities along E with r times the log discrepancy and the multiplicities along the image F.

2.3. **Intersections and multiplicities.**

Definition 2.7. Let X be a normal variety, $T = \sum_i n_i T_i$ a $\mathbb{Q}$-cycle on X, where the T_i are distinct irreducible closed subvarieties of X, and $C \subset X$ an irreducible closed subvariety. We

define $\operatorname{mult}_C T := \sum_i n_i e_C T_i$, where the sum runs over the indices i with $C \subset T_i$ and $e_C T_i$ is the multiplicity of T_i along C [Ful98, Section 4.3]. If $C = \{p\}$ for a closed point p, then we write $\operatorname{mult}_p T := \operatorname{mult}_C T$. For a $\mathbb{Q}$-Cartier Weil divisor D on X such that no T_i is contained in $\operatorname{Supp} D$, we define $D \cdot T := \frac{1}{m} \sum_i n_i (mD) \cdot T_i$, where m is a positive integer such that mD is Cartier and $(mD) \cdot T_i$ is the intersection cycle of mD with T_i [Ful98, Definition 2.3].

Lemma 2.8. *Let X be a normal projective variety, H an ample $\mathbb{Q}$-Cartier Weil divisor on X, $p \in X$ a smooth closed point, and $T \geq 0$ a $\mathbb{Q}$-cycle of pure dimension $d \geq 1$ on X. Let $D \geq 0$ be a $\mathbb{Q}$-Cartier Weil divisor on X with $D \sim_{\mathbb{Q}} \lambda H$ for a rational number $\lambda > 0$, and assume that no component of T is contained in $\operatorname{Supp} D$. Then the following hold.*

(1) $D \cdot T \geq 0$ is a $\mathbb{Q}$-cycle of pure dimension $d-1$ with $(H^{d-1} \cdot D \cdot T) = \lambda (H^d \cdot T)$.
(2) If a local equation of D at p belongs to $\mathfrak{m}_p^s$ for an integer $s \geq 1$, then $\operatorname{mult}_p (D \cdot T) \geq s \operatorname{mult}_p T$.

Proof. Write $T = \sum_i n_i T_i$ as in Definition 2.7 with $n_i > 0$, and let m be a positive integer such that mD is Cartier. Since $mD \geq 0$ and $T_i \not\subset \operatorname{Supp} D$, $(mD) \cdot T_i$ is the Weil divisor of the Cartier divisor $mD|_{T_i} \geq 0$ on T_i. Thus $D \cdot T \geq 0$ is a $\mathbb{Q}$-cycle of pure dimension $d-1$. Since $D|_{T_i} \sim_{\mathbb{Q}} \lambda H|_{T_i}$, the definition of intersection numbers and the projection formula imply

$$(H^{d-1} \cdot D \cdot T) = \sum_i n_i \big((H|_{T_i})^{d-1} \cdot D|_{T_i} \big)_{T_i} = \lambda \sum_i n_i \big((H|_{T_i})^d \big)_{T_i} = \lambda (H^d \cdot T).$$

This proves (1).

We prove (2). Since mult_p is $\mathbb{Q}$-linear and $n_i > 0$, it suffices to show that

$$\operatorname{mult}_p (D \cdot T_i) \geq s \operatorname{mult}_p T_i$$

for any i, and we may assume that $p \in T_i$. Since p is a smooth point of X, D is Cartier near p. Near p, $D \cdot T_i$ is the Weil divisor of the Cartier divisor $D|_{T_i}$, and a local equation of $D|_{T_i}$ at p belongs to $\mathfrak{m}_{T_i,p}^s$, where $\mathfrak{m}_{T_i,p}$ is the maximal ideal of $\mathcal{O}_{T_i,p}$. We use [Ful98, Example 12.4.8]. Let $D_1, \dots, D_k \geq 0$ be Cartier divisors on a variety Z' with $\dim Z' = k$ such that p is the only point of $D_1 \cap \cdots \cap D_k$ (which can be achieved by shrinking Z', since $i(p, D_1 \cdots D_k; Z')$ is local), let σ be the blow-up of Z' at p with exceptional divisor G, and let $s_1, \dots, s_k$ be positive integers such that $R_j := \sigma^* D_j - s_j G \geq 0$ for any j. Then

$$i(p, D_1 \cdots D_k; Z') \geq s_1 \cdots s_k e_p Z',$$

with equality if and only if $R_1 \cap \cdots \cap R_k = \emptyset$.

If $d = 1$, then

$$\operatorname{mult}_p (D \cdot T_i) = i(p, D|_{T_i}; T_i) \geq s e_p T_i = s \operatorname{mult}_p T_i.$$

Assume that $d \geq 2$. Fix an affine open neighborhood $X_0 \subset \mathbb{A}^N$ of p in X, and let $L_1, \dots, L_{d-1}$ be the zero loci on X_0 of $d-1$ general linear functions on $\mathbb{A}^N$ vanishing at p. Let Z be any component of $D \cdot T_i$ through p. Then $\dim Z = d-1$, p is an isolated point of $Z \cap L_1 \cap \cdots \cap L_{d-1}$, $\sigma_Z^*(L_j|_Z) - G_Z$ is the strict transform of $L_j|_Z$, where σ_Z is the blow-up of Z at p with exceptional divisor G_Z, and the strict transforms of $L_1|_Z, \dots, L_{d-1}|_Z$ on the blow-up of Z at p have no common point near the exceptional divisor, since they cut the exceptional divisor, of dimension $d-2$, by $d-1$ general hyperplanes. Thus, after shrinking Z to a neighborhood of p in which p is the only point of $Z \cap L_1 \cap \cdots \cap L_{d-1}$, the equality case implies $e_p Z = i(p, L_1 \cdots L_{d-1}; Z)$. Computing the intersection product first with D and then with $L_1, \dots, L_{d-1}$, we obtain

$$\operatorname{mult}_p (D \cdot T_i) = i(p, D \cdot L_1 \cdots L_{d-1}; T_i).$$

Since p is an isolated point of $\operatorname{Supp} D \cap L_1 \cap \cdots \cap L_{d-1} \cap T_i$, a local equation of $D|_{T_i}$ at p belongs to $\mathfrak{m}_{T_i,p}^s$, and a local equation of each $L_j|_{T_i}$ at p belongs to $\mathfrak{m}_{T_i,p}$, by [Ful98, Example 12.4.8] for $Z' = T_i$, $D_1 = D|_{T_i}$, $D_{j+1} = L_j|_{T_i}$, $s_1 = s$, and $s_2 = \cdots = s_d = 1$, we have

$$\operatorname{mult}_p (D \cdot T_i) \geq s e_p T_i = s \operatorname{mult}_p T_i.$$

□

2.4. **Weighted complete intersections.**

Definition 2.9. Let $a_0, \dots, a_N$ be positive integers. The *weighted projective space* $\mathbb{P}(a_0, \dots, a_N)$ is $\operatorname{Proj} \mathbb{C}[z_0, \dots, z_N]$, where $\deg z_i = a_i$ [IF00, Definition 5.1]. The weighted projective space $\mathbb{P}(a_0, \dots, a_N)$ is *well formed* if $\gcd\{a_j \mid j \neq l\} = 1$ for any $0 \leq l \leq N$. A closed subvariety $X \subset \mathbb{P}(a_0, \dots, a_N)$ is a *weighted complete intersection* if the weighted homogeneous ideal of X is generated by a regular sequence of weighted homogeneous polynomials. A weighted complete intersection X is *quasismooth* if the affine cone of X in $\mathbb{A}^{N+1}$ is smooth outside the origin, and X is *well formed* if $\mathbb{P}(a_0, \dots, a_N)$ is well formed and $X \cap \operatorname{Sing} \mathbb{P}(a_0, \dots, a_N)$ is of codimension at least two in X.

Definition 2.9 is consistent with [PST17, Definition 2.2] and [IF00, Definitions 5.11, 6.3 and 6.9].

Lemma 2.10. *Let $R = \bigoplus_{k \geq 0} R_k$ be a graded normal domain with $R_0 = \mathbb{C}$, generated over $\mathbb{C}$ by homogeneous elements $z_0, \dots, z_N$ of positive degrees $a_0, \dots, a_N$. Let $X := \operatorname{Proj} R$, $\widehat{X} := \operatorname{Spec} R \backslash \{0\}$, and $q : \widehat{X} \to X$ the natural morphism. Assume that $\widehat{X}$ is smooth, and that for any $0 \leq l \leq N$, the quotient morphism $\widehat{X} \cap (z_l = 1) \to X \cap (z_l \neq 0)$ by μ_{a_l} is étale in codimension one. Let Y be a smooth variety, $q_Y : \widehat{X} \times Y \to X \times Y$ the product of q and the identity morphism of Y, $D \geq 0$ a $\mathbb{Q}$-divisor on $X \times Y$, and $\Sigma \subset X \times Y$ a closed subset. Then $(X \times Y, D)$ is lc near Σ if and only if $(\widehat{X} \times Y, q_Y^* D)$ is lc near $q_Y^{-1}(\Sigma)$.*

Proof. The grading of R induces the action $t \cdot (z_0, \dots, z_N) = (t^{a_0} z_0, \dots, t^{a_N} z_N)$ of $\mathbb{G}_m$ on $\widehat{X}$. For any $0 \leq l \leq N$, let $\widehat{X}_l := \widehat{X} \cap (z_l = 1)$ be the slice on which $\mu_{a_l} \subset \mathbb{G}_m$ acts by restriction, and let $X_l := X \cap (z_l \neq 0)$. Let $\psi_l : \widehat{X}_l \to X_l$ be the quotient morphism by μ_{a_l}. The morphism

$$\varphi_l : \widehat{X}_l \times \mathbb{G}_m \to q^{-1}(X_l) = \widehat{X} \cap (z_l \neq 0), \quad (p, t) \to t \cdot p$$

is the quotient by the action $\zeta \cdot (p, t) = (\zeta \cdot p, \zeta^{-1} t)$ of μ_{a_l}. For a homogeneous element f of $R[z_l^{-1}]$ of degree k, we have $\varphi_l^* f = (f|_{\widehat{X}_l}) t^k$, and the elements $(f|_{\widehat{X}_l}) t^k$ span the ring of μ_{a_l}-invariants of $\mathcal{O}(\widehat{X}_l)[t, t^{-1}]$. Since every point of $\widehat{X} \cap (z_l \neq 0)$ is of the form $t \cdot p$ with $p \in \widehat{X}_l$, φ_l^* is injective. Since μ_{a_l} acts freely on the factor $\mathbb{G}_m$, φ_l is finite étale. In particular, since $\widehat{X}$ is smooth, $\widehat{X}_l$ is smooth. The group μ_{a_l} permutes the connected components of $\widehat{X}_l$ transitively, since X_l is irreducible. Replacing $\widehat{X}_l$ by one connected component C and μ_{a_l} by the stabilizer $\operatorname{Stab}(C) \subset \mu_{a_l}$ of C, we may assume that $\widehat{X}_l$ is a smooth variety: $C / \operatorname{Stab}(C) = X_l$, $C \to X_l$ is étale in codimension one, and $C \times \mathbb{G}_m \to q^{-1}(X_l)$ is finite étale and surjective.

Let $\Phi_l := \varphi_l \times \mathrm{id}_Y$ and $\Psi_l := \psi_l \times \mathrm{id}_Y$, and let

$$\mathrm{pr} : \widehat{X}_l \times \mathbb{G}_m \times Y \to \widehat{X}_l \times Y$$

be the projection. Since $q(t \cdot p) = q(p)$, we have

$$q_Y \circ \Phi_l = \Psi_l \circ \mathrm{pr}\,.$$

The morphism Ψ_l is the quotient of $\widehat{X}_l \times Y$ by μ_{a_l} acting trivially on Y, and the locus where Ψ_l is not étale is the product of Y with the locus where ψ_l is not étale. Thus Ψ_l is étale in codimension one. Since $X_l \times Y$ is the quotient of the smooth variety $\widehat{X}_l \times Y$ by a finite group, $X_l \times Y$ is normal and, by [KM98, Lemma 5.16], $\mathbb{Q}$-factorial. Since the open subsets $X_l \times Y$ cover $X \times Y$, $X \times Y$ is $\mathbb{Q}$-factorial. Hence D is $\mathbb{Q}$-Cartier, $(X \times Y, D)$ is a pair, and $(\widehat{X} \times Y, q_Y^* D)$ is a pair.

Let $D_l := \Psi_l^* D \geq 0$ on $\widehat{X}_l \times Y$. By the Hurwitz formula, $K_{\widehat{X}_l \times Y} = \Psi_l^* K_{X_l \times Y}$ since Ψ_l is étale in codimension one, and

$$K_{\widehat{X}_l \times \mathbb{G}_m \times Y} = \Phi_l^* K_{q^{-1}(X_l) \times Y}$$

since Φ_l is étale. Since

$$\Phi_l^* q_Y^* D = \mathrm{pr}^*\, \Psi_l^* D = \mathrm{pr}^*\, D_l,$$

we obtain

$$K_{\widehat{X}_l\times Y}+D_l=\Psi_l^*(K_{X_l\times Y}+D)\quad\text{and}\quad K_{\widehat{X}_l\times\mathbb{G}_m\times Y}+\operatorname{pr}^*D_l=\Phi_l^*(K_{q^{-1}(X_l)\times Y}+q_Y^*D).$$

We use the following consequence of Lemma 2.3. Let $g:X'\to X''$ be a finite surjective morphism between normal varieties, and $B''\geq 0$ and $B'\geq 0$ $\mathbb{Q}$-divisors on X'' and X' respectively such that $K_{X''}+B''$ is $\mathbb{Q}$-Cartier and $K_{X'}+B'=g^*(K_{X''}+B'')$. Let $\Gamma'\subset X'$ be the set of points near which (X',B') is not lc. Then Γ' is closed, and $g(\Gamma')$ is closed since g is finite. Lemma 2.3, applied to $g^{-1}(V)\to V$ for open subsets $V\subset X''$, shows that (X'',B'') is lc near a point $x\in X''$ if and only if $x\notin g(\Gamma')$. Thus for any closed subset $\Gamma\subset X''$, (X'',B'') is lc near Γ if and only if (X',B') is lc near $g^{-1}(\Gamma)$.

Moreover, for any closed subset $\Gamma\subset\widehat{X}_l\times Y$, $(\widehat{X}_l\times Y,D_l)$ is lc near Γ if and only if the pair

$$(\widehat{X}_l\times\mathbb{G}_m\times Y,\operatorname{pr}^*D_l)$$

is lc near $\operatorname{pr}^{-1}(\Gamma)$. Let $V\subset\widehat{X}_l\times Y$ be an open subset, $h:W\to V$ a log resolution of (V,D_l), and D_W the $\mathbb{Q}$-divisor on W with $K_W+D_W=h^*(K_V+D_l)$. Then $h\times\mathrm{id}_{\mathbb{G}_m}$ is a log resolution of $(V\times\mathbb{G}_m,\operatorname{pr}^*D_l)$, and the pullback D'_W of D_W to $W\times\mathbb{G}_m$ satisfies

$$K_{W\times\mathbb{G}_m}+D'_W=(h\times\mathrm{id}_{\mathbb{G}_m})^*(K_{V\times\mathbb{G}_m}+\operatorname{pr}^*D_l).$$

Each component of D'_W is $Q\times\mathbb{G}_m$ for a component Q of D_W, with coefficient $\operatorname{mult}_Q D_W$, and lc is read off from one log resolution [KM98, Lemma 2.30 and Corollary 2.31(3)]. Hence (V,D_l) is lc if and only if $(V\times\mathbb{G}_m,\operatorname{pr}^*D_l)$ is lc. Since the translations of the factor $\mathbb{G}_m$ preserve pr^*D_l, the set of points near which the pair

$$(\widehat{X}_l\times\mathbb{G}_m\times Y,\operatorname{pr}^*D_l)$$

is lc is $\operatorname{pr}^{-1}(V_0)$, where V_0 is the set of points near which $(\widehat{X}_l\times Y,D_l)$ is lc.

Now $(X\times Y,D)$ is lc near Σ if and only if $(X_l\times Y,D)$ is lc near $\Sigma\cap(X_l\times Y)$ for any $0\leq l\leq N$. Fix l. We apply the consequence of Lemma 2.3 above to Ψ_l and Φ_l, and the equivalence above for pr. Since

$$\Phi_l^{-1}\big(q_Y^{-1}(\Sigma)\big)=\operatorname{pr}^{-1}\left(\Psi_l^{-1}(\Sigma\cap(X_l\times Y))\right),$$

we obtain that $(X_l\times Y,D)$ is lc near $\Sigma\cap(X_l\times Y)$ if and only if $(q^{-1}(X_l)\times Y,q_Y^*D)$ is lc near $q_Y^{-1}(\Sigma)\cap(q^{-1}(X_l)\times Y)$. Since the open subsets $q^{-1}(X_l)\times Y$ cover $\widehat{X}\times Y$, Lemma 2.10 follows. □

Lemma 2.10 is applied to the prime divisor P of Set-up 3.3 and to V_∞ of Set-up 6.1, whose punctured affine cones are smooth, so lc thresholds can be computed on the cones and then descended (Proposition 7.4 and Proposition 9.5). The assumption that the quotient morphisms are étale in codimension one can fail on the exceptional divisors E_p of the weighted blow-ups ψ_p in Construction 6.5, which are used in Section 7. On E_p, Lemma 2.10 is not used, and the descent goes through Lemma 2.4 instead.

2.5. **Birational superrigidity and K-stability.**

Definition 2.11. A *Fano variety* is a normal projective variety X such that $-K_X$ is an ample $\mathbb{Q}$-Cartier divisor. A *Mori fiber space* is a contraction $f:Y\to T$ between normal projective varieties such that Y is $\mathbb{Q}$-factorial with terminal singularities, $\dim T<\dim Y$, $\rho(Y/T)=1$, and $-K_Y$ is ample over T. Let X be a $\mathbb{Q}$-factorial Fano variety of Picard number one with terminal singularities. We say that X is *birationally rigid* if for any Mori fiber space $f:Y\to T$ such that Y is birational to X, T is a point and $Y\cong X$. We say that X is *birationally superrigid* if X is birationally rigid and $\operatorname{Bir}(X)=\operatorname{Aut}(X)$.

Definition 2.11 is consistent with [KOW23, Definition 2.2]. In Section 11, we prove birational superrigidity in the sense of [SZ19, Section 2.3] by the Noether–Fano criterion [SZ19, Theorem 2.3], and deduce birational superrigidity in the sense of Definition 2.11.

Definition 2.12. Let X be a Fano variety with klt singularities. Let F be a prime divisor over X, which is a divisor on a higher model $h : Y \to X$. We define (cf. [Fuj19a, Definitions 1.1(2) and 1.3(4)])

$$S_X(F) := \frac{1}{\mathrm{vol}(-K_X)} \int_0^{+\infty} \mathrm{vol}(-h^*K_X - tF)dt \quad \text{and} \quad \beta_X(F) := a(F, X) - S_X(F).$$

We say that F is *dreamy* if for some positive integer r such that rK_X is Cartier, the bigraded algebra $\bigoplus_{k,j\geq 0} H^0(X, -krK_X - jF)$ is finitely generated, where $H^0(X, -krK_X - jF)$ is the space of sections $s \in H^0(X, \mathcal{O}_X(-krK_X))$ such that the divisor of zeros $D_s \geq 0$ of s satisfies $\mathrm{mult}_F D_s \geq j$ [Fuj19a, Definition 1.3(1)]. We say that X is *K-semistable* if $\beta_X(F) \geq 0$ for any prime divisor F over X. We say that X is *K-stable* if $\beta_X(F) > 0$ for any dreamy prime divisor F over X. A *log Fano pair* is a projective klt pair (X, B) such that B is a $\mathbb{Q}$-divisor and $-(K_X + B)$ is ample [Zhu21, Section 2.1]. For a log Fano pair (X, B) and a prime divisor F over X, we define $S_{(X,B)}(F)$ and $\beta_{(X,B)}(F)$ by the formulas defining $S_X(F)$ and $\beta_X(F)$, replacing K_X by $K_X + B$ and $a(F, X)$ by $a(F, X, B)$. We say that a log Fano pair (X, B) is *K-semistable* if $\beta_{(X,B)}(F) \geq 0$ for any prime divisor F over X.

In [Fuj19a], $A_X(F)$ denotes $a(F, X)$, and $\beta_X(F)$ in Definition 2.12 is $\beta(F)$ of [Fuj19a, Definition 1.3(4)] multiplied by $\mathrm{vol}(-K_X)^{-1}$. By [Fuj19a, Corollary 1.5(2) and Theorem 1.6] (see also [Li17, Theorem 3.7] for K-semistability), X is K-semistable (resp. K-stable) in the sense of Definition 2.12 if and only if X is K-semistable (resp. K-stable) in the sense of test configurations [Fuj19a, Definition 2.7]. By [Zhu21, Theorem 2.5], a log Fano pair (X, B) is K-semistable in the sense of Definition 2.12 if and only if it is K-semistable in the sense of test configurations, since for any prime divisor F over X, $S(F)$ of [Zhu21, Definition 2.3] equals $S_{(X,B)}(F)$ by [BJ20, Lemma 3.1, Lemma 3.5, Corollary 3.6, Lemma 3.7, and Corollary 3.9(ii)]. Thus the K-semistability in Theorem 2.13 and Theorem 2.14, which is defined by test configurations in [Fuj17, Definition 2.3(1)] and [Zhu21, Definition 2.1], coincides with the K-semistability of Definition 2.12.

Theorem 2.13 and Theorem 2.14 are used twice. In Section 4, they imply that X is K-semistable. In Section 7, they imply the K-semistability of a log Fano pair on the exceptional divisor E_p of the weighted blow-up ψ_p in Construction 6.5, a finite Galois cover of a projective space branched along hyperplanes, from which we bound the lc thresholds of arbitrary sections on E_p (Proposition 7.11).

Theorem 2.13 ([Fuj17, Corollary 1.6(1)])**.** *Let $m \geq 1$ be an integer, let $H_1, \dots, H_m \subset \mathbb{P}^n$ be distinct hyperplanes such that $\sum_{i=1}^m H_i$ is a simple normal crossing divisor, and let $b_1, \dots, b_m \in (0, 1) \cap \mathbb{Q}$ be rational numbers such that $(\mathbb{P}^n, \sum_{i=1}^m b_i H_i)$ is a log Fano pair. Then $(\mathbb{P}^n, \sum_{i=1}^m b_i H_i)$ is K-semistable if and only if*

$$k \sum_{i=1}^{m} b_i \geq (n+1) \sum_{l=1}^{k} b_{i_l}$$

for any $1 \leq k \leq n$ and any $1 \leq i_1 < \cdots < i_k \leq m$.

Theorem 2.14 ([Zhu21, Corollary 4.13(1)])**.** *Let (X, D) and (Y, B) be log Fano pairs and $g : X \to Y$ a finite surjective Galois morphism such that $K_X + D = g^*(K_Y + B)$. Then (X, D) is K-semistable if and only if (Y, B) is K-semistable.*

3. Construction of the example

In this section, we construct the weighted complete intersection X of Set-up 3.3 and prove the properties of X used in the rest of the paper.

3.1. **Diagonal complete intersections.** In this subsection, we prove that the coordinate rings of diagonal complete intersections whose coefficient vectors are in general position are normal domains. The homogeneous coordinate rings of X and of P, the ring R_∞ of the cone $h^{-1}(0)$ of the diagonal morphism h of Set-up 6.1, and the rings A_p of the residual cones at the singular points of the fibers of h (Construction 6.5) are rings of the form A in Lemma 3.2 below. Thus we state Lemma 3.2 for arbitrary r, N, and exponents.

Lemma 3.1. *Let $A = \bigoplus_{k\geq 0} A_k$ be a Noetherian graded ring with $A_0 = \mathbb{C}$. If A is normal, then A is a domain.*

Proof. Let $\epsilon \in A$ be an idempotent and $\epsilon_0 \in A_0 = \mathbb{C}$ the degree zero component of ϵ. Then $\epsilon_0^2 = \epsilon_0$, and replacing ϵ with $1-\epsilon$, we may assume that $\epsilon_0 = 0$. Then $\epsilon = \epsilon^n \in \bigoplus_{k\geq n} A_k$ for any $n \geq 1$, so $\epsilon = 0$. Since a Noetherian normal ring is a finite product of normal domains [Mat89, Section 9, Remark after the definition of normal ring] and A has no idempotents other than 0 and 1, A is a domain. □

Lemma 3.2. *Let $r \geq 1$ and $N \geq r+2$ be integers, $\mathbf{w}_1, \dots, \mathbf{w}_N \in \mathbb{C}^r$ vectors any r of which are linearly independent, and $e'_1, \dots, e'_N \geq 2$ integers. Let*

$$A := \mathbb{C}[z_1, \dots, z_N] \Big/ \Big(\sum_{j=1}^{N} z_j^{e'_j} \mathbf{w}_j \Big),$$

where the vector equation stands for r polynomial equations. Then the following hold.

1. *For any r distinct indices $j_1, \dots, j_r$, the ring A is a free module over $\mathbb{C}[z_j \mid j \notin \{j_1, \dots, j_r\}]$ with basis the monomials $z_{j_1}^{b_1} \cdots z_{j_r}^{b_r}$, $0 \leq b_l < e'_{j_l}$. In particular, A is a complete intersection ring of dimension $N - r$.*
2. *Any closed point of $\operatorname{Spec} A$ other than the origin has at least $r+1$ non-zero coordinates, and $\operatorname{Spec} A$ is smooth outside the origin.*
3. *A is a normal domain.*
4. *For any set I of at most $r-1$ indices, any $r - \#I$ of the images of $\mathbf{w}_j$, $j \notin I$, in $\mathbb{C}^r / \langle \mathbf{w}_l \mid l \in I \rangle$ are linearly independent.*

Proof. (1) Let $S := \mathbb{C}[z_j \mid j \notin \{j_1, \dots, j_r\}]$ and let M be the $r \times r$ matrix with columns $\mathbf{w}_{j_1}, \dots, \mathbf{w}_{j_r}$. Since M is invertible, the vector equation $\sum_j z_j^{e'_j} \mathbf{w}_j = 0$ is equivalent to the r equations $f_1 = \cdots = f_r = 0$, where

$$f_l := z_{j_l}^{e'_{j_l}} + g_l \quad \text{and} \quad (g_1, \dots, g_r)^\top := M^{-1} \sum_{j \notin \{j_1, \dots, j_r\}} z_j^{e'_j} \mathbf{w}_j.$$

Any g_l belongs to S, so f_l is a monic polynomial in z_{j_l} with coefficients in S. By induction on l, the ring

$$A_l := S[z_{j_1}, \dots, z_{j_l}]/(f_1, \dots, f_l)$$

is a free S-module with basis the monomials $z_{j_1}^{b_1} \cdots z_{j_l}^{b_l}$, $0 \leq b_i < e'_{j_i}$, and f_{l+1} is a nonzerodivisor in

$$\mathbb{C}[z_1, \dots, z_N]/(f_1, \dots, f_l) = A_l[z_{j_{l+1}}, \dots, z_{j_r}]$$

because f_{l+1} is monic in $z_{j_{l+1}}$. Thus $f_1, \dots, f_r$ is a regular sequence in $\mathbb{C}[z_1, \dots, z_N]$, and $A = A_r$ is a free S-module with the claimed basis. In particular, A is finite over S and contains S, so $\dim A = \dim S = N - r$, and A is a complete intersection ring.

(2) Let $p = (p_1, \dots, p_N) \neq 0$ be a closed point of $\operatorname{Spec} A$, and let $J_p := \{j \mid p_j \neq 0\}$. Then

$$\sum_{j \in J_p} p_j^{e'_j} \mathbf{w}_j = 0, \quad p_j^{e'_j} \neq 0 \text{ for any } j \in J_p.$$

Since $N \geq r$, any set of at most r of the vectors $\mathbf{w}_j$ is linearly independent, so $\#J_p \geq r+1$. Let $j_1, \dots, j_r \in J_p$ be distinct. The columns of the Jacobian matrix of the r equations at p corresponding to $z_{j_1}, \dots, z_{j_r}$ are

$$e'_{j_l} p_{j_l}^{e'_{j_l}-1} \mathbf{w}_{j_l}, \quad 1 \leq l \leq r,$$

which are non-zero multiples of linearly independent vectors. Thus the differentials of the r equations at p are linearly independent, the r equations are part of a regular system of parameters of the regular local ring $\mathcal{O}_{\mathbb{A}^N,p}$, and the local ring of $\operatorname{Spec} A$ at p is regular. Since the singular locus of $\operatorname{Spec} A$ is closed, $\operatorname{Spec} A$ is smooth outside the origin.

(3) By (1) and (2), $\operatorname{Spec} A$ is a local complete intersection in $\mathbb{A}^N$ of dimension $N - r \geq 2$ whose singular locus is contained in the origin. Thus A is regular in codimension one, and A is normal by Serre's criterion [Har77, Proposition II.8.23]. Let L' be the least common multiple of $e'_1, \dots, e'_N$, and grade $\mathbb{C}[z_1, \dots, z_N]$ by $\deg z_j := L'/e'_j$. The r equations are homogeneous of degree L', so $A = \bigoplus_{k \geq 0} A_k$ is positively graded with $A_0 = \mathbb{C}$. Thus A is a normal domain by Lemma 3.1.

(4) Let $W := \langle \mathbf{w}_l \mid l \in I \rangle$. Then $\dim W = \#I$. Let $j_1, \dots, j_{r-\#I} \notin I$ be distinct indices, and suppose that $\sum_i c_i \mathbf{w}_{j_i} \in W$ for some $c_1, \dots, c_{r-\#I} \in \mathbb{C}$, not all zero. Then we obtain a linear relation among the r vectors $\mathbf{w}_{j_1}, \dots, \mathbf{w}_{j_{r-\#I}}$ and $\mathbf{w}_l$, $l \in I$, in which some $c_i \neq 0$ occurs as a coefficient, a contradiction. Thus the images of $\mathbf{w}_{j_1}, \dots, \mathbf{w}_{j_{r-\#I}}$ in $\mathbb{C}^r/W$ are linearly independent. □

3.2. **The variety X.** In this subsection, we define X and prove that X is a terminal Fano variety of Picard number one with the properties used in later sections. For weighted projective spaces, an exponent indicates how many coordinates have that weight. For example, $\mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ has one coordinate of weight 1, three of weight 903, five of weight 602, three of weight 258, and eighteen of weight 42.

Set-up 3.3. Let $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$, $d_1, \dots, d_{11}$, $e_1, \dots, e_{11}$, x, $u_1, \dots, u_{11}$, $y_1, \dots, y_{18}$, X, R, H, P, R_P, U, and D_f be as follows.

(1) $\boldsymbol{v}_a := (1, a, a^2, a^3)^\top \in \mathbb{C}^4$ for $0 \leq a \leq 29$.
(2) $(d_1, \dots, d_{11}) := (21, 21, 21, 14, 14, 14, 14, 14, 6, 6, 6)$ and $e_i := 42/d_i$ for $1 \leq i \leq 11$.
(3) The weighted projective space $\mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ has coordinates x of weight 1, u_i of weight $43d_i$ for $1 \leq i \leq 11$, and y_j of weight 42 for $1 \leq j \leq 18$.
(4) $X \subset \mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ is the weighted complete intersection defined by

$$\boldsymbol{v}_0 x^{1806} + \sum_{i=1}^{11} \boldsymbol{v}_i u_i^{e_i} + \sum_{j=1}^{18} \boldsymbol{v}_{11+j} y_j^{43} = 0,$$

and R is the homogeneous coordinate ring of X.
(5) H is a Weil divisor on X with $\mathcal{O}_X(H) \cong \mathcal{O}_X(1)$.
(6) $P := (x = 0) \subset X$, $R_P := R/(x)$, and $U := X \setminus P$.
(7) For a non-zero homogeneous element f of R_P, $D_f \geq 0$ is the Weil divisor of zeros of f on P.

By Proposition 3.4(1) below, X and P are normal, so H and D_f are well defined. The proof of Proposition 3.4(1) does not use them.

Of the vectors $\boldsymbol{v}_a$, we only use that any four of them are linearly independent (Proposition 3.4(1)), which implies that the coordinate strata of X and of P have the expected dimensions and that the 30 branch hyperplanes of the finite morphism of Construction-Proposition 4.2 are in general position. The weights are $1806/m$ for $m = 1806, 2, 3, 7, 43$, and the identity $1 + 903 + 602 + 258 + 42 = 1806$ implies that, for any non-identity element of a local stabilizer of X, the sum, over one coordinate of each weight, of the fractional parts appearing in the age of the element is a positive integer. Any weight other than 1 occurs at least three times: two copies suffice for the terminality of X, and the third copy is used in Section 10.

Proposition 3.4. *Notation and conditions as in Set-up 3.3. Then the following hold.*

(1) *Any four of* $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$ *are linearly independent,* R *and* R_P *are normal domains, and* X *is a well-formed quasismooth weighted complete intersection of dimension* 25.
(2) X *has terminal singularities.*
(3) $\mathrm{Cl}(X) = \mathbb{Z}[H]$, $-K_X \sim 26H$, *and* $1806H$ *is Cartier. In particular,* X *is a* $\mathbb{Q}$*-factorial Fano variety of Picard number one.*
(4) $H^0(X, \mathcal{O}_X(kH)) = R_k$ *for any* $k \in \mathbb{Z}$. *In particular,* $H^0(X, \mathcal{O}_X(kH)) = \mathbb{C}x^k$ *for* $0 \le k \le 41$.
(5) P *is a prime divisor on* X *with* $P \sim H$, *and* $P = \operatorname{Proj} R_P$ *is a quasismooth weighted complete intersection.*
(6) U *is smooth.*

Proof. (1) For $0 \le a_1 < a_2 < a_3 < a_4 \le 29$, the determinant of the matrix with columns $\boldsymbol{v}_{a_1}, \dots, \boldsymbol{v}_{a_4}$ is the Vandermonde determinant $\prod_{k<l}(a_l - a_k) \neq 0$. Thus any four of $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$ are linearly independent. The ring R is the ring A of Lemma 3.2 with $r = 4$, $N = 30$, the vectors $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$, and the exponents 1806, $e_1, \dots, e_{11}$, and 43 (18 times), all of which are at least 2. By Lemma 3.2(3), R is a normal domain. Multiplying the vector equation of X by the inverse of the matrix with columns $\boldsymbol{v}_1, \dots, \boldsymbol{v}_4$, we obtain four weighted homogeneous polynomials of degree 1806 which generate the ideal generated by the four components of the vector equation of X and are monic in u_1, u_2, u_3, u_4, respectively, with coefficients in $\mathbb{C}[x, u_5, \dots, u_{11}, y_1, \dots, y_{18}]$. Thus the four polynomials form a regular sequence. Since R is a domain, the ideal they generate is prime, so it is the weighted homogeneous ideal of X. Thus X is a weighted complete intersection, $\dim X = \dim R - 1 = 25$ by Lemma 3.2(1), and X is quasismooth by Lemma 3.2(2). Similarly, $R_P = R/(x)$ is the ring A of Lemma 3.2 with $r = 4$, $N = 29$, the vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{29}$, and the exponents $e_1, \dots, e_{11}$ and 43 (18 times), so R_P is a normal domain by Lemma 3.2(3).

We show that X is well formed. We have

$$903 = 3 \cdot 7 \cdot 43, \quad 602 = 2 \cdot 7 \cdot 43, \quad 258 = 2 \cdot 3 \cdot 43, \quad 42 = 2 \cdot 3 \cdot 7,$$

so $\gcd(903, 602, 258, 42) = 1$. If we omit one weight other than 1, then the weight 1 remains, and if we omit the weight 1, then the weights 903, 602, 258, and 42 remain. Thus $\mathbb{P} := \mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ is well formed. For a point $q \in \mathbb{P}$, the stabilizer in $\mathbb{C}^*$ of a point of $\mathbb{A}^{30}\backslash\{0\}$ over q is μ_d, where d is the greatest common divisor of the weights of the coordinates which are non-zero at q. If $d = 1$, then $\mathbb{P}$ is smooth at q, since for a coordinate z of weight w with $z(q) \neq 0$, the chart $\mathbb{P} \cap (z \neq 0)$ is the quotient of the affine space $(z = 1) \subset \mathbb{A}^{30}$ by μ_w, and μ_w acts freely near the points over q. If $d > 1$, let ℓ be a prime dividing d. Then $\ell \in \{2, 3, 7, 43\}$, and x and the coordinates of the weight $m \in \{903, 602, 258, 42\}$ with $\ell \nmid m$ vanish at q. Let $t \in \{3, 5, 18\}$ be the number of coordinates of weight m. The affine cone of $X \cap (x = 0) \cap (\text{coordinates of weight } m = 0)$ is $\operatorname{Spec} A$ for the ring A of Lemma 3.2 with $r = 4$ and $N = 29 - t \ge 11$, and $\dim A = 25 - t$ by Lemma 3.2(1). Let $X^\circ \subset X$ be the open subset where the stabilizer is trivial. Then $X\backslash X^\circ$ is contained in a union of four closed subsets of X of dimension at most $24 - t \le 21$, so $X\backslash X^\circ$ is of codimension at least four in X. Since $X \cap \operatorname{Sing} \mathbb{P} \subset X\backslash X^\circ$, X is well formed.

(2) Let $p \in X$ be a closed point. By (1) and Lemma 3.2(2), at least five coordinates are non-zero at any point of $\operatorname{Spec} R$ other than the origin, so there exists a coordinate $z \neq x$ with $z(p) \neq 0$. Let w be the weight of z, and let $\widetilde{X}_z$ be the closed subscheme of the affine space $\mathbb{A}^{29}$ with the coordinates other than z defined by the equations of X together with $z = 1$, on which μ_w acts by $\zeta \cdot c = \zeta^m c$ for a coordinate c of weight m. By Lemma 3.7(1)(2) below, whose proof uses only Lemma 3.2 and (1), $\widetilde{X}_z$ is a smooth variety of dimension 25, and $X \cap (z \neq 0) = \widetilde{X}_z/\mu_w$. Let $\tilde{p} \in \widetilde{X}_z$ be a point over p and $\mu_d \subset \mu_w$ the stabilizer of $\tilde{p}$. Then d is the greatest common divisor of the weights of the coordinates which are non-zero at p, and $d \mid 1806$. If $d = 1$, then the quotient morphism is étale at $\tilde{p}$, and X is smooth at p. Thus we may assume that $d \ge 2$. In particular, $x(p) = 0$, since x has weight 1.

The group μ_d acts linearly on $\mathbb{A}^{29}$ and fixes $\tilde{p}$. Let $T \subset \mathbb{C}^{29}$ be the tangent space of $\widetilde{X}_z$ at $\tilde{p}$, which is a μ_d-stable linear subspace. Since μ_d is the stabilizer of $\tilde{p}$ in μ_w, the morphism

$$\widetilde{X}_z/\mu_d \to \widetilde{X}_z/\mu_w = X \cap (z \neq 0)$$

is étale at the image of $\tilde{p}$ [Kol13, 3.17]. Since $\widetilde{X}_z$ is smooth at the μ_d-fixed point $\tilde{p}$, the germ of $\widetilde{X}_z/\mu_d$ at the image of $\tilde{p}$ and the germ of T/μ_d at the origin have isomorphic Henselizations [Kol13, 3.17]. Thus the germ of X at p is étale-locally isomorphic to the germ of T/μ_d at the origin. The differentials at $\tilde{p}$ of the four equations of $\widetilde{X}_z$ are μ_d-invariant linear forms on $\mathbb{C}^{29}$ with kernel T, so $\mathbb{C}^{29} \cong T \oplus \mathbb{C}^4$ as representations of μ_d, where μ_d acts trivially on $\mathbb{C}^4$. Thus the characters of μ_d on T are those of the coordinates other than z with four trivial characters removed.

For $\zeta \in \mu_d$ whose eigenvalues on T are $e^{2\pi i c_l/d}$ with $0 \leq c_l < d$, the *age* of ζ is $\operatorname{age}(\zeta) := \sum_l c_l/d$. Let $\zeta = e^{2\pi i a/d}$ with $0 < a < d$, and denote by $\{\cdot\}$ the fractional part. Trivial characters contribute zero to the age, and ζ acts trivially on z since $d \mid w$. Since the characters of μ_d on T are those of the coordinates other than z with four trivial characters removed, the age is the sum of $\{ma/d\}$ over all 30 coordinates, where m is the weight of the coordinate:

$$\operatorname{age}(\zeta) = \left\{\frac{a}{d}\right\} + 3\left\{\frac{903a}{d}\right\} + 5\left\{\frac{602a}{d}\right\} + 3\left\{\frac{258a}{d}\right\} + 18\left\{\frac{42a}{d}\right\}.$$

Since $1 + 903 + 602 + 258 + 42 = 1806$ and $d \mid 1806$, the number

$$\left\{\frac{a}{d}\right\} + \left\{\frac{903a}{d}\right\} + \left\{\frac{602a}{d}\right\} + \left\{\frac{258a}{d}\right\} + \left\{\frac{42a}{d}\right\}$$

is congruent to $1806a/d$ modulo $\mathbb{Z}$, so it is an integer. It is positive because $\{a/d\} = a/d > 0$, so it is at least 1. Moreover, $\{903a/d\} + \{602a/d\} + \{258a/d\} + \{42a/d\} > 0$, since otherwise the integer $\{a/d\} + \{903a/d\} + \cdots + \{42a/d\}$ would be equal to $a/d \notin \mathbb{Z}$. Since any weight other than 1 occurs at least twice,

$$\operatorname{age}(\zeta) \geq 1 + 2\left\{\frac{903a}{d}\right\} + 4\left\{\frac{602a}{d}\right\} + 2\left\{\frac{258a}{d}\right\} + 17\left\{\frac{42a}{d}\right\} > 1.$$

In particular, any non-identity element of μ_d has at least two eigenvalues different from 1 on T, so μ_d contains no pseudo-reflections. The character of μ_d on the coordinate x is the inclusion $\mu_d \subset \mathbb{C}^*$, which is non-trivial, so it is one of the characters of μ_d on T, and the action of μ_d on T is faithful. By Reid's criterion in the form of [Ess25, Theorem 2.1], T/μ_d has a terminal singularity at the origin. Thus X is terminal at p (discrepancies are étale-local, and terminality near p is read off from the completion of $\mathcal{O}_{X,p}$ [Kol13, 2.16]).

(3) By (1), X is a quasismooth weighted complete intersection of dimension $25 \geq 3$ in the well-formed weighted projective space $\mathbb{P}$, so $\operatorname{Cl}(X)$ is freely generated by the class of $\mathcal{O}_X(1)$ [PST17, Proposition 2.3]. Thus $\operatorname{Cl}(X) = \mathbb{Z}[H]$. For $k \in \mathbb{Z}$, the sheaves $\mathcal{O}_X(kH)$ and $\mathcal{O}_X(k)$ are reflexive of rank one (on $X \cap (z \neq 0) = \widetilde{X}_z/\mu_w$, where z is a coordinate of weight w and $\widetilde{X}_z$ is defined by the equations of X together with $z = 1$, $\mathcal{O}_X(k)$ is the weight-k eigensheaf of the pushforward of $\mathcal{O}_{\widetilde{X}_z}$, hence reflexive, and it is invertible where μ_w acts freely). Recall that $X^\circ \subset X$ is the open subset where the stabilizer is trivial. Then $\mathcal{O}_X(1)|_{X^\circ}$ is invertible and

$$\mathcal{O}_X(k)|_{X^\circ} = \mathcal{O}_X(1)|_{X^\circ}^{\otimes k} \cong \mathcal{O}_X(kH)|_{X^\circ}.$$

We have seen in (1) that $X \backslash X^\circ$ is of codimension at least four in X. Since X is normal, $\mathcal{O}_X(kH) \cong \mathcal{O}_X(k)$. By [Dol82, Theorem 3.3.4] (see also [IF00, 6.14] and the formula after [PST17, Proposition 2.3]),

$$\omega_X \cong \mathcal{O}_X\Big(\sum \text{degrees} - \sum \text{weights}\Big),$$

where

$$\sum \text{weights} = 1 + 3 \cdot 903 + 5 \cdot 602 + 3 \cdot 258 + 18 \cdot 42 = 7250, \quad \sum \text{degrees} = 4 \cdot 1806 = 7224.$$

Thus $\omega_X \cong \mathcal{O}_X(-26)$, and $-K_X \sim 26H$. For any coordinate z of weight w, $w \mid 1806$, so $z^{1806/w}$ is a unit of degree 1806 in $R[z^{-1}]$. Thus $\mathcal{O}_X(1806)$ is trivialized by $z^{1806/w}$ on $X \cap (z \neq 0)$, $\mathcal{O}_X(1806)$ is invertible, and $1806H$ is a Cartier divisor. Since X is projective, there exists an ample Cartier divisor L on X, and $L \sim kH$ for some $k \in \mathbb{Z}$ since $\mathrm{Cl}(X) = \mathbb{Z}[H]$. Suppose that $k \leq 0$. Since R is a domain, $x^{-k} \in R_{-k}$ is a non-zero section of $\mathcal{O}_X(-k) \cong \mathcal{O}_X(-kH)$, so $-L \sim D$ for some Weil divisor $D \geq 0$ on X. Then

$$(L^{25}) = -(L^{24} \cdot D) \leq 0,$$

a contradiction. Thus $k \geq 1$, and $k \cdot 1806H \sim 1806L$ is ample, so $1806H$ is ample. Since $\mathrm{Cl}(X) = \mathbb{Z}[H]$, any Weil divisor on X is $\mathbb{Q}$-Cartier, and X has Picard number one. Since $-K_X \sim 26H$ is ample and X is terminal by (2), X is a $\mathbb{Q}$-factorial Fano variety of Picard number one.

(4) By (1) and [IF00, Lemma 7.1] (see also [PST17, Lemma 2.4]), $H^0(X, \mathcal{O}_X(k)) = R_k$ for any $k \in \mathbb{Z}$, since X is a well-formed quasismooth weighted complete intersection. By (3), $H^0(X, \mathcal{O}_X(kH)) = R_k$. The equations of X are of degree 1806, and any coordinate other than x has weight at least 42. Thus for $0 \leq k \leq 41$, R_k coincides with the degree k part of the polynomial ring, which is $\mathbb{C}x^k$.

(5) Since $R_P = R/(x)$ is a domain by (1), P is irreducible and reduced. The affine cone of P is $\operatorname{Spec} R_P$, which is smooth outside the origin by Lemma 3.2(2), and the weighted homogeneous ideal of P is generated by x and the four polynomials in (1). Thus $P = \operatorname{Proj} R_P$ is a quasismooth weighted complete intersection, and $\dim P = 24$ by Lemma 3.2(1). By (4), $x \in R_1 = H^0(X, \mathcal{O}_X(H))$, and the divisor of zeros of x is supported on P. Thus it is equal to nP for some integer $n \geq 1$, and $nP \sim H$. Since $\mathrm{Cl}(X) = \mathbb{Z}[H]$ by (3), $n = 1$ and $P \sim H$.

(6) Since x has weight one, $U = X \cap (x \neq 0)$ is isomorphic to the affine variety defined by the equations of X together with $x = 1$ [IF00, 5.3]. At any point of U, at least four of the coordinates u_i, y_j are non-zero by Lemma 3.2(2), and the corresponding columns of the Jacobian matrix are non-zero multiples of four linearly independent vectors. Thus U is smooth. □

Remark 3.5. The equality $|kP| = \{kP\}$ for $1 \leq k \leq 25$, which follows from Proposition 3.4(4)(5), is necessary for birational superrigidity [SZ19, Theorem 2.3]. Indeed, if $|kP|$ contained a pencil for some $k \leq 25$, let $\mathcal{M}$ be the movable part of the pencil. Since $\mathrm{Cl}(X) = \mathbb{Z}[P]$, the fixed part of the pencil is linearly equivalent to $(k-k')P$ and $\mathcal{M} \subset |k'P|$ for some $1 \leq k' \leq k$. Thus $\frac{26}{k'}\mathcal{M}$ is a movable boundary with $K_X + \frac{26}{k'}\mathcal{M} \sim_{\mathbb{Q}} 0$. Two general members of $\mathcal{M}$ intersect along a closed subset $V \subset X$ of codimension two, whose general points are smooth points of X since X is terminal [KM98, Corollary 5.18], and the exceptional divisor E of the blow-up of X along a component of V near the generic point of the component satisfies, by [KM98, Lemma 2.29],

$$a\Big(E, X, \frac{26}{k'}\mathcal{M}\Big) \leq 2 - \frac{26}{k'} \leq 2 - \frac{26}{k} < 1.$$

Thus $(X, \frac{26}{k'}\mathcal{M})$ would not be canonical.

Remark 3.6. The number of equations of X is at least four, and for four equations the numbers of coordinates of each weight are forced, as follows. Let X' be a well-formed quasismooth weighted complete intersection of c hypersurfaces of degree 1806 in $\mathbb{P}(1, 903^{t_2}, 602^{t_3}, 258^{t_7}, 42^{t_{43}})$. By [Dol82, Theorem 3.3.4] and [IF00, 6.14],

$$\omega_{X'} \cong \mathcal{O}_{X'}(1806c - 1 - 903t_2 - 602t_3 - 258t_7 - 42t_{43}),$$

and $\dim X' = t_2 + t_3 + t_7 + t_{43} - c$. Thus $\omega_{X'} \cong \mathcal{O}_{X'}(-\dim X' - 1)$, as for X in Proposition 3.4(3), if and only if

$$902t_2 + 601t_3 + 257t_7 + 41t_{43} = 1805c.$$

As explained after Set-up 3.3, we require $t_m \geq 3$ for $m = 2, 3, 7, 43$. Let $s_m := t_m - 3 \geq 0$. Then the left-hand side is $5403 + 902s_2 + 601s_3 + 257s_7 + 41s_{43}$. Since $1805c < 5403$ for $c \leq 2$, and

$902s_2 + 601s_3 + 257s_7 + 41s_{43} = 12$ has no solution, we have $c \geq 4$. If $c = 4$, then

$$902s_2 + 601s_3 + 257s_7 + 41s_{43} = 1817.$$

Modulo 41, we have $902 \equiv 0$, $601 \equiv 27$, $257 \equiv 11$, and $1817 \equiv 13$, so $27s_3 + 11s_7 \equiv 13$ modulo 41. Since $601s_3 \leq 1817$, we have $s_3 \leq 3$. For $s_3 = 0, 1, 2, 3$, the congruence $27s_3 + 11s_7 \equiv 13$ implies $s_7 \equiv 31, 36, 0, 5$ modulo 41, while $257s_7 \leq 1817 - 601s_3$ implies $s_7 \leq 7, 4, 2, 0$, respectively. Thus $s_3 = 2$ and $s_7 = 0$, so $902s_2 + 41s_{43} = 615$. Since $902 > 615$, we have $s_2 = 0$ and $s_{43} = 15$. Hence $(t_2, t_3, t_7, t_{43}) = (3, 5, 3, 18)$, which is the choice in Set-up 3.3. In particular, X has 11 coordinates u_i, the fibers of the diagonal polynomial map $\mathbb{A}^{11} \to \mathbb{A}^4$ in the coordinates $u_1, \dots, u_{11}$ (Section 6) are of dimension 7, and $d_1 + \cdots + d_{11} = 151$.

Lemma 3.7. *Notation and conditions as in Set-up 3.3. Let z be one of the coordinates $u_1, \dots, u_{11}, y_1, \dots, y_{18}$, let w be the weight of z, and let $\widetilde{X}_z$ and $\widetilde{P}_z$ be the affine varieties defined by the equations of X and of P, respectively, together with $z = 1$. Then the following hold.*

(1) $\widetilde{X}_z$ and $\widetilde{P}_z$ are smooth irreducible varieties of dimension 25 and 24, respectively, $\widetilde{P}_z = \widetilde{X}_z \cap (x = 0)$, and x is part of a regular system of parameters of $\widetilde{X}_z$ at any point of $\widetilde{P}_z$. Moreover, for any closed point $q \in \widetilde{X}_z$, at least four of the coordinates other than z do not vanish at q, and for any four such coordinates $c_1, \dots, c_4$, the 25 coordinates other than $z, c_1, \dots, c_4$, minus their values at q, form a regular system of parameters of $\widetilde{X}_z$ at q.

(2) The group μ_w acts on $\widetilde{X}_z$ and on $\widetilde{P}_z$ through the weights of the coordinates, and the quotient morphisms $\widetilde{X}_z \to X \cap (z \neq 0)$ and $\widetilde{P}_z \to P \cap (z \neq 0)$ by μ_w are étale in codimension one.

(3) P is normal with klt singularities, and $(K_X + P)|_P = K_P$.

Proof. (1) We regard $\widetilde{X}_z$ as a closed subscheme of the affine space $\mathbb{A}^{29}$ with the coordinates other than z, defined by four equations. Let $q \in \widetilde{X}_z$ be a closed point. Then q together with $z = 1$ is a non-zero point of the affine cone $\operatorname{Spec} R$, so at least five coordinates are non-zero at it by Lemma 3.2(2), and at least four of them are different from z. Let $c_1, \dots, c_4$ be four coordinates other than z that do not vanish at q. The columns of the Jacobian matrix of the four equations corresponding to $c_1, \dots, c_4$ are non-zero multiples of four of the vectors $\boldsymbol{v}_a$, which are linearly independent by Proposition 3.4(1), so the 4×4 minor in the columns of $c_1, \dots, c_4$ is invertible at q. Since any irreducible component of $\widetilde{X}_z$ is of dimension at least $29 - 4 = 25$, $\widetilde{X}_z$ is smooth of dimension 25 at q. Since the minor is invertible, the differentials at q of $c_1, \dots, c_4$ are linear combinations of the differentials of the other 25 coordinates and of the four equations, so the 25 coordinates other than $z, c_1, \dots, c_4$, minus their values at q, generate the maximal ideal of $\mathcal{O}_{\widetilde{X}_z, q}$, and they form a regular system of parameters since $\dim \mathcal{O}_{\widetilde{X}_z, q} = 25$. The equations of P together with $z = 1$ are the equations of X together with $x = 0$ and $z = 1$, so $\widetilde{P}_z = \widetilde{X}_z \cap (x = 0)$ as schemes. Similarly, at any closed point of $\widetilde{P}_z$ at least four coordinates other than x and z are non-zero by Lemma 3.2(2) applied to $\operatorname{Spec} R_P$, and the corresponding columns of the Jacobian matrix of the four equations of $\widetilde{P}_z$ in the affine space $\mathbb{A}^{28}$ with the coordinates other than x and z are non-zero multiples of four linearly independent vectors $\boldsymbol{v}_a$. Thus $\widetilde{P}_z$ is smooth of dimension $28 - 4 = 24$. For $q \in \widetilde{P}_z$, the local ring

$$\mathcal{O}_{\widetilde{X}_z, q}/(x) = \mathcal{O}_{\widetilde{P}_z, q}$$

is regular of dimension $\dim \mathcal{O}_{\widetilde{X}_z, q} - 1$, so x is part of a regular system of parameters of $\widetilde{X}_z$ at q. Equivalently, the partial derivatives of the equations of X with respect to x vanish along $x = 0$, so dx is linearly independent of the differentials of the equations at q.

We show that $\widetilde{X}_z$ is irreducible. Let s be a new coordinate of weight 1, and let X' be the weighted complete intersection defined by the equations of X with $z^{1806/w}$ replaced by s^{1806}, in the weighted projective space with the coordinates s and the coordinates other than z. The homogeneous coordinate ring of X' is the ring A of Lemma 3.2 with $r = 4$, $N = 30$, and the

vectors $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$, so X' is integral by Lemma 3.2(3). Since s has weight one, $X' \cap (s \neq 0)$ is isomorphic to the affine variety defined by the equations of X' together with $s = 1$, which is $\widetilde{X}_z$. Thus $\widetilde{X}_z$ is a non-empty open subset of X', and $\widetilde{X}_z$ is irreducible. Replacing X with P and the vectors $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$ with $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{29}$, we obtain that $\widetilde{P}_z$ is irreducible.

(2) For $\zeta \in \mu_w$, let ζ act on a coordinate c of weight m by $\zeta \cdot c = \zeta^m c$. Since $\zeta^w = 1$ and the equations of X are homogeneous of degree 1806 with $w \mid 1806$, the equations of $\widetilde{X}_z$ and of $\widetilde{P}_z$ are μ_w-invariant. By [IF00, 5.3], the open subset $(z \neq 0)$ of $\mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ is $\mathbb{A}^{29}/\mu_w$. Since taking μ_w-invariants is exact, $X \cap (z \neq 0) = \widetilde{X}_z/\mu_w$ and $P \cap (z \neq 0) = \widetilde{P}_z/\mu_w$.

Let $\zeta \in \mu_w$ be a non-identity element. Since $\gcd(903, 602, 258, 42) = 1$, there exists $m \in \{903, 602, 258, 42\}$ with $\zeta^m \neq 1$. Let C_m be the set of coordinates of weight m and $t := \#C_m \geq 3$. Since $\zeta^w = 1$, $z \notin C_m$. Any point of $\widetilde{X}_z$ fixed by ζ satisfies $c = 0$ for any $c \in C_m$. Let $A_m := R/(c \mid c \in C_m)$. Then A_m is the ring A of Lemma 3.2 with $r = 4$ and $N = 30 - t \geq 12$, so A_m is a domain of dimension $26 - t$ by Lemma 3.2(1)(3). The function $z - 1$ on $\operatorname{Spec} A_m$ is non-zero at the origin, so

$$\dim \left(\widetilde{X}_z \cap (c = 0 \text{ for any } c \in C_m)\right) = \dim A_m/(z-1) \leq 25 - t \leq 22.$$

Thus the fixed locus of ζ on $\widetilde{X}_z$ is of codimension at least three. Similarly, $R_P/(c \mid c \in C_m)$ is the ring A of Lemma 3.2 with $r = 4$ and $N = 29 - t \geq 11$, so it is a domain of dimension $25 - t$, and the fixed locus of ζ on $\widetilde{P}_z$ is of dimension at most $24 - t$. Thus it is of codimension at least three in $\widetilde{P}_z$. Since a quotient morphism by a finite group is étale at points with trivial stabilizer, the quotient morphisms $\widetilde{X}_z \to X \cap (z \neq 0)$ and $\widetilde{P}_z \to P \cap (z \neq 0)$ are étale outside closed subsets of codimension at least three. In particular, they are étale in codimension one.

(3) By Proposition 3.4(1)(5), R_P is a normal domain and $P = \operatorname{Proj} R_P$ is a prime divisor on X. For any coordinate z as in the lemma, the ring of degree zero elements of $R_P[z^{-1}]$ is normal (since the degree zero part of a $\mathbb{Z}$-graded normal domain is normal), so P is a normal prime divisor, which is a hypothesis of Lemma 2.4.

Let $v \in P$ be a closed point. Since $x(v) = 0$, there exists a coordinate $z \neq x$ with $z(v) \neq 0$. Let $\tilde{v} \in \widetilde{P}_z$ be a point over v. By (1), x is part of a regular system of parameters of $\widetilde{X}_z$ at $\tilde{v}$, so x is not identically zero on $\widetilde{X}_z$. Since any $\zeta \in \mu_w$ multiplies x by ζ, a non-identity element of μ_w moves any point of $\widetilde{X}_z$ where $x \neq 0$, so μ_w acts faithfully on $\widetilde{X}_z$. We check the hypotheses of Lemma 2.4 for $V := X$, $S := P$, $V^\circ := X \cap (z \neq 0)$, $\widetilde{V} := \widetilde{X}_z$, and $G := \mu_w$. The quotient morphism $q : \widetilde{X}_z \to X \cap (z \neq 0)$ is étale in codimension one by (2), and

$$q^{-1}(P \cap (z \neq 0)) = \widetilde{X}_z \cap (x = 0) = \widetilde{P}_z$$

is smooth by (1). By (2), $q|_{\widetilde{P}_z}$ is étale in codimension one. Thus $(K_X + P)|_P = K_P$ near v by Lemma 2.4. Since v is arbitrary, $(K_X + P)|_P = K_P$.

Since X is $\mathbb{Q}$-factorial by Proposition 3.4(3), $K_X + P$ is $\mathbb{Q}$-Cartier, so $K_P = (K_X + P)|_P$ is $\mathbb{Q}$-Cartier. Let $q_P := q|_{\widetilde{P}_z}$. Since q_P is étale in codimension one, $K_{\widetilde{P}_z} = q_P^* K_P$ by the Hurwitz formula. Since $\widetilde{P}_z$ is smooth, $(\widetilde{P}_z, 0)$ is klt, so $(P \cap (z \neq 0), 0)$ is klt by Lemma 2.3. Since the open subsets $P \cap (z \neq 0)$ cover P, P has klt singularities. □

By Lemma 3.7, whether a pair on X or on P is klt or log canonical can be checked on the smooth charts $\widetilde{X}_z$ and $\widetilde{P}_z$ (Lemma 2.3), and Lemma 3.7(3) is the starting point for inversion of adjunction in Section 10.

4. K-stability of X

In this section, we prove the K-stability part of Theorem 1.4.

Theorem 4.1. *Notation and conditions as in Set-up 3.3. Then $\beta_X(P) = 0$, and X is K-semistable but not K-stable.*

Construction-Proposition 4.2. Notation and conditions as in Set-up 3.3. Let $\Pi \subset \mathbb{P}^{29}$ be the linear subspace defined by $\sum_{a=0}^{29} t_a \boldsymbol{v}_a = 0$, where $t_0, \dots, t_{29}$ are the homogeneous coordinates of $\mathbb{P}^{29}$, and let $\Pi_a := \Pi \cap (t_a = 0)$ for $0 \le a \le 29$. Let $g : X \to \Pi$ be the morphism given by $t_0 = x^{1806}$, $t_i = u_i^{e_i}$ for $1 \le i \le 11$, and $t_{11+j} = y_j^{43}$ for $1 \le j \le 18$. Then the following hold.

(1) $\Pi \cong \mathbb{P}^{25}$, and $\sum_{a=0}^{29} \Pi_a$ is a simple normal crossing divisor on Π.
(2) g is a finite surjective Galois morphism.
(3) Let $m_0 := 1806$, $m_i := e_i$ for $1 \le i \le 11$, and $m_{11+j} := 43$ for $1 \le j \le 18$. Then

$$K_X = g^*\Big(K_\Pi + \sum_{a=0}^{29} \Big(1 - \frac{1}{m_a}\Big)\Pi_a\Big).$$

Proof. For $0 \le a \le 29$, let z_a be the coordinate x, u_i, or y_j according as $a = 0$, $a = i$ with $1 \le i \le 11$, or $a = 11 + j$ with $1 \le j \le 18$, and let w_a be the weight of z_a. Then $w_a m_a = 1806$ for any a, the four equations of X are

$$\sum_{a=0}^{29} z_a^{m_a} \boldsymbol{v}_a = 0,$$

and g is given by $t_a = z_a^{m_a}$ for $0 \le a \le 29$.

(1) Let M be the 4×30 matrix with columns $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$. By Proposition 3.4(1), any four columns of M are linearly independent, so $\operatorname{rank} M = 4$, and $\Pi = \mathbb{P}(\operatorname{Ker} M) \cong \mathbb{P}^{25}$. Let $S \subset \{0, \dots, 29\}$ be a subset with $k := \#S \le 26$, and let M_S be the $4 \times (30 - k)$ matrix with columns $\boldsymbol{v}_b$, $b \notin S$. Then $\bigcap_{a \in S} \Pi_a \cong \mathbb{P}(\operatorname{Ker} M_S)$, where $\operatorname{Ker} M_S$ is viewed as the subspace of $\operatorname{Ker} M$ defined by $t_a = 0$ for $a \in S$. Since $30 - k \ge 4$, $\operatorname{rank} M_S = 4$, so $\dim \operatorname{Ker} M_S = 26 - k$. Thus the restrictions of the linear forms t_a, $a \in S$, to $\operatorname{Ker} M$ are linearly independent for any subset S with $\#S \le 26 = \dim \operatorname{Ker} M$. Therefore $\Pi_0, \dots, \Pi_{29}$ are distinct hyperplanes of Π, any 26 of them have empty intersection, and for any point $p \in \Pi$, the equations of the hyperplanes Π_a containing p are linearly independent linear forms. In particular, $\sum_{a=0}^{29} \Pi_a$ is a simple normal crossing divisor on Π.

(2) Let

$$\widetilde{G} := \prod_{a=0}^{29} \mu_{m_a} = \mu_{1806} \times \prod_{i=1}^{11} \mu_{e_i} \times \mu_{43}^{18}$$

act on $\mathbb{C}[z_0, \dots, z_{29}]$ by $(\zeta_a)_a \cdot z_a := \zeta_a z_a$. A monomial $\prod_a z_a^{c_a}$ is $\widetilde{G}$-invariant if and only if $m_a \mid c_a$ for any a, so the invariant ring is $\mathbb{C}[z_0^{m_0}, \dots, z_{29}^{m_{29}}]$. The equations of X belong to $\mathbb{C}[z_0^{m_0}, \dots, z_{29}^{m_{29}}]$, so $\widetilde{G}$ acts on R by graded automorphisms. Taking $\widetilde{G}$-invariants is exact, and the Reynolds operator of $\widetilde{G}$ is $\mathbb{C}[z_0^{m_0}, \dots, z_{29}^{m_{29}}]$-linear, so the invariant elements of the ideal of X form the ideal of $\mathbb{C}[z_0^{m_0}, \dots, z_{29}^{m_{29}}]$ generated by the four equations of X. Hence

$$R^{\widetilde{G}} = \mathbb{C}[z_0^{m_0}, \dots, z_{29}^{m_{29}}] \Big/ \Big(\sum_{a=0}^{29} z_a^{m_a} \boldsymbol{v}_a \Big),$$

which is the homogeneous coordinate ring R_Π of Π, with $t_a = z_a^{m_a}$ placed in degree 1806.

Fix $0 \le a \le 29$. Since $z_a^{-1} = z_a^{m_a - 1} t_a^{-1}$ in $R[z_a^{-1}]$ and t_a is $\widetilde{G}$-invariant, we have $R[z_a^{-1}]^{\widetilde{G}} = R_\Pi[t_a^{-1}]$. Since $\widetilde{G}$ preserves degrees, taking degree zero parts, we have $(R_{(z_a)})^{\widetilde{G}} = (R_\Pi)_{(t_a)}$. Moreover, $g^{-1}(\Pi \cap (t_a \ne 0)) = X \cap (z_a \ne 0) = \operatorname{Spec} R_{(z_a)}$ and $\Pi \cap (t_a \ne 0) = \operatorname{Spec} (R_\Pi)_{(t_a)}$. Thus over $\Pi \cap (t_a \ne 0)$, g is the quotient morphism of $\operatorname{Spec} R_{(z_a)}$ by $\widetilde{G}$. Since the open subsets $\Pi \cap (t_a \ne 0)$, $0 \le a \le 29$, cover Π, g is the quotient morphism of X by $\widetilde{G}$. In particular, g is finite and surjective.

The homomorphism $\mu_{1806} \to \widetilde{G}$, $\lambda \to (\lambda^{w_a})_a$, is well defined since $\lambda^{w_a m_a} = \lambda^{1806} = 1$, and injective since $w_0 = 1$. The image of $\lambda \in \mu_{1806}$ acts on R_d by multiplication by λ^d for any d,

hence acts trivially on X. Let $G := \widetilde{G}/\mu_{1806}$. Then G acts on X, and g is the quotient morphism of X by G.

Let $\Pi^\circ := \Pi \setminus \bigcup_{a=0}^{29} \Pi_a$. Then $g^{-1}(\Pi^\circ) = X \cap (z_0 z_1 \cdots z_{29} \neq 0)$, which is non-empty since g is surjective. Let $p \in g^{-1}(\Pi^\circ)$ be a point with weighted homogeneous coordinates $(c_0, \dots, c_{29})$, and let $\zeta = (\zeta_a)_a \in \widetilde{G}$ be an element such that $\zeta \cdot p = p$. Then there exists $\lambda \in \mathbb{C}^*$ such that $\zeta_a c_a = \lambda^{w_a} c_a$ for any a. Since $c_a \neq 0$ for any a, $\zeta_a = \lambda^{w_a}$ for any a, and $\lambda = \zeta_0 \in \mu_{1806}$. Thus ζ belongs to the image of μ_{1806}, that is, G acts freely on $g^{-1}(\Pi^\circ)$. In particular, G acts faithfully on X. Hence g is Galois, with Galois group

$$G = \Big(\mu_{1806} \times \prod_{i=1}^{11} \mu_{e_i} \times \mu_{43}^{18}\Big) \Big/ \mu_{1806}.$$

(3) For $0 \leq a \leq 29$, let $P_a := (z_a = 0) \subset X$. Then $P_0 = P$, and $g^{-1}(\Pi_a) = P_a$ as sets. We show that P_a is a prime divisor on X and

$$g^*\Pi_a = m_a P_a. \tag{4.1}$$

First let $a = 0$. By Proposition 3.4(5), $P_0 = P$ is a prime divisor on X. Since x vanishes on P and the coordinates of X have no common zero on X, there exists $1 \leq b \leq 29$ such that $P \cap (z_b \neq 0) \neq \emptyset$. Let $\widetilde{X}_{z_b}$ and $\widetilde{P}_{z_b}$ be the affine varieties defined by the equations of X and of P, respectively, together with $z_b = 1$, and let $q : \widetilde{X}_{z_b} \to X \cap (z_b \neq 0)$ be the quotient morphism by μ_{w_b}. By Lemma 3.7(1), $\widetilde{P}_{z_b} = \widetilde{X}_{z_b} \cap (x = 0)$ is a smooth irreducible variety of dimension 24, and x is part of a regular system of parameters of $\widetilde{X}_{z_b}$ at any point of $\widetilde{P}_{z_b}$, so $\operatorname{mult}_{\widetilde{P}_{z_b}} x = 1$. By Lemma 3.7(2), $q(\widetilde{P}_{z_b}) = P \cap (z_b \neq 0)$, and q is étale at the generic point of the prime divisor $\widetilde{P}_{z_b}$. The rational function t_0/t_b is a local equation of Π_0 on $\Pi \cap (t_b \neq 0)$, and $q^*g^*(t_0/t_b) = x^{1806}$ on $\widetilde{X}_{z_b}$. Thus

$$\operatorname{mult}_P g^*\Pi_0 = \operatorname{mult}_{\widetilde{P}_{z_b}} q^*g^*\Pi_0 = 1806 \operatorname{mult}_{\widetilde{P}_{z_b}} x = 1806.$$

Next let $1 \leq a \leq 29$. Then

$$R/(z_a) \cong \mathbb{C}[z_b \mid b \neq a] \Big/ \Big(\sum_{b \neq a} z_b^{m_b} \boldsymbol{v}_b\Big).$$

By Proposition 3.4(1), any four of the 29 vectors $\boldsymbol{v}_b$, $b \neq a$, are linearly independent, and $m_b \geq 2$ for any b. Thus by Lemma 3.2(3) with $r = 4$ and $N = 29$, $R/(z_a)$ is a normal domain. By Lemma 3.2(1), applied to four indices different from 0 and a, $R/(z_a)$ is a free module over a polynomial ring one of whose variables is x, so $x \neq 0$ in $R/(z_a)$. Hence P_a is a prime divisor on X and $P_a \cap U \neq \emptyset$. Since x has weight one, $U = \operatorname{Spec} R/(x-1)$, and the regular function z_a/x^{w_a} on U corresponds to the class of z_a. The ring $R/(x-1, z_a)$ is the degree zero part of $(R/(z_a))[x^{-1}]$, which is a non-zero domain. Thus z_a/x^{w_a} generates the prime ideal of $P_a \cap U$ in the coordinate ring of the smooth variety U, so $\operatorname{mult}_{P_a}(z_a/x^{w_a}) = 1$. Since t_a/t_0 is a local equation of Π_a on $\Pi \cap (t_0 \neq 0)$, $g^{-1}(\Pi \cap (t_0 \neq 0)) = U$, and $g^*(t_a/t_0) = (z_a/x^{w_a})^{m_a}$ on U, we have $\operatorname{mult}_{P_a} g^*\Pi_a = m_a$.

Since $g^{-1}(\Pi_a) = P_a$, (4.1) holds for any $0 \leq a \leq 29$. In particular, the ramification index of g along P_a is m_a. By (2), over Π°, g is the quotient morphism of $g^{-1}(\Pi^\circ)$ by the free action of G, hence is étale [Mum74, Section 7, Theorem]. Since $X \setminus g^{-1}(\Pi^\circ) = \bigcup_{a=0}^{29} P_a$, the ramification index of g along any prime divisor on X other than $P_0, \dots, P_{29}$ is one. Thus by the Hurwitz formula [Kol13, (2.41.2)] and (4.1),

$$K_X = g^*K_\Pi + \sum_{a=0}^{29} (m_a - 1) P_a = g^*\Big(K_\Pi + \sum_{a=0}^{29} \Big(1 - \frac{1}{m_a}\Big)\Pi_a\Big).$$

□

Proof of Theorem 4.1. By Proposition 3.4(2)(3), X is a Fano variety with terminal singularities. In particular, X has klt singularities, and H is ample since $-K_X \sim 26H$.

We first show that $\beta_X(P) = 0$. By Proposition 3.4(5), P is a prime divisor on X with $P \sim H$, so $a(P, X) = 1$. Taking h in Definition 2.12 to be the identity morphism of X, we have

$$\mathrm{vol}(-K_X - tP) = \mathrm{vol}((26-t)H) = (26-t)^{25}(H^{25}) \quad \text{for } 0 \le t \le 26,$$

and $\mathrm{vol}(-K_X - tP) = 0$ for $t \ge 26$, since $-K_X - tP \sim_{\mathbb{R}} (26-t)H$ is not big. Thus

$$S_X(P) = \frac{1}{26^{25}(H^{25})} \int_0^{26} (26-t)^{25}(H^{25})dt = \frac{26^{26}}{26 \cdot 26^{25}} = 1,$$

and $\beta_X(P) = a(P, X) - S_X(P) = 0$.

Next we show that X is not K-stable. By [Fuj19a, Remark 1.7(1)], any prime divisor on a Fano variety with klt singularities is dreamy. Thus P is dreamy. Since $\beta_X(P) = 0$, X is not K-stable by Definition 2.12.

Finally we show that X is K-semistable. Recall the morphism $g : X \to \Pi$, the hyperplanes Π_a, and the integers m_a of Construction-Proposition 4.2. For $0 \le a \le 29$, let $b_a := 1 - 1/m_a$, and let $\Delta := \sum_{a=0}^{29} b_a \Pi_a$. Since $m_a \ge 2$, $b_a \in (0,1) \cap \mathbb{Q}$ for any a. In particular, Δ is a $\mathbb{Q}$-divisor. We have

$$\sum_{a=0}^{29} \frac{1}{m_a} = \frac{1}{1806} + \frac{3}{2} + \frac{5}{3} + \frac{3}{7} + \frac{18}{43} = \frac{7250}{1806}, \quad \text{so} \quad \sum_{a=0}^{29} b_a = 30 - \frac{7250}{1806} = 26\Big(1 - \frac{1}{1806}\Big). \tag{4.2}$$

By Construction-Proposition 4.2(1), $\Pi \cong \mathbb{P}^{25}$ is smooth and $\sum_{a=0}^{29} \Pi_a$ is a simple normal crossing divisor. Since $b_a < 1$ for any a, [KM98, Corollary 2.31(3)] implies that $\mathrm{discrep}(\Pi, \Delta) > -1$, so (Π, Δ) is klt [KM98, Definition 2.34]. Since $K_\Pi \sim -26\Pi_0$ and $\Pi_a \sim \Pi_0$ for any a, by (4.2),

$$-(K_\Pi + \Delta) \sim_{\mathbb{Q}} \Big(26 - \sum_{a=0}^{29} b_a\Big)\Pi_0 = \frac{26}{1806}\Pi_0,$$

which is ample. Thus (Π, Δ) is a log Fano pair.

We check the hypothesis of Theorem 2.13 for $n = 25$, the hyperplanes $\Pi_0, \dots, \Pi_{29}$, and the rational numbers $b_0, \dots, b_{29} \in (0,1)$. Let $S \subset \{0, \dots, 29\}$ be a subset with $1 \le k := \#S \le 25$. By (4.2),

$$k \sum_{a=0}^{29} b_a - 26 \sum_{a \in S} b_a = 26 \sum_{a \in S} \Big(\frac{1}{m_a} - \frac{1}{1806}\Big) \ge 0,$$

since $m_a \le 1806$ for any a. Since $m_a \le 43$ for $1 \le a \le 29$, equality holds if and only if $S = \{0\}$. Thus (Π, Δ) is K-semistable by Theorem 2.13.

Since X is projective with klt singularities, $-K_X$ is ample, and the boundary 0 is a $\mathbb{Q}$-divisor, $(X, 0)$ is a log Fano pair. By Construction-Proposition 4.2(2)(3), $g : X \to \Pi$ is a finite surjective Galois morphism such that $K_X + 0 = g^*(K_\Pi + \Delta)$. Thus by Theorem 2.14, $(X, 0)$ is K-semistable. Since $\beta_{(X,0)}(F) = \beta_X(F)$ for any prime divisor F over X, X is K-semistable. □

5. Canonicity away from P

In this section, we prove Theorem 5.1. For centers of dimension at most 14, which are of codimension at least three, we use the $4n^2$-inequality (Lemma 5.6): the multiplicity bound of Proposition 5.3 at points of U is smaller than the bound that Lemma 5.6 requires. Since Lemma 5.6 requires centers of codimension at least three, centers of dimension at least 15, in particular of codimension two, are treated by Proposition 5.5 and Lemma 5.4.

Theorem 5.1. *Notation and conditions as in Set-up 3.3. Let $\ell > 0$ be an integer, $\mathcal{M} \subset |\ell H|$ a movable linear system, and E an exceptional prime divisor over X with $\mathrm{center}_X E \cap U \neq \emptyset$. Then*

$$\frac{26}{\ell} \mathrm{mult}_E \mathcal{M} \le a(E, X) - 1.$$

We prove Theorem 5.1 at the end of this section, in Subsection 5.2. We first prove a multiplicity bound for cycles of codimension two.

5.1. **A multiplicity bound for cycles of codimension two.** In this subsection, we prove Proposition 5.3. In the proof, we cut a cycle through a closed point p of U by sections of $\mathcal{O}_X(42H)$ vanishing at p, and some components of the new cycle may be equal to the common zero set F_c of all such sections, a 7-dimensional subvariety. Definition-Lemma 5.2 computes the degree of F_c and bounds the multiplicity of F_c at p, which we use to treat such components.

Definition-Lemma 5.2. Notation and conditions as in Set-up 3.3. For $c = (c_1, \dots, c_{18}) \in \mathbb{C}^{18}$, let $F_c \subset X$ be the closed subscheme defined by $y_j = c_j x^{42}$ for $1 \le j \le 18$. Then the following hold.

(1) F_c is a projective variety of dimension 7, $F_c \cap U$ is the fiber over c of the morphism $U \to \mathbb{A}^{18}$ given by $y_1, \dots, y_{18}$, and $(H^7 \cdot F_c) = 42^{18}(H^{25})$.
(2) Identify U with the closed subvariety $X \cap (x = 1)$ of $\mathbb{A}^{29}$ with coordinates $u_1, \dots, u_{11}, y_1, \dots, y_{18}$. Let $p \in F_c \cap U$, let $\mathfrak{m}_p$ be the maximal ideal of $\mathcal{O}_{U,p}$, let k be the number of non-zero values among $u_1(p), \dots, u_{11}(p)$, and let $r := \max\{4-k, 0\}$. Then $\operatorname{mult}_p F_c \le 6 \cdot 2^{\max\{r-2,0\}}$, and the images of $y_1 - c_1, \dots, y_{18} - c_{18}$ in $\mathfrak{m}_p/\mathfrak{m}_p^2$ span a vector space of dimension $18 - r$.

Proof. By Proposition 3.4(1), any four of $\boldsymbol{v}_0, \dots, \boldsymbol{v}_{29}$ are linearly independent. Let

$$C(c) := \boldsymbol{v}_0 + \sum_{j=1}^{18} c_j^{43} \boldsymbol{v}_{11+j} \in \mathbb{C}^4.$$

Since R is the quotient of $\mathbb{C}[x, u_1, \dots, u_{11}, y_1, \dots, y_{18}]$ by the four entries of the equation in Set-up 3.3(4), the homogeneous coordinate ring of F_c is

$$R_{F_c} := R/(y_1 - c_1 x^{42}, \dots, y_{18} - c_{18} x^{42}) \cong \mathbb{C}[x, u_1, \dots, u_{11}] \Big/ \Big(C(c) x^{1806} + \sum_{i=1}^{11} \boldsymbol{v}_i u_i^{e_i} \Big), \tag{5.1}$$

where the vector equation stands for four polynomial equations.

(1) Let Q be the 4×4 matrix with columns $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4$. Multiplying the vector equation in (5.1) by Q^{-1}, we obtain four equations

$$u_l^{e_l} + f_l = 0, \quad f_l \in \mathbb{C}[x, u_5, \dots, u_{11}], \quad 1 \le l \le 4,$$

where $(e_1, e_2, e_3, e_4) = (2, 2, 2, 3)$. Thus R_{F_c} is a free module of rank 24 over $\mathbb{C}[x, u_5, \dots, u_{11}]$ (the l-th equation is monic in u_l with coefficients in $\mathbb{C}[x, u_5, \dots, u_{11}]$) with basis $u_1^{b_1} u_2^{b_2} u_3^{b_3} u_4^{b_4}$, $0 \le b_l < e_l$. In particular, $\dim R_{F_c} = 8$ and the image of x in R_{F_c} is non-zero. Let $A := \mathbb{C}[x, u_1, \dots, u_{11}]$ and $\mathfrak{a} \subset A$ the ideal generated by the four equations in (5.1). Any minimal prime ideal of $\mathfrak{a}$ has height at most 4 [Mat89, Theorem 13.5], so any irreducible component of $\operatorname{Spec} R_{F_c}$ is of dimension $12 - 4 = 8$. Thus for any maximal ideal $\mathfrak{q} \supset \mathfrak{a}$ of A, the regular local ring $A_\mathfrak{q}$ is Cohen–Macaulay [Har77, Theorem II.8.21A(a)] and

$$\dim A_\mathfrak{q}/\mathfrak{a}A_\mathfrak{q} = 8 = \dim A_\mathfrak{q} - 4,$$

so the four equations form a regular sequence in $A_\mathfrak{q}$ [Har77, Theorem II.8.21A(c)], and $(R_{F_c})_\mathfrak{q}$ is Cohen–Macaulay [Har77, Theorem II.8.21A(d)]. Hence R_{F_c} is Cohen–Macaulay [Har77, Theorem II.8.21A(b)].

We show that R_{F_c} is normal. Let $q \ne 0$ be a closed point of $\operatorname{Spec} R_{F_c} \subset \mathbb{A}^{12}$. If at least four of $u_1(q), \dots, u_{11}(q)$ are non-zero, then the corresponding columns $e_i u_i(q)^{e_i - 1} \boldsymbol{v}_i$ of the Jacobian matrix of the four equations in (5.1) are linearly independent. The local ring $\mathcal{O}_{\mathbb{A}^{12},q}$ is a regular local ring of dimension 12 with maximal ideal $\mathfrak{m}_q$, and the images of the four equations in $\mathfrak{m}_q/\mathfrak{m}_q^2$ are their differentials at q, which are linearly independent. Thus $(R_{F_c})_q$ is a regular local ring of dimension 8 [Mat89, Theorem 14.2]. If $x(q) = 0$, then $\sum_i \boldsymbol{v}_i u_i(q)^{e_i} = 0$ and $q \ne 0$, so at least five of the $u_i(q)$ are non-zero. The group $\mathbb{C}^*$ acts on $\operatorname{Spec} R_{F_c}$ by $t \cdot (x, u_i) = (tx, t^{43 d_i} u_i)$, and the

singular locus of $\operatorname{Spec} R_{F_c}$ is $\mathbb{C}^*$-stable. Any orbit with $x \neq 0$ contains exactly one point with $x = 1$. If $x(q) = 1$ and the set I of indices i with $u_i(q) \neq 0$ satisfies $\#I \leq 3$, then

$$\sum_{i \in I} \boldsymbol{v}_i u_i(q)^{e_i} = -C(c),$$

and since the $\boldsymbol{v}_i$, $i \in I$, are linearly independent, the values $u_i(q)^{e_i}$, $i \in I$, are determined by I. Thus there are only finitely many such points q, and the singular locus of $\operatorname{Spec} R_{F_c}$ is contained in the union of $\{0\}$ and finitely many $\mathbb{C}^*$-orbits. Hence the singular locus is of dimension at most $1 < \dim R_{F_c} - 1$, so R_{F_c} satisfies (R_1). Since R_{F_c} is Cohen–Macaulay, Serre's criterion [Mat89, Theorem 23.8] implies that R_{F_c} is normal.

Since R_{F_c} is positively graded with degree zero part $\mathbb{C}$, R_{F_c} is a normal domain by Lemma 3.1.

Therefore F_c is a projective variety with $\dim F_c = \dim R_{F_c} - 1 = 7$. Since the image of x in R_{F_c} is non-zero, $F_c \not\subset P$. On $U = X \cap (x = 1)$, the equations of F_c are $y_j = c_j$, so $F_c \cap U$ is the fiber over c of the morphism $U \to \mathbb{A}^{18}$ given by $y_1, \dots, y_{18}$.

We compute $(H^7 \cdot F_c)$. For $1 \leq j \leq 18$, let $D_j := (y_j - c_j x^{42} = 0) \geq 0$. By Proposition 3.4(3) and (4), D_j is a $\mathbb{Q}$-Cartier Weil divisor with $D_j \sim 42H$, and $\operatorname{Supp} D_1 \cap \dots \cap \operatorname{Supp} D_{18} = F_c$. Let $T_0 := X$, regarded as a cycle, and $T_i := D_i \cdot T_{i-1}$ for $1 \leq i \leq 18$. We show by induction on i that no component of T_{i-1} is contained in $\operatorname{Supp} D_i$. Assume that no component of $T_{i'-1}$ is contained in $\operatorname{Supp} D_{i'}$ for any $i' < i$. By Lemma 2.8(1), $T_{i-1} \geq 0$ is a $\mathbb{Q}$-cycle of pure dimension $26 - i$ with $\operatorname{Supp} T_{i-1} \subset \operatorname{Supp} D_1 \cap \dots \cap \operatorname{Supp} D_{i-1}$. Suppose that a component V of T_{i-1} is contained in $\operatorname{Supp} D_i$. Since a multiple of each D_j is an ample Cartier divisor, intersecting a closed subset of positive dimension with $\operatorname{Supp} D_j$ yields a non-empty set and lowers the dimension of any component by at most one [Mat89, Theorem 13.5]. Hence any component of

$$V \cap \operatorname{Supp} D_{i+1} \cap \dots \cap \operatorname{Supp} D_{18} \subset F_c$$

is of dimension at least $(26 - i) - (18 - i) = 8 > \dim F_c$, a contradiction. Thus Lemma 2.8(1) implies that $T_{18} \geq 0$ is a $\mathbb{Q}$-cycle of pure dimension 7 with $\operatorname{Supp} T_{18} \subset F_c$ and

$$(H^7 \cdot T_{18}) = 42^{18}(H^{25}).$$

Since F_c is a variety of dimension 7, T_{18} is a rational multiple of F_c. Let $\xi \in F_c \cap U$ be the generic point of F_c. Near ξ, any D_j is a Cartier divisor with local equation $y_j - c_j$, and $\mathcal{O}_{X,\xi}$ is regular by Proposition 3.4(6). Since the intersection with a $\mathbb{Q}$-Cartier Weil divisor is defined through a Cartier multiple (Definition 2.7), the coefficient of F_c in T_{18} is the coefficient of F_c in the intersection of the Cartier divisors $D_j|_U$, which is

$$\operatorname{length} \mathcal{O}_{X,\xi}/(y_1 - c_1, \dots, y_{18} - c_{18}) = \operatorname{length} \mathcal{O}_{F_c,\xi} = 1$$

by [Ful98, Example 7.1.10(a)], as F_c is integral. Thus $(H^7 \cdot F_c) = 42^{18}(H^{25})$.

(2) Let $I := \{i \mid u_i(p) \neq 0\}$, so $k = \#I$. By (1), $F_c \cap U$ is the closed subscheme of $\mathbb{A}^{11}$ with coordinates $u_1, \dots, u_{11}$ defined by

$$C(c) + \sum_{i=1}^{11} \boldsymbol{v}_i u_i^{e_i} = 0. \tag{5.2}$$

We first compute the images of $y_j - c_j$ in $\mathfrak{m}_p/\mathfrak{m}_p^2$. The variety $U \subset \mathbb{A}^{29}$ is defined by the four entries of $\boldsymbol{v}_0 + \sum_i \boldsymbol{v}_i u_i^{e_i} + \sum_j \boldsymbol{v}_{11+j} y_j^{43} = 0$. Let J be the 4×29 Jacobian matrix of the four entries at p. The columns of J are $e_i u_i(p)^{e_i - 1} \boldsymbol{v}_i$ and $43 c_j^{42} \boldsymbol{v}_{11+j}$. At least four of $u_1(p), \dots, u_{11}(p), c_1, \dots, c_{18}$ are non-zero, since otherwise $\boldsymbol{v}_0$ would be a linear combination of at most three other $\boldsymbol{v}_a$. Thus $\operatorname{rank} J = 4$, so $\lambda \to \lambda J$ is injective on row vectors $\lambda \in \mathbb{C}^4$. The space $\mathfrak{m}_p/\mathfrak{m}_p^2$ is the quotient of the space of linear forms in du_i, dy_j by the rows λJ, and the image of $y_j - c_j$ is the class of dy_j. A row λJ belongs to the span of $dy_1, \dots, dy_{18}$ if and only if $\lambda \boldsymbol{v}_i = 0$ for any $i \in I$. Since the $\boldsymbol{v}_i$, $i \in I$, span a vector space of dimension $\min\{k, 4\}$, such λ form a vector space of dimension $4 - \min\{k, 4\} = r$. Thus the images of $y_1 - c_1, \dots, y_{18} - c_{18}$ in $\mathfrak{m}_p/\mathfrak{m}_p^2$ span a vector space of dimension $18 - r$.

We now bound $\operatorname{mult}_p F_c$, which is the multiplicity of the local ring $\mathcal{O}_{F_c,p}$ [Ful98, Section 4.3]. If $k \geq 4$, then the Jacobian matrix of (5.2) at p is of rank 4, so the images of the four equations of (5.2) in $\mathfrak{n}_p/\mathfrak{n}_p^2$ are linearly independent, where $\mathfrak{n}_p$ is the maximal ideal of the 11-dimensional regular local ring of $\mathbb{A}^{11}$ at p. Thus $\mathcal{O}_{F_c,p}$ is a regular local ring of dimension 7 [Mat89, Theorem 14.2], and $\operatorname{mult}_p F_c = 1$.

Suppose that $k \leq 3$, so $r = 4 - k \geq 1$. Evaluating (5.2) at p, we have $C(c) = -\sum_{i \in I} \boldsymbol{v}_i u_i(p)^{e_i}$. Write $I = \{l_1, \dots, l_k\}$. Since $\boldsymbol{v}_{l_1}, \dots, \boldsymbol{v}_{l_k}$ are linearly independent, there exists an invertible 4×4 matrix Q_p such that $Q_p \boldsymbol{v}_{l_s}$ is the s-th standard basis vector for $1 \leq s \leq k$. For $i \notin I$, let $\bar{\boldsymbol{v}}_i \in \mathbb{C}^r$ be the vector of the last r entries of $Q_p \boldsymbol{v}_i$. Since $\bar{\boldsymbol{v}}_i$ is the image of $\boldsymbol{v}_i$ under the isomorphism

$$\mathbb{C}^4/\langle \boldsymbol{v}_l \mid l \in I \rangle \to \mathbb{C}^r$$

induced by Q_p, Lemma 3.2(4), applied to $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{11} \in \mathbb{C}^4$ and $e_1, \dots, e_{11}$, implies that any r of the vectors $\bar{\boldsymbol{v}}_i$, $i \notin I$, are linearly independent. Multiplying (5.2) by Q_p, we obtain k equations

$$h_s := u_{l_s}^{e_{l_s}} - u_{l_s}(p)^{e_{l_s}} + \sum_{i \notin I} q_{s,i} u_i^{e_i} = 0, \quad q_{s,i} \in \mathbb{C}, \quad 1 \leq s \leq k,$$

and r equations

$$g_1 = \cdots = g_r = 0, \quad (g_1, \dots, g_r)^\top := \sum_{i \notin I} \bar{\boldsymbol{v}}_i u_i^{e_i},$$

which only involve the $7 + r$ variables u_i, $i \notin I$, all vanishing at p. Since

$$\frac{\partial h_s}{\partial u_{l_t}}(p) = \delta_{st} e_{l_s} u_{l_s}(p)^{e_{l_s}-1} \quad (1 \leq s, t \leq k)$$

and $u_{l_s}(p) \neq 0$, the formal implicit function theorem implies that there exist power series $\psi_1, \dots, \psi_k$ in the ring $\mathbb{C}[[u_i \mid i \notin I]]$ such that, in the completion of the local ring of $\mathbb{A}^{11}$ at p, the ideal $(h_1, \dots, h_k)$ is equal to the ideal generated by $u_{l_s} - u_{l_s}(p) - \psi_s$, $1 \leq s \leq k$. Hence the completion of $\mathcal{O}_{F_c,p}$ is

$$\widehat{\mathcal{O}} := \mathbb{C}[[u_i \mid i \notin I]]/(g_1, \dots, g_r).$$

Let $\mathfrak{m}$ and $\widehat{\mathfrak{m}}$ be the maximal ideals of $\mathcal{O}_{F_c,p}$ and $\widehat{\mathcal{O}}$. Since

$$\mathcal{O}_{F_c,p}/\mathfrak{m}^n \cong \widehat{\mathcal{O}}/\widehat{\mathfrak{m}}^n \quad \text{for any } n \geq 1,$$

the multiplicity $\operatorname{mult}_p F_c$ is equal to the multiplicity of $\widehat{\mathcal{O}}$.

For a non-zero power series f, we denote by f^* the leading form of f, that is, the non-zero homogeneous part of f of lowest degree. The associated graded ring $\operatorname{gr} \widehat{\mathcal{O}}$ of $\widehat{\mathcal{O}}$ with respect to $\widehat{\mathfrak{m}}$ is the quotient of $\mathbb{C}[u_i \mid i \notin I]$ by the ideal generated by the leading forms of the non-zero elements of $(g_1, \dots, g_r)$. Thus for any non-zero linear combinations $g_1', \dots, g_r'$ of $g_1, \dots, g_r$, $\operatorname{gr} \widehat{\mathcal{O}}$ is a quotient of

$$S := \mathbb{C}[u_i \mid i \notin I]/(g_1'^*, \dots, g_r'^*).$$

We choose $g_1', \dots, g_r'$ such that S is a free module over a polynomial ring in 7 variables. Then S, $\widehat{\mathcal{O}}$, and $\operatorname{gr} \widehat{\mathcal{O}}$ are of dimension 7, the Hilbert polynomials of S and $\operatorname{gr} \widehat{\mathcal{O}}$ are of degree 6, and the Hilbert function of $\operatorname{gr} \widehat{\mathcal{O}}$ is bounded by the Hilbert function of S. Hence $\operatorname{mult}_p F_c$ is at most the multiplicity of S (the Hilbert–Samuel multiplicity is read off from the leading coefficient of the Hilbert polynomial of the associated graded ring). Recall that the indices i with $e_i = 2$, $e_i = 3$, and $e_i = 7$ are $1 \leq i \leq 3$, $4 \leq i \leq 8$, and $9 \leq i \leq 11$, respectively.

Suppose that $I \not\subset \{1, 2, 3\}$. Then $\#(I \cap \{1, 2, 3\}) \leq k - 1$, so at least $3 - (k - 1) = r$ of the indices $1, 2, 3$ are not in I, say $i_1, \dots, i_r$. Since $\bar{\boldsymbol{v}}_{i_1}, \dots, \bar{\boldsymbol{v}}_{i_r}$ are linearly independent, the $r \times r$ matrix with columns $\bar{\boldsymbol{v}}_{i_1}, \dots, \bar{\boldsymbol{v}}_{i_r}$ has an inverse Q_p'. Let $(\gamma_{1,i}, \dots, \gamma_{r,i})^\top := Q_p' \bar{\boldsymbol{v}}_i$ for $i \notin I$, and let g_s' be the s-th entry of $Q_p'(g_1, \dots, g_r)^\top$. Then

$$g_s' = u_{i_s}^2 + \sum_{i \notin I \cup \{i_1, \dots, i_r\}} \gamma_{s,i} u_i^{e_i}, \quad 1 \leq s \leq r,$$

and $g_s'^* = u_{i_s}^2 + \sum \gamma_{s,i} u_i^2$, where the sum runs over $i \in \{1,2,3\}\backslash(I \cup \{i_1,\dots,i_r\})$. Thus S is a free module over $\mathbb{C}[u_i \mid i \notin I \cup \{i_1,\dots,i_r\}]$ with basis $u_{i_1}^{b_1}\cdots u_{i_r}^{b_r}$, $b_s \in \{0,1\}$, the Hilbert series of S is $(1+t)^r/(1-t)^7$, and

$$\operatorname{mult}_p F_c \le 2^r.$$

Suppose that $I \subset \{1,2,3\}$. Then exactly $3-k = r-1$ of the indices $1,2,3$ are not in I, say $i_1,\dots,i_{r-1}$. Since $\bar{\boldsymbol{v}}_{i_1},\dots,\bar{\boldsymbol{v}}_{i_{r-1}}$ are linearly independent, there exists an invertible $r\times r$ matrix Q_p' with $Q_p'\bar{\boldsymbol{v}}_{i_s}$ equal to the s-th standard basis vector for $1 \le s \le r-1$. Let $(\gamma_{1,i},\dots,\gamma_{r,i})^\top := Q_p'\bar{\boldsymbol{v}}_i$ for $i \notin I$, and let g_s' be the s-th entry of $Q_p'(g_1,\dots,g_r)^\top$. Then

$$g_s' = u_{i_s}^2 + \sum_{i=4}^{11} \gamma_{s,i} u_i^{e_i} \quad (1 \le s \le r-1), \qquad g_r' = \sum_{i=4}^{11} \gamma_{r,i} u_i^{e_i}.$$

For $4 \le i \le 8$, $\gamma_{r,i} \ne 0$: otherwise $Q_p'\bar{\boldsymbol{v}}_i$ would belong to the span of $Q_p'\bar{\boldsymbol{v}}_{i_1},\dots,Q_p'\bar{\boldsymbol{v}}_{i_{r-1}}$, while any r of the vectors $\bar{\boldsymbol{v}}_i$, $i \notin I$, are linearly independent. Thus $g_s'^* = u_{i_s}^2$ for $1 \le s \le r-1$ and $g_r'^* = \sum_{i=4}^8 \gamma_{r,i} u_i^3$ with $\gamma_{r,4} \ne 0$. Hence S is a free module over $\mathbb{C}[u_i \mid i \notin I \cup \{i_1,\dots,i_{r-1},4\}]$ with basis $u_{i_1}^{b_1}\cdots u_{i_{r-1}}^{b_{r-1}} u_4^{b_r}$, $b_s \in \{0,1\}$ for $s \le r-1$ and $b_r \in \{0,1,2\}$, the Hilbert series of S is $(1+t)^{r-1}(1+t+t^2)/(1-t)^7$, and

$$\operatorname{mult}_p F_c \le 3\cdot 2^{r-1}.$$

In both cases, $\operatorname{mult}_p F_c \le 3\cdot 2^{r-1} \le 6\cdot 2^{\max\{r-2,0\}}$, as $3 \le 6$ for $r = 1$ and $3\cdot 2^{r-1} = 6\cdot 2^{r-2}$ for $r \ge 2$. □

The constant $\frac{1}{294} = \frac{6}{42^2}$ in Proposition 5.3 is determined by the components equal to F_c, with $c = (y_1(p),\dots,y_{18}(p))$, after cutting Z by 16 sections of degree 42 vanishing at p. By Definition-Lemma 5.2, such a component has multiplicity at most $6\cdot 2^{\max\{r-2,0\}}$ at p and degree $(H^7\cdot F_c) = 42^{18}(H^{25})$, where r is the integer of Definition-Lemma 5.2(2), and the multiplicity of the cut cycle at p is at least $2^{\max\{r-2,0\}}\operatorname{mult}_p Z$. The other components are further cut down to points by sections of degrees 42, 258, and 602, and for them the constant is only $\frac{2}{903} < \frac{1}{294}$.

Proposition 5.3. *Notation and conditions as in Set-up 3.3. Let $Z \ge 0$ be a $\mathbb{Q}$-cycle of pure codimension two on X and $p \in U$ a closed point. Then*

$$\operatorname{mult}_p Z \le \frac{(H^{23}\cdot Z)}{294(H^{25})}.$$

Proof. The proof is long, so we divide it into several steps. We identify U with $X \cap (x=1)$, and for $\varphi \in R_\delta = H^0(X,\mathcal{O}_X(\delta H))$ (Proposition 3.4(4)), we write $\varphi|_U$ for the regular function on U obtained by setting $x=1$, and $(\varphi=0) \ge 0$ for the divisor of zeros of φ. Note that $(\varphi=0) \sim \delta H$, and $(\varphi=0)|_U$ is the Cartier divisor with local equation $\varphi|_U$.

Step 1. In this step, we show that general sections of a linear system whose base set is small cut cycles through p properly. Let $\delta \ge 1$ and $s \ge 1$ be integers, and $V \subset R_\delta$ a non-zero linear subspace with $\varphi|_U \in \mathfrak{m}_p^s$ for any $\varphi \in V$. Let $\operatorname{Bs}_U(V) \subset U$ be the common zero set of the functions $\varphi|_U$, $\varphi \in V$. Let $T \ge 0$ be a $\mathbb{Q}$-cycle of pure dimension $d \ge 1$ on X whose components all contain p, and assume that $\dim(\operatorname{Bs}_U(V) \cap \operatorname{Supp} T) \le d-1$.

Let Γ be a component of T. Since $p \in \Gamma \cap U$, $\dim(\Gamma\cap U) = d > \dim(\operatorname{Bs}_U(V)\cap\operatorname{Supp} T)$, so $\Gamma\cap U \not\subset \operatorname{Bs}_U(V)$. Thus the elements $\varphi \in V$ with $\Gamma \subset \operatorname{Supp}(\varphi=0)$ form a proper linear subspace $V_\Gamma \subsetneq V$. Let $\varphi \in V\backslash\bigcup_\Gamma V_\Gamma$, where Γ runs over the components of T. By Lemma 2.8, $(\varphi=0)\cdot T \ge 0$ is a $\mathbb{Q}$-cycle of pure dimension $d-1$ with

$$(H^{d-1}\cdot(\varphi=0)\cdot T) = \delta(H^d\cdot T) \quad\text{and}\quad \operatorname{mult}_p((\varphi=0)\cdot T) \ge s\operatorname{mult}_p T.$$

Let T^φ be the sum of the terms of $(\varphi=0)\cdot T$ whose components contain p. Since H is ample, the other terms have non-negative degree, and they do not contribute to the multiplicity at p.

Thus

$$(H^{d-1}\cdot T^{\varphi})\le\delta(H^d\cdot T),\quad \operatorname{mult}_p T^{\varphi}\ge s\operatorname{mult}_p T,\\ \operatorname{Supp}T^{\varphi}\subset\operatorname{Supp}T\cap\operatorname{Supp}(\varphi=0). \tag{5.3}$$

In particular, if $\dim(\operatorname{Bs}_U(V)\cap\operatorname{Supp}T)\le d-j$ for an integer $1\le j\le d$, then $\dim(\operatorname{Bs}_U(V)\cap\operatorname{Supp}T^{\varphi})\le(d-1)-(j-1)$, and we may repeat the cut with T^{φ} in place of T. Hence for general $\varphi_1,\dots,\varphi_j\in V$, more precisely for φ_i outside a finite union of proper linear subspaces of V depending on $\varphi_1,\dots,\varphi_{i-1}$, we obtain a $\mathbb{Q}$-cycle $T^{(j)}\ge 0$ of pure dimension $d-j$ whose components all contain p, with

$$(H^{d-j}\cdot T^{(j)})\le\delta^j(H^d\cdot T),\quad \operatorname{mult}_p T^{(j)}\ge s^j\operatorname{mult}_p T,\\ \operatorname{Supp}T^{(j)}\subset\operatorname{Supp}T\cap\operatorname{Supp}(\varphi_1=0)\cap\cdots\cap\operatorname{Supp}(\varphi_j=0). \tag{5.4}$$

Step 2. In this step, we cut Z by 16 general sections of degree 42 vanishing at p. Let $c:=(y_1(p),\dots,y_{18}(p))$, so $p\in F_c$. Let k and r be as in Definition-Lemma 5.2(2), and let $g:=2^{\max\{r-2,0\}}$. Let $W\subset R_{42}$ be the linear subspace spanned by $y_j-c_jx^{42}$, $1\le j\le 18$. Then $\varphi|_U\in\mathfrak{m}_p$ for any $\varphi\in W$, and $\operatorname{Bs}_U(W)=F_c\cap U$ is of dimension 7 by Definition-Lemma 5.2(1).

Let Z' be the sum of the terms of Z whose components contain p. Then $Z'\ge 0$ is of pure dimension 23, $(H^{23}\cdot Z')\le(H^{23}\cdot Z)$, and $\operatorname{mult}_p Z'=\operatorname{mult}_p Z$. If $Z'=0$, then $\operatorname{mult}_p Z=0$ and there is nothing to prove, so we may assume that $Z'\ne 0$. We construct a $\mathbb{Q}$-cycle $T\ge 0$ of pure dimension 7 whose components all contain p, with

$$(H^7\cdot T)\le 42^{16}(H^{23}\cdot Z)\quad\text{and}\quad \operatorname{mult}_p T\ge g\operatorname{mult}_p Z. \tag{5.5}$$

Suppose that $r\le 2$, so $g=1$. Since $\dim(\operatorname{Bs}_U(W)\cap\operatorname{Supp}Z')\le 7=23-16$, **Step 1** with $V=W$, $s=1$, and $j=16$ applied to Z' yields T satisfying (5.5).

Suppose that $r\ge 3$, so $g=2^{r-2}$. Let $W_2\subset W$ be the linear subspace of the elements φ such that the image of $\varphi|_U$ in $\mathfrak{m}_p/\mathfrak{m}_p^2$ is zero. By Definition-Lemma 5.2(2), the image of W in $\mathfrak{m}_p/\mathfrak{m}_p^2$ is of dimension $18-r$, so $\dim W_2=r$, and $\varphi|_U\in\mathfrak{m}_p^2$ for any $\varphi\in W_2$. Since $\dim(\operatorname{Bs}_U(W)\cap\operatorname{Supp}Z')\le 7\le 23-(18-r)$, we may apply **Step 1** with $V=W$, $s=1$, and $j=18-r$ to Z'. When choosing φ_i, we may further require that φ_i is not in the linear subspace of the elements of W whose images in $\mathfrak{m}_p/\mathfrak{m}_p^2$ belong to the span of the images of $\varphi_1|_U,\dots,\varphi_{i-1}|_U$. The linear subspace of such elements is proper, as $i-1<18-r$. Thus we obtain $\varphi_1,\dots,\varphi_{18-r}\in W$ whose images in $\mathfrak{m}_p/\mathfrak{m}_p^2$ are linearly independent, and a $\mathbb{Q}$-cycle $Z''\ge 0$ of pure dimension $5+r$ whose components all contain p, with

$$(H^{5+r}\cdot Z'')\le 42^{18-r}(H^{23}\cdot Z),\quad \operatorname{mult}_p Z''\ge\operatorname{mult}_p Z,\\ \operatorname{Supp}Z''\subset\operatorname{Supp}(\varphi_1=0)\cap\cdots\cap\operatorname{Supp}(\varphi_{18-r}=0).$$

Since W is the direct sum of W_2 and the span of $\varphi_1,\dots,\varphi_{18-r}$, we have

$$\operatorname{Bs}_U(W_2)\cap\operatorname{Supp}Z''\subset\operatorname{Bs}_U(W)=F_c\cap U.$$

Hence $\dim(\operatorname{Bs}_U(W_2)\cap\operatorname{Supp}Z'')\le 7=(5+r)-(r-2)$, and **Step 1** with $V=W_2$, $s=2$, and $j=r-2$ applied to Z'' yields T with

$$(H^7\cdot T)\le 42^{r-2}\cdot 42^{18-r}(H^{23}\cdot Z)\quad\text{and}\quad \operatorname{mult}_p T\ge 2^{r-2}\operatorname{mult}_p Z,$$

that is, T satisfies (5.5).

Step 3. In this step, we separate the multiple of F_c in T and bound the multiplicity of the rest. Since F_c is a variety of dimension 7, we may write

$$T=mF_c+T',$$

where $m\ge 0$ is a rational number and $T'\ge 0$ is a $\mathbb{Q}$-cycle of pure dimension 7 with no component equal to F_c. We show that

$$\operatorname{mult}_p T'\le 42\cdot 258^2\cdot 602^4(H^7\cdot T'). \tag{5.6}$$

We may assume that $T' \neq 0$. Any component Γ of T' is a variety of dimension 7 other than F_c, so $\dim(\Gamma \cap F_c) \leq 6$. Thus $\dim(\mathrm{Bs}_U(W) \cap \operatorname{Supp} T') \leq 6$, and **Step 1** with $V = W$, $s = 1$, and $j = 1$ yields a $\mathbb{Q}$-cycle $T'_1 \geq 0$ of pure dimension 6 whose components all contain p, with $(H^6 \cdot T'_1) \leq 42(H^7 \cdot T')$ and $\operatorname{mult}_p T'_1 \geq \operatorname{mult}_p T'$.

Let $V_{258} \subset R_{258}$ be the linear subspace spanned by

$$x^{216}(y_j - c_j x^{42}) \quad (1 \leq j \leq 18) \quad \text{and} \quad u_i - u_i(p)x^{258} \quad (9 \leq i \leq 11),$$

where $43d_i = 258$ for $9 \leq i \leq 11$. On U, the sections above restrict to $y_j - c_j$ and $u_i - u_i(p)$, which vanish at p. Thus $\mathrm{Bs}_U(V_{258})$ is the set of points $q \in U$ with $y_j(q) = c_j$ for $1 \leq j \leq 18$ and $u_i(q) = u_i(p)$ for $9 \leq i \leq 11$. On $\mathrm{Bs}_U(V_{258})$, the coordinates $u_1, \dots, u_8$ satisfy

$$\sum_{i=1}^{8} \boldsymbol{v}_i u_i^{e_i} = -C(c) - \sum_{i=9}^{11} \boldsymbol{v}_i u_i(p)^7.$$

Since $\boldsymbol{v}_1, \dots, \boldsymbol{v}_4$ are linearly independent, multiplying by the inverse of the matrix with columns $\boldsymbol{v}_1, \dots, \boldsymbol{v}_4$ shows that $\mathrm{Bs}_U(V_{258})$ is the zero set of four polynomials $u_l^{e_l} + f_l$, $1 \leq l \leq 4$, with $f_l \in \mathbb{C}[u_5, \dots, u_8]$. Hence $\mathrm{Bs}_U(V_{258})$ is finite over $\mathbb{A}^4$ with coordinates $u_5, \dots, u_8$, and $\dim \mathrm{Bs}_U(V_{258}) \leq 4 = 6 - 2$. **Step 1** with $V = V_{258}$, $s = 1$, and $j = 2$ yields a $\mathbb{Q}$-cycle $T'_2 \geq 0$ of pure dimension 4 whose components all contain p, with $(H^4 \cdot T'_2) \leq 258^2(H^6 \cdot T'_1)$ and $\operatorname{mult}_p T'_2 \geq \operatorname{mult}_p T'_1$.

Let $V_{602} \subset R_{602}$ be the linear subspace spanned by

$$x^{560}(y_j - c_j x^{42}) \quad (1 \leq j \leq 18), \qquad x^{344}(u_i - u_i(p)x^{258}) \quad (9 \leq i \leq 11),$$
$$\text{and} \quad u_i - u_i(p)x^{602} \quad (4 \leq i \leq 8),$$

where $43d_i = 602$ for $4 \leq i \leq 8$. On U, the sections above restrict to $y_j - c_j$ and $u_i - u_i(p)$ for $4 \leq i \leq 11$, which vanish at p. On $\mathrm{Bs}_U(V_{602})$, all coordinates other than u_1, u_2, u_3 are constant, and

$$\boldsymbol{v}_1 u_1^2 + \boldsymbol{v}_2 u_2^2 + \boldsymbol{v}_3 u_3^2 = -C(c) - \sum_{i=4}^{11} \boldsymbol{v}_i u_i(p)^{e_i}.$$

Since $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ are linearly independent, the values u_1^2, u_2^2, u_3^2 are constant on $\mathrm{Bs}_U(V_{602})$, so $\mathrm{Bs}_U(V_{602})$ is finite and $\dim \mathrm{Bs}_U(V_{602}) \leq 0 = 4 - 4$. **Step 1** with $V = V_{602}$, $s = 1$, and $j = 4$ yields a $\mathbb{Q}$-cycle $T'_3 \geq 0$ of dimension 0 whose components all contain p, with $(H^0 \cdot T'_3) \leq 602^4(H^4 \cdot T'_2)$ and $\operatorname{mult}_p T'_3 \geq \operatorname{mult}_p T'_2$. Thus $T'_3 = \operatorname{mult}_p T'_3 \cdot p$, and

$$\operatorname{mult}_p T' \leq \operatorname{mult}_p T'_3 = (H^0 \cdot T'_3) \leq 602^4 \cdot 258^2 \cdot 42(H^7 \cdot T'),$$

which is (5.6).

Step 4. In this step, we compute the constants and conclude the proof. We first compute (H^{25}). Let $w_0, \dots, w_{29}$ be the weights of $x, u_1, \dots, u_{11}, y_1, \dots, y_{18}$, so that $X \subset \mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ and $\prod_a w_a = 903^3 \cdot 602^5 \cdot 258^3 \cdot 42^{18}$. Since 1806 is divisible by any w_a, $\mathcal{O}(1806)$ is an invertible sheaf on $\mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$. The variety X is the proper intersection of four sections of $\mathcal{O}(1806)$, R is a domain (Proposition 3.4(1)), and the local ring of $\mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18})$ along X is regular since X is well formed (Proposition 3.4(1)), so the intersection multiplicity along X is one [Ful98, Example 7.1.10(a)]. Since $1806H$ is a Cartier divisor with $\mathcal{O}_X(1806H) \cong \mathcal{O}(1806)|_X$ (Set-up 3.3(5) and Proposition 3.4(3)), we have

$$(H^{25}) = 1806^{-25}(\mathcal{O}(1806)^{25} \cdot X) = 1806^{-25}(\mathcal{O}(1806)^{29}).$$

The morphism

$$\mathbb{P}^{29} \to \mathbb{P}(1, 903^3, 602^5, 258^3, 42^{18}), \quad [z_0 : \cdots : z_{29}] \to [z_0^{w_0} : \cdots : z_{29}^{w_{29}}],$$

is a finite morphism of degree $\prod_a w_a$ (a general fiber consists of $\prod_a w_a$ points), and it pulls back $\mathcal{O}(1806)$ to $\mathcal{O}_{\mathbb{P}^{29}}(1806)$. Thus $(\mathcal{O}(1806)^{29}) = 1806^{29}/\prod_a w_a$, and

$$(H^{25}) = \frac{1806^4}{903^3 \cdot 602^5 \cdot 258^3 \cdot 42^{18}}.$$

Since $1806 = 2 \cdot 903 = 3 \cdot 602$, we have

$$42 \cdot 258^2 \cdot 602^4 \cdot 42^{16}(H^{25}) = \frac{1806^4}{903^3 \cdot 602 \cdot 258 \cdot 42} = \frac{2^3 \cdot 3}{258 \cdot 42} = \frac{2}{903}. \tag{5.7}$$

By Definition-Lemma 5.2, $\operatorname{mult}_p F_c \le 6g$ and $(H^7 \cdot F_c) = 42^{18}(H^{25})$. Let

$$\kappa := \max\left\{\frac{6g}{42^{18}(H^{25})}, 42 \cdot 258^2 \cdot 602^4\right\}.$$

By (5.5), (5.6), and Definition 2.7,

$$\begin{aligned} g \operatorname{mult}_p Z &\le \operatorname{mult}_p T = m \operatorname{mult}_p F_c + \operatorname{mult}_p T' \\ &\le 6gm + 42 \cdot 258^2 \cdot 602^4 (H^7 \cdot T') \\ &\le \kappa\big(m(H^7 \cdot F_c) + (H^7 \cdot T')\big) = \kappa(H^7 \cdot T). \end{aligned}$$

By (5.5) and (5.7),

$$g \operatorname{mult}_p Z \le 42^{16}\kappa(H^{23} \cdot Z) = \max\left\{\frac{6g}{42^2}, \frac{2}{903}\right\}\frac{(H^{23} \cdot Z)}{(H^{25})}.$$

Since $6/42^2 = 1/294$, $g \ge 1$, and $2/903 < 1/294$, we have $\max\{6g/42^2, 2/903\} = g/294$. Dividing by g, we obtain

$$\operatorname{mult}_p Z \le \frac{(H^{23} \cdot Z)}{294(H^{25})}.$$

□

5.2. **Centers meeting U.** In this subsection, we prove Theorem 5.1.

The following lemma replaces a bound m on the order of a function along a center C by the bound $m(a(E,Y)-1)$, and not $ma(E,Y)$, on the multiplicity of the function along any prime divisor E over Y with center C, since canonicity compares multiplicities with $a(E,Y)-1$.

Lemma 5.4. *Let Y be a smooth variety, E a prime divisor over Y whose center $C := \operatorname{center}_Y E$ has codimension $r \ge 2$, $m \ge 0$ an integer, and φ a regular function near the generic point of C which does not belong to $\mathcal{I}_C^{m+1}$ at the generic point of C. Then $\operatorname{mult}_E \varphi \le m(a(E,Y)-1)$.*

Proof. Let $n := \dim Y$, and let m' be the largest integer such that $\varphi \in \mathcal{I}_C^{m'}$ at the generic point of C. Then $m' \le m$. Let p be a general closed point of C. Since C is smooth at p, there exist an open neighborhood $V \subset Y$ of p and regular functions $z_1, \dots, z_n$ on V such that $dz_1, \dots, dz_n$ form a basis of Ω_V and $\mathcal{I}_C|_V = (z_1, \dots, z_r)$ [Har77, Theorem II.8.17]. Possibly shrinking V, we may assume that φ is regular on V. Let $\partial_1, \dots, \partial_n$ be the derivations of $\mathcal{O}_V$ dual to $dz_1, \dots, dz_n$. Then

$$d\psi = \sum_{i=1}^{n} \partial_i \psi dz_i$$

for any regular function ψ on V, and the derivations ∂_i extend to the local ring $\mathcal{O}_{Y,C}$. For $\beta = (\beta_1, \dots, \beta_r) \in \mathbb{Z}^r_{\ge 0}$, we write $|\beta| := \sum_i \beta_i$, $\beta! := \prod_i \beta_i!$, $z^\beta := \prod_i z_i^{\beta_i}$, and

$$\partial^\beta := \partial_1^{\beta_1} \circ \cdots \circ \partial_r^{\beta_r}.$$

We first find a derivative of φ of order m' in the directions $z_1, \dots, z_r$ which is a unit at the generic point of C. Since $z_1, \dots, z_r$ is a regular sequence generating the maximal ideal $\mathcal{I}_C\mathcal{O}_{Y,C}$ of the regular local ring $\mathcal{O}_{Y,C}$, the quotient

$$\mathcal{I}_C^{m'}\mathcal{O}_{Y,C}/\mathcal{I}_C^{m'+1}\mathcal{O}_{Y,C}$$

is a free $\mathcal{O}_{Y,C}/\mathcal{I}_C\mathcal{O}_{Y,C}$-module with basis z^β, $|\beta| = m'$ [Har77, Theorem II.8.21A(e)]. Thus there exist $c_\beta \in \mathcal{O}_{Y,C}$ for $|\beta| = m'$ and $\chi \in \mathcal{I}_C^{m'+1}\mathcal{O}_{Y,C}$ such that

$$\varphi = \sum_{|\beta|=m'} c_\beta z^\beta + \chi.$$

Since $\varphi \notin \mathcal{I}_C^{m'+1}$ at the generic point of C, there exists β with $|\beta| = m'$ and $c_\beta \notin \mathcal{I}_C\mathcal{O}_{Y,C}$. We fix such β. By the Leibniz rule,

$$\partial_i(\mathcal{I}_C^{j+1}\mathcal{O}_{Y,C}) \subset \mathcal{I}_C^j\mathcal{O}_{Y,C} \quad \text{for any } i \text{ and any } j \geq 0.$$

In particular, $\partial^\beta\chi \in \mathcal{I}_C\mathcal{O}_{Y,C}$, and for any β' with $|\beta'| = m'$, the Leibniz rule implies

$$\partial^\beta(c_{\beta'}z^{\beta'}) - c_{\beta'}\partial^\beta(z^{\beta'}) \in \mathcal{I}_C\mathcal{O}_{Y,C},$$

because in any other term $z^{\beta'}$ is differentiated fewer than m' times. Since $\partial_i z_j = 1$ if $i = j$ and $\partial_i z_j = 0$ if $i \neq j$, we have $\partial^\beta(z^\beta) = \beta!$, and $\partial^\beta(z^{\beta'}) = 0$ for $\beta' \neq \beta$ with $|\beta'| = m'$. Thus

$$\partial^\beta\varphi - \beta! c_\beta \in \mathcal{I}_C\mathcal{O}_{Y,C},$$

so $\partial^\beta\varphi$ is a unit in $\mathcal{O}_{Y,C}$. Since $C = \operatorname{center}_Y E$, we have $\operatorname{mult}_E(\partial^\beta\varphi) = 0$.

We next bound the loss of multiplicity along E under one derivative. Let $h : Y' \to Y$ be a birational morphism from a normal variety Y' on which E is a divisor. Then Y' is smooth at the generic point of E, and h maps the generic point of E to the generic point of C, which belongs to V. Let $T, w_2, \dots, w_n$ be local coordinates of Y' at a general point of E such that $E = (T = 0)$. Since the coefficient of E in $K_{Y'} - h^*K_Y$ is $a(E, Y) - 1$ (Definition 2.1), near a general point of E we have

$$h^*(dz_1 \wedge \cdots \wedge dz_n) = T^{a(E,Y)-1}\epsilon dT \wedge dw_2 \wedge \cdots \wedge dw_n$$

for some unit ϵ. Let ψ be a regular function near the generic point of C and $b := \operatorname{mult}_E \psi$. Writing $\psi \circ h = T^b\epsilon'$ with ϵ' a unit near a general point of E, we have

$$h^*d\psi = \big(bT^{b-1}\epsilon' + T^b\partial\epsilon'/\partial T\big)dT + \sum_{l=2}^{n} T^b\partial\epsilon'/\partial w_l dw_l,$$

so the coefficient of dT in $h^*d\psi$ has multiplicity at least $b-1$ along E, and the coefficient of any dw_l has multiplicity at least b along E. For regular functions $\psi_1, \dots, \psi_n$ near the generic point of C, write

$$h^*(d\psi_1 \wedge \cdots \wedge d\psi_n) = \Theta dT \wedge dw_2 \wedge \cdots \wedge dw_n.$$

Then Θ is the determinant of the matrix of the coefficients of $h^*d\psi_1, \dots, h^*d\psi_n$, and any term in the expansion of the determinant contains exactly one coefficient of dT. Thus

$$\operatorname{mult}_E \Theta \geq \sum_{j=1}^{n} \operatorname{mult}_E \psi_j - 1. \tag{5.8}$$

Let $a_j := \operatorname{mult}_E z_j$ for $1 \leq j \leq n$. Then $a_j \geq 0$ for any j, and $a_j \geq 1$ for $j \leq r$ since $z_j \in \mathcal{I}_C$. Applying (5.8) to $z_1, \dots, z_n$, we obtain $a(E, Y) - 1 \geq \sum_j a_j - 1$, so

$$\delta := a(E, Y) - \sum_{j=1}^{n} a_j \geq 0 \quad \text{and} \quad a(E, Y) - 1 \geq r - 1 \geq 1.$$

Now let φ' be a regular function on V and $1 \leq i \leq n$ an integer such that $\partial_i\varphi' \neq 0$. Since

$$d\varphi' \wedge dz_1 \wedge \cdots \wedge \widehat{dz_i} \wedge \cdots \wedge dz_n = (-1)^{i-1}\partial_i\varphi' dz_1 \wedge \cdots \wedge dz_n,$$

applying (5.8) to the left-hand side and taking multiplicities along E of both sides of the pullback, we obtain

$$\operatorname{mult}_E(\partial_i\varphi') + a(E, Y) - 1 \geq \operatorname{mult}_E \varphi' + \sum_{j\neq i} a_j - 1,$$

that is,

$$\operatorname{mult}_E(\partial_i\varphi') \geq \operatorname{mult}_E \varphi' - a_i - \delta.$$

If $i \leq r$, then since $r \geq 2$, there exists $j \leq r$ with $j \neq i$, and $a_j \geq 1$. Thus

$$a_i + \delta = a(E,Y) - \sum_{j\neq i} a_j \leq a(E,Y) - 1 \quad \text{for any } 1 \leq i \leq r.$$

Finally, let $i_1, \ldots, i_{m'} \in \{1, \ldots, r\}$ be indices such that

$$\partial^\beta = \partial_{i_{m'}} \circ \cdots \circ \partial_{i_1},$$

and let $\varphi_0 := \varphi$ and $\varphi_t := \partial_{i_t}\varphi_{t-1}$ for $1 \leq t \leq m'$. Then $\varphi_{m'} = \partial^\beta\varphi \neq 0$, so $\varphi_t \neq 0$ for any t. For $1 \leq t \leq m'$, the inequalities $\operatorname{mult}_E(\partial_i\varphi') \geq \operatorname{mult}_E \varphi' - a_i - \delta$ and $a_i + \delta \leq a(E,Y) - 1$, applied with $\varphi' = \varphi_{t-1}$ and $i = i_t$, imply

$$\operatorname{mult}_E \varphi_{t-1} \leq \operatorname{mult}_E \varphi_t + a(E,Y) - 1.$$

Summing over t and using $\operatorname{mult}_E \varphi_{m'} = 0$, we obtain

$$\operatorname{mult}_E \varphi \leq m'(a(E,Y) - 1) \leq m(a(E,Y) - 1).$$

□

Lemma 5.6 below holds only for centers of codimension at least three, and along a center of codimension two the non-canonicity of $\mathcal{M}$ is too weak to contradict Proposition 5.3. We therefore bound directly the order of a divisor in $|\ell H|$ along any subvariety of U of dimension at least 15, which covers all subvarieties of U of codimension two, using [Suz17, Proposition 2.1]. Since [Suz17, Proposition 2.1] only considers subvarieties which do not meet the singular locus of the complete intersection, and the closure of U in $\mathbb{P}^{29}$ is singular along the linear subspace spanned by the u-coordinates, we first cut the closure of U with a general linear subspace of codimension 11.

Proposition 5.5. *Notation and conditions as in Set-up 3.3. Let $\ell > 0$ be an integer, $D \geq 0$ a Weil divisor on X with $D \sim \ell H$, and $Z \subset U$ an irreducible closed subvariety with $\dim Z \geq 15$. Then a local equation of $D|_U$ does not belong to $\mathcal{I}_Z^{\lfloor \ell/42 \rfloor + 1}$ at the generic point of Z.*

Proof. Since x has weight one, $U = X \cap (x \neq 0)$ is the closed subvariety of $\mathbb{A}^{29}$ with coordinates $u_1, \ldots, u_{11}, y_1, \ldots, y_{18}$ defined by

$$\boldsymbol{v}_0 + \sum_{i=1}^{11} \boldsymbol{v}_i u_i^{e_i} + \sum_{j=1}^{18} \boldsymbol{v}_{11+j} y_j^{43} = 0.$$

By Proposition 3.4(1), any four of $\boldsymbol{v}_0, \ldots, \boldsymbol{v}_{29}$ are linearly independent.

Step 1. In this step we show that $D|_U$ is the divisor of a polynomial in u_i, y_j of total degree at most $\lfloor \ell/42 \rfloor$.

By Proposition 3.4(4), D is the divisor of zeros of a non-zero element $f \in R_\ell$. Let $\tilde{f}$ be a weighted homogeneous polynomial in $x, u_1, \ldots, u_{11}, y_1, \ldots, y_{18}$ of weighted degree ℓ representing f, and let $g := \tilde{f}(1, u_1, \ldots, u_{11}, y_1, \ldots, y_{18})$. Since x is a unit on U, we have $D|_U = \operatorname{div}(g)$, and $g \neq 0$ on U since R is a domain (Proposition 3.4(1)). For any monomial $x^\alpha \prod_i u_i^{\beta_i} \prod_j y_j^{\gamma_j}$ of $\tilde{f}$, we have

$$\alpha + \sum_{i=1}^{11} 43 d_i \beta_i + 42 \sum_{j=1}^{18} \gamma_j = \ell.$$

Since $43d_i \geq 42$ for any i, we have $\sum_i \beta_i + \sum_j \gamma_j \leq \lfloor \ell/42 \rfloor$. Thus the total degree b of g as a polynomial in u_i, y_j satisfies $b \leq \lfloor \ell/42 \rfloor$.

Step 2. In this step we construct a projective closure W of U and show that W is smooth outside a linear subspace $\Pi \cong \mathbb{P}^{10}$.

Let $s, u_1, \dots, u_{11}, y_1, \dots, y_{18}$ be homogeneous coordinates of $\mathbb{P}^{29}$, so that U is a closed subvariety of $\mathbb{A}^{29}$, where $\mathbb{A}^{29} = \mathbb{P}^{29} \cap (s \neq 0)$. Let $W \subset \mathbb{P}^{29}$ be the closed subscheme defined by the homogenizations of the four equations of U with respect to the total degree of a polynomial in u_i, y_j, that is,

$$\sum_{j=1}^{18} \boldsymbol{v}_{11+j} y_j^{43} + \sum_{i=1}^{11} \boldsymbol{v}_i s^{43-e_i} u_i^{e_i} + \boldsymbol{v}_0 s^{43} = 0.$$

Then $W \cap (s \neq 0) = U$. Multiplying the four equations by the inverse of the matrix with columns $\boldsymbol{v}_{12}, \dots, \boldsymbol{v}_{15}$, we express $y_1^{43}, \dots, y_4^{43}$ as polynomials in the other variables. Thus the homogeneous coordinate ring of W is a free module over $\mathbb{C}[s, u_1, \dots, u_{11}, y_5, \dots, y_{18}]$ with basis $y_1^{b_1} \cdots y_4^{b_4}$, $0 \leq b_l \leq 42$, so W is a complete intersection of four hypersurfaces of degree 43 and $\dim W = 25$. Thus the ideal $I_W \subset \mathbb{C}[s, u_1, \dots, u_{11}, y_1, \dots, y_{18}]$ generated by the four homogenized equations has height 4, and since the polynomial ring is Cohen–Macaulay [Mat89, Theorem 17.7], by the unmixedness theorem [Mat89, Theorem 17.6], I_W has no embedded associated primes and any minimal prime of I_W has height 4. In particular, W has no embedded components and any irreducible component of W is of dimension 25. Setting $s = 0$, the homogeneous coordinate ring of $W \cap (s = 0)$ is a free module over $\mathbb{C}[u_1, \dots, u_{11}, y_5, \dots, y_{18}]$ with basis $y_1^{b_1} \cdots y_4^{b_4}$, $0 \leq b_l \leq 42$, so $\dim(W \cap (s = 0)) = 24$ and U is dense in W. Since U is smooth and irreducible (Proposition 3.4(1)(6)), W is integral and W is the closure of U in $\mathbb{P}^{29}$.

Let $\Pi := (s = y_1 = \cdots = y_{18} = 0) \subset \mathbb{P}^{29}$ be the linear subspace spanned by the u-coordinates. Then $\Pi \subset W$, and since $43 - e_i \geq 36$ for any i, all partial derivatives of the four equations vanish along Π. Let $q \in W \backslash \Pi$ be a closed point. If $s(q) \neq 0$, then $q \in U$ and W is smooth at q. If $s(q) = 0$, then some $y_j(q) \neq 0$ and $\sum_j \boldsymbol{v}_{11+j} y_j(q)^{43} = 0$. Since any four of the $\boldsymbol{v}_a$ are linearly independent, at least five of the $y_j(q)$ are non-zero, and for four such indices j, the partial derivatives $43 y_j(q)^{42} \boldsymbol{v}_{11+j}$ of the four equations with respect to y_j are linearly independent. By the Jacobian criterion [Har77, Theorem I.5.1], W is smooth at q. Thus $\operatorname{Sing} W = \Pi$.

Step 3. In this step we cut W with a general linear subspace of codimension 11 and conclude by [Suz17, Proposition 2.1].

Let $d := \dim Z \geq 15$, let $\overline{Z}$ be the closure of Z in W, and let m be the largest integer such that $g \in \mathcal{I}_Z^m$ at the generic point of Z. Then there exists an open subset $V \subset U$ such that $V \cap Z \neq \emptyset$ and $g|_V \in H^0(V, \mathcal{I}_{Z \cap V}^m)$. We show $m \leq b$, which implies the proposition by **Step 1**. If $b = 0$, then g is a non-zero constant and $m = 0$. Thus we may assume that $b \geq 1$. Let $\bar{g}$ be the homogenization of g of degree b, and let $\Gamma \geq 0$ be the Cartier divisor on W defined by $\bar{g}$, which is well defined since W is integral and $\bar{g}|_U = g \neq 0$. Then $\mathcal{O}_W(\Gamma) \cong \mathcal{O}_W(b)$, and g is a local equation of Γ on U.

Let $\Lambda \subset \mathbb{P}^{29}$ be a general linear subspace of codimension 11 and $Y := W \cap \Lambda$. Since $\dim \Pi = 10$, $\Lambda \cap \Pi = \emptyset$. By Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], applied 11 times to the base point free linear system $|\mathcal{O}(1)|$ on the smooth variety $W \backslash \Pi$, Y is smooth. Moreover, $\dim Y = 14$ [Har77, Theorem I.7.2 and Exercise I.1.8], so Y is a smooth complete intersection of four hypersurfaces of degree 43 in $\Lambda \cong \mathbb{P}^{18}$. Since $\dim Y \geq 1$, Y is connected [Har77, Exercise III.5.5(b)], hence integral. Since $\dim \operatorname{Supp} \Gamma = 24$ and Λ is general, $Y \not\subset \operatorname{Supp} \Gamma$, so $\Gamma|_Y \geq 0$ is a Cartier divisor on Y with $\mathcal{O}_Y(\Gamma|_Y) \cong \mathcal{O}_Y(b)$. Since Λ is general, $\overline{Z} \cap \Lambda$ is non-empty of pure dimension $d - 11 \geq 4$, and since $\dim(\overline{Z} \backslash V) \leq d - 1$, no irreducible component of $\overline{Z} \cap \Lambda$ is contained in $\overline{Z} \backslash V$ [Har77, Theorem I.7.2 and Exercise I.1.8].

Let S be an irreducible component of $\overline{Z} \cap \Lambda$. Then $S \subset Y$, $\dim S \geq 4$, and the generic point of S belongs to $V \cap Y$. Since $S \cap V \subset Z \cap V$, the local equation $g|_{V \cap Y}$ of $\Gamma|_Y$ belongs to $\mathcal{I}_S^m$ at the generic point of S. Since Y is smooth, the multiplicity along S of the cycle $[\Gamma|_Y]$ of $\Gamma|_Y$ is the order of a local equation of $\Gamma|_Y$ in the regular local ring $\mathcal{O}_{Y,S}$ [Ful98, Examples 4.3.4 and 4.3.9]. Thus

$$\operatorname{mult}_S [\Gamma|_Y] \geq m.$$

The number e_S in [Suz17, Definition 1.2] is the minimum over the closed points $p \in S$ of the multiplicity e_p at p, extended linearly to cycles, and differs from mult_S in Definition 2.7. Write $[\Gamma|_Y] = \sum_j c_j \Gamma_j$, where $c_j > 0$ and the Γ_j are distinct prime divisors on Y. Let $p \in S$ be a closed point. For any j with $S \subset \Gamma_j$, by [Ben70, Theorem (2)], applied to the local ring $\mathcal{O}_{\Gamma_j,p}$ and the prime ideal of S, the leading coefficients of the Hilbert–Samuel functions satisfy $\mathrm{mult}_p \Gamma_j \geq \mathrm{mult}_S \Gamma_j$ [Ful98, Example 4.3.4]. For any j with $S \not\subset \Gamma_j$, we have $\mathrm{mult}_S \Gamma_j = 0$. Thus

$$e_S([\Gamma|_Y]) \geq \mathrm{mult}_S[\Gamma|_Y] \geq m.$$

On the other hand, $[\Gamma|_Y]$ is an effective cycle of pure codimension one on Y which is rationally equivalent to $bc_1(\mathcal{O}_Y(1)) \cap [Y]$, and $\dim S \geq 4 = 1 \cdot 4$. [Suz17, Proposition 2.1] states that if $Y \subset \mathbb{P}^N$ is a complete intersection of t hypersurfaces and ξ is an effective cycle on Y of pure codimension k rationally equivalent to $bc_1(\mathcal{O}_Y(1))^k \cap [Y]$, and either Y is smooth or $kt + \dim \mathrm{Sing}\, Y + 1 < N$, then $e_S(\xi) \leq b$ for any closed subvariety $S \subset Y$ of dimension at least kt not meeting $\mathrm{Sing}\, Y$. By [Suz17, Proposition 2.1], applied with $k = 1$ and $t = 4$ to the smooth complete intersection $Y \subset \Lambda$, we have $e_S([\Gamma|_Y]) \leq b$. Thus $m \leq b \leq \lfloor \ell/42 \rfloor$. □

The $4n^2$-inequality [Puk17, Section 1, Theorem and Remark 1(iii)] is formulated for a movable linear system (mobile in the terminology of [Puk17]) on a smooth germ of dimension at least three which is not canonical at the closed point and canonical outside it (see the paragraph after [Puk17, Remark 1]). The following lemma reduces a non-canonical center of codimension at least three to a general transverse germ at a general point of the center.

Lemma 5.6. *Let Y be a smooth variety, $\mathcal{M}$ a movable linear system on Y, $\lambda > 0$ a rational number, and $C \subset Y$ an irreducible closed subvariety of codimension at least three. Assume that $\lambda \,\mathrm{mult}_E \mathcal{M} > a(E, Y) - 1$ for some prime divisor E over Y with $\mathrm{center}_Y E = C$, and that $\lambda \,\mathrm{mult}_{E'} \mathcal{M} \leq a(E', Y) - 1$ for any exceptional prime divisor E' over Y with $C \subsetneq \mathrm{center}_Y E'$. Then for general members $D_1, D_2 \in \mathcal{M}$, the multiplicity of the cycle $D_1 \cdot D_2$ at the generic point of C is greater than $4/\lambda^2$.*

Proof. The proof is long, so we divide it into several steps. Let $n := \dim Y$ and $c := \mathrm{codim}\, C \geq 3$. The assumptions and the conclusion only depend on a neighborhood of the generic point of C. Thus possibly replacing Y with an affine open subset intersecting C, we may assume that Y is affine. The conclusion is $\mathrm{mult}_C(D_1 \cdot D_2) > 4/\lambda^2$ (Definition 2.7).

Step 1. In this step we show that, after shrinking Y, the centers of the exceptional prime divisors G over Y with $\lambda \,\mathrm{mult}_G \mathcal{M} > a(G, Y) - 1$ are contained in C, and that C is the image of a prime divisor on a log resolution.

Let $g : Y'' \to Y$ be a log resolution of the base ideal of $\mathcal{M}$ which is an isomorphism over the complement of the base locus of $\mathcal{M}$. Write $g^*\mathcal{M} = \mathcal{M}'' + F$, where $F \geq 0$ is the fixed part of $g^*\mathcal{M}$ and $\mathcal{M}''$ is base point free, and let $E_1, \ldots, E_q$ be the prime divisors contained in $\mathrm{Exc}(g) \cup \mathrm{Supp}\, F$. Then $\sum_k E_k$ is an snc divisor, and since $\mathcal{M}$ is movable, the base locus of $\mathcal{M}$ is of codimension at least two, so any E_k is g-exceptional. For any prime divisor G over Y, a general member of $\mathcal{M}''$ does not contain $\mathrm{center}_{Y''} G$, so $\mathrm{mult}_G \mathcal{M} = \mathrm{mult}_G F$. Let

$$b_k := 1 - a(E_k, Y) + \lambda \,\mathrm{mult}_{E_k} F \quad \text{and} \quad \Delta'' := \sum_{k=1}^{q} b_k E_k.$$

Since $K_{Y''} = g^*K_Y + \sum_k (a(E_k, Y) - 1)E_k$, we have $K_{Y''} + \Delta'' = g^*K_Y + \lambda F$. Thus for any prime divisor G over Y,

$$a(G, Y) - 1 - \lambda \,\mathrm{mult}_G \mathcal{M} = a(G, Y'', \Delta'') - 1. \tag{5.9}$$

Let $N := \bigcup_{b_k > 0} g(E_k)$. By (5.9), $\lambda \,\mathrm{mult}_{E_k} \mathcal{M} - (a(E_k, Y) - 1) = b_k$, so any E_k with $b_k > 0$ is an exceptional prime divisor over Y with $\lambda \,\mathrm{mult}_{E_k} \mathcal{M} > a(E_k, Y) - 1$. Conversely, let G be an exceptional prime divisor over Y with $\lambda \,\mathrm{mult}_G \mathcal{M} > a(G, Y) - 1$, and suppose that $\mathrm{center}_{Y''} G$ is not contained in any E_k with $b_k > 0$. Then $\Delta'' \leq 0$ near the generic point of

$\operatorname{center}_{Y''} G$, so $a(G, Y'', \Delta'') \geq a(G, Y'')$. If G is exceptional over Y'', then $a(G, Y'') \geq 2$ since Y'' is smooth [KM98, Corollary 2.31(3)]. Otherwise G is a g-exceptional prime divisor on Y'', so $G = E_k$ for some k with $b_k \leq 0$, and $a(G, Y'', \Delta'') = 1 - b_k \geq 1$. In both cases (5.9) implies $\lambda \operatorname{mult}_G \mathcal{M} \leq a(G, Y) - 1$, a contradiction. Thus $\operatorname{center}_{Y''} G \subset E_k$ for some k with $b_k > 0$, and $\operatorname{center}_Y G \subset N$.

In particular, $C = \operatorname{center}_Y E \subset N$, so $C \subset g(E_k)$ for some k with $b_k > 0$. For any such k, E_k is an exceptional prime divisor over Y with $\lambda \operatorname{mult}_{E_k} \mathcal{M} > a(E_k, Y) - 1$, so the second assumption implies $g(E_k) = C$. Let N' be the union of the $g(E_k)$ with $b_k > 0$ and $C \not\subset g(E_k)$. Then $C \not\subset N'$. Possibly replacing Y with $Y \backslash N'$ and Y'' with $g^{-1}(Y \backslash N')$, we may assume that $N = C$. Moreover, there exists k_0 such that $b_{k_0} > 0$ and $g(E_{k_0}) = C$.

Step 2. In this step we construct a smooth germ (S, p) of dimension c transverse to C such that the restriction of $\mathcal{M}$ to S is non-canonical at p and canonical outside p.

If $\dim C = 0$, then let $S := Y$ and $\{p\} := C$. Otherwise, let A be a very ample divisor on Y, let $Y \subset \mathbb{P}^{N_0}$ be the embedding defined by A, let $H_1, \dots, H_{n-c} \in |A|$ be the restrictions of general hyperplanes $L_1, \dots, L_{n-c} \subset \mathbb{P}^{N_0}$, and let $S := H_1 \cap \dots \cap H_{n-c}$. By Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], applied on Y, on the smooth locus of C, on Y'' to the base point free linear system $g^*|A|$, and on the strata of $\sum_k E_k$, the following hold: S is smooth with $\dim S = c$; $S \cap C$ consists of smooth points of C at which S and C intersect transversally; $S'' := g^{-1}(S)$ is smooth; and $\sum_k E_k|_{S''}$ is an snc divisor on S'' whose prime components are the irreducible components of the $E_k \cap S''$, each of which is a component of $E_k \cap S''$ for exactly one k. Let $\overline{C}$ be the closure of C in $\mathbb{P}^{N_0}$. Since $\dim \overline{C} = n - c$, the set $\overline{C} \cap L_1 \cap \dots \cap L_{n-c}$ is non-empty and finite, and since $\dim(\overline{C} \backslash C) \leq n - c - 1$ and the L_t are general, we have

$$(\overline{C} \backslash C) \cap L_1 \cap \dots \cap L_{n-c} = \emptyset$$

[Har77, Theorem I.7.2 and Exercise I.1.8]. Thus $S \cap C = \overline{C} \cap L_1 \cap \dots \cap L_{n-c}$ is a non-empty finite set. Let $p \in S \cap C$. Possibly shrinking S to a neighborhood of p, we may assume that S is irreducible and $S \cap C = \{p\}$. Since S is general, p is a general closed point of C.

Let $g_S : S'' \to S$ be the morphism induced by g, and let

$$\mathcal{M}|_S := \{D|_S \mid D \in \mathcal{M},\ S \not\subset \operatorname{Supp} D\}.$$

Since S is general, S is not contained in the base locus of $\mathcal{M}$, the base locus of $\mathcal{M}|_S$ is of codimension at least two in S, and g_S is birational. Thus $\mathcal{M}|_S$ is movable, and $g_S^*(\mathcal{M}|_S) = \mathcal{M}''|_{S''} + F|_{S''}$ with $\mathcal{M}''|_{S''}$ base point free. By adjunction, $K_{S''} = (K_{Y''} + \sum_t g^* H_t)|_{S''}$ and $K_S = (K_Y + \sum_t H_t)|_S$, so

$$K_{S''} + \Delta''|_{S''} = g_S^* K_S + \lambda F|_{S''},$$

and the coefficient of any component of $E_k \cap S''$ in $\Delta''|_{S''}$ is b_k. Possibly replacing Y, Y'', $\mathcal{M}$, and the E_k with S, S'', $\mathcal{M}|_S$, and the irreducible components of the $E_k \cap S''$, the argument of **Step 1** implies that $\operatorname{center}_S G \subset N \cap S = \{p\}$ for any exceptional prime divisor G over S with $\lambda \operatorname{mult}_G(\mathcal{M}|_S) > a(G, S) - 1$. On the other hand, since $p \in C = g(E_{k_0})$, there exists an irreducible component G_0 of $E_{k_0} \cap S''$ with $p \in g(G_0)$. Since $g(G_0) \subset C \cap S = \{p\}$ and $\dim G_0 = c - 1 \geq 2$, G_0 is an exceptional prime divisor over S with $\operatorname{center}_S G_0 = p$, and the analogue of (5.9) on S implies

$$\lambda \operatorname{mult}_{G_0}(\mathcal{M}|_S) - (a(G_0, S) - 1) = b_{k_0} > 0.$$

Thus, in the sense of [Puk17, Remark 1(i)], the pair $(S, \lambda \mathcal{M}|_S)$ is not canonical at p and is canonical outside p.

Step 3. In this step we compare multiplicities and conclude.

Let $D_1, D_2 \in \mathcal{M}$ be general members and write $D_1 \cdot D_2 = \sum_i n_i Z_i$, where $n_i > 0$ and the Z_i are distinct irreducible closed subvarieties of codimension two. The conditions on Z_i used below hold for $(L_1, \dots, L_{n-c}, p, D_1, D_2)$ in a constructible subset of $\widehat{G} \times \mathcal{M} \times \mathcal{M}$, where $\widehat{G}$ is the irreducible variety of $(L_1, \dots, L_{n-c}, p)$ with $p \in C \cap L_1 \cap \dots \cap L_{n-c}$, whose fibers over a

dense open subset of $\mathcal{M}\times\mathcal{M}$ contain dense open subsets of $\widehat{G}$. Thus it contains a dense open subset of $\widehat{G}\times\mathcal{M}\times\mathcal{M}$, and since $(L_1,\dots,L_{n-c},p)$ is general ($\widehat{G}$ is a $(\mathbb{P}^{N_0-1})^{n-c}$-bundle over C, and $\widehat{G}\to((\mathbb{P}^{N_0})^\vee)^{n-c}$ is a dominant morphism with finite general fibers between varieties of dimension $(n-c)N_0$, so for general $L_1,\dots,L_{n-c}$ and any $p\in S\cap C$, $(L_1,\dots,L_{n-c},p)$ is a general point of $\widehat{G}$), we may choose D_1,D_2 general such that the conditions on Z_i used below hold. Then $D_1|_S$ and $D_2|_S$ are general members of $\mathcal{M}|_S$. Since S is smooth, $\dim S=c\geq 3$, and $\mathcal{M}|_S$ is movable, [Puk17, Section 1, Theorem and Remark 1(iii)] and the paragraph after [Puk17, Remark 1], applied to the pair $(S,\lambda\mathcal{M}|_S)$, imply

$$\operatorname{mult}_p\big((D_1|_S)\cdot(D_2|_S)\big)>\frac{4}{\lambda^2}$$

(the pair $(X,\frac{1}{n}\Sigma)$ of [Puk17] is $(S,\lambda\mathcal{M}|_S)$ here, so $4n^2=4/\lambda^2$). Thus it suffices to show $\operatorname{mult}_p((D_1|_S)\cdot(D_2|_S))=\operatorname{mult}_C(D_1\cdot D_2)$.

For any i, let $[Z_i\cap S]$ be the cycle of the scheme $Z_i\cap S$. Since $H_1,\dots,H_{n-c}$ are general, for any i and any $1\leq t\leq n-c$, H_t contains no associated point of $Z_i\cap H_1\cap\cdots\cap H_{t-1}$, so

$$[Z_i\cap S]=H_1\cdots H_{n-c}\cdot[Z_i]$$

[Ful98, Definition 2.3 and Lemma 1.7.2]. By the commutativity of intersections with Cartier divisors [Ful98, Corollary 2.4.2], we have

$$(D_1|_S)\cdot(D_2|_S)=\sum_i n_i[Z_i\cap S]\quad\text{in } A_{c-2}(\operatorname{Supp}D_1\cap\operatorname{Supp}D_2\cap S).$$

Near p, $\dim(\operatorname{Supp}D_1\cap\operatorname{Supp}D_2\cap S)=c-2$, so near p the cycles $(D_1|_S)\cdot(D_2|_S)$ and $\sum_i n_i[Z_i\cap S]$ are equal. If $C\not\subset Z_i$, then $p\notin Z_i$ since p is a general point of C. Fix i with $C\subset Z_i$, and let $\mathcal{C}_CZ_i$ be the normal cone of C in Z_i. By [Ful98, Section 4.3], $\operatorname{mult}_C Z_i$ is the degree of a general fiber of $\mathbb{P}(\mathcal{C}_CZ_i)\to C$, and $\operatorname{mult}_p[Z_i\cap S]$ is the degree of the projectivized tangent cone of $Z_i\cap S$ at p. Let $\sigma:Y_C\to Y$ be the blow-up of Y along C. Since S and C intersect transversally at p and $S\cap C=\{p\}$, the strict transform of S in Y_C is the blow-up of S at p, and the strict transform of Z_i is the blow-up of Z_i along C, whose exceptional divisor is $\mathbb{P}(\mathcal{C}_CZ_i)$. Since p is a general point of C, $\mathbb{P}(\mathcal{C}_CZ_i)\to C$ is flat over a neighborhood of p, and since S is general, the projectivized tangent cone of $Z_i\cap S$ at p is the fiber of $\mathbb{P}(\mathcal{C}_CZ_i)$ over p in $\mathbb{P}(T_pS)$. As cycles on Y_C, the exceptional divisor of the blow-up of $Z_i\cap S$ at p is

$$\sigma^*H_1\cdots\sigma^*H_{n-c}\cdot[\mathbb{P}(\mathcal{C}_CZ_i)]$$

by [Ful98, Corollary 2.4.2], that is, the fiber of $\mathbb{P}(\mathcal{C}_CZ_i)$ over p, of degree $\operatorname{mult}_C Z_i$ by flatness [Ful98, Section 4.3]. Thus $\operatorname{mult}_p[Z_i\cap S]=\operatorname{mult}_C Z_i$, and

$$\operatorname{mult}_p\big((D_1|_S)\cdot(D_2|_S)\big)=\sum_{C\subset Z_i}n_i\operatorname{mult}_C Z_i=\operatorname{mult}_C(D_1\cdot D_2).$$

□

Proof of Theorem 5.1. Let E be an exceptional prime divisor over X with $\operatorname{center}_X E\cap U\neq\emptyset$. Since U is open in X, we have $a(E,X)=a(E,U)$, and for any $D\in\mathcal{M}$, $\operatorname{mult}_E D=\operatorname{mult}_E\varphi_D$, where φ_D is a local equation of the Cartier divisor $D|_U$ near the generic point of $\operatorname{center}_X E\cap U$ (Definition 2.2). Since X has terminal singularities (Proposition 3.4(2)), $a(E,X)>1$.

Step 1. In this step we prove the theorem when $\dim\operatorname{center}_X E\geq 15$.

Let $Z:=\operatorname{center}_X E\cap U$. Then Z is an irreducible closed subvariety of U with $\dim Z\geq 15$, and $\operatorname{codim}_U Z\geq 2$ since E is exceptional. Let $D\in\mathcal{M}$ be a member such that $\operatorname{mult}_E D=\operatorname{mult}_E\mathcal{M}$. By Proposition 5.5, $\varphi_D\notin\mathcal{I}_Z^{\lfloor\ell/42\rfloor+1}$ at the generic point of Z. By Lemma 5.4, applied with $Y=U$ and $m=\lfloor\ell/42\rfloor$, we have $\operatorname{mult}_E\varphi_D\leq\lfloor\ell/42\rfloor(a(E,X)-1)$. Thus

$$\frac{26}{\ell}\operatorname{mult}_E\mathcal{M}=\frac{26}{\ell}\operatorname{mult}_E\varphi_D\leq\frac{26}{\ell}\Big\lfloor\frac{\ell}{42}\Big\rfloor(a(E,X)-1)\leq\frac{26}{42}(a(E,X)-1)<a(E,X)-1.$$

Step 2. In this step we prove the theorem when $\dim \operatorname{center}_X E \le 14$.

Suppose that the theorem does not hold. Among the exceptional prime divisors G over X with $\operatorname{center}_X G \cap U \ne \emptyset$ and $\frac{26}{\ell} \operatorname{mult}_G \mathcal{M} > a(G,X) - 1$, we pick one, denoted by E_0, with $\dim \operatorname{center}_X E_0$ maximal, and let $C := \operatorname{center}_X E_0$. By **Step 1**, applied to E_0 in place of E, $\dim C \le 14$, so $\operatorname{codim}_U(C \cap U) \ge 11 \ge 3$. Let E' be an exceptional prime divisor over U with $C \cap U \subsetneq \operatorname{center}_U E'$. Then $\operatorname{center}_X E'$ is the closure of $\operatorname{center}_U E'$ in X, $\operatorname{center}_X E' \cap U \ne \emptyset$, and $\dim \operatorname{center}_X E' > \dim C$. By the maximality of $\dim C$, we have $\frac{26}{\ell} \operatorname{mult}_{E'} \mathcal{M} \le a(E',X) - 1 = a(E',U) - 1$. The restriction of $\mathcal{M}$ to U is a movable linear system on the smooth variety U (Proposition 3.4(6)). By Lemma 5.6, applied with $Y = U$, with $C \cap U$ in place of C, and with $\lambda = 26/\ell$, for general members $D_1, D_2 \in \mathcal{M}$, the multiplicity of $(D_1|_U) \cdot (D_2|_U)$ at the generic point of $C \cap U$, which is $\operatorname{mult}_C(D_1 \cdot D_2)$, satisfies

$$\operatorname{mult}_C(D_1 \cdot D_2) > \frac{4\ell^2}{26^2} = \frac{\ell^2}{169}.$$

On the other hand, since $\mathcal{M}$ is movable, $D_1 \cdot D_2 = \sum_i n_i Z_i$ is an effective $\mathbb{Q}$-cycle of pure codimension two on X (Definition 2.7). Since $D_1 \sim D_2 \sim \ell H$, we have $(H^{23} \cdot D_1 \cdot D_2) = \ell^2 (H^{25})$ by Lemma 2.8(1), applied first with $T = X$ and $D = D_1$, and then with $T = D_1$ and $D = D_2$, where no component of D_1 is contained in $\operatorname{Supp} D_2$ since $\mathcal{M}$ is movable and D_1, D_2 are general. Let $p \in C \cap U$ be a closed point. For any i with $C \subset Z_i$, by [Ben70, Theorem (2)], applied to the local ring $\mathcal{O}_{Z_i,p}$ and the prime ideal of C, the leading coefficients of the Hilbert–Samuel functions satisfy $\operatorname{mult}_p Z_i \ge \operatorname{mult}_C Z_i$ [Ful98, Example 4.3.4]. Since $n_i \ge 0$ for any i, we have $\operatorname{mult}_p(D_1 \cdot D_2) \ge \operatorname{mult}_C(D_1 \cdot D_2)$. By Proposition 5.3, applied to the cycle $D_1 \cdot D_2$,

$$\frac{\ell^2}{169} < \operatorname{mult}_C(D_1 \cdot D_2) \le \operatorname{mult}_p(D_1 \cdot D_2) \le \frac{(H^{23} \cdot D_1 \cdot D_2)}{294 (H^{25})} = \frac{\ell^2}{294}.$$

Thus $\ell^2/169 < \ell^2/294$, a contradiction. □

6. Fibers of a diagonal morphism

In this section, we study a single fiber of the morphism h in Set-up 6.1 and the compactification of the fiber. Sections 8 and 9 consider families of varieties of the form $h(u) = c(y)$ on the charts $(y_j \ne 0)$ of P, and Section 7 continues the study of a single fiber of h.

Set-up 6.1. Let $\mathbf{w}_1, \dots, \mathbf{w}_{11}$, h, V_c, R_c, $\overline{V}_c$, R_∞, V_∞, and D_f be as follows, where $d_1, \dots, d_{11}$ and $e_1, \dots, e_{11}$ are as in Set-up 3.3.

(1) $\mathbf{w}_1, \dots, \mathbf{w}_{11} \in \mathbb{C}^4$ are vectors any four of which are linearly independent.
(2) $h : \mathbb{A}^{11} \to \mathbb{A}^4$ is the morphism given by $h(u) = \sum_{i=1}^{11} u_i^{e_i} \mathbf{w}_i$, and $V_c := h^{-1}(c)$ for $c \in \mathbb{C}^4$.
(3) For $c \in \mathbb{C}^4$, R_c is the graded ring

$$R_c := \mathbb{C}[W, U_1, \dots, U_{11}] \Big/ \Big(\sum_{i=1}^{11} U_i^{e_i} \mathbf{w}_i - cW^{42} \Big)$$

with $\deg W = 1$ and $\deg U_i = d_i$, and $\overline{V}_c := \operatorname{Proj} R_c$, which is a closed subscheme of $\mathbb{P}(1, 21^3, 14^5, 6^3)$.
(4) R_∞ is the graded ring

$$R_\infty := \mathbb{C}[U_1, \dots, U_{11}] \Big/ \Big(\sum_{i=1}^{11} U_i^{e_i} \mathbf{w}_i \Big)$$

with $\deg U_i = d_i$, and $V_\infty := \operatorname{Proj} R_\infty$, which is the closed subscheme $(W = 0)$ of $\overline{V}_c$ for any $c \in \mathbb{C}^4$, so that $\overline{V}_c \backslash V_\infty = \operatorname{Spec}(R_c[W^{-1}])_0$, which is $\operatorname{Spec} R_c/(W-1) = V_c$ since $\deg W = 1$ [IF00, 5.3].
(5) For a non-zero homogeneous element f of R_c or of R_∞, $D_f \ge 0$ is the Weil divisor of zeros of f on $\overline{V}_c$ or on V_∞, respectively.

We compactify the fibers of h in $\mathbb{P}(1, 21^3, 14^5, 6^3)$ because, for the rescaled family of compactified fibers in Construction 8.4, the center of a prime divisor over the total space of the family can be contained in the union of the copies of V_∞ in the fibers. Thus the lc conditions in Section 8 have to hold on all of $\overline{V}_c$, and in particular near V_∞.

6.1. **The fibers of h.** In this subsection, we prove that the fibers of h and their compactifications are normal varieties, that the different of V_∞ on $\overline{V}_c$ is zero, and that the fiber V_0 is not lc.

Proposition 6.2. *Notation and conditions as in Set-up 6.1. Then the following hold.*

1. *The morphism h is flat. For any $c \in \mathbb{C}^4$, R_c is a normal domain of dimension 8, and V_c and $\overline{V}_c$ are normal varieties of dimension 7.*
2. *For any $c \in \mathbb{C}^4$, V_c has only finitely many singular points, and any singular point of V_c has at most three non-zero coordinates.*
3. *R_∞ is a normal domain, and $\operatorname{Spec} R_\infty$ is smooth outside the origin. For any $1 \le i \le 11$ and any $\zeta \in \mu_{d_i} \backslash \{1\}$, the fixed locus of ζ on $\operatorname{Spec} R_\infty \cap (U_i = 1)$ is of dimension at most 3, where ζ multiplies U_j by ζ^{d_j}. In particular, the quotient morphism*
$$\operatorname{Spec} R_\infty \cap (U_i = 1) \to V_\infty \cap (U_i \neq 0)$$
by μ_{d_i} is étale in codimension one. For any $c \in \mathbb{C}^4$, V_∞ is a normal prime divisor on $\overline{V}_c$ with $(K_{\overline{V}_c} + V_\infty)|_{V_\infty} = K_{V_\infty}$.
4. *V_0 is not lc. More precisely, the exceptional divisor E of the weighted blow-up of the origin of V_0 with weights $d_1, \dots, d_{11}$ satisfies $a(E, V_0) = \sum_{j=1}^{11} d_j - 4 \cdot 42 = -17$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove that h is flat, that V_c is a complete intersection in $\mathbb{A}^{11}$ of pure dimension 7, and that $\operatorname{Spec} R_c$ is a complete intersection in $\mathbb{A}^{12}$ of pure dimension 8.

Fix four distinct indices $i_1, \dots, i_4$, for example $1, 2, 3, 4$, and let M be the 4×4 matrix with columns $\mathbf{w}_{i_1}, \dots, \mathbf{w}_{i_4}$. Then M is invertible. Let $A := \mathbb{C}[u_j \mid j \notin \{i_1, \dots, i_4\}]$, let $h_1, \dots, h_4$ be the components of h, and define $g_1, \dots, g_4 \in \mathbb{C}[u_1, \dots, u_{11}]$ by
$$(g_1, \dots, g_4)^\top := M^{-1}(h_1, \dots, h_4)^\top.$$
Then $g_l - u_{i_l}^{e_{i_l}}$ is the l-th entry of $M^{-1} \sum_{j \notin \{i_1, \dots, i_4\}} u_j^{e_j} \mathbf{w}_j$, so $g_l - u_{i_l}^{e_{i_l}} \in A$ for $1 \le l \le 4$. Since $\mathbb{C}[u_1, \dots, u_{11}] = A[u_{i_1}, \dots, u_{i_4}]$, the ring $\mathbb{C}[u_1, \dots, u_{11}]$ is a free module over the polynomial ring $A[g_1, \dots, g_4]$ with basis
$$u_{i_1}^{b_1} u_{i_2}^{b_2} u_{i_3}^{b_3} u_{i_4}^{b_4}, \quad 0 \le b_l < e_{i_l} \text{ for } 1 \le l \le 4.$$
Since M is invertible, $A[g_1, \dots, g_4] = A[h_1, \dots, h_4]$, and $A[h_1, \dots, h_4]$ is a polynomial ring over $\mathbb{C}[h_1, \dots, h_4]$. Hence $\mathbb{C}[u_1, \dots, u_{11}]$ is a free module over $\mathbb{C}[h_1, \dots, h_4]$, so h is flat. Moreover, for any $c \in \mathbb{C}^4$, the coordinate ring
$$\mathbb{C}[u_1, \dots, u_{11}]/(h_1 - c_1, \dots, h_4 - c_4) = \mathbb{C}[u_1, \dots, u_{11}]/\big(g_l - (M^{-1}c)_l \mid 1 \le l \le 4\big)$$
of V_c is a free A-module with basis $u_{i_1}^{b_1} \cdots u_{i_4}^{b_4}$, $0 \le b_l < e_{i_l}$. Thus V_c is a complete intersection in $\mathbb{A}^{11}$ of pure dimension 7.

Similarly, for $c \in \mathbb{C}^4$, define $G_1, \dots, G_4 \in \mathbb{C}[W, U_1, \dots, U_{11}]$ by
$$(G_1, \dots, G_4)^\top := M^{-1}\Big(\sum_{i=1}^{11} U_i^{e_i} \mathbf{w}_i - cW^{42}\Big).$$
Let $A' := \mathbb{C}[W, U_j \mid j \notin \{i_1, \dots, i_4\}]$. Then $G_l - U_{i_l}^{e_{i_l}} \in A'$ for $1 \le l \le 4$, so the ring
$$R_c = \mathbb{C}[W, U_1, \dots, U_{11}]/(G_1, \dots, G_4)$$
is a free module over the polynomial ring A' in 8 variables with basis $U_{i_1}^{b_1} \cdots U_{i_4}^{b_4}$, $0 \le b_l < e_{i_l}$. Thus $\dim R_c = 8$, $\operatorname{Spec} R_c$ is a complete intersection in $\mathbb{A}^{12}$ of pure dimension 8, and W is a nonzerodivisor in R_c.

Step 2. In this step, we prove (2).

Let $Y \subset \mathbb{A}^n$ be the closed subscheme defined by $f_1, \dots, f_r$ and $y \in Y$ a closed point at which the Jacobian matrix $(\partial f_l/\partial x_k)(y)$ has rank r. Then the images of $f_1, \dots, f_r$ in $\mathfrak{m}/\mathfrak{m}^2$, where $\mathfrak{m}$ is the maximal ideal of $\mathbb{A}^n$ at y, are linearly independent, so $\mathcal{O}_{Y,y}$ has embedding dimension $n-r$. Since $\dim \mathcal{O}_{Y,y} \geq n-r$ [Mat89, Theorem 13.6(ii)], applied to the maximal ideal of $\mathcal{O}_{\mathbb{A}^n,y}$, which has height n, $\mathcal{O}_{Y,y}$ is a regular local ring of dimension $n-r$, that is, Y is smooth at y of dimension $n-r$.

Let $c \in \mathbb{C}^4$ and $u \in V_c$ a closed point. The Jacobian matrix of h at u has columns

$$\frac{\partial h}{\partial u_i}(u) = e_i u_i^{e_i-1}\mathbf{w}_i, \quad 1 \leq i \leq 11.$$

If at least four coordinates of u are non-zero, then four of the columns are non-zero multiples of four of the vectors $\mathbf{w}_1, \dots, \mathbf{w}_{11}$, so the Jacobian matrix has rank 4. Since V_c is defined by four equations in $\mathbb{A}^{11}$, V_c is smooth at u by the argument at the beginning of **Step 2**. Hence any singular point of V_c has at most three non-zero coordinates.

Let u be a singular point of V_c and $I := \{i \mid u_i \neq 0\}$. Then $\#I \leq 3$ and

$$c = \sum_{i \in I} u_i^{e_i}\mathbf{w}_i.$$

Since the vectors $\mathbf{w}_i$, $i \in I$, are linearly independent, c determines the numbers $u_i^{e_i}$, $i \in I$. Thus for any subset $I \subset \{1, \dots, 11\}$ with $\#I \leq 3$, there exist at most $\prod_{i \in I} e_i$ singular points u of V_c such that $\{i \mid u_i \neq 0\} = I$. Since there exist only finitely many such subsets I, V_c has only finitely many singular points.

Step 3. In this step, we prove (1).

By **Step 1**, h is flat and V_c is a complete intersection in $\mathbb{A}^{11}$ of pure dimension 7. By **Step 2**, V_c is smooth outside finitely many points, so V_c is normal by [Har77, Proposition II.8.23].

Since $d_i e_i = 42$ for any i, we have

$$\sum_{i=1}^{11} (u_i W^{d_i})^{e_i}\mathbf{w}_i - cW^{42} = W^{42}(h(u) - c).$$

Hence the homomorphism

$$R_c[W^{-1}] \to \mathbb{C}[u_1, \dots, u_{11}, W, W^{-1}]/(h_1 - c_1, \dots, h_4 - c_4)$$

given by $U_i \to u_i W^{d_i}$ and $W \to W$ is an isomorphism, with inverse given by $u_i \to U_i W^{-d_i}$. Thus $\operatorname{Spec} R_c \cap (W \neq 0) \cong V_c \times \mathbb{G}_m$, and by **Step 2** the singular locus Σ of $\operatorname{Spec} R_c \cap (W \neq 0)$ satisfies $\dim \Sigma \leq 1$. Let $(0, a)$ be a closed point of $\operatorname{Spec} R_c \cap (W = 0)$ other than the origin, where $a = (a_1, \dots, a_{11})$. Then a is a closed point of $\operatorname{Spec} R_\infty$ other than the origin, so at least five coordinates of a are non-zero by Lemma 3.2(2), applied with $r = 4$, $N = 11$, and $e'_j = e_j$. The partial derivatives of $\sum_{i=1}^{11} U_i^{e_i}\mathbf{w}_i - cW^{42}$ with respect to $U_1, \dots, U_{11}$ at $(0, a)$ are $e_i a_i^{e_i-1}\mathbf{w}_i$, so the Jacobian matrix at $(0, a)$ has rank 4 as in **Step 2**, and $\operatorname{Spec} R_c$ is smooth at $(0, a)$ by the argument at the beginning of **Step 2**. Therefore the singular locus of $\operatorname{Spec} R_c$ is contained in the union of Σ and the origin. Since $\dim R_c = 8$ and $\operatorname{Spec} R_c$ is a complete intersection in $\mathbb{A}^{12}$, R_c is normal by [Har77, Proposition II.8.23].

Since R_c is positively graded with $(R_c)_0 = \mathbb{C}$, R_c is a normal domain by Lemma 3.1. Since $W \neq 0$ in R_c, the ring $R_c[W^{-1}]$ is a domain, and the inverse isomorphism maps the coordinate ring of V_c isomorphically onto the subring of $R_c[W^{-1}]$ generated by $U_1 W^{-d_1}, \dots, U_{11} W^{-d_{11}}$. Thus V_c is a normal variety of dimension 7.

For a homogeneous element $f \in R_c$ of positive degree, the ring $(R_c[f^{-1}])_0$ is the intersection of the normal domain $R_c[f^{-1}]$ with the fraction field of $(R_c[f^{-1}])_0$, so $(R_c[f^{-1}])_0$ is a normal domain. Since $\overline{V}_c$ is covered by the open subsets $\operatorname{Spec}(R_c[f^{-1}])_0$, $\overline{V}_c$ is a normal variety, and $\dim \overline{V}_c = \dim R_c - 1 = 7$. This proves (1).

Step 4. In this step, we prove all statements of (3) except the last one.

By Lemma 3.2(2)(3), applied with $r = 4$, $N = 11$, and $e'_j = e_j \geq 2$, R_∞ is a normal domain and $\operatorname{Spec} R_\infty$ is smooth outside the origin.

Fix $1 \leq i \leq 11$, and let $\widetilde{S}_i := \operatorname{Spec} R_\infty \cap (U_i = 1)$. The affine piece $(U_i \neq 0)$ of the ambient weighted projective space $\operatorname{Proj} \mathbb{C}[U_1, \dots, U_{11}]$ is the quotient of $(U_i = 1) \cong \mathbb{A}^{10}$ by μ_{d_i} acting by $\zeta \cdot U_j = \zeta^{d_j} U_j$ [IF00, 5.3]. Since taking μ_{d_i}-invariants is exact, the closed subscheme $V_\infty \cap (U_i \neq 0)$ is the quotient of $\widetilde{S}_i$ by μ_{d_i}, that is,

$$(R_\infty[U_i^{-1}])_0 = (R_\infty/(U_i - 1))^{\mu_{d_i}}.$$

The scheme $\widetilde{S}_i$ is defined in $\mathbb{A}^{10}$ with coordinates U_j, $j \neq i$, by the four equations

$$\mathbf{w}_i + \sum_{j \neq i} U_j^{e_j} \mathbf{w}_j = 0.$$

A closed point $s \in \widetilde{S}_i$ is a closed point of $\operatorname{Spec} R_\infty$ other than the origin, so at least five coordinates of s are non-zero by Lemma 3.2(2). Thus $U_j(s) \neq 0$ for at least four indices $j \neq i$, and the Jacobian matrix at s, whose columns are $e_j U_j(s)^{e_j - 1} \mathbf{w}_j$, $j \neq i$, has rank 4. Hence, by the argument at the beginning of **Step 2**, $\widetilde{S}_i$ is smooth with $\dim \widetilde{S}_i = 6$, and the tangent space $T_s \widetilde{S}_i \subset T_s \mathbb{A}^{10}$ is the kernel of the Jacobian matrix at s.

Let $\zeta \in \mu_{d_i} \backslash \{1\}$, $m > 1$ the order of ζ, and $I_\zeta := \{j \mid \zeta^{d_j} \neq 1\}$. Since $\gcd(21, 14, 6) = 1$, m does not divide at least one of 21, 14, and 6, so I_ζ contains at least one of the sets $\{1, 2, 3\}$, $\{4, \dots, 8\}$, and $\{9, 10, 11\}$. In particular, $\#I_\zeta \geq 3$, and $i \notin I_\zeta$ because $\zeta^{d_i} = 1$. Let $s \in \widetilde{S}_i$ be a closed point fixed by ζ. Then $U_j(s) = 0$ for any $j \in I_\zeta$. Since $e_j \geq 2$, the column $e_j U_j(s)^{e_j - 1} \mathbf{w}_j$ of the Jacobian matrix at s is zero for any $j \in I_\zeta$, so $\partial/\partial U_j \in T_s \widetilde{S}_i$ for any $j \in I_\zeta$. The differential of ζ at s multiplies $\partial/\partial U_j$ by $\zeta^{d_j} \neq 1$. The tangent space at s of the fixed locus $\widetilde{S}_i^\zeta$ of ζ is contained in the ζ-invariant subspace of $T_s \widetilde{S}_i$, whose dimension is at most $6 - \#I_\zeta \leq 3$. Hence $\dim \widetilde{S}_i^\zeta \leq 3$ for any $\zeta \in \mu_{d_i} \backslash \{1\}$. In particular, μ_{d_i} acts freely on the complement in $\widetilde{S}_i$ of a closed subset of codimension at least 3. Since the quotient morphism by a free action of a finite group is étale, the quotient morphism $\widetilde{S}_i \to V_\infty \cap (U_i \neq 0)$ by μ_{d_i} is étale in codimension one.

Step 5. In this step, we prove the last statement of (3).

Fix $c \in \mathbb{C}^4$. By **Step 4** and Lemma 3.2(1), R_∞ is a normal domain with $\dim R_\infty = 7$. By the argument for $\overline{V}_c$ in the last paragraph of **Step 3**, V_∞ is a normal variety of dimension 6. Since V_∞ is the closed subscheme $(W = 0)$ of $\overline{V}_c$ and $\dim \overline{V}_c = 7$, V_∞ is a normal prime divisor on $\overline{V}_c$.

Let $v \in V_\infty$ be a closed point. Since $W(v) = 0$, there exists $1 \leq i \leq 11$ such that $U_i(v) \neq 0$. Let $\widetilde{V}_i := \operatorname{Spec} R_c \cap (U_i = 1)$. The affine piece $(U_i \neq 0)$ of $\mathbb{P}(1, 21^3, 14^5, 6^3)$ is the quotient of $(U_i = 1) \cong \mathbb{A}^{11}$ by μ_{d_i} acting by $\zeta \cdot W = \zeta W$ and $\zeta \cdot U_j = \zeta^{d_j} U_j$ [IF00, 5.3]. Since taking μ_{d_i}-invariants is exact, the natural morphism $q_i : \widetilde{V}_i \to \overline{V}_c \cap (U_i \neq 0)$ is the quotient morphism by μ_{d_i}. Since W^{d_i} is μ_{d_i}-invariant and $V_\infty \cap (U_i \neq 0)$ is the zero locus of W^{d_i}/U_i, we have $q_i^{-1}(V_\infty) = \widetilde{V}_i \cap (W = 0) = \widetilde{S}_i$ as sets. The scheme $\widetilde{V}_i$ is defined in $\mathbb{A}^{11}$ with coordinates W and U_j, $j \neq i$, by the four equations

$$\mathbf{w}_i + \sum_{j \neq i} U_j^{e_j} \mathbf{w}_j - cW^{42} = 0.$$

At a closed point $t \in \widetilde{S}_i$, at least four coordinates $U_j(t)$ with $j \neq i$ are non-zero (**Step 4**), so the Jacobian matrix of the four equations at t has rank 4, and $\widetilde{V}_i$ is smooth at t with $\dim_t \widetilde{V}_i = 7$ by the argument at the beginning of **Step 2**.

Let $\widetilde{V}$ be the connected component of the smooth locus of $\widetilde{V}_i$ that contains a point of $q_i^{-1}(v)$, $G \subset \mu_{d_i}$ the stabilizer of $\widetilde{V}$, $q := q_i|_{\widetilde{V}}$, and $V^\circ := q(\widetilde{V})$. Then $\widetilde{V}$ is a smooth variety, V° is an open neighborhood of v, and q induces an isomorphism $\widetilde{V}/G \cong V^\circ$ (since q_i is open and $q_i^{-1}(V^\circ) = \mu_{d_i} \cdot \widetilde{V}$ is the disjoint union of the translates of $\widetilde{V}$, which are connected components

of the smooth locus of $\widetilde{V}_i$). Moreover, the subset $\widetilde{S} := q^{-1}(V_\infty \cap V^\circ)$ is equal to $\widetilde{V} \cap \widetilde{S}_i$ and is open in $\widetilde{S}_i$, so $\widetilde{S}$ is smooth, and $q|_{\widetilde{S}}$ is the quotient morphism $\widetilde{S} \to V_\infty \cap V^\circ$ by G.

Let $\zeta \in G\backslash\{1\}$ and $t \in \widetilde{V}$ a closed point fixed by ζ. Then $W(t) = 0$ because $\zeta \cdot W = \zeta W$, so $t \in \widetilde{S}$, and $U_j(t) = 0$ for any $j \in I_\zeta$. The column of the Jacobian matrix at t corresponding to W is $-42cW(t)^{41} = 0$, and the columns corresponding to U_j, $j \in I_\zeta$, are zero as in **Step 4**. Thus $T_t\widetilde{V}$ contains $\partial/\partial W$ and $\partial/\partial U_j$, $j \in I_\zeta$, which the differential of ζ multiplies by $\zeta \neq 1$ and by $\zeta^{d_j} \neq 1$, respectively. Hence the fixed locus of ζ in $\widetilde{V}$ is of dimension at most $7 - 1 - \#I_\zeta \leq 3$, and the fixed locus of ζ in $\widetilde{S}$ is of dimension at most 3 by **Step 4**. Therefore G acts faithfully on $\widetilde{V}$, q is étale outside a closed subset of codimension at least 4, and $q|_{\widetilde{S}}$ is étale outside a closed subset of codimension at least 3. In particular, q and $q|_{\widetilde{S}}$ are étale in codimension one.

By Lemma 2.4, applied to $\overline{V}_c$, V_∞, v, V°, $\widetilde{V}$, and G, we have $(K_{\overline{V}_c} + V_\infty)|_{V_\infty} = K_{V_\infty}$ near v. Since $v \in V_\infty$ is an arbitrary closed point, $(K_{\overline{V}_c} + V_\infty)|_{V_\infty} = K_{V_\infty}$. This proves (3).

Step 6. In this step, we prove (4) (cf. Lemma 6.8(3)).

We regard $\operatorname{Spec} R_\infty$ as V_0 via $u_i = U_i$, so that V_0 is the affine cone over $V_\infty = \operatorname{Proj} R_\infty$. By (1) and **Step 4**, V_0 is a normal variety of dimension 7, R_∞ is a domain, and $V_0\backslash\{0\}$ is smooth. Let ω be the 7-form on $V_0\backslash\{0\}$ with

$$\omega \wedge dh_1 \wedge \cdots \wedge dh_4 = du_1 \wedge \cdots \wedge du_{11}$$

along $V_0\backslash\{0\}$, where $h_1, \dots, h_4$ are the components of h as in **Step 1**. Since R_∞ is a domain, $h_1, \dots, h_4$ generate the ideal $\mathcal{J}$ of V_0 in $\mathbb{A}^{11}$. Since $V_0\backslash\{0\}$ is a smooth variety of codimension 4 in $\mathbb{A}^{11}\backslash\{0\}$, the images of $h_1, \dots, h_4$ form a basis of $\mathcal{J}/\mathcal{J}^2$ over $V_0\backslash\{0\}$, and ω is nowhere vanishing [Har77, Theorem II.8.17, Proposition II.8.20]. Since V_0 is normal, $K_{V_0} = \operatorname{div}(\omega) = 0$.

Let $\psi : V_0' \to V_0$ be the weighted blow-up of the origin of V_0 with weights $d_1, \dots, d_{11}$, and let $R_\infty[\xi]$ be graded by $\deg U_j = d_j$ and $\deg \xi = -1$. By the definition of the weighted blow-up,

$$V_0' = \operatorname{Proj} \bigoplus_{m \geq 0} (R_\infty)_{\geq m},$$

where $(R_\infty)_{\geq m}$ is the ideal of R_∞ generated by the homogeneous elements of degree at least m. For $m \geq 0$, setting $\xi = 1$ maps $R_\infty[\xi]_m$ bijectively onto $(R_\infty)_{\geq m}$, compatibly with multiplication. Thus $V_0' = \operatorname{Proj} R_\infty[\xi]_{\geq 0}$ and $R_\infty[\xi]_0 \cong R_\infty$, under which the function u_j on V_0 corresponds to $U_j\xi^{d_j}$. The exceptional divisor E is the closed subscheme $(\xi = 0)$ of V_0', so $E \cong \operatorname{Proj} R_\infty = V_\infty$, which is a normal variety of dimension 6 by **Step 5**. For $1 \leq l \leq 11$, the ideal $(R_\infty)_{\geq m}$ contains u_l^m for any $m \geq 0$, so ψ is an isomorphism over $V_0 \cap (u_l \neq 0)$. Since the open subsets $V_0 \cap (u_l \neq 0)$, $1 \leq l \leq 11$, cover $V_0\backslash\{0\}$, ψ is an isomorphism over $V_0\backslash\{0\}$. Since $R_\infty[\xi]_{\geq 0}$ is a domain, V_0' is a variety, so ψ is birational and E is a prime divisor on V_0'.

Fix $1 \leq l \leq 11$. By **Step 4**, $\widetilde{S}_l = \operatorname{Spec} R_\infty \cap (U_l = 1)$ is smooth of pure dimension 6, and it is the closed subscheme of $\mathbb{A}^{10}$ with coordinates U_j, $j \neq l$, defined by $F_1 = \cdots = F_4 = 0$, where

$$(F_1, \dots, F_4) := \mathbf{w}_l + \sum_{j \neq l} U_j^{e_j} \mathbf{w}_j.$$

Setting $U_l = 1$ induces an isomorphism from $(R_\infty[\xi][U_l^{-1}])_0$ onto the ring of invariants of $(R_\infty/(U_l - 1))[\xi]$ under the action of μ_{d_l} given by $\zeta \cdot U_j = \zeta^{d_j} U_j$ and $\zeta \cdot \xi = \zeta^{-1}\xi$, whose inverse sends the image of a homogeneous element b of degree nd_l, $n \in \mathbb{Z}$, to b/U_l^n. The inclusion of the ring of invariants defines the quotient morphism

$$\varrho_l : \widetilde{S}_l \times \mathbb{A}^1 \to D_+(U_l) = \operatorname{Spec}(R_\infty[\xi][U_l^{-1}])_0$$

by μ_{d_l}, where $D_+(U_l)$ is an open subset of V_0'. Since $\widetilde{S}_l \times \mathbb{A}^1$ is smooth, $D_+(U_l)$ is normal. Let $\Phi_l := \psi \circ \varrho_l$. Since u_j corresponds to $U_j\xi^{d_j}$, Φ_l is the restriction of the morphism $\mathbb{A}^{10} \times \mathbb{A}^1 \to \mathbb{A}^{11}$ given by $u_l = \xi^{d_l}$ and $u_j = \xi^{d_j} U_j$ for $j \neq l$. Moreover, $\varrho_l^{-1}(E) = \widetilde{S}_l \times \{0\}$, since the image of the ideal $(\xi R_\infty[\xi][U_l^{-1}])_0$ of $E \cap D_+(U_l)$ is contained in (ξ) and contains ξ^{d_l}, the image of $U_l\xi^{d_l}$. Let

$\zeta \in \mu_{d_l}\backslash\{1\}$. A point of $\widetilde{S}_l \times \mathbb{A}^1$ fixed by ζ is of the form $(x, 0)$ for a point $x \in \widetilde{S}_l$ fixed by ζ, so the fixed locus of ζ on $\widetilde{S}_l \times \mathbb{A}^1$ is of dimension at most 3 by **Step 4**. Thus μ_{d_l} acts freely outside a closed subset of $\widetilde{S}_l \times \mathbb{A}^1$ of codimension at least 4, and ϱ_l is étale in codimension one.

Let ω_l be the 6-form on $\widetilde{S}_l$ with

$$\omega_l \wedge dF_1 \wedge \cdots \wedge dF_4 = \bigwedge_{j\neq l} dU_j$$

along $\widetilde{S}_l$. Any connected component C of $\widetilde{S}_l$ is a smooth irreducible closed subvariety of codimension 4 of the open subset $\mathbb{A}^{10}\backslash(\widetilde{S}_l\backslash C)$, whose ideal there is generated by $F_1, \ldots, F_4$. Thus ω_l is nowhere vanishing [Har77, Theorem II.8.17, Proposition II.8.20]. Since u_l pulls back to ξ^{d_l} and u_j pulls back to $\xi^{d_j}U_j$ for $j \neq l$, we obtain

$$\Phi_l^*(du_1 \wedge \cdots \wedge du_{11}) = \pm d_l \xi^{\sum_{j=1}^{11} d_j - 1} d\xi \wedge \bigwedge_{j\neq l} dU_j,$$

and h_k pulls back to $\xi^{42}F_k$ for $1 \leq k \leq 4$. Since F_k vanishes on $\widetilde{S}_l$, $d(\xi^{42}F_k) = \xi^{42}dF_k$ along $\widetilde{S}_l \times \mathbb{A}^1$. Thus, as $\sum_{j=1}^{11} d_j = 151$,

$$\Phi_l^*\omega = \pm d_l \xi^{\sum_{j=1}^{11} d_j - 1 - 4\cdot 42} d\xi \wedge \omega_l = \pm d_l \xi^{-18} d\xi \wedge \omega_l.$$

Since $d\xi \wedge \omega_l$ is nowhere vanishing on $\widetilde{S}_l \times \mathbb{A}^1$, we obtain

$$K_{\widetilde{S}_l \times \mathbb{A}^1} = \Phi_l^* K_{V_0} - 18(\xi = 0).$$

Since ϱ_l is étale in codimension one, $K_{\widetilde{S}_l \times \mathbb{A}^1} = \varrho_l^* K_{V_0'}$ and $\varrho_l^* E = (\xi = 0)$. Hence $K_{V_0'} = \psi^* K_{V_0} - 18E$ on $D_+(U_l)$, and $a(E, V_0) = 1 - 18 = -17$ [Kol13, Definition 2.4]. Since $\sum_{j=1}^{11} d_j - 4 \cdot 42 = 151 - 168 = -17$, this proves (4). □

By Proposition 6.2(1)(3), $\overline{V}_c$ and V_∞ are normal, so the divisors D_f in Set-up 6.1(5) are well defined. The following local equations of D_g hold for any $c \in \mathbb{C}^4$, including $c = 0$.

Lemma 6.3. *Notation and conditions as in Set-up 6.1, and let $c \in \mathbb{C}^4$. Let g be a non-zero homogeneous element of R_c of degree J. Then for any $1 \leq i \leq 11$, $g^{42}/U_i^{42J/d_i}$ is a regular function on $\overline{V}_c \cap (U_i \neq 0)$ whose divisor is the restriction of $42D_g$, and g^{42}/W^{42J} is a regular function on V_c whose divisor is the restriction of $42D_g$. In particular, $42D_g$ is a Cartier divisor on $\overline{V}_c$.*

Proof. Since $d_i \mid 42$ for any i, the elements $g^{42}/U_i^{42J/d_i}$ and g^{42}/W^{42J} are homogeneous of degree zero in $R_c[U_i^{-1}]$ and in $R_c[W^{-1}]$, so they are regular functions on $\overline{V}_c \cap (U_i \neq 0) = \operatorname{Spec}(R_c[U_i^{-1}])_0$ and on $V_c = \operatorname{Spec}(R_c[W^{-1}])_0$, respectively. The divisor of zeros of the section g^{42} of $\mathcal{O}_{\overline{V}_c}(42J)$ is $42D_g$ (the multiplication map from $\mathcal{O}_{\overline{V}_c}(J)^{\otimes 42}$ to $\mathcal{O}_{\overline{V}_c}(42J)$ is an isomorphism at the generic point of any prime divisor, since W is a unit of degree one on V_c and, by Proposition 6.2(3), μ_{d_i} acts freely at the generic points of $\operatorname{Spec} R_\infty \cap (U_i = 1)$ for any i), and U_i^{42J/d_i} and W^{42J} are nowhere vanishing sections of $\mathcal{O}_{\overline{V}_c}(42J)$ on $\overline{V}_c \cap (U_i \neq 0)$ and on V_c, respectively. Hence the divisors of $g^{42}/U_i^{42J/d_i}$ and of g^{42}/W^{42J} are the restrictions of $42D_g$. Since the open subsets $\overline{V}_c \cap (U_i \neq 0)$, $1 \leq i \leq 11$, and V_c cover $\overline{V}_c$, $42D_g$ is a Cartier divisor. □

6.2. **Residual cones at singular points.** In this subsection, we describe a fiber of a diagonal morphism near a singular point by a weighted homogeneous cone and the weighted blow-up of the vertex of the cone.

Set-up 6.4. Let N, $d_1', \ldots, d_N'$, $e_1', \ldots, e_N'$, $\mathbf{w}_1', \ldots, \mathbf{w}_N'$, c', R', and V' be as follows, where $d_1, \ldots, d_{11}$ are as in Set-up 3.3.

(1) $N \in \{10, 11\}$, $(d_1', \ldots, d_N')$ is $(d_1, \ldots, d_{11})$ if $N = 11$ and is $(d_1, \ldots, d_{11})$ with one entry equal to 6 deleted if $N = 10$, and $e_j' := 42/d_j'$ for $1 \leq j \leq N$.

(2) $\mathbf{w}'_1, \dots, \mathbf{w}'_N \in \mathbb{C}^4$ are vectors any four of which are linearly independent, and $c' \in \mathbb{C}^4 \backslash \{0\}$.
(3) R' is the graded ring

$$R' := \mathbb{C}[W, U_1, \dots, U_N] \Big/ \Big(\sum_{j=1}^{N} U_j^{e'_j} \mathbf{w}'_j - c' W^{42} \Big)$$

with $\deg W = 1$ and $\deg U_j = d'_j$, and $V' := \{u \in \mathbb{A}^N \mid \sum_{j=1}^{N} u_j^{e'_j} \mathbf{w}'_j = c'\}$.
(4) For a homogeneous element f of R', $f|_{V'}$ is the regular function on V' given by $f|_{V'}(u) = f(1, u_1, \dots, u_N)$.

If $N = 11$, $\mathbf{w}'_j = \mathbf{w}_j$ for $1 \le j \le 11$, and $c' = c \ne 0$, then $R' = R_c$ and $V' = V_c$ in the notation of Set-up 6.1. The case $N = 10$ is used in the proof of Proposition 7.13 for a Cartier slice of V_c.

At a singular point p of V', taking e'_i-th roots of the coordinates u_i with $u_i(p) \ne 0$ shows that the germ (V', p) is étale-locally isomorphic to the vertex of a cone Z_p. Thus questions at p reduce to a graded ring A_p and to the exceptional divisor E_p of a weighted blow-up of Z_p.

Construction 6.5. Notation and conditions as in Set-up 6.4. Let $p \in V'$ be a singular point of V'.

(1) $I := \{j \mid u_j(p) \ne 0\}$, $k := \#I$, and $a_i := u_i(p)$ for $i \in I$.
(2) For $j \notin I$, $\overline{\mathbf{w}}'_j$ is the image of $\mathbf{w}'_j$ in $\mathbb{C}^4 / \langle \mathbf{w}'_i \mid i \in I \rangle$, and

$$A_p := \mathbb{C}[x_j \mid j \notin I] \Big/ \Big(\sum_{j \notin I} x_j^{e'_j} \overline{\mathbf{w}}'_j \Big)$$

is graded by $\deg x_j = d'_j$.
(3) $Z_p := \operatorname{Spec} A_p$, which we call the *residual cone* of V' at p, and $\varpi_p : V' \to Z_p$ is the morphism given by $x_j = u_j$ for $j \notin I$.
(4) For $i \in I$, $L_i \in A_p$ is the homogeneous element of degree 42 such that $u_i^{e'_i} = a_i^{e'_i} + \varpi_p^* L_i$ on V'.
(5) $\gamma_p := \gcd\{d'_j \mid j \notin I\}$, $\delta_p := \sum_{j \notin I} d'_j - 42(4 - k)$, and

$$\kappa_p := \min \Big(\Big\{ \frac{\delta_p}{N - 4} \Big\} \cup \{d'_j \mid j \notin I\} \Big).$$

(6) $\psi_p : Z'_p \to Z_p$ is the weighted blow-up of the origin of Z_p with weights d'_j / γ_p for $j \notin I$, $E_p \cong \operatorname{Proj} A_p$ is the exceptional divisor of ψ_p, $q_p : Z_p \backslash \{0\} \to E_p$ is the natural morphism, and Δ_p is the $\mathbb{Q}$-divisor on E_p with $(K_{Z'_p} + E_p)|_{E_p} = K_{E_p} + \Delta_p$.
(7) For a non-zero homogeneous element g of A_p, D_g is the $\mathbb{Q}$-divisor on E_p such that $q_p^* D_g$ is the divisor of g on $Z_p \backslash \{0\}$.
(8) $r_i := U_i - a_i W^{d'_i} \in R'$ for $i \in I$.
(9) $e_* := 1$ if $k = 1$, and e_* is the second largest number among e'_i, $i \in I$, counted with multiplicity, if $k \ge 2$.

Lemma 6.6. *Notation and conditions as in Construction 6.5. Then the following hold.*

(1)

$$Z'_p = \operatorname{Proj} \bigoplus_{m \ge 0} (A_p)_{\ge m\gamma_p},$$

where $(A_p)_{\ge m\gamma_p}$ is the ideal of A_p generated by the homogeneous elements of degree at least $m\gamma_p$, and

$$E_p = \operatorname{Proj} \bigoplus_{m \ge 0} (A_p)_{m\gamma_p} \cong \operatorname{Proj} A_p.$$

(2) ϖ_p maps V' into Z_p, and for any $i \in I$, there exists a unique homogeneous element $L_i \in A_p$ of degree 42 with $u_i^{e'_i} = a_i^{e'_i} + \varpi_p^* L_i$ on V'.

(3) *$K_{Z'_p} + E_p$ is $\mathbb{Q}$-Cartier near E_p, and Δ_p is the different of (Z'_p, E_p) on E_p* [FST11, 11.1].
(4) *For any non-zero homogeneous element g of A_p, there exists a unique $\mathbb{Q}$-divisor D_g on E_p such that $q_p^* D_g$ is the divisor of g on $Z_p \backslash \{0\}$.*

Proof. The proof of Lemma 6.7(1) uses only part (1) of the present lemma, whose proof uses only the definition of the weighted blow-up, and the proofs of Lemma 6.8(1)(2) do not use the present lemma.

(1) Since any d'_j is divisible by γ_p, a homogeneous element of A_p of degree a has weight a/γ_p with respect to the weights d'_j/γ_p, and $(A_p)_a = 0$ unless γ_p divides a. Thus (1) follows from the definition of the weighted blow-up.

(2) Since $p \in V'$ and $u_j(p) = 0$ for $j \notin I$, we have $c' = \sum_{i\in I} a_i^{e'_i} \mathbf{w}'_i$, and projecting the equation of V' modulo $\langle \mathbf{w}'_i \mid i \in I\rangle$ we obtain $\sum_{j\notin I} u_j^{e'_j} \overline{\mathbf{w}}'_j = 0$ on V', so ϖ_p maps V' into Z_p. Since $k \le 3$ (Lemma 6.8(1)), there exist linear functions λ_i on $\mathbb{C}^4$, $i \in I$, with $\lambda_i(\mathbf{w}'_i) = 1$ and $\lambda_i(\mathbf{w}'_l) = 0$ for $l \in I\backslash\{i\}$. Applying λ_i to the equation of V', we obtain

$$u_i^{e'_i} = a_i^{e'_i} - \sum_{j\notin I} \lambda_i(\mathbf{w}'_j) u_j^{e'_j},$$

so L_i is the image of $-\sum_{j\notin I} \lambda_i(\mathbf{w}'_j) x_j^{e'_j}$ in A_p. The uniqueness of L_i follows from the injectivity of ϖ_p^*, since A_p is a domain and ϖ_p is étale at p, so that the image of ϖ_p contains a non-empty open subset of Z_p (Lemma 6.8(1)(2)).

(3) By Lemma 6.7(1), [KM98, Proposition 5.15], and Lemma 2.4, $K_{Z'_p} + E_p$ is $\mathbb{Q}$-Cartier near E_p, so Δ_p is well defined. The equality in Construction 6.5(6) is an equality of divisors, with the convention of Lemma 2.4 and Theorem 2.5, that is, Δ_p is the different of (Z'_p, E_p) on E_p [FST11, 11.1].

(4) Since g is homogeneous, the divisor of g on $Z_p\backslash\{0\}$ is invariant under the $\mathbb{G}_m$-action given by the grading, any $\mathbb{G}_m$-invariant prime divisor on $Z_p\backslash\{0\}$ is $q_p^{-1}(T)$ for a prime divisor T on E_p, and $q_p^* T$ is a positive integer multiple of $q_p^{-1}(T)$ (over the smooth locus of E_p, whose complement has preimage of codimension at least two since q_p has one-dimensional fibers, T is Cartier and $q_p^* T \ge 0$ is a Cartier divisor supported on the prime divisor $q_p^{-1}(T)$). □

The number e_* enters through Lemma 7.8, and κ_p is the threshold constant on E_p in Proposition 7.11.

Lemma 6.7. *Notation and conditions as in Construction 6.5. For $j \notin I$, let $w_j := d'_j/\gamma_p$, $s_j := \gcd\{w_i \mid i \notin I,\ i \ne j\}$, $Z_{p,j} := Z_p \cap (x_j = 1)$, and $T_j := q_p\big((Z_p \cap (x_j = 0))\backslash\{0\}\big)$. Then the following hold.*

(1) *For any $l \notin I$, $Z_{p,l}$ is a smooth irreducible variety, and Z'_p has an open subset $Z'_{p,l}$ containing $E_p \cap (x_l \ne 0)$ with a quotient morphism $\varrho_l : Z_{p,l} \times \mathbb{A}^1 \to Z'_{p,l}$ by μ_{w_l}, where $\zeta \in \mu_{w_l}$ multiplies x_j by ζ^{w_j} for $j \notin I$ and multiplies the coordinate ξ of $\mathbb{A}^1$ by ζ^{-1}. Moreover, $\psi_p \circ \varrho_l(x, \xi) = (\xi^{w_j} x_j)_{j\notin I}$ for $x = (x_j)_{j\notin I} \in Z_{p,l}$, the morphism ϱ_l is étale in codimension one, $\varrho_l^{-1}(E_p) = Z_{p,l} \times \{0\}$, and $\varrho_l|_{Z_{p,l}\times\{0\}}$ is the restriction $q_p|_{Z_{p,l}}$ of q_p to $Z_{p,l}$, which is the quotient morphism $Z_{p,l} \to E_p \cap (x_l \ne 0)$ by μ_{w_l}.*
(2) *For any $j \notin I$, T_j is a prime divisor on E_p and $D_{x_j} = \frac{1}{s_j} T_j$. Moreover,*

$$\Delta_p = \sum_{j\notin I} \Big(1 - \frac{1}{s_j}\Big) T_j,$$

and $K_{Z_{p,l}} = (q_p|_{Z_{p,l}})^*(K_{E_p} + \Delta_p)$ over $E_p \cap (x_l \ne 0)$ for any $l \notin I$.
(3) *E_p has quotient singularities and is $\mathbb{Q}$-factorial.*

Proof. By Lemma 6.8(1), whose proof does not use the present lemma, $1 \le k \le 3$, A_p is a normal domain, and $Z_p\backslash\{0\}$ is smooth. Let $r := 4 - k$. By Lemma 3.2(4), any r of the vectors $\overline{\mathbf{w}}'_j$, $j \notin I$,

in $\mathbb{C}^4/\langle \mathbf{w}'_i \mid i \in I\rangle \cong \mathbb{C}^r$ are linearly independent. For $j \notin I$,

$$A_p/(x_j) \cong \mathbb{C}[x_i \mid i \notin I \cup \{j\}]\Big/\Big(\sum_{i\notin I\cup\{j\}} x_i^{e'_i}\overline{\mathbf{w}}'_i\Big),$$

where the number of variables is $N-k-1 \geq r+2$ and any r of the vectors $\overline{\mathbf{w}}'_i$, $i \notin I\cup\{j\}$, are linearly independent. Thus by Lemma 3.2, $A_p/(x_j)$ is a normal domain with $\dim A_p/(x_j) = N-5$, $\operatorname{Spec} A_p/(x_j)$ is smooth outside the origin, and any x_i, $i \notin I\cup\{j\}$, is non-zero in $A_p/(x_j)$, since we may take the r indices in Lemma 3.2(1) different from i. In particular, $(x_j = 0)$ is a prime divisor on Z_p and $\operatorname{mult}_{(x_j=0)} x_j = 1$.

(1) Fix $l \notin I$. The morphism

$$Z_{p,l} \times \mathbb{G}_m \to (Z_p \cap (x_l \neq 0)) \times \mathbb{G}_m, \quad (x,\eta) \to ((\eta^{w_j}x_j)_{j\notin I}, \eta),$$

is an isomorphism onto the closed subscheme $(\eta^{w_l} = x_l)$, with inverse $(y,\eta) \to ((\eta^{-w_j}y_j)_{j\notin I}, \eta)$. The projection from $(\eta^{w_l} = x_l)$ to $Z_p \cap (x_l \neq 0)$ is finite étale, as a base change of the w_l-th power map of $\mathbb{G}_m$. Thus the morphism

$$\beta_l : Z_{p,l} \times \mathbb{G}_m \to Z_p \cap (x_l \neq 0), \quad (x,\eta) \to (\eta^{w_j}x_j)_{j\notin I},$$

is finite étale, and $Z_{p,l}$ is smooth. The fibers of β_l are the orbits of the free action of μ_{w_l} given by

$$\zeta \cdot (x,\eta) = ((\zeta^{w_j}x_j)_{j\notin I}, \zeta^{-1}\eta).$$

Moreover, $Z_{p,l}$ is irreducible. Let K be the fraction field of A_p, and let $B_l := A_p[x_l^{-1}][\eta]/(\eta^{w_l} - x_l)$. Then $Z_{p,l} \times \mathbb{G}_m \cong \operatorname{Spec} B_l$, and B_l is a free $A_p[x_l^{-1}]$-module with basis $1, \eta, \dots, \eta^{w_l-1}$, so B_l is a subring of $K[\eta]/(\eta^{w_l} - x_l)$. If $\eta^{w_l} - x_l$ is reducible in $K[\eta]$, then by [Lan02, Chapter VI, Theorem 9.1], $x_l \in K^d$ for a prime number d dividing w_l, or $4 \mid w_l$ and $x_l \in -4K^4$. Since $-4 = (1+\sqrt{-1})^4$, we have $-4K^4 = K^4 \subset K^2$. Thus $x_l \in K^d$ for a prime number d dividing w_l. Since A_p is normal, $(x_l = 0)$ is a prime divisor on Z_p, and $\operatorname{mult}_{(x_l=0)} x_l = 1$ is not divisible by d, a contradiction. Thus $K[\eta]/(\eta^{w_l} - x_l)$ is a field, $Z_{p,l} \times \mathbb{G}_m$ is irreducible, and so is $Z_{p,l}$.

Let $A_p[\xi]$ be graded by $\deg x_j = w_j$ and $\deg \xi = -1$. For $m \geq 0$, setting $\xi = 1$ maps the degree-m part of $A_p[\xi]$ bijectively onto $(A_p)_{\geq m\gamma_p}$, compatibly with multiplication. Thus $Z'_p = \operatorname{Proj} A_p[\xi]_{\geq 0}$, where $A_p[\xi]_0 \cong A_p$, and E_p is the closed subscheme $(\xi = 0)$ of Z'_p. Since any monomial of positive degree is divisible by some x_j, Z'_p is covered by the open subsets $Z'_{p,l} := D_+(x_l) = \operatorname{Spec}(A_p[\xi][x_l^{-1}])_0$, $l \notin I$. Setting $x_l = 1$ induces an isomorphism from $(A_p[\xi][x_l^{-1}])_0$ onto the ring of invariants of $(A_p/(x_l-1))[\xi]$ under the action of μ_{w_l} in the statement, whose inverse sends the image of a homogeneous element b of degree mw_l, $m \in \mathbb{Z}$, to b/x_l^m. The inclusion of the ring of invariants defines ϱ_l. An element $f \in (A_p)_{a\gamma_p}$ corresponds to $f\xi^a \in A_p[\xi]_0$, so

$$\psi_p \circ \varrho_l(x,\xi) = (\xi^{w_j}x_j)_{j\notin I},$$

where $x_l = 1$. Moreover, $\varrho_l^{-1}(E_p) = Z_{p,l} \times \{0\}$, and $\varrho_l|_{Z_{p,l}\times\{0\}}$ is the quotient morphism $Z_{p,l} \to E_p \cap (x_l \neq 0)$ by μ_{w_l}, which is the restriction of q_p to $Z_{p,l}$.

Let $\zeta \in \mu_{w_l}$ with $\zeta \neq 1$. A point of $Z_{p,l}\times\mathbb{A}^1$ fixed by ζ satisfies $\xi = 0$. Since $\gcd\{w_j \mid j \notin I\} = 1$, there exists $j \notin I$ with $\zeta^{w_j} \neq 1$, and $x_j = 0$ at the point. Since β_l is flat and $\beta_l^* x_j = \eta^{w_j}x_j$, the function x_j is a nonzerodivisor on $Z_{p,l}$. Thus the fixed locus of ζ has codimension at least two, and ϱ_l is étale in codimension one.

(2) Since $A_p/(x_j)$ is a domain with $\dim A_p/(x_j) = N-5$, $T_j = \operatorname{Proj} A_p/(x_j)$ is a prime divisor on E_p. Fix $l \notin I$ with $l \neq j$, and let

$$g_l := \varrho_l|_{Z_{p,l}\times\{0\}} : Z_{p,l} \to E_p \cap (x_l \neq 0).$$

Since $(Z_p \cap (x_j = 0)) \cap (x_l \neq 0)$ is a non-empty open subset of the irreducible variety $\operatorname{Spec} A_p/(x_j)$, on which any x_i, $i \notin I\cup\{j\}$, is non-zero, any component of $Z_{p,l} \cap (x_j = 0)$ has a dense open subset on which $x_i \neq 0$ for all $i \notin I\cup\{j\}$. At such a point, the stabilizer in μ_{w_l} is

$$\{\zeta \mid \zeta^{w_i} = 1 \text{ for } i \notin I\cup\{j\}\} = \mu_{s_j},$$

which acts on x_j by ζ^{w_j}, faithfully since $\gcd(s_j, w_j) = 1$. Thus g_l is ramified along $Z_{p,l} \cap (x_j = 0)$ with index s_j, and $g_l^* T_j = s_j(x_j = 0)$, where $(x_j = 0)$ is reduced on $Z_{p,l}$ since $\beta_l^{-1}(x_j = 0)$ is étale over the smooth variety $(Z_p \cap (x_j = 0)) \cap (x_l \neq 0)$. Let pr_1 be the projection from $Z_{p,l} \times \mathbb{G}_m$ to $Z_{p,l}$. Since $q_p \circ \beta_l = g_l \circ \mathrm{pr}_1$, we obtain

$$\beta_l^* q_p^* T_j = \mathrm{pr}_1^* g_l^* T_j = s_j \,\mathrm{pr}_1^*(x_j = 0) = s_j \beta_l^* \operatorname{div}(x_j).$$

As β_l is étale and surjective, $q_p^* T_j = s_j \operatorname{div}(x_j)$ on $Z_p \cap (x_l \neq 0)$. On $Z_p \cap (x_j \neq 0)$ both sides are zero, so $q_p^* T_j = s_j \operatorname{div}(x_j)$ on $Z_p \backslash \{0\}$ (since $l \notin I \cup \{j\}$ is arbitrary, the open subsets $Z_p \cap (x_l \neq 0)$, $l \notin I \cup \{j\}$, and $Z_p \cap (x_j \neq 0)$ cover $Z_p \backslash \{0\}$) and $D_{x_j} = \frac{1}{s_j} T_j$.

For a prime divisor T on E_p different from all T_i, at a point of $g_l^{-1}(T)$ over the generic point of T all coordinates x_i are non-zero, so the stabilizer is trivial as $\gcd_i w_i = 1$, and g_l is unramified over the generic point of T. The variety $E_p \cap (x_l \neq 0) = Z_{p,l}/\mu_{w_l}$ is normal, $Z_{p,l} \times \mathbb{A}^1$ is a smooth variety on which μ_{w_l} acts faithfully, and $\varrho_l^{-1}(E_p) = Z_{p,l} \times \{0\}$ is smooth. By Lemma 2.4 applied to ϱ_l,

$$\Delta_p = \sum_{i \notin I, i \neq l} \Big(1 - \frac{1}{s_i}\Big) T_i \quad \text{on } E_p \cap (x_l \neq 0),$$

and $E_p \cap (x_l \neq 0)$ does not intersect T_l. Since the open subsets $E_p \cap (x_l \neq 0)$, $l \notin I$, cover E_p, $\Delta_p = \sum_{j \notin I} \big(1 - \frac{1}{s_j}\big) T_j$.

Finally, fix $l \notin I$. By (1), $E_p \cap (x_l \neq 0)$ is the quotient of the smooth variety $Z_{p,l}$ by μ_{w_l}, so it has quotient singularities and is $\mathbb{Q}$-factorial [KM98, Proposition 5.15]. Since the open subsets $E_p \cap (x_l \neq 0)$, $l \notin I$, cover E_p, this proves (3). By the computation above, the ramification index of $g_l = q_p|_{Z_{p,l}}$ along a prime divisor of $Z_{p,l}$ over a prime divisor T of $E_p \cap (x_l \neq 0)$ is s_T, where $\Delta_p = \sum_T (1 - \frac{1}{s_T}) T$ on $E_p \cap (x_l \neq 0)$ (Lemma 2.4). Hence $K_{Z_{p,l}} = g_l^*(K_{E_p} + \Delta_p)$ over $E_p \cap (x_l \neq 0)$ by the Hurwitz formula [Kol13, 2.41]. □

Lemma 6.8. *Notation and conditions as in Construction 6.5. Then the following hold.*

1. *R' is a normal domain, V' is a normal variety of dimension $N - 4$, and $1 \leq k \leq 3$. Moreover, A_p is a normal domain of dimension $N - 4$, and Z_p is smooth outside the origin.*
2. *ϖ_p is étale at p and $\varpi_p(p) = 0$. For any $i \in I$, r_i is a prime element of R', and $r_i|_{V'} = \epsilon_i \cdot \varpi_p^* L_i$ for a regular function ϵ_i near p with $\epsilon_i(p) \neq 0$.*
3. *ψ_p is an isomorphism over $Z_p \backslash \{0\}$, Z_p' has quotient singularities near E_p, and $a(E_p, Z_p) = \delta_p/\gamma_p$. If $N = 11$, then $\delta_p \geq 4$, 25, or 46 if $k = 1$, 2, or 3, respectively.*
4. *For any $i \in I$, the pair $(E_p, \Delta_p + D_{L_i})$ is lc.*

Proof. (1) Fix four indices $j_1, \ldots, j_4$. Since $\mathbf{w}'_{j_1}, \ldots, \mathbf{w}'_{j_4}$ form a basis of $\mathbb{C}^4$, the defining equations of R' are equivalent to four equations expressing $U_{j_l}^{e'_{j_l}}$, $1 \leq l \leq 4$, as linear combinations of W^{42} and $U_j^{e'_j}$, $j \notin \{j_1, \ldots, j_4\}$. Thus R' is a free module over $\mathbb{C}[W, U_j \mid j \notin \{j_1, \ldots, j_4\}]$ with basis the monomials $\prod_{l=1}^4 U_{j_l}^{b_l}$, $0 \leq b_l < e'_{j_l}$, and R' is a complete intersection ring with $\dim R' = N - 3$. The Jacobian matrix of the equations of $\operatorname{Spec} R'$ has columns $e'_j U_j^{e'_j - 1} \mathbf{w}'_j$ and $-42 W^{41} c'$, so it has rank four at any point with at least four non-zero coordinates U_j. A point of $\operatorname{Spec} R'$ with $W = 0$ other than the origin has at least five non-zero coordinates U_j, since any four of the $\mathbf{w}'_j$ are linearly independent. At a point with $W \neq 0$ whose non-zero coordinates U_j are indexed by a set I' with $\#I' \leq 3$, the equation $\sum_{i \in I'} (U_i^{e'_i}/W^{42}) \mathbf{w}'_i = c'$ determines the numbers $U_i^{e'_i}/W^{42}$. Thus the singular locus of $\operatorname{Spec} R'$ is a finite union of $\mathbb{G}_m$-orbits, so $\dim \operatorname{Sing}(\operatorname{Spec} R') \leq 1$. Since $\dim R' = N - 3 \geq 7$, $\operatorname{Spec} R'$ is regular in codimension one, so R' is normal [Har77, Proposition II.8.23]. Since R' is positively graded with $R'_0 = \mathbb{C}$, R' is a normal domain by Lemma 3.1. The argument so far does not use $c' \neq 0$. Since W has degree one, $\operatorname{Spec} R' \cap (W \neq 0) \cong V' \times \mathbb{G}_m$, so V' is a normal variety with $\dim V' = N - 4$.

The Jacobian matrix of V' at p has columns $e'_j u_j(p)^{e'_j-1}\mathbf{w}'_j$. Since V' is reduced, $\dim V' = N-4$, and p is a singular point, the Jacobian matrix of V' at p has rank less than four [Har77, Theorem I.5.1], so $k \leq 3$. Since $c' \neq 0$, $k \geq 1$. Let $r := 4-k$. By Lemma 3.2(4), any r of the vectors $\overline{\mathbf{w}}'_j$, $j \notin I$, are linearly independent, and $N-k \geq r+2$. Thus by Lemma 3.2(1)(2)(3), A_p is a normal domain of dimension $N-k-r = N-4$, and Z_p is smooth outside the origin.

(2) Since $u_j(p) = 0$ for $j \notin I$, we have $\varpi_p(p) = 0$ and $c' = \sum_{i\in I} a_i^{e'_i}\mathbf{w}'_i$. Let λ_i, $i \in I$, be linear functions on $\mathbb{C}^4$ with $\lambda_i(\mathbf{w}'_i) = 1$ and $\lambda_i(\mathbf{w}'_l) = 0$ for $l \in I\backslash\{i\}$. Together with the linear functions vanishing on $\langle \mathbf{w}'_i \mid i \in I\rangle$, they span the dual space of $\mathbb{C}^4$. Thus the equations of V' are equivalent to the equations of Z_p in the coordinates u_j, $j \notin I$, together with

$$u_i^{e'_i} = a_i^{e'_i} - \sum_{j\notin I} \lambda_i(\mathbf{w}'_j) u_j^{e'_j}, \quad i \in I.$$

Hence V' is the closed subscheme of $Z_p \times \mathbb{A}^k$ defined by $u_i^{e'_i} = a_i^{e'_i} + L_i$, $i \in I$, where L_i is the image of $-\sum_{j\notin I} \lambda_i(\mathbf{w}'_j) x_j^{e'_j}$ in A_p, and ϖ_p is the projection. Thus ϖ_p is the base change of the morphism $\mathbb{A}^k \to \mathbb{A}^k$, $(u_i)_{i\in I} \to (u_i^{e'_i})_{i\in I}$, by the morphism $Z_p \to \mathbb{A}^k$ given by $(a_i^{e'_i} + L_i)_{i\in I}$. The power morphism is étale where all u_i are non-zero, and $u_i(p) = a_i \neq 0$, so ϖ_p is étale at p. In particular, the image of ϖ_p contains a non-empty open subset of the irreducible variety Z_p, so ϖ_p^* is injective.

Let $i \in I$. Substituting $U_i = a_i W^{d'_i}$, we obtain

$$R'/(r_i) \cong \mathbb{C}[W, U_j \mid j \neq i] \Big/ \Big(\sum_{j\neq i} U_j^{e'_j}\mathbf{w}'_j - (c' - a_i^{e'_i}\mathbf{w}'_i) W^{42} \Big).$$

Replacing $U_1, \dots, U_N$ with U_j, $j \neq i$, and c' with $c' - a_i^{e'_i}\mathbf{w}'_i$, the argument of (1) shows that $R'/(r_i)$ is a normal domain with $\dim R'/(r_i) = N-4$, also when $c' - a_i^{e'_i}\mathbf{w}'_i = 0$. Thus r_i is a prime element of R'. Moreover, $r_i|_{V'} = u_i - a_i$ and

$$\varpi_p^* L_i = u_i^{e'_i} - a_i^{e'_i} = (u_i - a_i) \sum_{m=0}^{e'_i-1} u_i^m a_i^{e'_i-1-m},$$

where the last factor takes the value $e'_i a_i^{e'_i-1} \neq 0$ at p. Thus $r_i|_{V'} = \epsilon_i \cdot \varpi_p^* L_i$, where ϵ_i is the inverse of the last factor near p.

(3) For $l \notin I$, any ideal $(A_p)_{\geq m\gamma_p}$ contains x_l^m, so it becomes the unit ideal over $Z_p \cap (x_l \neq 0)$, and ψ_p is an isomorphism over $Z_p \cap (x_l \neq 0)$. Since the open subsets $Z_p \cap (x_l \neq 0)$, $l \notin I$, cover $Z_p\backslash\{0\}$, ψ_p is an isomorphism over $Z_p\backslash\{0\}$, and E_p, which is irreducible as A_p is a domain, is the only exceptional divisor of ψ_p. By Lemma 6.7(1), Z'_p is covered near E_p by the quotients $Z'_{p,l}$ of the smooth varieties $Z_{p,l} \times \mathbb{A}^1$ by μ_{w_l}, the morphisms ϱ_l have no pseudo-reflections, and $\varrho_l^{-1}(E_p) = Z_{p,l} \times \{0\}$ is smooth. In particular, Z'_p has quotient singularities near E_p.

Let $F_1, \dots, F_r$ be the coordinates of $\sum_{j\notin I} x_j^{e'_j}\overline{\mathbf{w}}'_j$ with respect to a basis of $\mathbb{C}^4/\langle \mathbf{w}'_i \mid i \in I\rangle$, where $r = 4-k$, and let ω be the $(N-4)$-form on $Z_p\backslash\{0\}$ with

$$\omega \wedge dF_1 \wedge \cdots \wedge dF_r = \bigwedge_{j\notin I} dx_j$$

along $Z_p\backslash\{0\}$. Since A_p is a domain, $F_1, \dots, F_r$ generate the ideal $\mathcal{I}$ of $Z_p\backslash\{0\}$ in $\mathbb{A}^{N-k}\backslash\{0\}$. Since $Z_p\backslash\{0\}$ is a smooth irreducible variety of codimension r in $\mathbb{A}^{N-k}\backslash\{0\}$, $\mathcal{I}/\mathcal{I}^2$ is locally free of rank r and

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{\mathbb{A}^{N-k}}|_{Z_p\backslash\{0\}} \to \Omega_{Z_p\backslash\{0\}} \to 0$$

is exact [Har77, Theorem II.8.17], so $\mathcal{I}/\mathcal{I}^2$ has basis the images of $F_1, \dots, F_r$. Taking the highest exterior powers [Har77, Proposition II.8.20], we see that ω is nowhere vanishing, and since Z_p is

normal, $K_{Z_p} = \mathrm{div}(\omega) = 0$. Fix $l \notin I$, let ϱ_l be the quotient morphism in Lemma 6.7(1), and let $\Phi_l := \psi_p \circ \varrho_l$. By Lemma 6.7(1), $\Phi_l(x,\xi) = (\xi^{w_j} x_j)_{j\notin I}$, where ξ is the coordinate of $\mathbb{A}^1$. Each F_m is homogeneous of degree $42/\gamma_p$ for $\deg x_j = w_j$, so $\Phi_l^* F_m = \xi^{42/\gamma_p} F_m|_{x_l=1}$, and

$$\Phi_l^* \bigwedge_{j\notin I} dx_j = \pm w_l \xi^{\sum_{j\notin I} w_j - 1} d\xi \wedge \bigwedge_{j\notin I, j\neq l} dx_j.$$

Let ω_l be the form on $Z_{p,l}$ with

$$\omega_l \wedge d(F_1|_{x_l=1}) \wedge \cdots \wedge d(F_r|_{x_l=1}) = \bigwedge_{j\notin I, j\neq l} dx_j$$

along $Z_{p,l}$. Since $Z_{p,l}$ is a smooth irreducible variety (Lemma 6.7(1)) of dimension $N-5$, as a non-empty hypersurface section of Z_p, the functions $F_1|_{x_l=1}, \ldots, F_r|_{x_l=1}$ generate the ideal $\mathcal{I}_l$ of $Z_{p,l}$ in $\mathbb{A}^{N-k-1}$, and their images form a basis of $\mathcal{I}_l/\mathcal{I}_l^2$. Thus ω_l is nowhere vanishing [Har77, Theorem II.8.17, Proposition II.8.20]. Since $F_m|_{x_l=1}$ vanishes on $Z_{p,l}$, $d(\Phi_l^* F_m) = \xi^{42/\gamma_p} d(F_m|_{x_l=1})$ along $Z_{p,l} \times \mathbb{G}_m$. Thus

$$\Phi_l^* \omega = \pm w_l \xi^{\sum_{j\notin I} w_j - 42r/\gamma_p - 1} d\xi \wedge \omega_l = \pm w_l \xi^{\delta_p/\gamma_p - 1} d\xi \wedge \omega_l,$$

so

$$K_{Z_{p,l}\times\mathbb{A}^1} = \Phi_l^* K_{Z_p} + \Big(\frac{\delta_p}{\gamma_p} - 1\Big)(\xi = 0).$$

Since ϱ_l is étale in codimension one, $K_{Z_{p,l}\times\mathbb{A}^1} = \varrho_l^* K_{Z_p'}$ and $\varrho_l^* E_p = (\xi = 0)$. Hence

$$K_{Z_p'} = \psi_p^* K_{Z_p} + \Big(\frac{\delta_p}{\gamma_p} - 1\Big) E_p$$

near E_p, and $a(E_p, Z_p) = \delta_p/\gamma_p$.

If $N = 11$, then $\sum_{j=1}^{11} d_j = 151$ and

$$\delta_p = 151 - \sum_{i\in I} d_i - 42(4-k) = -17 + \sum_{i\in I}(42 - d_i).$$

Since $42 - d_i \geq 21$ for any i, $\delta_p \geq 4$, 25, or 46 if $k = 1$, 2, or 3, respectively.

(4) Let $i \in I$, and let λ_i be the linear function in the proof of (2), so that L_i is the image of $-\sum_{j\notin I} \lambda_i(\mathbf{w}_j') x_j^{e_j'}$. The linear map

$$\mathbb{C}^4 \to \mathbb{C}^4/\langle \mathbf{w}_l' \mid l \in I\rangle \oplus \mathbb{C}, \quad v \to (\overline{v}, \lambda_i(v)),$$

has kernel $\langle \mathbf{w}_l' \mid l \in I\backslash\{i\}\rangle$. Thus

$$A_p/(L_i) \cong \mathbb{C}[x_j \mid j \notin I] \Big/ \Big(\sum_{j\notin I} x_j^{e_j'} \mathbf{w}_j''\Big),$$

where $\mathbf{w}_j''$ is the image of $\mathbf{w}_j'$ in

$$\mathbb{C}^4/\langle \mathbf{w}_l' \mid l \in I\backslash\{i\}\rangle \cong \mathbb{C}^{5-k}.$$

By Lemma 3.2(4), any $5-k$ of the $\mathbf{w}_j''$, $j \notin I$, are linearly independent, and $N - k \geq (5-k) + 2$. By Lemma 3.2(1)(2)(3), $A_p/(L_i)$ is a normal domain of dimension $N-5$ whose spectrum is smooth outside the origin. In particular, $L_i \neq 0$, and $\mathrm{div}(L_i)$ is a reduced smooth divisor on $Z_p\backslash\{0\}$.

Fix $l \notin I$ and let

$$g_l := \varrho_l|_{Z_{p,l}\times\{0\}} : Z_{p,l} \to E_p \cap (x_l \neq 0),$$

which is the restriction of q_p to $Z_{p,l}$ by Lemma 6.7(1). Let

$$\beta_l : Z_{p,l} \times \mathbb{G}_m \to Z_p \cap (x_l \neq 0), \quad (x,\eta) \to (\eta^{w_j} x_j)_{j\notin I},$$

which is finite étale as the base change of the w_l-th power map of $\mathbb{G}_m$ by x_l, and let pr_1 be the projection from $Z_{p,l} \times \mathbb{G}_m$ to $Z_{p,l}$. Then $q_p \circ \beta_l = g_l \circ \mathrm{pr}_1$ and $\beta_l^* L_i = \eta^{42/\gamma_p} L_i|_{Z_{p,l}}$, so

$$\mathrm{pr}_1^* g_l^* D_{L_i} = \beta_l^* \operatorname{div}(L_i) = \mathrm{pr}_1^* \operatorname{div}(L_i|_{Z_{p,l}}).$$

Hence $g_l^* D_{L_i} = \operatorname{div}(L_i|_{Z_{p,l}})$, which is a reduced smooth divisor as β_l is étale. By Lemma 6.7(2), $K_{Z_{p,l}} = g_l^*(K_{E_p} + \Delta_p)$ over $E_p \cap (x_l \neq 0)$. The pair $(Z_{p,l}, \operatorname{div}(L_i|_{Z_{p,l}}))$ is lc, since $Z_{p,l}$ is smooth and $\operatorname{div}(L_i|_{Z_{p,l}})$ is a reduced smooth divisor [KM98, Corollary 2.31(3)]. By Lemma 6.7(3), $K_{E_p} + \Delta_p + D_{L_i}$ is $\mathbb{Q}$-Cartier. By Lemma 2.3 applied to g_l, $(E_p, \Delta_p + D_{L_i})$ is lc on $E_p \cap (x_l \neq 0)$. Since the open subsets $E_p \cap (x_l \neq 0)$, $l \notin I$, cover E_p, (4) follows. □

6.3. **Singularities of the compactified fibers.** In this subsection, we prove that the compactification $\overline{V}_c$ of a fiber of h over $c \neq 0$ is klt, and we compute the canonical class of $\overline{V}_c$.

The assumption $c \neq 0$ cannot be removed: by Proposition 6.2(4), $V_0 = h^{-1}(0)$ is not lc, and the exceptional divisor of the weighted blow-up of the vertex of V_0 with weights $d_1, \dots, d_{11}$ has log discrepancy $151 - 168 = -17$, the sum of the weights minus the sum of the degrees of the four equations. For $c \neq 0$, Lemma 6.8(3) shows that $\delta_p = -17 + \sum_{i \in I}(42 - d_i) \geq 4$ at any singular point p of V_c. This is why the relative discrepancy estimate in Section 8 (Proposition 8.6) has a loss term with coefficient $\frac{17}{42} = \frac{4 \cdot 42 - 151}{42}$ over $c = 0$ and needs a rescaling.

Proposition 6.9. *Notation and conditions as in Set-up 6.1, and let $c \in \mathbb{C}^4 \backslash \{0\}$. Then $\overline{V}_c$ has klt singularities, $K_{\overline{V}_c} \sim_{\mathbb{Q}} 16 V_\infty$, and D_f is $\mathbb{Q}$-Cartier for any non-zero homogeneous element f of R_c.*

Proof. We first consider $\overline{V}_c$ near V_∞. Fix $1 \leq l \leq 11$, and let $\widetilde{V}_l := \operatorname{Spec} R_c \cap (U_l = 1)$, with coordinates W and U_j, $j \neq l$, defined by $G_1 = \cdots = G_4 = 0$, where

$$(G_1, \dots, G_4) := \sum_{j \neq l} U_j^{e_j} \mathbf{w}_j + \mathbf{w}_l - cW^{42}.$$

The natural morphism $\widetilde{V}_l \to \overline{V}_c \cap (U_l \neq 0)$ is the quotient by μ_{d_l}, where $\zeta \in \mu_{d_l}$ multiplies W by ζ and U_j by ζ^{d_j}. A point of $\widetilde{V}_l$ with $W = 0$ is a non-zero point of $\operatorname{Spec} R_\infty$, which has at least five non-zero coordinates by Lemma 3.2(2). Thus there exist indices $j_1, \dots, j_4 \neq l$ with $U_{j_1}, \dots, U_{j_4}$ non-zero at the point, and the partial derivatives of $G_1, \dots, G_4$ with respect to $U_{j_1}, \dots, U_{j_4}$ have rank four. Hence $\widetilde{V}_l$ is smooth along $(W = 0)$, and $(W = 0)$ is a smooth divisor on $\widetilde{V}_l$ near $W = 0$. Since the open subsets $\overline{V}_c \cap (U_l \neq 0)$ cover V_∞, $\overline{V}_c$ has quotient singularities near V_∞. In particular, $\overline{V}_c$ is $\mathbb{Q}$-factorial near V_∞ [KM98, Proposition 5.15]. Moreover, near $(W = 0)$, the quotient morphism $q_l : \widetilde{V}_l \to \overline{V}_c \cap (U_l \neq 0)$ satisfies $K_{\widetilde{V}_l} = q_l^*(K_{\overline{V}_c} + \Delta_l)$ by the Hurwitz formula [Kol13, 2.41], where $\Delta_l \geq 0$ is the $\mathbb{Q}$-divisor on $\overline{V}_c \cap (U_l \neq 0)$ with coefficient $1 - 1/m$ along each prime divisor over which q_l has ramification index m. Since $\widetilde{V}_l$ is smooth there, $(\overline{V}_c \cap (U_l \neq 0), \Delta_l)$ is klt near $V_\infty \cap (U_l \neq 0)$ [KM98, Proposition 5.20(4)], and hence so is $\overline{V}_c \cap (U_l \neq 0)$, as $\Delta_l \geq 0$. Since l is arbitrary, $\overline{V}_c$ is klt near V_∞.

We now show that V_c is klt. By Proposition 6.2(1), V_c is normal, and K_{V_c} is Cartier since V_c is a complete intersection. Let p be a singular point of V_c. We use Construction 6.5 for $N = 11$, $\mathbf{w}'_j = \mathbf{w}_j$, and $c' = c$, so that $V' = V_c$. By Lemma 6.8(3),

$$K_{Z'_p} + \Big(1 - \frac{\delta_p}{\gamma_p}\Big) E_p = \psi_p^* K_{Z_p}$$

with $\delta_p / \gamma_p > 0$. For $l \notin I$, Lemma 6.7(1) implies that

$$\varrho_l^* \Big(K_{Z'_p} + \Big(1 - \frac{\delta_p}{\gamma_p}\Big) E_p \Big) = K_{Z_{p,l} \times \mathbb{A}^1} + \Big(1 - \frac{\delta_p}{\gamma_p}\Big)(Z_{p,l} \times \{0\}).$$

The sub-pair

$$\left(Z_{p,l}\times\mathbb{A}^1,\left(1-\frac{\delta_p}{\gamma_p}\right)(Z_{p,l}\times\{0\})\right)$$

consists of a smooth variety and a smooth divisor with coefficient less than one, so it is sub-klt [KM98, Corollary 2.31(3)]. By Lemma 2.3, $(Z'_p,(1-\delta_p/\gamma_p)E_p)$ is sub-klt near E_p. Since ψ_p is an isomorphism over the smooth variety $Z_p\backslash\{0\}$, for any prime divisor F over Z_p we have $a(F,Z_p)=a(F,Z'_p,(1-\delta_p/\gamma_p)E_p)>0$ [KM98, Lemma 2.30]. Thus Z_p is klt. By Lemma 6.8(2), there exists an open neighborhood V° of p such that $\varpi_p|_{V^\circ}$ is étale, and $\varpi_p(V^\circ)$ is open in Z_p. By [Kol13, Proposition 2.15], V° is klt. Since p is an arbitrary singular point of V_c and smooth points are klt, V_c is klt. Together with the first paragraph, $\overline{V}_c$ has klt singularities.

Next we compute $K_{\overline{V}_c}$. Let $h_1,\dots,h_4$ and $c_1,\dots,c_4$ be the coordinates of h and c, and let ω be the 7-form on the smooth locus of V_c with

$$\omega\wedge dh_1\wedge\cdots\wedge dh_4=du_1\wedge\cdots\wedge du_{11}$$

along V_c. Since

$$\mathbb{C}[u_1,\dots,u_{11}]/(h_1-c_1,\dots,h_4-c_4)\cong R_c/(W-1)$$

is a domain (Proposition 6.2(1)), $h_1-c_1,\dots,h_4-c_4$ generate the ideal $\mathcal{J}$ of V_c in $\mathbb{A}^{11}$. Since V_c is a variety of dimension 7 (Proposition 6.2(1)), the images of $h_1-c_1,\dots,h_4-c_4$ form a basis of $\mathcal{J}/\mathcal{J}^2$ over the smooth locus of V_c, and ω generates the canonical sheaf of the smooth locus of V_c [Har77, Theorem II.8.17, Proposition II.8.20]. Since V_c is normal and has finitely many singular points (Proposition 6.2(1)(2)), $\operatorname{div}(\omega)|_{V_c}=0$. Fix $1\le l\le 11$. The composite of $\widetilde{V}_l\cap(W\neq 0)\to\overline{V}_c$ with $V_c\subset\overline{V}_c$ is given by $u_l=W^{-d_l}$ and $u_j=U_jW^{-d_j}$ for $j\neq l$. Since $\sum_{j=1}^{11}d_j=151$, we obtain

$$du_1\wedge\cdots\wedge du_{11}=\pm d_lW^{-152}dW\wedge\bigwedge_{j\neq l}dU_j,$$

and $h_m-c_m=W^{-42}G_m$, so $dh_m=W^{-42}dG_m$ along $\widetilde{V}_l\cap(W\neq 0)$ and

$$dh_1\wedge\cdots\wedge dh_4=W^{-168}dG_1\wedge\cdots\wedge dG_4.$$

Let ω_l be the form on $\widetilde{V}_l$ near $(W=0)$ with

$$\omega_l\wedge dG_1\wedge\cdots\wedge dG_4=dW\wedge\bigwedge_{j\neq l}dU_j,$$

which is nowhere vanishing since $\widetilde{V}_l$ is a smooth complete intersection there. Thus the pullback of ω to $\widetilde{V}_l$ is

$$\pm d_lW^{168-152}\omega_l=\pm d_lW^{16}\omega_l,$$

whose divisor near $(W=0)$ is $16(W=0)$. Since $\zeta\in\mu_{d_l}$ multiplies W by ζ, any point of $\widetilde{V}_l$ fixed by some $\zeta\neq 1$ is in $\widetilde{V}_l\cap(W=0)=\operatorname{Spec}R_\infty\cap(U_l=1)$, where the fixed locus of ζ is of dimension at most 3 by Proposition 6.2(3). Since $(W=0)$ is a Cartier divisor on $\widetilde{V}_l$, which is smooth of dimension 7 along $(W=0)$, any component of $\widetilde{V}_l\cap(W=0)$ is of dimension 6, so a general point of it has trivial stabilizer. Thus $\widetilde{V}_l\to\overline{V}_c\cap(U_l\neq 0)$ is étale over the generic point of the prime divisor V_∞ (Proposition 6.2(3)), the pullback of V_∞ is $(W=0)$, and $\operatorname{mult}_{V_\infty}\operatorname{div}(\omega)=16$. Since $\overline{V}_c\backslash V_c=V_\infty$, we obtain $K_{\overline{V}_c}\sim\operatorname{div}(\omega)=16V_\infty$. In particular, $K_{\overline{V}_c}\sim_{\mathbb{Q}}16V_\infty$.

Finally, D_f is $\mathbb{Q}$-Cartier for any non-zero homogeneous element f of R_c by Lemma 6.3. □

7. Log canonicity on the fibers of h

The goal of this section is to prove Theorem 7.1.

Theorem 7.1. *Notation and conditions as in Set-up 6.1. Let $c\in\mathbb{C}^4\backslash\{0\}$, $J>0$ an integer, and f a non-zero element of $(R_c)_J$. Then $(V_c,\frac{7}{6J}D_f|_{V_c})$ is lc and $(\overline{V}_c,\frac{1}{J}D_f)$ is lc.*

Section 9 uses Theorem 7.1 only through the pair $(\overline{V}_c, \frac{1}{J}D_f)$, for c ranging over $\mathbb{C}^4\backslash\{0\}$ and for vectors $\mathbf{w}_i$ that change under row operations. Thus Theorem 7.1 allows any such c and any vectors $\mathbf{w}_1, \dots, \mathbf{w}_{11}$ any four of which are linearly independent. The coefficient $\frac{7}{6J}$ in the pair $(V_c, \frac{7}{6J}D_f|_{V_c})$ is the coefficient that Proposition 7.3 allows at the smooth points of V_c, and the proof keeps the coefficient $\frac{7}{6J}$ at the singular points of V_c. We prove Theorem 7.1 at the end of this section, treating the smooth points of V_c in Subsection 7.1, the points of V_∞ in Subsection 7.2, and the singular points of V_c in Subsections 7.3–7.6.

7.1. **Smooth points.** In this subsection, we bound the multiplicity of D_f at the smooth points of V_c for any $c \in \mathbb{C}^4$ (Proposition 7.3).

At the smooth points of V_c, Theorem 7.1 reduces to a bound on the multiplicity of D_f [Laz04, Proposition 9.5.13]. To bound $\operatorname{mult}_p D_f$, we intersect D_f with the zero loci of sections of degree 6 and 14 vanishing at p, and Definition-Lemma 7.2 proves that the components that may occur are integral and computes the degree of F_b. The subvariety F_b of Definition-Lemma 7.2 plays the role of the subvariety F_c of Definition-Lemma 5.2 in Section 5.

Definition-Lemma 7.2. Notation and conditions as in Set-up 6.1, and let $c \in \mathbb{C}^4$. Then the following hold.

(1) Any homogeneous prime ideal of height one of R_c is generated by one homogeneous element, and any homogeneous prime element of R_c of degree at most 5 is a non-zero multiple of W.

(2) For $b = (b_1, b_2, b_3) \in \mathbb{C}^3$, the closed subscheme $F_b \subset \overline{V}_c$ defined by $U_{8+l} = b_l W^6$ for $l = 1, 2, 3$ is a normal projective variety of dimension 4, and

$$(\mathcal{O}_{\overline{V}_c}(1)^4 \cdot F_b) = 6^3(\mathcal{O}_{\overline{V}_c}(1)^7).$$

(3) For $b \in \mathbb{C}^3$ and $a_0, \dots, a_5 \in \mathbb{C}$ with $(a_1, \dots, a_5) \neq 0$, the divisor of zeros on F_b of $\sum_{j=1}^5 a_j U_{3+j} + a_0 W^{14}$ is a prime divisor. Here the divisor of zeros on F_b of a non-zero homogeneous element of the graded ring $R_c/(U_9 - b_1W^6, U_{10} - b_2W^6, U_{11} - b_3W^6)$, whose Proj is F_b, is defined as in Set-up 6.1(5), with F_b in place of $\overline{V}_c$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we study three complete intersection rings and their singular loci. Let (N, m) be one of $(11, 42)$, $(8, 42)$, and $(8, 6)$, let $c' \in \mathbb{C}^4$, and let

$$A := \mathbb{C}[\sigma, U_1, \dots, U_N] \Big/ \Big(\sum_{i=1}^N U_i^{e_i} \mathbf{w}_i - c'\sigma^m \Big)$$

be the graded ring with $\deg \sigma = 1$ and $\deg U_i = md_i/42$. Recall that $e_1 = e_2 = e_3 = 2$ and $e_4 = 3$. Since $\mathbf{w}_1, \dots, \mathbf{w}_4$ are linearly independent, the four equations defining A are equivalent to four equations $U_i^{e_i} = P_i$, $1 \le i \le 4$, with $P_i \in \mathbb{C}[\sigma, U_5, \dots, U_N]$. Thus A is a free $\mathbb{C}[\sigma, U_5, \dots, U_N]$-module with basis the 24 monomials $U_1^{\beta_1}U_2^{\beta_2}U_3^{\beta_3}U_4^{\beta_4}$, $0 \le \beta_i < e_i$. In particular, $\dim A = N - 3$, and A is a complete intersection ring, since A is the quotient of a polynomial ring in $N + 1$ variables by 4 elements.

Let (s, u) be a closed point of $\operatorname{Spec} A$, where $s \in \mathbb{C}$ and $u = (u_1, \dots, u_N) \in \mathbb{C}^N$ are the values of σ and $U_1, \dots, U_N$. The column of the Jacobian matrix of the four equations with respect to U_i at (s, u) is $e_i u_i^{e_i-1}\mathbf{w}_i$. Since any four of the $\mathbf{w}_i$ are linearly independent, the Jacobian matrix has rank $4 = (N + 1) - \dim A$ whenever at least four u_i are non-zero, and then the images of the four equations in $\mathfrak{n}/\mathfrak{n}^2$ are linearly independent, where $\mathfrak{n}$ is the maximal ideal of the local ring of $\mathbb{A}^{N+1}$ at (s, u), so the local ring of $\operatorname{Spec} A$ at (s, u) is a regular local ring of dimension $N - 3$ [Mat89, Theorem 14.2]. If $s = 0$ and $u \neq 0$, then $\sum_{i=1}^N u_i^{e_i}\mathbf{w}_i = 0$, so at least five u_i are non-zero by Lemma 3.2(2) with $r = 4$. If $s \neq 0$ and $u_i = 0$ for $i \notin I$, where I is a set of at most three

indices, then

$$\sum_{i\in I} u_i^{e_i}\mathbf{w}_i = c's^m, \tag{7.1}$$

and since $\mathbf{w}_i$, $i \in I$, are linearly independent, the values $u_i^{e_i}$, $i \in I$, are determined by s and I. Let $\Sigma_A \subset \operatorname{Spec} A$ be the closed subset defined by the products $U_{i_1}U_{i_2}U_{i_3}U_{i_4}$ with $i_1 < i_2 < i_3 < i_4$. Then the set of closed points of $\operatorname{Spec} A$ at which the local ring is not regular is contained in Σ_A, and Σ_A has finite fibers over the σ-line, so $\dim \Sigma_A \leq 1$. If $c' = 0$, then (7.1) implies $u_i = 0$ for $i \in I$, so Σ_A is the σ-axis ($U_1 = \cdots = U_N = 0$).

Step 2. In this step we show that A is a normal domain and that any homogeneous prime ideal of height one of A is generated by one homogeneous element. By **Step 1**, $\operatorname{Spec} A$ is a local complete intersection in $\mathbb{A}^{N+1}$. Let $\mathfrak{q} \subset A$ be a prime ideal with $V(\mathfrak{q}) \not\subset \Sigma_A$. Then $V(\mathfrak{q})\backslash\Sigma_A$ is a non-empty open subset of $V(\mathfrak{q})$, so it contains a closed point q, and $A_\mathfrak{q}$ is a localization of the regular local ring A_q, hence regular [Mat89, Theorem 19.3]. If $\operatorname{ht}\mathfrak{q} \leq N-5$, then

$$\dim V(\mathfrak{q}) = \dim A - \operatorname{ht}\mathfrak{q} \geq 2 > \dim \Sigma_A,$$

where the equality $\dim A/\mathfrak{q} + \operatorname{ht}\mathfrak{q} = \dim A$ holds since A is equidimensional, being a finite free module over a polynomial ring by **Step 1**. Thus $A_\mathfrak{q}$ is regular. Since $N - 5 \geq 3$, A is normal [Har77, Proposition II.8.23]. Since A is positively graded with $A_0 = \mathbb{C}$, A is a normal domain by Lemma 3.1.

Let $\mathfrak{m} \subset A$ be the ideal generated by $\sigma, U_1, \ldots, U_N$. For a prime ideal $\mathfrak{q} \subset A$ with $\operatorname{ht}\mathfrak{q} \leq 3 \leq N-5$, $A_\mathfrak{q}$ is regular as shown above, hence factorial [Mat89, Theorem 20.3]. Since $A_\mathfrak{m}$ is a local complete intersection ring, $A_\mathfrak{m}$ is factorial [Gro05, Exp. XI, Corollaire 3.14].

Let $\mathfrak{p} \subset A$ be a homogeneous prime ideal of height one. Then $\mathfrak{p} \subset \mathfrak{m}$, and $\mathfrak{p}A_\mathfrak{m}$ is a prime ideal of height one of the factorial ring $A_\mathfrak{m}$, so $\mathfrak{p}A_\mathfrak{m} = \pi' A_\mathfrak{m}$ for some non-zero $\pi' \in A_\mathfrak{m}$ [Mat89, Theorem 20.1]. Since A is a positively graded $\mathbb{C}$-algebra, $(A, \mathfrak{m})$ is a *local ring [BH98, Examples 1.5.14(b)]. Since $\mathfrak{p}A_\mathfrak{m}$ is generated by one element, $\mathfrak{p}$ is generated by one homogeneous element [BH98, Proposition 1.5.15(a)].

Step 3. In this step we prove (1). The ring R_c is the ring A of **Step 1** for $(N, m) = (11, 42)$, $c' = c$, and $\sigma = W$. Thus the first part of (1) follows from **Step 2**. **Steps 1 and 2** cover the case $c = 0$: the singular locus of $\operatorname{Spec} R_0$ is contained in the W-axis by **Step 1**, and the local ring of $\operatorname{Spec} R_0$ at the vertex is factorial by **Step 2**. Since $d_i \geq 6$ for any i, the only monomial in $W, U_1, \ldots, U_{11}$ of degree $j \leq 5$ is W^j, so $(R_c)_j = \mathbb{C}W^j$ for $0 \leq j \leq 5$. Since the non-zero elements of $(R_c)_0 = \mathbb{C}$ are units, any homogeneous prime element of R_c of degree at most 5 is a non-zero constant multiple of W^j for some $1 \leq j \leq 5$, hence a non-zero multiple of W.

Step 4. In this step we prove (2). Let $b \in \mathbb{C}^3$ and $c_b := c - \sum_{l=1}^3 b_l^7 \mathbf{w}_{8+l}$. Since $U_{8+l}^7 \equiv b_l^7 W^{42}$ modulo $U_{8+l} - b_l W^6$, the homogeneous coordinate ring of F_b is

$$S_b := R_c/(U_9 - b_1W^6, U_{10} - b_2W^6, U_{11} - b_3W^6) = \mathbb{C}[W, U_1, \ldots, U_8]\Big/\Big(\sum_{i=1}^8 U_i^{e_i}\mathbf{w}_i - c_bW^{42}\Big),$$

which is the ring A of **Step 1** for $(N, m) = (8, 42)$, $c' = c_b$, and $\sigma = W$. By **Steps 1 and 2**, S_b is a normal domain with $\dim S_b = 5$, also when $c_b = 0$. For any homogeneous $y \in S_b$, the ring $(S_b)_{(y)}$ is the degree-zero part of the normal graded domain $(S_b)_y$, and an element of the fraction field of $(S_b)_{(y)}$ integral over $(S_b)_{(y)}$ belongs to $(S_b)_y$ and is of degree zero. Thus $(S_b)_{(y)}$ is normal, and $F_b = \operatorname{Proj} S_b$ is a normal projective variety of dimension 4.

We compute the degree of F_b. Since 42 is divisible by each weight of $\mathbb{P}(1, 21^3, 14^5, 6^3)$, the sheaf $\mathcal{O}_{\overline{V}_c}(42)$ is a line bundle, and intersection numbers of four and of seven factors $\mathcal{O}_{\overline{V}_c}(1)$ are 42^{-4} and 42^{-7} times the corresponding intersection numbers with $\mathcal{O}_{\overline{V}_c}(42)$. By **Step 1**, S_b/WS_b is a free $\mathbb{C}[U_5, \ldots, U_8]$-module of finite rank, so $\dim(F_b \cap V_\infty) = 3$, and the generic point η of F_b belongs to V_c. Since V_c has only finitely many singular points (Proposition 6.2(2)), the local

ring $B := \mathcal{O}_{\overline{V}_c,\eta}$ is a regular local ring of dimension 3. Let $t_l := u_{8+l} - b_l$ for $l = 1,2,3$. Since F_b is integral, $B/(t_1,t_2,t_3) = \mathcal{O}_{F_b,\eta}$ is a field, so t_1,t_2,t_3 is a regular system of parameters of B. The sections $(U_{8+l} - b_l W^6)^7$ of $\mathcal{O}_{\overline{V}_c}(42)$ have local equations t_l^7 at η, and the common zero locus of the three sections is F_b, with $\dim F_b = 7-3$. Hence by [Ful98, Example 7.1.10(a)], the intersection cycle of the three divisors of zeros is $e[F_b]$ with

$$e = \operatorname{length} B/(t_1^7, t_2^7, t_3^7) = 7^3,$$

where the last equality holds since the completion of B is a power series ring in t_1,t_2,t_3 over a field. Thus

$$7^3(\mathcal{O}_{\overline{V}_c}(42)^4 \cdot F_b) = (\mathcal{O}_{\overline{V}_c}(42)^7),$$

that is,

$$7^3 \cdot 42^4(\mathcal{O}_{\overline{V}_c}(1)^4 \cdot F_b) = 42^7(\mathcal{O}_{\overline{V}_c}(1)^7).$$

Since $42^3/7^3 = 6^3$, we obtain

$$(\mathcal{O}_{\overline{V}_c}(1)^4 \cdot F_b) = 6^3(\mathcal{O}_{\overline{V}_c}(1)^7).$$

Step 5. In this step we prove (3). By **Step 4**, the graded ring in (3) is S_b, and $F_b = \operatorname{Proj} S_b$. We first pass to the seventh Veronese ring of S_b, in which the elements of degree one are the multiples of W^7.

Let $S_b^{(7)} := \bigoplus_{k\geq 0}(S_b)_{7k}$ be the graded ring with $(S_b^{(7)})_k := (S_b)_{7k}$. Since 7 divides $d_1,\dots,d_8$, a monomial $W^a U_1^{\beta_1}\cdots U_8^{\beta_8}$ has degree divisible by 7 if and only if $7 \mid a$. Moreover,

$$\mathbb{C}[W,U_1,\dots,U_8] = \bigoplus_{k=0}^{6} W^k \mathbb{C}[W^7,U_1,\dots,U_8],$$

and the four equations defining S_b belong to $\mathbb{C}[W^7,U_1,\dots,U_8]$. Thus

$$S_b^{(7)} = \mathbb{C}[W^7,U_1,\dots,U_8]\Big/\Big(\sum_{i=1}^{8} U_i^{e_i}\mathbf{w}_i - c_b(W^7)^6\Big),$$

which is the ring A of **Step 1** for $(N,m) = (8,6)$, $c' = c_b$, and $\sigma = W^7$, with $\deg W^7 = 1$, $\deg U_i = 3$ for $i \leq 3$, and $\deg U_i = 2$ for $4 \leq i \leq 8$. By **Step 2**, $S_b^{(7)}$ is a normal domain, and any homogeneous prime ideal of height one of $S_b^{(7)}$ is generated by one homogeneous element, also when $c_b = 0$.

For a homogeneous element y of S_b, we have $D_+(y) = D_+(y^7)$, and if $7 \mid \deg y$, then $(S_b)_{(y)} = (S_b^{(7)})_{(y)}$. The open subsets $D_+(y)$ with $y \in \{W^7,U_1,\dots,U_8\}$ cover F_b. Thus $F_b = \operatorname{Proj} S_b$ is equal to $\operatorname{Proj} S_b^{(7)}$, the sheaf $\mathcal{O}(1)$ of $\operatorname{Proj} S_b^{(7)}$ is $\mathcal{O}_{F_b}(7) = \widetilde{S_b(7)}$, and for a homogeneous element g of S_b of degree divisible by 7, the divisor of zeros of g on F_b with respect to the grading of S_b equals the divisor of zeros of g on F_b with respect to the grading of $S_b^{(7)}$.

Let $a_0,\dots,a_5 \in \mathbb{C}$ with $(a_1,\dots,a_5) \neq 0$, and let $x := \sum_{j=1}^{5} a_j U_{3+j} + a_0 W^{14}$, an element of degree 2 of $S_b^{(7)}$. Since any U_i is of degree at least 2 in $S_b^{(7)}$, we have $(S_b^{(7)})_1 = \mathbb{C}W^7$. We show that $x \notin W^7 S_b$. With respect to the basis of **Step 1** over $\mathbb{C}[W,U_5,\dots,U_8]$, the coefficient of x at U_4 is a_1, and the coefficient of x at 1 is $\sum_{j=2}^{5} a_j U_{3+j} + a_0 W^{14}$. If $x \in W^7 S_b$, then both coefficients are divisible by W^7 in $\mathbb{C}[W,U_5,\dots,U_8]$, so $a_1 = \cdots = a_5 = 0$, a contradiction.

Let $\mathfrak{p}$ be a minimal prime ideal over $xS_b^{(7)}$. Then $\mathfrak{p}$ is a minimal element of $\operatorname{Supp}(S_b^{(7)}/xS_b^{(7)})$, hence an associated prime of $S_b^{(7)}/xS_b^{(7)}$ [Mat89, Theorem 6.5(iii)], hence homogeneous [BH98, Lemma 1.5.6(b)(ii)], and $\mathfrak{p}$ has height one [Mat89, Theorem 13.5], so $\mathfrak{p} = \pi S_b^{(7)}$ for a homogeneous element π by **Step 2**. Write $x = \pi z$ with $z \in S_b^{(7)}$, which is homogeneous since x and π are homogeneous and $S_b^{(7)}$ is a domain. Since π is not a unit and the non-zero elements of $(S_b^{(7)})_0 = \mathbb{C}$

are units, $\deg \pi \in \{1,2\}$. If $\deg \pi = 1$, then $\pi \in \mathbb{C}^* W^7$ and $x \in W^7 S_b$, which is impossible. Thus $\deg z = 0$, $z \in \mathbb{C}^*$, and x is a prime element of $S_b^{(7)}$.

Since $\deg W^7 = 1$, we have $S_b^{(7)}[W^{-7}] = (S_b^{(7)})_{(W^7)}[W^7, W^{-7}]$, and

$$S_b^{(7)}[W^{-7}]/(x) = \Big((S_b^{(7)})_{(W^7)}/(xW^{-14})\Big)[W^7, W^{-7}].$$

The left-hand side is a non-zero localization of the domain $S_b^{(7)}/(x)$, since $W^7 \notin xS_b^{(7)}$. Thus xW^{-14} is a prime element of the normal domain $(S_b^{(7)})_{(W^7)}$, and the divisor of zeros of x on $F_b \cap V_c = D_+(W^7)$ is the prime divisor $V_+(x) \cap V_c$ with coefficient one. Moreover, $V_+(x)$ is irreducible with $\dim V_+(x) = 3$, and $V_+(x) \not\subset V_+(W^7)$, since otherwise W^7 would belong to the prime ideal $xS_b^{(7)}$, which is impossible as $\deg x = 2 > \deg W^7$. Hence the divisor of zeros of x on F_b with respect to the grading of $S_b^{(7)}$, which equals the divisor of zeros of x on F_b with respect to the grading of S_b, is the prime divisor $V_+(x)$. □

Proposition 7.3 gives the coefficient $\frac{7}{6J}$ of Theorem 7.1 at the smooth points of V_c. In its proof, after the cuts described before Definition-Lemma 7.2, we bound the multiplicities of the integral component (F_b or a prime divisor on F_b) and of the remaining cycle separately, both against the bound B on the degree in (7.4).

Proposition 7.3. *Notation and conditions as in Set-up 6.1. Let $c \in \mathbb{C}^4$, $J > 0$ an integer, f a non-zero element of $(R_c)_J$, and $p \in V_c$ a smooth point of V_c. Then* $\operatorname{mult}_p D_f \le \frac{6}{7}J$. *In particular, $(V_c, \frac{7}{6J} D_f|_{V_c})$ is lc near p.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the polarization and the rules for cutting cycles.

Each weight of $\mathbb{P}(1, 21^3, 14^5, 6^3)$ divides 42, so $\mathcal{O}(42)$ is an ample invertible sheaf on $\mathbb{P}(1, 21^3, 14^5, 6^3)$. Let g be a non-zero homogeneous element of R_c. By Lemma 6.3, applied to g and to W, D_g is a $\mathbb{Q}$-Cartier Weil divisor, and

$$42 D_g - 42(\deg g) D_W = \operatorname{div}(g^{42}/W^{42 \deg g}).$$

Let $H := D_W$. Then H is an ample $\mathbb{Q}$-Cartier Weil divisor, $D_g \sim_{\mathbb{Q}} (\deg g) H$, and

$$\mathcal{O}_{\overline{V}_c}(42H) \cong \mathcal{O}_{\overline{V}_c}(42).$$

On $V_c = \overline{V}_c \cap (W \neq 0)$, D_g is the Cartier divisor of the regular function $g|_{V_c} := g(1, u_1, \dots, u_{11})$. The intersection number $(H^7) = (\mathcal{O}_{\overline{V}_c}(1)^7)$ satisfies

$$(H^7) = \frac{42^4}{21^3 \cdot 14^5 \cdot 6^3}, \qquad 6^3 \cdot 14^3 \cdot (H^7) = \frac{12}{7}. \tag{7.2}$$

Since $\mathcal{O}(42)$ is invertible, the zero loci of the four sections of $\mathcal{O}(42)$ cutting out $\overline{V}_c$ are Cartier divisors, and since $\dim \overline{V}_c = 7$, they intersect properly. The general point of $\overline{V}_c$ is contained in the chart $(W \neq 0) \cong \mathbb{A}^{11}$, so the local ring of $\mathbb{P}(1, 21^3, 14^5, 6^3)$ along $\overline{V}_c$ is regular. On $(W \neq 0)$, the local equations of the four sections generate the prime ideal $(h(u) - c)$, because $\mathbb{C}[u_1, \dots, u_{11}]/(h(u) - c) = R_c/(W - 1)$ is a domain (Proposition 6.2(1)). Hence the intersection multiplicity along $\overline{V}_c$ is the length 1 [Ful98, Example 7.1.10(a)], so $(H^7) = 42^4(\mathcal{O}(1)^{11})$, where $(\mathcal{O}(1)^{11})$ is computed on $\mathbb{P}(1, 21^3, 14^5, 6^3)$. Write $d_0 := 1$. The morphism $\mathbb{P}^{11} \to \mathbb{P}(1, 21^3, 14^5, 6^3)$ given by

$$[x_0 : \cdots : x_{11}] \to [x_0^{d_0} : \cdots : x_{11}^{d_{11}}]$$

is finite of degree $d_0 \cdots d_{11} = 21^3 \cdot 14^5 \cdot 6^3$, and the pullback of $\mathcal{O}(42)$ is $\mathcal{O}_{\mathbb{P}^{11}}(42)$. Hence $(\mathcal{O}(1)^{11}) = 1/(21^3 \cdot 14^5 \cdot 6^3)$, which is (7.2). Lemma 2.8 is used below for the normal projective variety $\overline{V}_c$ (Proposition 6.2(1)), the divisor H, and the smooth point p of $\overline{V}_c$. By Lemma 2.8(1) applied to the cycle $\overline{V}_c$, $(H^6 \cdot D_g) = (\deg g)(H^7)$.

For a $\mathbb{Q}$-cycle $\Gamma \geq 0$ of pure dimension k, let Γ_p be the sum of the components of Γ containing p, with their coefficients. Then $\operatorname{mult}_p \Gamma_p = \operatorname{mult}_p \Gamma$ and $(H^k \cdot \Gamma_p) \leq (H^k \cdot \Gamma)$. Let g be a non-zero homogeneous element of R_c such that $g|_{V_c} \in \mathfrak{m}_p^s$ for an integer $s \geq 1$ and no component of Γ_p is contained in $\operatorname{Supp} D_g$. By Lemma 2.8, $\Gamma^g := (D_g \cdot \Gamma_p)_p$ is a $\mathbb{Q}$-cycle of pure dimension $k-1$ with

$$(H^{k-1} \cdot \Gamma^g) \leq (\deg g)(H^k \cdot \Gamma), \qquad \operatorname{mult}_p \Gamma^g \geq s \operatorname{mult}_p \Gamma. \tag{7.3}$$

If Σ is a vector space of homogeneous elements of one degree vanishing at p and no component of Γ_p is contained in the common zero locus of Σ, then a general $g \in \Sigma$ satisfies $g|_{V_c} \in \mathfrak{m}_p$, and no component of Γ_p is contained in $\operatorname{Supp} D_g$, so (7.3) holds for g with $s = 1$. If $k = 0$, then $\operatorname{mult}_p \Gamma \leq \deg \Gamma$.

Finally, let $\Gamma \geq 0$ be a $\mathbb{Q}$-cycle of pure dimension k with $\Gamma = m\Theta + \Gamma'$, where $m \geq 0$, Θ is an integral subvariety of dimension k containing p, and Θ is not a component of Γ'. Assume that $(H^k \cdot \Gamma) \leq B$, $\operatorname{mult}_p \Theta \leq \vartheta$, and $\operatorname{mult}_p \Gamma' \leq \rho(H^k \cdot \Gamma')$ for real numbers $B, \vartheta, \rho > 0$. Then

$$\begin{aligned} \operatorname{mult}_p \Gamma = m \operatorname{mult}_p \Theta + \operatorname{mult}_p \Gamma' &\leq \frac{\vartheta}{(H^k \cdot \Theta)} m (H^k \cdot \Theta) + \rho (H^k \cdot \Gamma') \\ &\leq \max\left\{ \frac{\vartheta}{(H^k \cdot \Theta)}, \rho \right\} B. \end{aligned} \tag{7.4}$$

Table 1 lists the values of Θ, $(H^k \cdot \Theta)$, ϑ, ρ, and B in the three applications of (7.4), where the gain is the factor by which the cuts multiply the lower bound on the multiplicity of the divisor that is cut.

Step	Θ	$(H^k \cdot \Theta)$	ϑ	ρ	B	gain	$\frac{1}{\text{gain}} \max\left\{ \frac{\vartheta B}{(H^k \cdot \Theta)}, \rho B \right\}$
3	F_b	$6^3(H^7)$	$3 \cdot 2^{3-n}$	$6 \cdot 14^3$	$6^2 J(H^7)$	2^{3-n}	$\max\{\frac{1}{2}, \frac{12}{7 \cdot 2^{3-n}}\} J \leq \frac{6}{7} J$
5	G	$14 \cdot 6^3(H^7)$	6	$6 \cdot 14^2$	$14 \cdot 6^2 J'(H^7)$	2	$\max\{\frac{1}{2}, \frac{6}{7}\} J' = \frac{6}{7} J'$
7	F_b	$6^3(H^7)$	3	$6 \cdot 14^3$	$6^2 J''(H^7)$	2	$\max\{\frac{1}{4}, \frac{6}{7}\} J'' = \frac{6}{7} J''$

TABLE 1. The bounds (7.4) for mult_p of the divisor that is cut, computed with (7.2) and Definition-Lemma 7.2(2).

Step 2. In this step we describe V_c and F_b near p.

The ideal of V_c in $\mathbb{C}[u_1, \dots, u_{11}]$ is generated by the four equations $h(u) = c$, since $\mathbb{C}[u_1, \dots, u_{11}]/(h(u) - c) = R_c/(W - 1)$ is a domain (Proposition 6.2(1)), and $\dim V_c = 7$. The i-th column of the Jacobian matrix of h at u is $e_i u_i^{e_i - 1} \mathbf{w}_i$, and any four of the $\mathbf{w}_i$ are linearly independent, so the rank of the Jacobian matrix at u is $\min\{\#\{i \mid u_i \neq 0\}, 4\}$. By the Jacobian criterion [Har77, Theorem I.5.1], a point of V_c is smooth if and only if at least four of the coordinates u_i of the point are non-zero. Let n_2 and n_3 be the numbers of indices $i \leq 3$ and $4 \leq i \leq 8$, respectively, with $u_i(p) \neq 0$, and let $n := n_2 + n_3$. Since only three indices satisfy $9 \leq i \leq 11$, $n \geq 1$, and if $n_2 = 3$ and $n_3 = 0$, then $u_{8+l}(p) \neq 0$ for some $1 \leq l \leq 3$.

Let $b := (u_9(p), u_{10}(p), u_{11}(p))$ and $c_b := c - \sum_{l=1}^3 b_l^7 \mathbf{w}_{8+l}$. Then $p \in F_b$, and since F_b is a variety (Definition-Lemma 7.2(2)), $F_b \cap V_c$ is the closed subvariety of $\mathbb{A}^8$ with coordinates $u_1, \dots, u_8$ defined by $\sum_{i=1}^8 u_i^{e_i} \mathbf{w}_i = c_b$. Let Σ_b be the linear system spanned by $U_{8+l} - b_l W^6$, $l = 1, 2, 3$, and let $\Sigma_b' \subset \Sigma_b$ be the subspace of the elements g with $g|_{V_c} \in \mathfrak{m}_p^2$. The common zero locus of Σ_b is F_b. We show that

$$\dim \Sigma_b' = \max\{4 - n, 0\}.$$

An element $\sum_l \lambda_l (U_{8+l} - b_l W^6)$ belongs to Σ_b' if and only if $\sum_l \lambda_l du_{8+l}$ vanishes on the tangent space of V_c at p, that is, if and only if there exists $\xi \in (\mathbb{C}^4)^*$ with $\xi(\mathbf{w}_i) e_i u_i(p)^{e_i - 1} = 0$ for $i \leq 8$ and $\lambda_l = 7\xi(\mathbf{w}_{8+l}) b_l^6$ for $l = 1, 2, 3$. The element ξ is unique, because the Jacobian matrix at p has rank 4. Thus Σ_b' is isomorphic to the space of ξ with $\xi(\mathbf{w}_i) = 0$ for any $i \leq 8$ with $u_i(p) \neq 0$, and the dimension of the space of such ξ is $\max\{4 - n, 0\}$.

Let Σ_p be the linear system spanned by $U_j - u_j(p)W^{14}$, $4 \le j \le 8$, and $W^8(U_{8+l} - b_l W^6)$, $l = 1, 2, 3$. The elements of Σ_p have degree 14 and vanish at p. On V_c, the common zeros of Σ_p have fixed $u_4, \dots, u_{11}$, and then u_1^2, u_2^2, u_3^2 are determined by the equations $h(u) = c$, because $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are linearly independent. Hence the intersection of V_c with the common zero locus of Σ_p is finite. If $Z \subset \overline{V}_c$ is an irreducible closed subset with $\dim Z \ge 1$ and $p \in Z$, then $\dim(Z \cap V_c) \ge 1$, so by **Step 1** a cycle Γ of pure dimension $k \ge 1$ can be cut k times by general elements of Σ_p. Each cut multiplies the bound on the degree by 14, and the cycle Γ_k after the last cut satisfies $\dim \Gamma_k = 0$. Thus

$$\operatorname{mult}_p \Gamma \le 14^k (H^k \cdot \Gamma). \tag{7.5}$$

We compute multiplicities of diagonal complete intersections. Let $M \subset \{1, \dots, 8\}$ be a subset such that $\{1, \dots, 8\} \backslash M$ consists of at most one index $j \ge 4$, let $c_M \in \mathbb{C}^4$, and let $V_M \subset \mathbb{A}^M$ be the subscheme defined by $\sum_{i \in M} u_i^{e_i} \mathbf{w}_i = c_M$. Let $q \in V_M$ be a point such that V_M is integral near q, and assume that the set I_q of indices $i \in M$ with $u_i(q) \ne 0$ satisfies $1 \le \#I_q \le 3$. We show that $\dim V_M = \#M - 4$ near q and

$$\operatorname{mult}_q V_M \le 2^{4 - \#I_q} \text{ if } I_q \not\subset \{1,2,3\}, \qquad \operatorname{mult}_q V_M \le 3 \cdot 2^{3 - \#I_q} \text{ if } I_q \subset \{1,2,3\}. \tag{7.6}$$

Choose $\eta_i \in (\mathbb{C}^4)^*$, $i \in I_q$, with $\eta_i(\mathbf{w}_i) = 1$ and $\eta_i(\mathbf{w}_{i'}) = 0$ for $i' \in I_q \backslash \{i\}$, and a basis $\eta'_1, \dots, \eta'_{4 - \#I_q}$ of the space of $\eta \in (\mathbb{C}^4)^*$ with $\eta(\mathbf{w}_i) = 0$ for any $i \in I_q$. Applying η_i, the equation of V_M expresses $u_i^{e_i}$ as $u_i(q)^{e_i} \ne 0$ plus a polynomial in u_j, $j \in M \backslash I_q$, vanishing at q, so u_i is a power series in u_j, $j \in M \backslash I_q$. Since $u_j(q) = 0$ for $j \notin I_q$, applying η'_t to the equation at q shows $\eta'_t(c_M) = 0$. Thus the complete local ring of V_M at q is

$$\mathbb{C}[[u_j \mid j \in M \backslash I_q]] \Big/ \Big(\sum_{j \in M \backslash I_q} \eta'_t(\mathbf{w}_j) u_j^{e_j} \;\Big|\; 1 \le t \le 4 - \#I_q \Big).$$

By Lemma 3.2(4) (with $r = 4$), any $4 - \#I_q$ of the images of $\mathbf{w}_j$, $j \in M \backslash I_q$, in $\mathbb{C}^4 / \langle \mathbf{w}_i \mid i \in I_q \rangle$ are linearly independent. If $I_q \not\subset \{1,2,3\}$, then $M \backslash I_q$ contains $3 - \#(I_q \cap \{1,2,3\}) \ge 4 - \#I_q$ indices $j \le 3$. After a change of the basis η'_t, the t-th equation becomes $u_{j_t}^2$ plus a combination of the other $u_j^{e_j}$ for distinct indices $j_t \le 3$. If $I_q \subset \{1,2,3\}$, then $M \backslash I_q$ contains the $3 - \#I_q$ indices $j \le 3$ outside I_q and at least four indices $j \ge 4$. After a change of the basis η'_t, the equations become $u_{j_t}^2$ plus terms of degree at least 3 for the $3 - \#I_q$ indices $j_t \le 3$ outside I_q, and $u_{j'}^3$ plus terms $u_j^{e_j}$ with $j \notin \{1,2,3\} \cup \{j'\}$ for one index $j' \ge 4$. In both cases the homogeneous parts of lowest degree of the equations are monic in distinct variables u_{j_t} (and $u_{j'}$), so they form a regular sequence in the polynomial ring, of degrees $2, \dots, 2$ in the first case and $2, \dots, 2, 3$ in the second case.

In particular, $\dim V_M = \#M - \#I_q - (4 - \#I_q) = \#M - 4$ near q, and V_M is Cohen–Macaulay near q (the tangent cone of V_M at q is cut out by an ideal containing the homogeneous parts of lowest degree above, so $\dim_q V_M \le \#M - 4$, with equality by [Mat89, Theorem 13.6(ii)], applied to the maximal ideal of $\mathcal{O}_{\mathbb{A}^M, q}$, which has height $\#M$, and V_M is Cohen–Macaulay near q by [Har77, Proposition II.8.23(a)]). Let $\ell_1, \dots, \ell_{\#M-4}$ be general linear forms in u_j, $j \in M \backslash I_q$. Consider the $\#M$ divisors on the non-singular variety $\mathbb{A}^M$ defined by the four equations $\eta_i\big(\sum_{i' \in M} u_{i'}^{e_{i'}} \mathbf{w}_{i'} - c_M\big)$, $i \in I_q$, and $\eta'_t\big(\sum_{i' \in M} u_{i'}^{e_{i'}} \mathbf{w}_{i'} - c_M\big)$, $1 \le t \le 4 - \#I_q$, of V_M and by $\ell_1, \dots, \ell_{\#M-4}$. Their homogeneous parts of lowest degree at q are non-zero multiples of $u_i - u_i(q)$ for $i \in I_q$, the forms of degrees $2, \dots, 2$ (and 3) above, and $\ell_1, \dots, \ell_{\#M-4}$. The general linear forms $\ell_1, \dots, \ell_{\#M-4}$ extend the regular sequence of the forms of degrees $2, \dots, 2$ (and 3) in $\mathbb{C}[u_j \mid j \in M \backslash I_q]$, so the $\#M$ homogeneous parts of lowest degree form a regular sequence in the polynomial ring. Thus the intersection multiplicity of the $\#M$ divisors at q is the product of the degrees [Ful98, Example 12.4.9]. Since a regular local ring is Cohen–Macaulay, the intersection multiplicity of the $\#M$ divisors at q is equal to the length of $\mathcal{O}_{\mathbb{A}^M, q}$ modulo the $\#M$ equations [Ful98, Example 7.1.10(a)], that is, the length of $\mathcal{O}_{V_M, q} / (\ell_1, \dots, \ell_{\#M-4})$. Since V_M is Cohen–Macaulay near q, the length of $\mathcal{O}_{V_M, q} / (\ell_1, \dots, \ell_{\#M-4})$ is equal to the intersection

multiplicity of $\ell_1, \dots, \ell_{\#M-4}$ on V_M at q [Ful98, Example 7.1.10(a)], which is at least $\operatorname{mult}_q V_M$ [Ful98, Example 12.4.8], since V_M is integral near q. Thus (7.6) holds.

We apply (7.6) to $V_M = F_b \cap V_c$, $M = \{1, \dots, 8\}$, $c_M = c_b$, and $q = p$, where $F_b \cap V_c$ is integral by Definition-Lemma 7.2(2). If $n \le 3$, then

$$\operatorname{mult}_p F_b \le 3 \cdot 2^{3-n}, \tag{7.7}$$

since $2^{4-n} \le 3 \cdot 2^{3-n}$. If $n \ge 4$, then the Jacobian matrix of the equations of $F_b \cap V_c$ at p has rank 4, so F_b is smooth at p by the Jacobian criterion [Har77, Theorem I.5.1].

Let $\xi_0 \in (\mathbb{C}^4)^*$ be a non-zero element with $\xi_0(\mathbf{w}_1) = \xi_0(\mathbf{w}_2) = \xi_0(\mathbf{w}_3) = 0$, and let $\alpha_j := \xi_0(\mathbf{w}_{3+j})$ and $\beta_l := \xi_0(\mathbf{w}_{8+l})$. Since any four of the $\mathbf{w}_i$ are linearly independent, $\alpha_j \ne 0$ and $\beta_l \ne 0$ for $1 \le j \le 5$ and $1 \le l \le 3$. Let $Q \subset \mathbb{A}^8$, with coordinates $u_4, \dots, u_{11}$, be the hypersurface defined by

$$\Phi := \sum_{j=1}^{5} \alpha_j u_{3+j}^3 + \sum_{l=1}^{3} \beta_l u_{8+l}^7 - \xi_0(c) = 0,$$

and let $\varphi : V_c \to Q$ be the projection $u \to (u_4, \dots, u_{11})$. Since $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are linearly independent, $\mathbb{C}[u_1, \dots, u_{11}]/(h(u) - c)$ is generated by u_1, u_2, u_3 over $\mathbb{C}[u_4, \dots, u_{11}]/(\Phi)$ with relations $u_i^2 = A_i(u_4, \dots, u_{11})$, $i = 1, 2, 3$, for polynomials A_i. Hence φ is finite.

If $n \ge 3$ and $n_3 = 0$, then $n_2 = n = 3$. Thus one of the following holds: $n \le 2$; $n \ge 3$, $n_3 \ge 1$, and $n_2 \le 2$; $n_2 = 3$ and $n_3 \ge 1$; $n_2 = 3$ and $n_3 = 0$.

Step 3. In this step we prove the proposition when $n \le 2$.

Since $n \le 2$, $\dim \Sigma_b' = 4 - n \ge 2$ by **Step 2**. Choose $n - 1$ general elements of Σ_b and then $3 - n$ general elements of Σ_b'. When $n = 1$, we take no element of Σ_b. In both cases we take two elements, and together with Σ_b' they span Σ_b. We cut $(D_f)_p$ first by the element of Σ_b if $n = 2$, and then by the elements of Σ_b'. The components of $(D_f)_p$ and of the cycle after the first cut have dimension 6 and 5, so they are not contained in F_b. Moreover, the cycle after the first cut is supported in the zero locus of the first element, and the common zero locus of the first element and Σ_b' is F_b (or the common zero locus of $\Sigma_b' = \Sigma_b$ is F_b if $n = 1$). Hence by **Step 1** both cuts satisfy the condition of (7.3), with $s = 1$ for an element of Σ_b and $s = 2$ for an element of Σ_b'. We obtain a $\mathbb{Q}$-cycle $\Gamma \ge 0$ of pure dimension 4 with

$$(H^4 \cdot \Gamma) \le 6^2 J(H^7), \qquad \operatorname{mult}_p \Gamma \ge 2^{3-n} \operatorname{mult}_p D_f.$$

Since F_b is integral by Definition-Lemma 7.2(2), we can write $\Gamma = mF_b + \Gamma'$, where F_b is not a component of Γ'. No component of Γ_p' is contained in F_b, so a general element of Σ_b cuts Γ', and by (7.5) with $k = 3$,

$$\operatorname{mult}_p \Gamma' \le 6 \cdot 14^3 (H^4 \cdot \Gamma').$$

By (7.7), $\operatorname{mult}_p F_b \le 3 \cdot 2^{3-n}$, and $(H^4 \cdot F_b) = 6^3(H^7)$ by Definition-Lemma 7.2(2). By (7.4) and the first row of Table 1,

$$\operatorname{mult}_p D_f \le \max\left\{\frac{1}{2}, \frac{12}{7 \cdot 2^{3-n}}\right\} J \le \frac{6}{7} J,$$

since $2^{3-n} \ge 2$.

Step 4. In this step we assume that $n \ge 3$ and $n_3 \ge 1$, and we construct sections of degree 14 vanishing to order 2 at p.

Let Λ_p be the space of $\xi \in (\mathbb{C}^4)^*$ such that $\xi(\mathbf{w}_i) = 0$ for any $i \le 3$ with $u_i(p) \ne 0$, so $\dim \Lambda_p = 4 - n_2$. The element ξ_0 of **Step 2** belongs to Λ_p. For $\xi \in \Lambda_p$, let

$$T_\xi := \sum_{j=4}^{8} 3\xi(\mathbf{w}_j) u_j(p)^2 \big(U_j - u_j(p) W^{14}\big) + \sum_{l=1}^{3} 7\xi(\mathbf{w}_{8+l}) b_l^6 W^8 \big(U_{8+l} - b_l W^6\big) \in \Sigma_p.$$

The linear part at p of $\xi(h(u))-\xi(c)$ is $T_\xi|_{V_c}$, because $\xi(\mathbf{w}_i)u_i(p)=0$ for $i\le 3$. Since $\xi(h(u))-\xi(c)$ vanishes on V_c, $T_\xi|_{V_c}\in\mathfrak{m}_p^2$. If all coefficients of T_ξ vanish, then $\xi(\mathbf{w}_i)=0$ for the at least four indices i with $u_i(p)\neq 0$, so $\xi=0$. Thus $\xi\to T_\xi$ is injective. Let $j_0\ge 4$ be an index with $u_{j_0}(p)\neq 0$. The vectors $\mathbf{w}_{j_0}$ and $\mathbf{w}_i$, $i\le 3$ with $u_i(p)\neq 0$, are linearly independent, so $\xi(\mathbf{w}_{j_0})\neq 0$ for a general $\xi\in\Lambda_p$. On F_b we have $U_{8+l}=b_lW^6$, so T_ξ restricts to F_b as $\sum_{j=4}^8 3\xi(\mathbf{w}_j)u_j(p)^2U_j+a_0W^{14}$ for some $a_0\in\mathbb{C}$. Assume that $\xi(\mathbf{w}_{j_0})\neq 0$. By Definition-Lemma 7.2(3), the divisor of zeros G_ξ of T_ξ on F_b is a prime divisor. Hence $T_\xi|_{F_b}\neq 0$, so $F_b\not\subset\operatorname{Supp}D_{T_\xi}$. On the charts $(U_i\neq 0)$ and $(W\neq 0)$, the local equations of $42D_{T_\xi}$ are $T_\xi^{42}/U_i^{42\cdot 14/d_i}$ and $T_\xi^{42}/W^{42\cdot 14}$, and their divisors on F_b are $42G_\xi$ by the definition of the divisor of zeros in Set-up 6.1(5). Thus

$$D_{T_\xi}\cdot F_b=\frac{1}{42}(42D_{T_\xi})\cdot F_b=G_\xi$$

by Definition 2.7 [Ful98, Definition 2.3], and $(H^3\cdot G_\xi)=14\cdot 6^3(H^7)$ by Lemma 2.8(1).

We show that Q contains no affine linear subspace L of dimension 6. Suppose that $L=x_0+L_0$ is such a subspace, where L_0 is a linear subspace. The homogeneous part of degree 7 of $v\to\Phi(x_0+v)$ is $\sum_l\beta_lu_{8+l}(v)^7$, so $\sum_l\beta_lu_{8+l}^7$ vanishes on the projection of L_0 to the coordinates u_9,u_{10},u_{11}. The plane curve $(\sum_l\beta_lz_l^7=0)\subset\mathbb{P}^2$ is smooth and irreducible, so it contains no line, and the projection of L_0 is of dimension at most 1. Hence the kernel $L_0'\subset L_0$ of the projection satisfies $\dim L_0'\ge 5$, and L_0' is contained in the space $\mathbb{C}^5$ of the coordinates $u_4,\dots,u_8$, so $L_0'=\mathbb{C}^5$. Then the homogeneous part of degree 3 of $v\to\Phi(x_0+v)$ on this $\mathbb{C}^5$ is $\sum_j\alpha_ju_{3+j}^3\neq 0$, a contradiction.

We show that if $n_2\le 2$, then a general $\xi\in\Lambda_p$ satisfies: no component of D_f containing p is contained in $\operatorname{Supp}D_{T_\xi}$. For a prime divisor S on $\overline{V}_c$ containing p, the set of $\xi\in\Lambda_p$ with $T_\xi|_S=0$ is a linear subspace. Suppose that the subspace is equal to Λ_p. Since $\dim\Lambda_p=4-n_2\ge 2$, there exist linearly independent $\xi,\xi'\in\Lambda_p$, and $T_\xi|_{V_c}$ and $T_{\xi'}|_{V_c}$ are pullbacks by φ of linearly independent affine linear functions on $\mathbb{A}^8$ vanishing at $\varphi(p)$. Their common zero locus L is an affine linear subspace of dimension 6, so $\dim(Q\cap L)\le 5$. Since φ is finite, $\dim\varphi^{-1}(Q\cap L)\le 5$, which contradicts $S\cap V_c\subset\varphi^{-1}(Q\cap L)$ and $\dim(S\cap V_c)=6$. Thus the subspace is proper, and the claim follows since D_f has finitely many components.

We show that $\operatorname{mult}_pG_\xi\le 6$ for a general $\xi\in\Lambda_p$, and $\operatorname{mult}_pG_{\xi_0}\le 3$ if $n_2=3$. First assume that $n\ge 4$. Then F_b is smooth at p by **Step 2**. Let $Q_b\subset\mathbb{A}^5$, with coordinates $u_4,\dots,u_8$, be the hypersurface defined by $\Phi_b:=\sum_j\alpha_ju_{3+j}^3-\xi_0(c_b)=0$, and $p_b:=(u_4(p),\dots,u_8(p))$. Then $F_b\cap V_c$ is defined by $\Phi_b=0$ and $u_i^2=A_{b,i}(u_4,\dots,u_8)$, $i=1,2,3$, for polynomials $A_{b,i}$, and $T_{\xi_0}|_{F_b\cap V_c}$ is the pullback of

$$\tau:=\sum_{j=1}^5 3\alpha_ju_{3+j}(p)^2\big(u_{3+j}-u_{3+j}(p)\big).$$

Since $u_{j_0}(p)\neq 0$, Q_b is smooth at p_b, and τ is the differential of Φ_b at p_b. We show that $\tau|_{Q_b}\notin\mathfrak{m}_{p_b}^4$. If Φ_b vanishes on the hyperplane $(\tau=0)$, then τ divides Φ_b, so the linear part of τ divides $\sum_j\alpha_ju_{3+j}^3$, which is irreducible since $(\sum_j\alpha_jz_j^3=0)\subset\mathbb{P}^4$ is smooth, a contradiction. Thus there exists a vector v with $\tau(p_b+v)=0$ such that $\Phi_b(p_b+tv)$ is a non-zero polynomial in t of degree at most 3, vanishing at $t=0$ to order $m\le 3$. Let v' be a vector with $\tau(p_b+v')=1$, and $\Psi(t,t'):=\Phi_b(p_b+tv+t'v')$. Since $\partial\Psi/\partial t'(0,0)\neq 0$, $(\Psi=0)$ is a smooth curve germ $C_0\subset Q_b$ at p_b, given by $t'=\psi(t)$ with $\Psi(t,0)$ equal to a unit times $\psi(t)$. Hence $\tau|_{C_0}=t'=\psi(t)$ vanishes to order $m\le 3$, and $\tau|_{Q_b}\notin\mathfrak{m}_{p_b}^4$.

Let I_2 be the set of indices $i\le 3$ with $u_i(p)=0$, so $\#I_2=3-n_2$. Near p, $F_b\cap V_c$ is étale over the variety $\{(y,(u_i)_{i\in I_2})\mid y\in Q_b,\ u_i^2=A_{b,i}(y)\}$, whose tangent space at $(p_b,0)$ consists of the vectors $(\dot y,\dot u)$ with $\dot y$ tangent to Q_b and $dA_{b,i}(\dot y)=0$ for $i\in I_2$. Since F_b is smooth of dimension 4 at p, the differentials $dA_{b,i}$, $i\in I_2$, are linearly independent on the tangent space of Q_b at p_b. Thus there exist local coordinates $(A_{b,i})_{i\in I_2},t_1,\dots,t_{1+n_2}$ of Q_b at p_b and local coordinates $(u_i)_{i\in I_2},t_1,\dots,t_{1+n_2}$ of F_b at p in which the projection is $u_i\to u_i^2=A_{b,i}$. Write τ as a power series in $(A_{b,i})_{i\in I_2},t_1,\dots,t_{1+n_2}$. Since $\tau|_{Q_b}\notin\mathfrak{m}_{p_b}^4$, a non-zero monomial of degree at most 3

occurs in the expansion of τ. The pullback replaces each $A_{b,i}$ with u_i^2, which sends distinct monomials to distinct monomials and multiplies the degree by at most 2. Thus $\operatorname{mult}_p G_{\xi_0} \le 6$, and $\operatorname{mult}_p G_{\xi_0} \le 3$ if $I_2 = \emptyset$, that is, if $n_2 = 3$. Since F_b is smooth at p, $\operatorname{mult}_p G_\xi$ is the largest s with $T_\xi|_{F_b} \in \mathfrak{m}_p^s$, and T_ξ depends linearly on ξ. Hence the set of $\xi \in \Lambda_p$ with $\operatorname{mult}_p G_\xi \ge 7$ is contained in a linear subspace not containing ξ_0, so $\operatorname{mult}_p G_\xi \le 6$ for a general ξ.

Next assume that $n = 3$, so $n_2 \le 2$. Let I' be the set of indices $i \le 8$ with $i \ne j_0$ and $u_i(p) \ne 0$, so $\#I' = 2$. Choose $\xi_1 \in (\mathbb{C}^4)^*$ with $\xi_1(\mathbf{w}_i) = 0$ for $i \in I'$ and $\xi_1(\mathbf{w}_{j_0}) \ne 0$. Then $\xi_1 \in \Lambda_p$, and T_{ξ_1} restricts to F_b as $3\xi_1(\mathbf{w}_{j_0})u_{j_0}(p)^2(U_{j_0} - u_{j_0}(p)W^{14})$. Thus $G_{\xi_1} \cap V_c$ is the subscheme $(u_{j_0} = u_{j_0}(p))$ of $F_b \cap V_c$, which is reduced since it is Cohen–Macaulay and G_{ξ_1} is a prime divisor. Apply (7.6) with $M = \{1, \dots, 8\} \setminus \{j_0\}$, $c_M = c_b - u_{j_0}(p)^3 \mathbf{w}_{j_0}$, and $q = p$, so that $I_q = I'$. Here V_M is isomorphic to $G_{\xi_1} \cap V_c$ by the projection forgetting u_{j_0}, and $G_{\xi_1} \cap V_c$ is integral, since G_{ξ_1} is a prime divisor and $G_{\xi_1} \cap V_c$ is reduced. If $n_2 = 2$, then $I' \subset \{1, 2, 3\}$ and $\operatorname{mult}_p G_{\xi_1} \le 6$. If $n_2 \le 1$, then $I' \not\subset \{1, 2, 3\}$ and $\operatorname{mult}_p G_{\xi_1} \le 4$.

Let $\Lambda^\circ \subset \Lambda_p$ be the open subset of the elements ξ with $T_\xi|_{F_b} \ne 0$, let $\mathcal{G} \subset (F_b \cap V_c) \times \Lambda^\circ$ be the closed subscheme defined by $T_\xi|_{V_c}(u) = 0$, so that the fiber of $\mathcal{G}$ over ξ is $G_\xi \cap V_c$, and let $\mathcal{I} \subset \mathcal{O}_\mathcal{G}$ be the ideal of $\{p\} \times \Lambda^\circ$. For an integer $s \ge 1$, $\mathcal{O}_\mathcal{G}/\mathcal{I}^s$ is supported on $\{p\} \times \Lambda^\circ$, so it is finite over Λ°, and the fiber of $\mathcal{O}_\mathcal{G}/\mathcal{I}^s$ over ξ is $\mathcal{O}_{G_\xi,p}/\mathfrak{m}_p^s$. Thus the function

$$\xi \to \operatorname{length}(\mathcal{O}_{G_\xi,p}/\mathfrak{m}_p^s)$$

is upper semicontinuous on Λ° [Har77, Example III.12.7.2]. By [Mat89, Theorem 24.1] for the finitely generated graded algebra $\bigoplus_{s\ge0} \mathcal{I}^s/\mathcal{I}^{s+1}$ over the coordinate ring of an affine open subset of Λ° containing ξ_1, there exists a dense open subset $\Lambda^1 \subset \Lambda^\circ$ over which any $\mathcal{I}^s/\mathcal{I}^{s+1}$, and hence any $\mathcal{O}_\mathcal{G}/\mathcal{I}^s$, is locally free. Hence for $\xi \in \Lambda^1$ and any $s \ge 1$, the length of $\mathcal{O}_{G_\xi,p}/\mathfrak{m}_p^s$ is independent of ξ, and

$$\operatorname{length}(\mathcal{O}_{G_\xi,p}/\mathfrak{m}_p^s) \le \operatorname{length}(\mathcal{O}_{G_{\xi_1},p}/\mathfrak{m}_p^s).$$

For any $\xi \in \Lambda^\circ$, $\dim \mathcal{O}_{G_\xi,p} = 3$, since $F_b \cap V_c$ is a complete intersection of dimension 4 and $G_\xi \cap V_c$ is a Cartier divisor on $F_b \cap V_c$. Comparing the leading coefficients of the Hilbert–Samuel polynomials, we obtain $\operatorname{mult}_p G_\xi \le \operatorname{mult}_p G_{\xi_1} \le 6$ for $\xi \in \Lambda^1$, where mult_p is the Samuel multiplicity [Ful98, Example 4.3.4]. Thus $\operatorname{mult}_p G_\xi \le 6$ for a general $\xi \in \Lambda_p$.

Step 5. In this step, for a homogeneous element T of R_c of degree 14 with $T|_{V_c} \in \mathfrak{m}_p^2$ whose divisor of zeros G on F_b is a prime divisor with $D_T \cdot F_b = G$ and $\operatorname{mult}_p G \le 6$, we bound $\operatorname{mult}_p D_g$ when no component of D_g containing p is contained in $\operatorname{Supp} D_T$, and we prove the proposition when $n \ge 3$, $n_3 \ge 1$, and $n_2 \le 2$.

Let T be a homogeneous element of R_c of degree 14 such that $T|_{V_c} \in \mathfrak{m}_p^2$, the divisor of zeros G of T on F_b is a prime divisor with $D_T \cdot F_b = G$, and $\operatorname{mult}_p G \le 6$. Let $J' > 0$ be an integer and g a non-zero element of $(R_c)_{J'}$ such that no component of D_g containing p is contained in $\operatorname{Supp} D_T$. We show that

$$\operatorname{mult}_p D_g \le \frac{6}{7} J'. \tag{7.8}$$

We cut D_g by T, with $s = 2$ in (7.3), and then by two general elements of Σ_b. The cycle after the first cut has pure dimension 5, so the components are not contained in F_b. The cycle after the second cut has pure dimension 4 and is supported in $\operatorname{Supp} D_T$. A component of the cycle after the second cut contained in F_b would be contained in $F_b \cap \operatorname{Supp} D_T = \operatorname{Supp} G$, and $\dim G = 3$. Hence by **Step 1** we obtain a $\mathbb{Q}$-cycle $\Gamma \ge 0$ of pure dimension 3 with

$$(H^3 \cdot \Gamma) \le 14 \cdot 6^2 J'(H^7), \qquad \operatorname{mult}_p \Gamma \ge 2 \operatorname{mult}_p D_g.$$

Write $\Gamma = mG + \Gamma'$, where G is not a component of Γ'. A component of Γ'_p is contained in $\operatorname{Supp} D_T$, so if it were contained in F_b, then it would be equal to G. Thus a general element of Σ_b cuts Γ', and by (7.5) with $k = 2$,

$$\operatorname{mult}_p \Gamma' \le 6 \cdot 14^2 (H^3 \cdot \Gamma').$$

Since $(H^3 \cdot G) = 14 \cdot 6^3 (H^7)$ by Lemma 2.8(1) and Definition-Lemma 7.2(2), (7.4) and the second row of Table 1 imply (7.8).

Now assume that $n \geq 3$, $n_3 \geq 1$, and $n_2 \leq 2$. By **Step 4**, a general $\xi \in \Lambda_p$ satisfies $\xi(\mathbf{w}_{j_0}) \neq 0$, G_ξ is a prime divisor with $D_{T_\xi} \cdot F_b = G_\xi$, $\operatorname{mult}_p G_\xi \leq 6$, and no component of D_f containing p is contained in $\operatorname{Supp} D_{T_\xi}$. Applying (7.8) to $T = T_\xi$ and $g = f$, we obtain $\operatorname{mult}_p D_f \leq \frac{6}{7} J$.

Step 6. In this step we prove the proposition when $n_2 = 3$ and $n_3 \geq 1$.

We first factor homogeneous elements of R_c. Let g be a non-zero homogeneous element of R_c of positive degree. The minimal primes of (g) are associated primes of $R_c/(g)$, hence homogeneous [BH98, Lemma 1.5.6(b)(ii)], and they have height one by Krull's principal ideal theorem [Mat89, Theorem 13.5], since R_c is a domain and $g \neq 0$. By Definition-Lemma 7.2(1), they are $(\pi_1), \dots, (\pi_t)$ for homogeneous prime elements π_i. Let v_i be the largest integer with $\pi_i^{v_i} \mid g$, and $g' := g / \prod_i \pi_i^{v_i}$. A height-one prime containing g' contains g, so it is some (π_i), which is impossible since $\pi_i \nmid g'$. Thus g' is contained in no height-one prime, so g' is a homogeneous unit of R_c, that is, a non-zero constant. Moreover, $D_g = \sum_i v_i D_{\pi_i}$. Let i be an index with $p \in D_{\pi_i}$. Since $p \in V_c$ and D_W is supported on V_∞, π_i is not a multiple of W, so $\deg \pi_i \geq 6$ by Definition-Lemma 7.2(1), and in particular $W \notin (\pi_i)$. Thus $V_+(\pi_i)$ is irreducible and not contained in V_∞, and $R_c[W^{-1}]/(\pi_i)$ is a non-zero localization of the domain $R_c/(\pi_i)$. Since $R_c[W^{-1}] = (R_c)_{(W)}[W, W^{-1}]$, $\pi_i W^{-\deg \pi_i}$ is a prime element of the normal domain $(R_c)_{(W)}$, the coordinate ring of V_c (Proposition 6.2(1)). On V_c, D_{π_i} is the divisor of $\pi_i|_{V_c} = \pi_i W^{-\deg \pi_i}$ (**Step 1**), and $\operatorname{Supp} D_{\pi_i} \subset V_+(\pi_i)$, so D_{π_i} is the prime divisor $V_+(\pi_i)$.

Since $n_2 = 3$, $\Lambda_p = \mathbb{C}\xi_0$. Let $T := T_{\xi_0}$. By **Step 4**, G_{ξ_0} is a prime divisor with $D_T \cdot F_b = G_{\xi_0}$, since $\xi_0(\mathbf{w}_{j_0}) = \alpha_{j_0 - 3} \neq 0$, and $\operatorname{mult}_p G_{\xi_0} \leq 3$. Since V_c is smooth at p, $\operatorname{mult}_p D_T$ is the largest s with $T|_{V_c} \in \mathfrak{m}_p^s$, so $\operatorname{mult}_p D_T \leq \operatorname{mult}_p G_{\xi_0} \leq 3$. Let $\pi_1, \dots, \pi_t$ be the prime factors of T as in the first paragraph of this step, numbered so that $\pi_1, \dots, \pi_{t'}$ are those with $p \in D_{\pi_i}$. Then

$$\operatorname{mult}_p D_{\pi_i} \leq \operatorname{mult}_p D_T \leq 3 \leq \frac{1}{2} \deg \pi_i \qquad (1 \leq i \leq t').$$

Write $f = \prod_{i=1}^{t'} \pi_i^{m_i} \cdot f''$, where $m_i \geq 0$ and $\pi_i \nmid f''$ for $1 \leq i \leq t'$, and let $J'' := J - \sum_i m_i \deg \pi_i$. The components of D_T containing p are $D_{\pi_1}, \dots, D_{\pi_{t'}}$, which are not components of $D_{f''}$. If $J'' > 0$, then (7.8) applied to T and $g = f''$ implies $\operatorname{mult}_p D_{f''} \leq \frac{6}{7} J''$. If $J'' = 0$, then $D_{f''} = 0$. Hence

$$\operatorname{mult}_p D_f = \sum_{i=1}^{t'} m_i \operatorname{mult}_p D_{\pi_i} + \operatorname{mult}_p D_{f''} \leq \frac{1}{2} \sum_{i=1}^{t'} m_i \deg \pi_i + \frac{6}{7} J'' \leq \frac{6}{7} J.$$

Step 7. In this step we prove the proposition when $n_2 = 3$ and $n_3 = 0$.

By **Step 2**, $n = 3$ and $b_{l_0} \neq 0$ for some $1 \leq l_0 \leq 3$. Let

$$T := \sum_{l=1}^{3} 7 \beta_l b_l^6 \left(U_{8+l} - b_l W^6 \right) \in \Sigma_b.$$

Since $\xi_0(\mathbf{w}_i) = 0$ for $i \leq 3$ and $u_j(p) = 0$ for $4 \leq j \leq 8$, $T|_{V_c}$ is the linear part at p of $\xi_0(h(u)) - \xi_0(c)$, so $T|_{V_c} \in \mathfrak{m}_p^2$. We show that $T|_{V_c} \notin \mathfrak{m}_p^4$. Since $u_1(p), u_2(p), u_3(p) \neq 0$, the morphism $\varphi : V_c \to Q$ of **Step 2** is étale at p. Since $\Phi(\varphi(p)) = 0$, $\xi_0(c) = \sum_l \beta_l b_l^7$. Let $C_1 \subset Q$ be the curve germ at $\varphi(p)$ given by $u_4 = t$, $u_j = 0$ for $5 \leq j \leq 8$, $u_{8+l} = b_l$ for $l \neq l_0$, and

$$u_{8+l_0} = b_{l_0} \left(1 - \frac{\alpha_1}{\beta_{l_0} b_{l_0}^7} t^3 \right)^{1/7} = b_{l_0} - \frac{\alpha_1}{7 \beta_{l_0} b_{l_0}^6} t^3 + (\text{terms of degree} \geq 6 \text{ in } t).$$

Then C_1 is a smooth curve germ, and T restricts to C_1 as $-\alpha_1 t^3$ plus terms of higher degree. Since φ is étale at p, C_1 lifts to a smooth curve germ in V_c through p, on which $T|_{V_c}$ vanishes to order 3. Thus $T|_{V_c} \notin \mathfrak{m}_p^4$, and $\operatorname{mult}_p D_T \leq 3$.

Factor T as in **Step 6**. Since $T|_{V_c}(p) = 0$, some prime factor π_i of T satisfies $p \in D_{\pi_i}$ and $\deg \pi_i \geq 6 = \deg T$. Hence T is a prime element, and D_T is a prime divisor. Write $f = T^\nu f''$,

where $\nu \geq 0$ and $T \nmid f''$, and let $J'' := J - 6\nu$. If $J'' = 0$, then $\operatorname{mult}_p D_f = \nu \operatorname{mult}_p D_T \leq 3\nu = \frac{1}{2}J$. Assume that $J'' > 0$. We cut $D_{f''}$ by T, with $s = 2$ in (7.3), which is possible since D_T is not a component of $D_{f''}$, and then by a general element of Σ_b. The components of the cycle after the first cut have dimension 5, so they are not contained in F_b. We obtain a $\mathbb{Q}$-cycle $\Gamma \geq 0$ of pure dimension 4 with

$$(H^4 \cdot \Gamma) \leq 6^2 J''(H^7), \qquad \operatorname{mult}_p \Gamma \geq 2 \operatorname{mult}_p D_{f''}.$$

Write $\Gamma = mF_b + \Gamma'$, where F_b is not a component of Γ'. As in **Step 3**, a general element of Σ_b cuts Γ', and by (7.5) with $k = 3$, $\operatorname{mult}_p \Gamma' \leq 6 \cdot 14^3 (H^4 \cdot \Gamma')$. Since $n = 3$ and $n_3 = 0$, $\operatorname{mult}_p F_b \leq 3$ by (7.7). By (7.4) and the third row of Table 1, $\operatorname{mult}_p D_{f''} \leq \frac{6}{7}J''$. Hence

$$\operatorname{mult}_p D_f = \nu \operatorname{mult}_p D_T + \operatorname{mult}_p D_{f''} \leq 3\nu + \frac{6}{7}J'' \leq \frac{6}{7}(6\nu + J'') = \frac{6}{7}J.$$

Step 8. In this step we finish the proof.

By the case division at the end of **Step 2**, **Steps 3, 5, 6, and 7** imply $\operatorname{mult}_p D_f \leq \frac{6}{7}J$. The smooth locus of V_c is an open neighborhood of p, on which $D_f|_{V_c}$ is the Cartier divisor of $f|_{V_c}$ and $\operatorname{mult}_q D_f \leq \frac{6}{7}J$ for any point q. For $0 < \epsilon < 1$ and q in the smooth locus of V_c near p,

$$\operatorname{mult}_q \left((1-\epsilon)\frac{7}{6J} D_f|_{V_c} \right) \leq 1 - \epsilon < 1,$$

so the multiplier ideal of $(1-\epsilon)\frac{7}{6J}D_f|_{V_c}$ is trivial near p [Laz04, Proposition 9.5.13]. Hence $(V_c, \frac{7}{6J}D_f|_{V_c})$ is lc near p [Laz04, Definition 9.3.9]. □

7.2. **The divisor at infinity.** In this subsection, we prove the bound $\frac{7}{6J}$ on V_∞ and near V_∞. Since $V_\infty = \operatorname{Proj} R_\infty$ and the punctured affine cone $\operatorname{Spec} R_\infty \backslash \{0\}$ is V_0 minus the origin, which consists of smooth points by Proposition 6.2(3), Proposition 7.3 with $c = 0$ implies the bound on V_∞, and inversion of adjunction extends it to a neighborhood of V_∞. For this reason, Proposition 7.3 includes the case $c = 0$.

Proposition 7.4. *Notation and conditions as in Set-up 6.1. Then the following hold.*

(1) For any integer $J > 0$ and any non-zero element g of $(R_\infty)_J$, the pair $(V_\infty, \frac{7}{6J}D_g)$ is lc.

(2) Let $c \in \mathbb{C}^4 \backslash \{0\}$, $J > 0$ an integer, and f a non-zero element of $(R_c)_J$. Then $(\overline{V}_c, \frac{1}{J}D_f)$ is lc near V_∞. If moreover W does not divide f, then $(\overline{V}_c, V_\infty + \frac{7}{6J}D_f)$ is lc near V_∞.

Proof. (1) Let $\widehat{V}_\infty := \operatorname{Spec} R_\infty \backslash \{0\}$ and let $q : \widehat{V}_\infty \to V_\infty$ be the natural morphism. Since $R_0 = R_\infty[W]$, we may regard g as a non-zero element of $(R_0)_J$. We regard $\operatorname{Spec} R_\infty$ as V_0 via $u_i = U_i$. Then $\widehat{V}_\infty = V_0 \backslash \{0\}$, and the restriction to V_0 of the divisor of zeros of $g \in (R_0)_J$ on $\overline{V}_0$ is $\operatorname{div}(g)$. By Proposition 6.2(3), any point of $\widehat{V}_\infty$ is a smooth point of V_0. Thus Proposition 7.3, applied with $c = 0$ to $g \in (R_0)_J$ and to any point of $\widehat{V}_\infty$, implies that $(\widehat{V}_\infty, \frac{7}{6J}\operatorname{div}(g))$ is lc.

The degree zero part of R_∞ is $\mathbb{C}$, and R_∞ is generated by $U_1, \dots, U_{11}$ of degrees $d_1, \dots, d_{11}$. By Proposition 6.2(3), R_∞ is a normal domain, $\widehat{V}_\infty$ is smooth, and for any $1 \leq i \leq 11$, the quotient morphism

$$\operatorname{Spec} R_\infty \cap (U_i = 1) \to V_\infty \cap (U_i \neq 0)$$

by μ_{d_i} is étale in codimension one. Fix $1 \leq i \leq 11$. Since the quotient morphism by μ_{d_i} is étale in codimension one, the multiplication map from $\mathcal{O}_{V_\infty}(J)^{\otimes d_i}$ to $\mathcal{O}_{V_\infty}(d_i J)$ is an isomorphism in codimension one on $V_\infty \cap (U_i \neq 0)$, so $d_i D_g$ is the divisor of the regular function g^{d_i}/U_i^J there (Set-up 6.1(5)). Since $q^*(g^{d_i}/U_i^J) = g^{d_i}/U_i^J$ and U_i is invertible on $\widehat{V}_\infty \cap (U_i \neq 0)$, the pullback of $d_i D_g$ under q is $d_i \operatorname{div}(g)$ there, so $q^* D_g = \operatorname{div}(g)$ on $\widehat{V}_\infty \cap (U_i \neq 0)$. Since the open subsets $\widehat{V}_\infty \cap (U_i \neq 0)$, $1 \leq i \leq 11$, cover $\widehat{V}_\infty$, we have $q^* D_g = \operatorname{div}(g)$ on $\widehat{V}_\infty$. By Lemma 2.10, applied to $R = R_\infty$ with Y a point and $\Sigma = V_\infty$, the pair $(V_\infty, \frac{7}{6J}D_g)$ is lc if and only if $(\widehat{V}_\infty, \frac{7}{6J}\operatorname{div}(g))$ is lc. This proves (1).

(2) We first show that $D_W = V_\infty$, and that $D_f|_{V_\infty} = D_{\bar f}$ when W does not divide f, where $\bar f$ is the image of f in $R_\infty = R_c/(W)$. Fix $1 \le i \le 11$. Let $\widehat V_{c,i} := \operatorname{Spec} R_c \cap (U_i = 1)$ and $\widehat V_{\infty,i} := \operatorname{Spec} R_\infty \cap (U_i = 1)$, and let $q_i : \widehat V_{c,i} \to \overline V_c \cap (U_i \ne 0)$ be the quotient morphism by μ_{d_i}, where $\zeta \in \mu_{d_i}$ multiplies W by ζ and U_l by ζ^{d_l}. Let f' be a non-zero homogeneous element of R_c of degree e. On $\overline V_c \cap (U_i \ne 0)$, $d_i D_{f'}$ is the divisor of the regular function f'^{d_i}/U_i^e (Lemma 6.3 and $(f'^{d_i}/U_i^e)^{42/d_i} = f'^{42}/U_i^{42e/d_i}$), whose pullback under q_i is $(f'|_{\widehat V_{c,i}})^{d_i}$, since $U_i = 1$ on $\widehat V_{c,i}$. Hence wherever q_i is étale, $q_i^* D_{f'} = \operatorname{div}(f'|_{\widehat V_{c,i}})$, and similarly for R_∞.

Since $R_c/(W) = R_\infty$, we have $\widehat V_{\infty,i} = \widehat V_{c,i} \cap (W = 0)$ as schemes. The morphism

$$\beta_i : \widehat V_{\infty,i} \times \mathbb{G}_m \to \operatorname{Spec} R_\infty \cap (U_i \ne 0)$$

defined by the $\mathbb{G}_m$-action is finite étale, so $\widehat V_{\infty,i}$ is smooth by Proposition 6.2(3). Hence $\widehat V_{c,i}$ is smooth along the Cartier divisor $(W = 0) = \widehat V_{\infty,i}$, and $\operatorname{div}(W) = \widehat V_{\infty,i}$ on $\widehat V_{c,i}$. By Proposition 6.2(3), the fixed locus F_ζ of any $\zeta \in \mu_{d_i} \backslash \{1\}$ on $\widehat V_{\infty,i}$ satisfies $\dim F_\zeta \le 3$, while $\dim \Gamma = 6$ for any component Γ of $\widehat V_{\infty,i}$, since β_i is finite étale onto an open subset of the 7-dimensional variety $\operatorname{Spec} R_\infty$. Therefore, q_i is étale at the generic point of any component and of any prime divisor of $\widehat V_{\infty,i}$, and the coefficient of V_∞ in D_W is one. Since W does not vanish on V_c, we have $D_W = V_\infty$.

Assume that W does not divide f. Then $\bar f \ne 0$, and D_f does not contain V_∞. Near the generic point of any prime divisor of $\widehat V_{\infty,i}$, the divisor $q_i^* D_f = \operatorname{div}(f|_{\widehat V_{c,i}})$ is Cartier, and the restriction of $q_i^* D_f$ to $\widehat V_{\infty,i}$ is $\operatorname{div}(\bar f|_{\widehat V_{\infty,i}})$, which is the pullback of $D_{\bar f}$. Since i is arbitrary and V_∞ is normal, $D_f|_{V_\infty} = D_{\bar f}$.

Now assume that W does not divide f. By Proposition 6.9, $\overline V_c$ is klt, and $V_\infty = D_W$ and D_f are $\mathbb{Q}$-Cartier. Thus $K_{\overline V_c} + V_\infty + \frac{7}{6J} D_f$ is $\mathbb{Q}$-Cartier. By Proposition 6.2(3), V_∞ is a normal prime divisor on $\overline V_c$ and $(K_{\overline V_c} + V_\infty)|_{V_\infty} = K_{V_\infty}$. By [FST11, 11.1(v)] and $D_f|_{V_\infty} = D_{\bar f}$, we have

$$\Big(K_{\overline V_c} + V_\infty + \frac{7}{6J} D_f\Big)\Big|_{V_\infty} = K_{V_\infty} + \frac{7}{6J} D_{\bar f}.$$

Since $\bar f$ is a non-zero element of $(R_\infty)_J$, the pair $(V_\infty, \frac{7}{6J} D_{\bar f})$ is lc by (1). The boundary $\frac{7}{6J} D_f \ge 0$ has no common components with V_∞. By Theorem 2.5, $(\overline V_c, V_\infty + \frac{7}{6J} D_f)$ is lc near V_∞. This proves the second statement of (2).

Now let f be arbitrary. Since $R_c/(W) = R_\infty$ is a domain, W is a prime element of R_c, and we may write $f = W^a f_0$, where $a \ge 0$ is an integer and $f_0 \in (R_c)_{J_0}$ is not divisible by W, $J_0 := J - a$. Then $D_f = a V_\infty + D_{f_0}$. If $J_0 = 0$, then f_0 is a non-zero constant and $\frac{1}{J} D_f = V_\infty$. Since $U_{11} \in (R_c)_6$ is not divisible by W (the relations of $R_\infty = R_c/(W)$ have degree $42 > 6$), the pair $(\overline V_c, V_\infty + \frac{7}{36} D_{U_{11}})$ is lc near V_∞ by the previous paragraph, hence so is $(\overline V_c, V_\infty)$. If $J_0 > 0$, then $(\overline V_c, V_\infty + \frac{7}{6J_0} D_{f_0})$ is lc near V_∞ by the previous paragraph. Since $\frac{a}{J} \le 1$ and $\frac{1}{J} \le \frac{1}{J_0} < \frac{7}{6J_0}$, we have

$$\frac{1}{J} D_f = \frac{a}{J} V_\infty + \frac{1}{J} D_{f_0} \le V_\infty + \frac{7}{6J_0} D_{f_0},$$

so $(\overline V_c, \frac{1}{J} D_f)$ is lc near V_∞. □

7.3. **Weighted blow-ups at singular points.** In this subsection, we prove four results near a singular point p of V'. They hold in both cases $N = 10$ and $N = 11$ of Construction 6.5.

At a singular point p, the multiplicity bound of Proposition 7.3 is not available. The following lemma reduces log canonicity near p to log canonicity on E_p with the lowest-degree component g. The condition $tM \le \delta_p$ ensures that, in the pullback of $K_{V'} + t \operatorname{div}(\varphi)$ to the weighted blow-up, the exceptional divisor has coefficient at most one, so that we may first apply inversion of adjunction to the pair in which E_p has coefficient one, and then decrease the coefficient of E_p.

Lemma 7.5. *Notation and conditions as in Construction 6.5. Let φ be a non-zero regular function near p on V', let $t \geq 0$ be a rational number, and let $g \in A_p$ be the non-zero homogeneous component of lowest degree of the image of φ under the isomorphism $\widehat{\mathcal{O}}_{V',p} \cong \widehat{A_p}$ induced by ϖ_p, where $\widehat{A_p}$ is the completion of A_p at the origin. Assume that $M := \deg g > 0$, that $tM \leq \delta_p$, and that $(E_p, \Delta_p + tD_g)$ is lc. Then $(V', t\operatorname{div}(\varphi))$ is lc near p.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct the base change $\widetilde{V}_p$ of ψ_p near p.

By Lemma 6.8(2), ϖ_p is étale at p and $\varpi_p(p) = 0$. Let V'° be an open neighborhood of p such that φ is regular on V'°, $\varpi_p|_{V'^{\circ}}$ is étale, and $\varpi_p^{-1}(0) \cap V'^{\circ} = \{p\}$. Let

$$\widetilde{V}_p := Z'_p \times_{Z_p} V'^{\circ},$$

let $\psi'_p : \widetilde{V}_p \to V'^{\circ}$ be the base change of ψ_p, and let $\varpi'_p : \widetilde{V}_p \to Z'_p$ be the projection, which is étale. By Lemma 6.8(3), ψ'_p is a proper birational morphism which is an isomorphism over $V'^{\circ}\backslash\{p\}$. Let $\widetilde{E}_p := \varpi'^{-1}_p(E_p)$. Since the fiber of $\varpi_p|_{V'^{\circ}}$ over 0 is the reduced point p, ϖ'_p maps $\widetilde{E}_p$ isomorphically onto E_p, and $\widetilde{E}_p$ is the exceptional divisor of ψ'_p with $\psi'^{-1}_p(p) = \widetilde{E}_p$ as sets. By Lemma 6.8(3), Z'_p has quotient singularities near E_p, hence so does $\widetilde{V}_p$ near $\widetilde{E}_p$. Since ψ'_p is proper and $\psi'^{-1}_p(p) = \widetilde{E}_p$, possibly shrinking V'°, we may assume that $\widetilde{V}_p$ has quotient singularities. Then $\widetilde{V}_p$ is normal and $\mathbb{Q}$-factorial [KM98, Proposition 5.15].

Step 2. In this step, we show that

$$\operatorname{mult}_{\widetilde{E}_p}(\varphi \circ \psi'_p) = \frac{M}{\gamma_p},$$

and that the strict transform D' of $\operatorname{div}(\varphi)$ on $\widetilde{V}_p$ satisfies $D'|_{\widetilde{E}_p} = D_g$, where we regard D_g as a $\mathbb{Q}$-divisor on $\widetilde{E}_p$ via $\widetilde{E}_p \cong E_p$.

For an integer m, denote by $(A_p)_{\geq m}$ the ideal of A_p generated by the homogeneous elements of degree at least m. Since A_p is generated by the x_j, $j \notin I$, of degrees d'_j, the degree of any non-zero homogeneous element of A_p is divisible by γ_p, so M/γ_p is an integer and $(A_p)_{\geq M+1} = (A_p)_{\geq M+\gamma_p}$. The image of $\varphi - \varpi_p^* g$ in $\widehat{A_p}$ has no non-zero homogeneous component of degree at most M, so it belongs to $(A_p)_{\geq M+1}\widehat{A_p}$. Since $(A_p)_{\geq M+1}$ contains a power of the maximal ideal of the origin, and

$$\mathcal{O}_{V',p}/\mathfrak{m}^n \cong \widehat{\mathcal{O}}_{V',p}/\mathfrak{m}^n\widehat{\mathcal{O}}_{V',p}$$

for the maximal ideal $\mathfrak{m}$ of $\mathcal{O}_{V',p}$ and any n, we have $\varphi - \varpi_p^* g \in (A_p)_{\geq M+1}\mathcal{O}_{V',p}$. Possibly shrinking V'°, we may write

$$\varphi = \varpi_p^* g + \sum_{\alpha} b_\alpha \varpi_p^* h_\alpha \quad \text{on } V'^{\circ},$$

where the sum is finite, b_α are regular functions on V'°, and $h_\alpha \in A_p$ are homogeneous of degrees at least $M + \gamma_p$.

Fix $l \notin I$. By Lemma 6.7(1), Z'_p has an open subset $Z'_{p,l}$ containing $E_p \cap (x_l \neq 0)$ with a quotient morphism $\varrho_l : Z_{p,l} \times \mathbb{A}^1 \to Z'_{p,l}$ by μ_{w_l}, and $\psi_p \circ \varrho_l(x, \xi) = (\xi^{w_j} x_j)_{j \notin I}$, where ξ is the coordinate of $\mathbb{A}^1$. Since $d'_j = \gamma_p w_j$, for any homogeneous $h \in A_p$ of degree m, we have

$$h \circ \psi_p \circ \varrho_l = \xi^{m/\gamma_p} h|_{Z_{p,l}}.$$

Let

$$\widetilde{V}_{p,l} := (Z_{p,l} \times \mathbb{A}^1) \times_{Z'_p} \widetilde{V}_p,$$

and let $\varrho'_l : \widetilde{V}_{p,l} \to \widetilde{V}_p$ be the base change of ϱ_l. We denote the pullbacks to $\widetilde{V}_{p,l}$ of ξ and of the functions $h|_{Z_{p,l}}$ again by ξ and $h|_{Z_{p,l}}$. Since ϖ'_p is étale and $Z_{p,l}$ is smooth, $\widetilde{V}_{p,l}$ is smooth.

Moreover, ϱ'_l is finite, surjective onto $\varpi'^{-1}_p(Z'_{p,l})$, and étale in codimension one, and $\varrho'^{-1}_l(\widetilde{E}_p)$ is the smooth divisor $(\xi = 0)$. Since $\varpi_p \circ \psi'_p = \psi_p \circ \varpi'_p$, the expression of φ above implies that

$$\varphi \circ \psi'_p \circ \varrho'_l = \xi^{M/\gamma_p}\big(g|_{Z_{p,l}} + \xi\beta\big)$$

for a regular function β on $\widetilde{V}_{p,l}$.

By Lemma 6.7(1), $Z_{p,l}$ is irreducible, and $g|_{Z_{p,l}} \neq 0$, since $g \neq 0$ is homogeneous and Z_p is irreducible. Hence $g|_{Z_{p,l}}$ does not vanish identically on any component of $(\xi = 0) \subset \widetilde{V}_{p,l}$, which is étale over $Z_{p,l} \times \{0\}$. Since ϱ'_l is étale at the generic points of $(\xi = 0) = \varrho'^{-1}_l(\widetilde{E}_p)$, we obtain

$$\operatorname{mult}_{\widetilde{E}_p}(\varphi \circ \psi'_p) = \frac{M}{\gamma_p}.$$

Thus

$$D' = \operatorname{div}(\varphi \circ \psi'_p) - \frac{M}{\gamma_p}\widetilde{E}_p, \quad \varrho'^*_l D' = \operatorname{div}\big(g|_{Z_{p,l}} + \xi\beta\big),$$

and

$$(\varrho'^*_l D')|_{(\xi=0)} = \operatorname{div}\big(g|_{Z_{p,l}}\big|_{(\xi=0)}\big).$$

By Lemma 6.7(1), $\varrho_l|_{Z_{p,l}\times\{0\}}$ is the restriction of q_p to $Z_{p,l}$, and by Construction 6.5(7), the pullback $q_p^* D_g$ is $\operatorname{div}(g)$ on $Z_p \backslash \{0\}$. Hence $D'|_{\widetilde{E}_p}$ and D_g both pull back to $\operatorname{div}(g|_{Z_{p,l}}|_{(\xi=0)})$ on $(\xi = 0)$, which is finite and surjective over $\widetilde{E}_p \cap (x_l \neq 0)$. Therefore, $D'|_{\widetilde{E}_p} = D_g$ on $\widetilde{E}_p \cap (x_l \neq 0)$. Since $l \notin I$ is arbitrary, $D'|_{\widetilde{E}_p} = D_g$.

Step 3. In this step, we conclude the proof.

By Lemma 6.8(3), $a(E_p, Z_p) = \frac{\delta_p}{\gamma_p}$, that is, $K_{Z'_p} + (1 - \frac{\delta_p}{\gamma_p})E_p = \psi_p^* K_{Z_p}$. Since ϖ'_p and $\varpi_p|_{V'^\circ}$ are étale, $K_{\widetilde{V}_p} = \varpi'^*_p K_{Z'_p}$, $K_{V'^\circ} = \varpi_p^* K_{Z_p}$, and $\varpi'^*_p E_p = \widetilde{E}_p$. Thus $a(\widetilde{E}_p, V') = \frac{\delta_p}{\gamma_p}$, and by Construction 6.5(6),

$$(K_{\widetilde{V}_p} + \widetilde{E}_p)|_{\widetilde{E}_p} = K_{\widetilde{E}_p} + \Delta_p.$$

By **Step 2**, we have

$$\psi'^*_p \operatorname{div}(\varphi) = D' + \frac{M}{\gamma_p}\widetilde{E}_p,$$

and hence

$$K_{\widetilde{V}_p} + tD' + \Big(1 - \frac{\delta_p - tM}{\gamma_p}\Big)\widetilde{E}_p = \psi'^*_p\big(K_{V'^\circ} + t\operatorname{div}(\varphi)\big).$$

Let $c_p := 1 - \frac{\delta_p - tM}{\gamma_p}$. Since $tM \leq \delta_p$, we have $c_p \leq 1$.

We check the conditions of Theorem 2.5 for $\widetilde{V}_p$, the reduced divisor $\widetilde{E}_p$, and tD'. By **Step 1**, $\widetilde{V}_p$ is normal and $\mathbb{Q}$-factorial, so $K_{\widetilde{V}_p} + \widetilde{E}_p + tD'$ is $\mathbb{Q}$-Cartier. The $\mathbb{Q}$-divisor $tD' \geq 0$ has no common components with $\widetilde{E}_p$. Since A_p is a normal domain (Lemma 6.8(1)), $\widetilde{E}_p \cong E_p \cong \operatorname{Proj} A_p$ is normal. By [FST11, 11.1(v)] and **Step 2**,

$$(K_{\widetilde{V}_p} + \widetilde{E}_p + tD')|_{\widetilde{E}_p} = K_{\widetilde{E}_p} + \Delta_p + tD_g.$$

Since $(E_p, \Delta_p + tD_g)$ is lc, Theorem 2.5 implies that $(\widetilde{V}_p, \widetilde{E}_p + tD')$ is lc near $\widetilde{E}_p$. Since $c_p \leq 1$, the sub-pair $(\widetilde{V}_p, c_p\widetilde{E}_p + tD')$ is sub-lc near $\widetilde{E}_p$.

Since ψ'_p is proper and $\psi'^{-1}_p(p) = \widetilde{E}_p$, there exists an open neighborhood V'' of p in V'° such that $(\widetilde{V}_p, c_p\widetilde{E}_p + tD')$ is sub-lc over V''. Since

$$\psi'_{p*}(c_p\widetilde{E}_p + tD') = t\operatorname{div}(\varphi),$$

by the crepant equality above and [KM98, Lemma 2.30], we have

$$a(F, V', t\operatorname{div}(\varphi)) = a(F, \widetilde{V}_p, c_p\widetilde{E}_p + tD') \geq 0$$

for any prime divisor F over V''. Thus $(V', t\operatorname{div}(\varphi))$ is lc near p. □

We treat the factors r_i, $i \in I$, separately. The assumption $6\delta_p > 7e'_i$ in Proposition 7.6 allows us to use Lemma 7.5 for $r_i|_{V'}$ with the coefficient $\frac{7}{6d'_i}$.

Proposition 7.6. *Notation and conditions as in Construction 6.5. Let $i \in I$, and assume that $6\delta_p > 7e'_i$. Then the pair $(V', \frac{7}{6d'_i}\operatorname{div}(r_i|_{V'}))$ is lc near p.*

Proof. By Lemma 6.8(2), $r_i|_{V'} = \epsilon_i \cdot \varpi_p^* L_i$ for a regular function ϵ_i near p with $\epsilon_i(p) \neq 0$. Since r_i is a non-zero homogeneous element of the domain R', $r_i|_{V'} \neq 0$, so $L_i \neq 0$, and L_i is homogeneous of degree 42. Under the isomorphism $\widehat{\mathcal{O}}_{V',p} \cong \widehat{A_p}$ induced by ϖ_p, the image of $r_i|_{V'}$ is the product of L_i and a unit whose degree zero component is $\epsilon_i(p)$. Thus the non-zero homogeneous component of lowest degree of the image of $r_i|_{V'}$ is $\epsilon_i(p)L_i$, of degree 42, and $D_{\epsilon_i(p)L_i} = D_{L_i}$.

Let $t := \frac{7}{6d'_i}$. Then

$$42t = \frac{7e'_i}{6} < \delta_p.$$

Since $d'_i \geq 6$, we have $t \leq \frac{7}{36} < 1$. By Lemma 6.7(3), E_p is $\mathbb{Q}$-factorial. Thus $\Delta_p + tD_{L_i} \leq \Delta_p + D_{L_i}$ and Lemma 6.8(4) imply that $(E_p, \Delta_p + tD_{L_i})$ is lc. By Lemma 7.5, applied to $\varphi = r_i|_{V'}$, t, $g = \epsilon_i(p)L_i$, and $M = 42$, the pair $(V', t\operatorname{div}(r_i|_{V'}))$ is lc near p. □

The following lemma reduces log canonicity near p to sections not divisible by any r_i. It is used in Subsections 7.4 and 7.6.

Lemma 7.7. *Notation and conditions as in Construction 6.5, and assume that $6\delta_p > 7e'_i$ for any $i \in I$. Assume moreover that $(V', \frac{7}{6J_1}\operatorname{div}(f_1|_{V'}))$ is lc near p for any integer $J_1 > 0$ and any non-zero element f_1 of R'_{J_1} which is not divisible by r_i for any $i \in I$. Then $(V', \frac{7}{6J}\operatorname{div}(f|_{V'}))$ is lc near p for any integer $J > 0$ and any non-zero element f of R'_J.*

Proof. Since $V' \subset \mathbb{A}^N$ is defined by four equations and $\dim V' = N - 4$ (Lemma 6.8(1)), V' is a complete intersection, so $K_{V'}$ is Cartier. For any prime divisor E over V', the log discrepancy $a(E, V', B)$ is an affine function of the $\mathbb{R}$-divisor B with $K_{V'} + B$ $\mathbb{R}$-Cartier. Thus if (V', B_1) and (V', B_2) are lc near p, then $(V', sB_1 + (1-s)B_2)$ is lc near p for any $s \in [0, 1]$.

Let $J > 0$ be an integer and f a non-zero element of R'_J. By Lemma 6.8(2), for any $i \in I$, r_i is a homogeneous prime element of R' of degree d'_i. We divide f by the elements r_i, $i \in I$, as long as possible. Since the degree drops by d'_i at each division, the process stops, and we obtain

$$f = \prod_{i \in I} r_i^{b_i} \cdot f_1,$$

where $b_i \geq 0$ are integers, $J_1 := J - \sum_{i \in I} b_i d'_i \geq 0$, and $f_1 \in R'_{J_1}$ is not divisible by r_i for any $i \in I$. Then

$$\frac{7}{6J}\operatorname{div}(f|_{V'}) = \sum_{i \in I} \frac{b_i d'_i}{J} \cdot \frac{7}{6d'_i}\operatorname{div}(r_i|_{V'}) + \frac{J_1}{J} \cdot \frac{7}{6J_1}\operatorname{div}(f_1|_{V'}),$$

where the last term is omitted if $J_1 = 0$, and the weights $b_i d'_i/J \geq 0$ ($i \in I$) and $J_1/J \geq 0$ have sum 1. By Proposition 7.6, $(V', \frac{7}{6d'_i}\operatorname{div}(r_i|_{V'}))$ is lc near p for any $i \in I$, and if $J_1 > 0$, then $(V', \frac{7}{6J_1}\operatorname{div}(f_1|_{V'}))$ is lc near p by assumption. Hence $(V', \frac{7}{6J}\operatorname{div}(f|_{V'}))$ is lc near p by the convexity above. □

After the factors r_i are removed, the lowest degree M of a section at p can still be larger than the degree J of the section. For instance, if $i, j \in I$ satisfy $e'_i = e'_j = 7$ and $d'_i = d'_j = 6$, then for $\lambda \in \mathbb{C}\backslash\{0\}$ the section $f = r_i - \lambda r_j$ has degree $J = 6$, and for all but at most one λ, the lowest-degree component $\epsilon_i(p)L_i - \lambda\epsilon_j(p)L_j$ of the image of $f|_{V'}$ is non-zero, of degree $42 = e_*J$. The following lemma shows that $M \leq e_*J$ in general.

Lemma 7.8. *Notation and conditions as in Construction 6.5. Let $J > 0$ be an integer and f a non-zero element of R'_J which is not divisible by r_i for any $i \in I$. Let M be the degree of the non-zero homogeneous component of lowest degree of the image of $f|_{V'}$ under the isomorphism $\widehat{\mathcal{O}}_{V',p} \cong \widehat{A_p}$ induced by ϖ_p. Then $M \le e_* J$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we write the coordinate ring $\mathcal{O}(V')$ of V' in normal form with respect to four of the coordinates u_j, $j \notin I$.

Since $p \in V'$ and $u_j(p) = 0$ for $j \notin I$, we have $c' = \sum_{i\in I} a_i^{e'_i}\mathbf{w}'_i$. Let $\mathcal{P} := \mathbb{C}[v_i, u_j \mid i \in I,\ j \notin I]$ be a polynomial ring. Then $\mathcal{O}(V') = \mathcal{P}/\mathfrak{g}$, where v_i maps to $u_i - a_i = r_i|_{V'}$ and $\mathfrak{g}$ is generated by the four components of

$$\sum_{i\in I} \big((a_i + v_i)^{e'_i} - a_i^{e'_i}\big)\mathbf{w}'_i + \sum_{j\notin I} u_j^{e'_j}\mathbf{w}'_j.$$

By Lemma 6.8(1), $k \le 3$, so $N - k \ge 7$, and we may fix a set Λ of four indices not in I. By Set-up 6.4(2), the vectors $\mathbf{w}'_l$, $l \in \Lambda$, form a basis of $\mathbb{C}^4$. For $j \notin \Lambda$, let β_{lj}, $l \in \Lambda$, be the coordinates of $\mathbf{w}'_j$ in the basis $(\mathbf{w}'_l)_{l\in\Lambda}$. Then $\mathfrak{g}$ is generated by

$$G_l := u_l^{e'_l} + \sum_{j\notin I\cup\Lambda} \beta_{lj} u_j^{e'_j} + \sum_{i\in I} \beta_{li}\big((a_i + v_i)^{e'_i} - a_i^{e'_i}\big), \quad l \in \Lambda.$$

Let

$$\mathcal{Q} := \mathbb{C}[v_i, u_j \mid i \in I,\ j \notin I \cup \Lambda].$$

Then $G_l - u_l^{e'_l} \in \mathcal{Q}$ for any $l \in \Lambda$. For a monomial $\mu = \prod_{i\in I} v_i^{b_i} \prod_{j\notin I} u_j^{b_j}$, let

$$\deg \mu := \sum_{i\in I} d'_i b_i + \sum_{j\notin I} d'_j b_j, \qquad \operatorname{wt} \mu := 42 \sum_{i\in I} b_i + \sum_{j\notin I} d'_j b_j,$$

and we say that μ is *standard* if $b_l < e'_l$ for any $l \in \Lambda$.

Since $\mathcal{P}/\mathfrak{g}$ is the tensor product over $\mathcal{Q}$ of the rings $\mathcal{Q}[u_l]/(G_l)$, $l \in \Lambda$, and G_l is monic of degree e'_l in u_l, the ring $\mathcal{O}(V')$ is a free $\mathcal{Q}$-module with basis the standard monomials in u_l, $l \in \Lambda$. Thus any $\chi \in \mathcal{O}(V')$ is the image of a unique $\mathbb{C}$-linear combination $\chi^\star$ of standard monomials, which we call the *normal form* of χ. The monomials of G_l have $\deg \le 42 = \deg u_l^{e'_l}$, since $\deg v_i^b = bd'_i \le 42$ for $b \le e'_i$. Hence replacing $u_l^{e'_l}$ with $u_l^{e'_l} - G_l$ does not increase deg. Since f is the image of a homogeneous polynomial of degree J in $W, U_1, \dots, U_N$, the function $f|_{V'}$ is the image of a polynomial in $\mathcal{P}$ whose monomials have $\deg \le J$, after substituting $u_i = a_i + v_i$. Therefore, any monomial of $(f|_{V'})^\star$ has $\deg \le J$.

Step 2. In this step, we show that for any non-zero $\chi \in \mathcal{O}(V')$, the lowest degree of a non-zero homogeneous component of the image of χ in $\widehat{A_p}$ is the minimum $m(\chi)$ of $\operatorname{wt}\mu$ over the monomials μ of $\chi^\star$. In particular, $M = m(f|_{V'})$.

Let $\Phi : \mathcal{O}(V') \to \widehat{A_p}$ be the composite of $\mathcal{O}(V') \to \mathcal{O}_{V',p}$ and the isomorphism $\widehat{\mathcal{O}}_{V',p} \cong \widehat{A_p}$ induced by ϖ_p. For $i \in I$, by Lemma 6.8(2), $r_i|_{V'} = \epsilon_i \cdot \varpi_p^* L_i$ for a regular function ϵ_i near p with $\epsilon_i(p) \ne 0$. Let $\Phi_0 : \mathcal{P} \to A_p$ be the homomorphism defined by $\Phi_0(u_j) = x_j$ for $j \notin I$ and $\Phi_0(v_i) = \epsilon_i(p) L_i$ for $i \in I$. Then Φ_0 maps a monomial μ to a homogeneous element of degree $\operatorname{wt}\mu$. By Construction 6.5(4) and Lemma 6.8(2), near p,

$$(u_i - a_i) \sum_{b=0}^{e'_i-1} u_i^b a_i^{e'_i-1-b} = u_i^{e'_i} - a_i^{e'_i} = \varpi_p^* L_i = \epsilon_i^{-1}(u_i - a_i).$$

Since V' is integral and $r_i|_{V'} = u_i - a_i$ is non-zero as r_i is a non-zero homogeneous element, we obtain $\epsilon_i(p) e'_i a_i^{e'_i-1} = 1$. Moreover, $\Phi(u_j) = x_j$ and $\Phi(v_i)$ is the product of L_i and a unit whose degree zero component is $\epsilon_i(p)$. Hence for any monomial μ, $\Phi(\mu)$ has no non-zero homogeneous

component of degree less than $\operatorname{wt}\mu$, and the homogeneous component of $\Phi(\mu)$ of degree $\operatorname{wt}\mu$ is $\Phi_0(\mu)$. Let $\chi \neq 0$, $m := m(\chi)$, and let $\chi^\star_m$ be the sum of the terms of $\chi^\star$ of weight m. Then $\Phi(\chi)$ has no non-zero homogeneous component of degree less than m, and the homogeneous component of $\Phi(\chi)$ of degree m is $\Phi_0(\chi^\star_m)$. It suffices to show that $\Phi_0(\chi^\star_m) \neq 0$.

For $l \in \Lambda$, let G^0_l be the sum of the terms of G_l of weight 42, that is,

$$G^0_l = u_l^{e'_l} + \sum_{j \notin I \cup \Lambda} \beta_{lj} u_j^{e'_j} + \sum_{i \in I} \beta_{li} e'_i a_i^{e'_i - 1} v_i.$$

By Construction 6.5(4), the equations of V' read

$$\varpi_p^*\Big(\sum_{i \in I} L_i \mathbf{w}'_i + \sum_{j \notin I} x_j^{e'_j} \mathbf{w}'_j \Big) = 0.$$

Since ϖ_p is étale at p and Z_p is irreducible, ϖ_p is dominant, and thus

$$\sum_{i \in I} L_i \mathbf{w}'_i + \sum_{j \notin I} x_j^{e'_j} \mathbf{w}'_j = 0 \quad \text{in } A_p^{\oplus 4}.$$

Since $\epsilon_i(p) e'_i a_i^{e'_i - 1} = 1$, we have $\Phi_0(G^0_l) = 0$ for any $l \in \Lambda$. Let $\operatorname{gr} := \mathcal{P}/(G^0_l \mid l \in \Lambda)$. The relations $G^0_l = 0$, $l \in \Lambda$, are equivalent to

$$\sum_{i \in I} e'_i a_i^{e'_i - 1} v_i \mathbf{w}'_i + \sum_{j \notin I} u_j^{e'_j} \mathbf{w}'_j = 0.$$

Since the vectors $\mathbf{w}'_i$, $i \in I$, are linearly independent, the relations $G^0_l = 0$ express each v_i as a polynomial in the u_j, $j \notin I$, and the remaining relations are $\sum_{j \notin I} u_j^{e'_j} \overline{\mathbf{w}}'_j = 0$. Hence Φ_0 induces an isomorphism $\operatorname{gr} \cong A_p$. On the other hand, $G^0_l - u_l^{e'_l} \in \mathcal{Q}$ for any $l \in \Lambda$, so as in **Step 1**, gr is a free $\mathcal{Q}$-module with basis the standard monomials in u_l, $l \in \Lambda$. Since $\chi^\star_m$ is a non-zero linear combination of standard monomials, the image of $\chi^\star_m$ in gr is non-zero, and thus $\Phi_0(\chi^\star_m) \neq 0$.

Step 3. In this step, we fix $i_0 \in I$ with $e'_{i_0} = \max_{i \in I} e'_i$, and we show that $(f|_{V'})^\star$ has a monomial which is not divisible by v_{i_0}.

We first show that $f|_{V'} \notin r_{i_0}|_{V'} \cdot \mathcal{O}(V')$. Suppose that $f|_{V'} = r_{i_0}|_{V'} \cdot b$ for some $b \in \mathcal{O}(V')$. Since $\mathcal{O}(V') = R'/(W-1)$, we may write $b = b'|_{V'}$ for a homogeneous element b' of R'. Since a homogeneous element of R' whose restriction to V' is zero is zero, $W^s f = W^{s'} r_{i_0} b'$ in R' for some integers $s, s' \geq 0$. By Lemma 6.8(2), r_{i_0} is a prime element of R', and r_{i_0} does not divide W since $\deg r_{i_0} = d'_{i_0} > 1 = \deg W$. Thus r_{i_0} divides f, a contradiction.

Since $G_l - u_l^{e'_l} \in \mathcal{Q}$, the ring $\mathcal{O}(V')/(v_{i_0}) = \mathcal{P}/(\mathfrak{g} + (v_{i_0}))$ is a free $\mathcal{Q}/(v_{i_0})$-module with basis the standard monomials in u_l, $l \in \Lambda$, and the image of $f|_{V'}$ in $\mathcal{O}(V')/(v_{i_0})$ is represented by the polynomial $(f|_{V'})^\star|_{v_{i_0}=0}$. Since $(f|_{V'})^\star|_{v_{i_0}=0}$ is a linear combination of standard monomials representing a non-zero element, $(f|_{V'})^\star|_{v_{i_0}=0} \neq 0$.

Step 4. In this step, we conclude the proof.

By **Step 3**, there exists a monomial $\mu = \prod_{i \in I} v_i^{b_i} \prod_{j \notin I} u_j^{b_j}$ of $(f|_{V'})^\star$ with $b_{i_0} = 0$. By Construction 6.5(9), $e_* \geq 1$, and $e_* \geq e'_i$ for any $i \in I \backslash \{i_0\}$. Since $42 = e'_i d'_i$, by **Step 1** we have

$$\operatorname{wt}\mu = \sum_{i \in I \backslash \{i_0\}} e'_i d'_i b_i + \sum_{j \notin I} d'_j b_j \leq e_* \deg \mu \leq e_* J.$$

By **Step 2**, $M = m(f|_{V'}) \leq \operatorname{wt}\mu \leq e_* J$. □

7.4. **Singular points with one non-zero coordinate.** In this subsection, we prove Proposition 7.10, which treats the singular points of V_c at which exactly one of the coordinates $u_1, \dots, u_{11}$ is non-zero.

When $N = 11$, Lemma 7.9 gives the coefficient $\frac{7}{6M}$ on E_p, deduced from Proposition 7.3 at the smooth points of V' near p. At a singular point of V_c with exactly one non-zero coordinate, $k = 1$ and $e_* = 1$, so for a section of degree J' not divisible by r_i, Lemma 7.8 gives $M \le J'$, and the coefficient $\frac{7}{6M}$ is at least the coefficient $\frac{7}{6J'}$ needed in Proposition 7.10. Since Proposition 7.3 is a statement on the fibers of h, Lemma 7.9 assumes that $N = 11$.

Lemma 7.9. *Notation and conditions as in Construction 6.5, and assume that $N = 11$. Then for any integer $M > 0$ and any non-zero element g of $(A_p)_M$, the pair $(E_p, \Delta_p + \frac{7}{6M} D_g)$ is lc.*

Proof. Since $N = 11$, $d'_j = d_j$ for any j. Set-up 6.1 only requires that any four of the vectors are linearly independent. Thus we may use Set-up 6.1, Proposition 6.2(2), and Proposition 7.3 for the vectors $\mathbf{w}'_1, \dots, \mathbf{w}'_{11}$ and for c' in place of $\mathbf{w}_1, \dots, \mathbf{w}_{11}$ and c, and then $R_{c'} = R'$ and $V_{c'} = V'$. By Proposition 6.2(2), V' has only finitely many singular points. By Lemma 6.8(2), ϖ_p is étale at p and $\varpi_p(p) = 0$. Thus there exists an open neighborhood V'° of p in V' such that $\varpi_p|_{V'^{\circ}}$ is étale and any point of $V'^{\circ} \backslash \{p\}$ is a smooth point of V'. Since étale morphisms are open, $Z^{\circ} := \varpi_p(V'^{\circ})$ is an open neighborhood of 0 in Z_p.

Fix an integer $M > 0$ and a non-zero element g of $(A_p)_M$. Choose a homogeneous polynomial $\tilde{g} \in R'_M$ of degree M in the variables U_j, $j \notin I$, such that substituting x_j for U_j for any $j \notin I$ gives a polynomial representing g. Then $\tilde{g}|_{V'} = \varpi_p^* g$. Since ϖ_p is étale at p, the homomorphism $\mathcal{O}_{Z_p,0} \to \mathcal{O}_{V',p}$ induced by ϖ_p is flat and local, hence injective. Since A_p is a domain and $g \neq 0$, the image of g in $\mathcal{O}_{Z_p,0}$ is non-zero, and thus $\varpi_p^* g \neq 0$. In particular, $\tilde{g} \neq 0$, and $D_{\tilde{g}}|_{V'} = \operatorname{div}(\varpi_p^* g)$ since W is invertible on V'.

We show that $(Z_p \backslash \{0\}, \frac{7}{6M} \operatorname{div}(g))$ is lc. Let $z \in Z^{\circ} \backslash \{0\}$, and let $v \in V'^{\circ}$ be a point with $\varpi_p(v) = z$. Then $v \neq p$, so v is a smooth point of V'. By Proposition 7.3 applied to $\tilde{g}$, the pair $(V', \frac{7}{6M} \operatorname{div}(\varpi_p^* g))$ is lc near v. Let $Y \to Z_p$ be a log resolution of $(Z_p, \operatorname{div}(g))$ over a neighborhood of z. Since ϖ_p is étale at v, $Y \times_{Z_p} V'^{\circ} \to V'^{\circ}$ is a log resolution of $(V', \operatorname{div}(\varpi_p^* g))$ over a neighborhood of v, the crepant pullback of $(V', \frac{7}{6M} \operatorname{div}(\varpi_p^* g))$ to $Y \times_{Z_p} V'^{\circ}$ is the pullback of the crepant pullback of $(Z_p, \frac{7}{6M} \operatorname{div}(g))$ to Y, and any prime divisor on Y whose image on Z_p contains z is the image of a prime divisor on $Y \times_{Z_p} V'^{\circ}$ whose image on V' contains v. Thus $(Z_p, \frac{7}{6M} \operatorname{div}(g))$ is lc near z [KM98, Lemma 2.30 and Corollary 2.31(3)].

For $\lambda \in \mathbb{C}^*$ and $x \in Z_p$, let $\lambda \cdot x := (\lambda^{d'_j} x_j)_{j \notin I}$. Since A_p is graded, the map $(\lambda, x) \to \lambda \cdot x$ is an action of $\mathbb{C}^*$ on Z_p, and $g(\lambda \cdot x) = \lambda^M g(x)$. Let $z \in Z_p \backslash \{0\}$. Since $d'_j > 0$ for any j, $\lambda \cdot z$ tends to 0 as λ tends to 0, so there exists $\lambda \in \mathbb{C}^*$ such that $\lambda \cdot z \in Z^{\circ} \backslash \{0\}$. The automorphism $x \to \lambda \cdot x$ of Z_p maps z to $\lambda \cdot z$ and pulls back $\operatorname{div}(g)$ to $\operatorname{div}(g)$. Thus $(Z_p, \frac{7}{6M} \operatorname{div}(g))$ is lc near z. Hence $(Z_p \backslash \{0\}, \frac{7}{6M} \operatorname{div}(g))$ is lc.

Now fix $l \notin I$. By Lemma 6.7(1)(3), $Z_{p,l}$ is a smooth irreducible variety, $q_p|_{Z_{p,l}} : Z_{p,l} \to E_p \cap (x_l \neq 0)$ is the quotient morphism by μ_{w_l}, in particular finite and surjective, and E_p is $\mathbb{Q}$-factorial.

By Lemma 6.7(2), $K_{Z_{p,l}} = (q_p|_{Z_{p,l}})^*(K_{E_p} + \Delta_p)$ over $E_p \cap (x_l \neq 0)$. By Construction 6.5(7), $q_p^* D_g = \operatorname{div}(g)$ on $Z_p \backslash \{0\}$, so $(q_p|_{Z_{p,l}})^* D_g = \operatorname{div}(g|_{Z_{p,l}})$. Here $g|_{Z_{p,l}} \neq 0$ since g does not vanish identically on the dense open subset $Z_p \cap (x_l \neq 0)$ of Z_p and $g(\lambda \cdot x) = \lambda^M g(x)$. Hence

$$K_{Z_{p,l}} + \frac{7}{6M} \operatorname{div}(g|_{Z_{p,l}}) = (q_p|_{Z_{p,l}})^* \Big(K_{E_p} + \Delta_p + \frac{7}{6M} D_g \Big) \tag{7.9}$$

over $E_p \cap (x_l \neq 0)$. In (7.9), $\operatorname{div}(g|_{Z_{p,l}}) \ge 0$ and $\Delta_p + \frac{7}{6M} D_g \ge 0$. By Lemma 2.3, it suffices to show that $(Z_{p,l}, \frac{7}{6M} \operatorname{div}(g|_{Z_{p,l}}))$ is lc.

Let $\Phi : Z_{p,l} \times \mathbb{G}_m \to Z_p \cap (x_l \neq 0)$ be the morphism given by $\Phi(x, \lambda) = \lambda \cdot x$. If $\lambda \cdot x = \lambda' \cdot x'$ for $x, x' \in Z_{p,l}$, then $\lambda^{d'_l} = \lambda'^{d'_l}$, so $\lambda' = \zeta^{-1} \lambda$ and $x' = \zeta \cdot x$ for some $\zeta \in \mu_{d'_l}$. Thus Φ is the

quotient morphism by the free action of $\mu_{d'_l}$ on $Z_{p,l} \times \mathbb{G}_m$ given by $\zeta \cdot (x, \lambda) = (\zeta \cdot x, \zeta^{-1}\lambda)$, and Φ is finite, étale, and surjective, since Φ is the base change of the d'_l-th power map of $\mathbb{G}_m$ along x_l. Set $\Gamma := \operatorname{div}(g|_{Z_{p,l}})$. Since $g(\lambda \cdot x) = \lambda^M g(x)$ and λ^M is a unit on $\mathbb{G}_m$, $\Phi^* \operatorname{div}(g) = \operatorname{pr}_1^* \Gamma$, where pr_1 denotes the first projection of a product with $\mathbb{G}_m$. Since Φ is étale, $K_{Z_{p,l} \times \mathbb{G}_m} = \Phi^* K_{Z_p}$, and by Lemma 2.3, the pair

$$\Big(Z_{p,l} \times \mathbb{G}_m, \frac{7}{6M} \operatorname{pr}_1^* \Gamma\Big)$$

is lc. Let $\psi : Y' \to Z_{p,l}$ be a log resolution of $(Z_{p,l}, \Gamma)$, and let $\Gamma_{Y'}$ be the $\mathbb{Q}$-divisor on Y' with $K_{Y'} + \Gamma_{Y'} = \psi^*(K_{Z_{p,l}} + \frac{7}{6M}\Gamma)$ and $\psi_* \Gamma_{Y'} = \frac{7}{6M}\Gamma$. Then $\psi \times \operatorname{id}_{\mathbb{G}_m}$ is a log resolution of $(Z_{p,l} \times \mathbb{G}_m, \operatorname{pr}_1^* \Gamma)$, and

$$K_{Y' \times \mathbb{G}_m} + \operatorname{pr}_1^* \Gamma_{Y'} = (\psi \times \operatorname{id}_{\mathbb{G}_m})^* \Big(K_{Z_{p,l} \times \mathbb{G}_m} + \frac{7}{6M} \operatorname{pr}_1^* \Gamma\Big).$$

The coefficient of $\operatorname{pr}_1^* Q$ in $\operatorname{pr}_1^* \Gamma_{Y'}$ equals the coefficient of Q in $\Gamma_{Y'}$ for any prime divisor Q on Y'. Thus $(Z_{p,l}, \frac{7}{6M}\Gamma)$ is lc [KM98, Lemma 2.30 and Corollary 2.31(3)].

Hence $(E_p, \Delta_p + \frac{7}{6M} D_g)$ is lc over $E_p \cap (x_l \neq 0)$ for any $l \notin I$. Since the open subsets $E_p \cap (x_l \neq 0)$, $l \notin I$, cover E_p, $(E_p, \Delta_p + \frac{7}{6M} D_g)$ is lc. □

Proposition 7.10. *Notation and conditions as in Set-up 6.1. Let $c \in \mathbb{C}^4 \backslash \{0\}$, let p be a singular point of V_c at which exactly one of the coordinates $u_1, \dots, u_{11}$ is non-zero, let $J > 0$ be an integer, and let f be a non-zero element of $(R_c)_J$. Then $(V_c, \frac{7}{6J} D_f|_{V_c})$ is lc near p.*

Proof. We use Set-up 6.4 with $N := 11$, $\mathbf{w}'_j := \mathbf{w}_j$ for $1 \le j \le 11$, and $c' := c$. Then $d'_j = d_j$, $e'_j = e_j$, $R' = R_c$, and $V' = V_c$. Since W is invertible on V_c, $D_f|_{V_c} = \operatorname{div}(f|_{V'})$. We use Construction 6.5 at p. Then $I = \{i\}$ for some $1 \le i \le 11$, $k = 1$, and $e_* = 1$. Since $d_1 + \cdots + d_{11} = 151$,

$$\delta_p = \sum_{j \neq i} d_j - 42 \cdot 3 = 25 - d_i.$$

Thus (d_i, e_i, δ_p) is one of $(21, 2, 4)$, $(14, 3, 11)$, and $(6, 7, 19)$, and respectively

$$6\delta_p - 7e_i = 24 - 14 = 10, \quad 66 - 21 = 45, \quad \text{or} \quad 114 - 49 = 65. \tag{7.10}$$

In particular, $6\delta_p > 7e_i$. By Proposition 7.6,

$$\Big(V_c, \frac{7}{6d_i} \operatorname{div}(r_i|_{V'})\Big) \text{ is lc near } p. \tag{7.11}$$

By Lemma 7.7, it suffices to show that $(V_c, \frac{7}{6J'} \operatorname{div}(f'|_{V'}))$ is lc near p for any integer $J' > 0$ and any non-zero $f' \in (R_c)_{J'}$ which is not divisible by r_i. Let $g \in A_p$ be the non-zero homogeneous component of lowest degree of the image of $f'|_{V'}$ under the isomorphism $\widehat{\mathcal{O}}_{V',p} \cong \widehat{A_p}$ induced by ϖ_p, and let $M := \deg g$. If $M = 0$, then $f'|_{V'}(p) \neq 0$, so $\operatorname{div}(f'|_{V'}) = 0$ near p, and $(V_c, 0)$ is lc near p by (7.11).

Assume that $M > 0$. Since $k = 1$ and $e_* = 1$, Lemma 7.8 implies that $M \le J'$. By Lemma 7.9, $(E_p, \Delta_p + \frac{7}{6M} D_g)$ is lc, so $(E_p, \Delta_p + \frac{7}{6J'} D_g)$ is lc. Moreover,

$$\frac{7}{6J'} M \le \frac{7}{6} < 4 \le \delta_p.$$

By Lemma 7.5 applied to $f'|_{V'}$ and $t := \frac{7}{6J'}$, $(V_c, \frac{7}{6J'} \operatorname{div}(f'|_{V'}))$ is lc near p. □

7.5. **Weighted blow-ups of diagonal cones.** In this subsection, we prove Proposition 7.11, a lower bound for the log canonical thresholds of the $\mathbb{Q}$-divisors D_g with respect to (E_p, Δ_p), which holds for both $N = 10$ and $N = 11$. We use Proposition 7.11 at the singular points of V_c with two or three non-zero coordinates, where the lowest degree in Lemma 7.8 can exceed the degree of the section and the coefficient in Lemma 7.9 is too small, and in the case $N = 10$, where Proposition 7.3 is not available.

The proof of Proposition 7.11 uses K-semistability as a tool to bound log canonical thresholds. The power map $E_p \to \mathbb{P}^{N-5}$ is a Galois cover branched along the coordinate hyperplanes. After raising the coefficients of the hyperplane arrangement to a K-semistable boundary, K-semistability ascends to E_p by Theorem 2.13 and Theorem 2.14, which are also used in Section 4, and the inequality $S \geq \frac{1}{N-4}T$ of [BJ20, (3.1)] bounds the multiplicities of D_g.

Proposition 7.11. *Notation and conditions as in Construction 6.5, and assume that $\delta_p > 0$. Then for any integer $M > 0$ and any non-zero element g of $(A_p)_M$, the pair $(E_p, \Delta_p + \frac{\kappa_p}{M} D_g)$ is lc.*

Proof. The proof is long, so we divide it into several steps. We use the notation w_j, s_j, $Z_{p,j}$, and T_j of Lemma 6.7.

Step 1. In this step we construct the power map $\pi : E_p \to \mathbb{P}^{N-5}$ and show that π is a finite surjective Galois morphism.

By Lemma 6.8(1), $1 \leq k \leq 3$, and A_p is a normal domain of dimension $N-4$. Since any four of $\mathbf{w}'_1, \dots, \mathbf{w}'_N$ are linearly independent, $\dim\langle \mathbf{w}'_i \mid i \in I\rangle = k$, and by Lemma 3.2(4), any $4-k$ of the vectors $\overline{\mathbf{w}}'_j$, $j \notin I$, form a basis of $\mathbb{C}^4/\langle \mathbf{w}'_i \mid i \in I\rangle$. Let

$$\Lambda := \Big\{ (\xi_j)_{j\notin I} \in \mathbb{C}^{N-k} \;\Big|\; \sum_{j\notin I} \xi_j \overline{\mathbf{w}}'_j = 0 \Big\}.$$

Then $\dim \Lambda = (N-k) - (4-k) = N-4$. Moreover, for any set I'' of $N-4$ indices not in I, the linear forms $\xi_j|_\Lambda$, $j \in I''$, form a basis of the dual space of Λ: if $\xi \in \Lambda$ and $\xi_j = 0$ for any $j \in I''$, then $\sum_{j\notin I\cup I''} \xi_j \overline{\mathbf{w}}'_j = 0$, where the $4-k$ vectors $\overline{\mathbf{w}}'_j$, $j \notin I \cup I''$, are linearly independent, so $\xi = 0$. Let $\mathbb{P}^{N-5} := \mathbb{P}(\Lambda)$, and for $j \notin I$, let $H_j \subset \mathbb{P}^{N-5}$ be the hyperplane $(\xi_j = 0)$.

Let $\mathbb{C}[\Lambda]$ be the coordinate ring of Λ, graded by $\deg \xi_j = 42$. Since $\sum_{j\notin I} x_j^{e'_j} \overline{\mathbf{w}}'_j = 0$ in A_p and $\deg x_j^{e'_j} = 42$, sending ξ_j to $x_j^{e'_j}$ defines a graded homomorphism $\mathbb{C}[\Lambda] \to A_p$. The ring A_p is finite over the image, since x_j is integral over $\mathbb{C}[x_j^{e'_j}]$ for any $j \notin I$. Since $\mathbb{C}[\Lambda]$ and A_p are domains of dimension $N-4$, the homomorphism is injective. Thus the morphism $Z_p \to \Lambda$ sending x to $(x_j^{e'_j})_{j\notin I}$ is finite and surjective, and the preimage of 0 is $\{0\}$. Hence it induces a finite surjective morphism $\pi : E_p \to \mathbb{P}^{N-5}$ sending $[x]$ to $[(x_j^{e'_j})_{j\notin I}]$.

Let $G := \prod_{j\notin I} \mu_{e'_j}$ act on Z_p by $(\zeta \cdot x)_j := \zeta_j x_j$. Since the equations of Z_p only involve the powers $x_j^{e'_j}$, G acts on A_p by graded automorphisms. Thus G acts on E_p, and $\pi \circ \zeta = \pi$ for any $\zeta \in G$. We show that the fibers of π are the G-orbits. Let $x, x' \in Z_p \backslash \{0\}$ with $\pi([x]) = \pi([x'])$. Then there exists $\alpha \in \mathbb{C}^*$ such that $x_j'^{e'_j} = \alpha x_j^{e'_j}$ for any $j \notin I$. Let $\lambda \in \mathbb{C}^*$ with $\lambda^{42} = \alpha$. Then $(\lambda^{d'_j} x_j)^{e'_j} = \alpha x_j^{e'_j} = x_j'^{e'_j}$, so $x'_j = \zeta_j \lambda^{d'_j} x_j$ for some $\zeta_j \in \mu_{e'_j}$, and thus $[x'] = \zeta \cdot [x]$ with $\zeta := (\zeta_j)_{j\notin I}$. Hence the finite morphism $E_p/G \to \mathbb{P}^{N-5}$ induced by π is bijective. Since a general fiber of $E_p/G \to \mathbb{P}^{N-5}$ is one point, it is birational, and since $\mathbb{P}^{N-5}$ is normal, it is an isomorphism: over any affine open subset of $\mathbb{P}^{N-5}$ it corresponds to a finite extension of a normal domain inside its field of fractions, and a normal domain is integrally closed in its field of fractions. Thus π is the quotient morphism by the image of G in $\mathrm{Aut}(E_p)$, and π is a finite surjective Galois morphism.

Let

$$E_p^\circ := \pi^{-1}\Big(\mathbb{P}^{N-5} \backslash \bigcup_{j\notin I} H_j\Big),$$

the open subset of E_p where $x_j \neq 0$ for any $j \notin I$. If $[x] \in E_p^\circ$ and $\zeta \cdot [x] = [x]$, then there exists $\lambda \in \mathbb{C}^*$ such that $\zeta_j x_j = \lambda^{d'_j} x_j$, so $\zeta_j = \lambda^{d'_j}$, for any $j \notin I$, and ζ acts on E_p as $\lambda \in \mathbb{C}^*$, that is, trivially. Thus the image of G in $\mathrm{Aut}(E_p)$ acts freely on E_p°, and π is étale over $\mathbb{P}^{N-5} \backslash \bigcup_{j\notin I} H_j$.

Step 2. In this step we show that $\sum_{j\notin I} H_j$ is a simple normal crossing divisor and compute $K_{E_p}+\Delta_p$ as a pullback under π.

By **Step 1**, any $N-4$ of the linear forms $\xi_j|_\Lambda$, $j\notin I$, form a basis of the dual space of Λ. Thus the $N-k$ hyperplanes H_j are distinct, and at any point of $\mathbb{P}^{N-5}$, the linear forms defining the hyperplanes H_j through the point are linearly independent. Hence $\sum_{j\notin I} H_j$ is a simple normal crossing divisor.

Let $j\notin I$. The composition $\pi\circ q_p$ is the composition of $Z_p\backslash\{0\}\to\Lambda\backslash\{0\}$ with the projection $\Lambda\backslash\{0\}\to\mathbb{P}^{N-5}$, so by Construction 6.5(7),

$$q_p^*\pi^*H_j=\operatorname{div}(x_j^{e'_j})=q_p^*D_{x_j^{e'_j}}.$$

Since q_p is surjective, q_p^* is injective on $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors. By Lemma 6.7(2),

$$\pi^*H_j=D_{x_j^{e'_j}}=e'_jD_{x_j}=\frac{e'_j}{s_j}T_j. \tag{7.12}$$

Thus T_j is the only prime divisor on E_p over H_j, and the ramification index of π along T_j is $\frac{e'_j}{s_j}$. By **Step 1**, π is étale outside $\bigcup_{j\notin I}T_j$. By the Hurwitz formula [Kol13, 2.41],

$$K_{E_p}=\pi^*K_{\mathbb{P}^{N-5}}+\sum_{j\notin I}\Big(\frac{e'_j}{s_j}-1\Big)T_j.$$

Let $B:=\sum_{j\notin I}(1-\frac{1}{e'_j})H_j$. By (7.12) and Lemma 6.7(2),

$$\begin{aligned}\pi^*(K_{\mathbb{P}^{N-5}}+B)&=K_{E_p}+\sum_{j\notin I}\Big(1-\frac{e'_j}{s_j}+\Big(1-\frac{1}{e'_j}\Big)\frac{e'_j}{s_j}\Big)T_j\\&=K_{E_p}+\sum_{j\notin I}\Big(1-\frac{1}{s_j}\Big)T_j=K_{E_p}+\Delta_p.\end{aligned} \tag{7.13}$$

Step 3. In this step we raise the coefficients of B and show that E_p carries a K-semistable log Fano pair (E_p,Δ') with $\Delta'\geq\Delta_p$.

Since $\delta_p=\sum_{j\notin I}d'_j-42(4-k)$, $\frac{1}{e'_j}=\frac{d'_j}{42}$, and $\kappa_p\leq\frac{\delta_p}{N-4}$,

$$\sum_{j\notin I}\Big(1-\frac{1}{e'_j}\Big)=(N-k)-\frac{1}{42}\sum_{j\notin I}d'_j=N-4-\frac{\delta_p}{42}\leq(N-4)\Big(1-\frac{\kappa_p}{42}\Big). \tag{7.14}$$

Since $\kappa_p\leq d'_j$, $1-\frac{1}{e'_j}\leq 1-\frac{\kappa_p}{42}$ for any $j\notin I$. Moreover, $\kappa_p>0$ since $\delta_p>0$, and $\kappa_p\leq d'_j\leq 21$, so $0<1-\frac{\kappa_p}{42}<1$. Also $\kappa_p\in\mathbb{Q}$. For a rational number $\tau\in[0,1]$, the sum

$$\sum_{j\notin I}\Big((1-\tau)\Big(1-\frac{1}{e'_j}\Big)+\tau\Big(1-\frac{\kappa_p}{42}\Big)\Big)$$

is an affine function of τ with rational coefficients. By (7.14), the sum at $\tau=0$ is at most $(N-4)(1-\frac{\kappa_p}{42})$, and the sum at $\tau=1$ is

$$(N-k)\Big(1-\frac{\kappa_p}{42}\Big)\geq(N-4)\Big(1-\frac{\kappa_p}{42}\Big).$$

Thus there exists a rational number $\tau_0\in[0,1]$ such that the sum at $\tau=\tau_0$ equals $(N-4)(1-\frac{\kappa_p}{42})$. For $j\notin I$, let

$$\beta'_j:=(1-\tau_0)\Big(1-\frac{1}{e'_j}\Big)+\tau_0\Big(1-\frac{\kappa_p}{42}\Big).$$

Then $\beta'_j \in \mathbb{Q}$,

$$0 < 1 - \frac{1}{e'_j} \le \beta'_j \le 1 - \frac{\kappa_p}{42} < 1, \quad \text{and} \quad \sum_{j\notin I} \beta'_j = (N-4)\Big(1 - \frac{\kappa_p}{42}\Big). \tag{7.15}$$

Let $B' := \sum_{j\notin I} \beta'_j H_j$, a $\mathbb{Q}$-divisor. Since $\deg K_{\mathbb{P}^{N-5}} = -(N-4)$, by (7.15),

$$-(K_{\mathbb{P}^{N-5}} + B') \sim_{\mathbb{Q}} \frac{(N-4)\kappa_p}{42}\mathcal{O}_{\mathbb{P}^{N-5}}(1),$$

which is ample since $\kappa_p > 0$. Since $\sum_{j\notin I} H_j$ is a simple normal crossing divisor and $\beta'_j \in (0,1)$, $(\mathbb{P}^{N-5}, B')$ is klt [KM98, Corollary 2.31(3)]. Thus $(\mathbb{P}^{N-5}, B')$ is a log Fano pair. We use Theorem 2.13 with $n := N-5$, the $N-k$ hyperplanes H_j, $j \notin I$, and the coefficients $\beta'_j \in (0,1)\cap\mathbb{Q}$. For any $1 \le l \le N-5$ and any subset $I' \subset \{j \mid j \notin I\}$ with $\#I' = l$, since $\beta'_j \le 1 - \frac{\kappa_p}{42}$ for any $j \in I'$,

$$l\sum_{j\notin I} \beta'_j = l(N-4)\Big(1 - \frac{\kappa_p}{42}\Big) \ge (N-4)\sum_{j\in I'} \beta'_j.$$

Thus $(\mathbb{P}^{N-5}, B')$ is K-semistable by Theorem 2.13.

Let $\Delta' := \Delta_p + \pi^*(B' - B)$. By (7.12) and Lemma 6.7(2),

$$\Delta' = \sum_{j\notin I}\Big(1 - \frac{1}{s_j} + \frac{e'_j}{s_j}\Big(\beta'_j - 1 + \frac{1}{e'_j}\Big)\Big)T_j = \sum_{j\notin I}\Big(1 - \frac{e'_j}{s_j}(1-\beta'_j)\Big)T_j.$$

By (7.15), the coefficient of T_j in Δ' is a rational number in $[1 - \frac{1}{s_j}, 1)$. Thus Δ' is a $\mathbb{Q}$-divisor and $\Delta' \ge \Delta_p \ge 0$. By (7.13), $K_{E_p} + \Delta' = \pi^*(K_{\mathbb{P}^{N-5}} + B')$. Since $(\mathbb{P}^{N-5}, B')$ is klt and $B', \Delta' \ge 0$, (E_p, Δ') is klt by Lemma 2.3. Since π is finite and $-(K_{\mathbb{P}^{N-5}} + B')$ is ample, $-(K_{E_p} + \Delta') = \pi^*(-(K_{\mathbb{P}^{N-5}} + B'))$ is ample, and

$$-(K_{E_p} + \Delta') \sim_{\mathbb{Q}} \frac{(N-4)\kappa_p}{42}\pi^*\mathcal{O}_{\mathbb{P}^{N-5}}(1). \tag{7.16}$$

Since E_p is projective, (E_p, Δ') is a log Fano pair. Thus (E_p, Δ') and $(\mathbb{P}^{N-5}, B')$ are log Fano pairs with $\mathbb{Q}$-divisor boundaries, π is a finite surjective Galois morphism by **Step 1**, and $K_{E_p} + \Delta' = \pi^*(K_{\mathbb{P}^{N-5}} + B')$. By Theorem 2.14, (E_p, Δ') is K-semistable. By Definition 2.12,

$$a(F, E_p, \Delta') \ge S_{(E_p,\Delta')}(F) \tag{7.17}$$

for any prime divisor F over E_p.

Step 4. In this step we bound $\operatorname{mult}_F D_g$ for any prime divisor F over E_p and conclude the proof.

Fix $j_0 \notin I$. The quotient $g^{42}/x_{j_0}^{Me'_{j_0}}$ of two homogeneous elements of A_p of degree $42M$ defines a rational function ϕ on E_p with $q_p^*\phi = g^{42}/x_{j_0}^{Me'_{j_0}}$. By Construction 6.5(7) and (7.12),

$$q_p^*(42D_g - M\pi^*H_{j_0}) = \operatorname{div}(g^{42}) - \operatorname{div}(x_{j_0}^{Me'_{j_0}}) = q_p^*\operatorname{div}(\phi).$$

Since q_p^* is injective on $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors, $42D_g - M\pi^*H_{j_0} = \operatorname{div}(\phi)$, so

$$D_g \sim_{\mathbb{Q}} \frac{M}{42}\pi^*\mathcal{O}_{\mathbb{P}^{N-5}}(1). \tag{7.18}$$

For a prime divisor F over E_p, let

$$T_{(E_p,\Delta')}(F) := \sup\{\operatorname{mult}_F D \mid D \ge 0 \text{ a } \mathbb{Q}\text{-divisor},\ D \sim_{\mathbb{Q}} -(K_{E_p} + \Delta')\}.$$

The invariant $T_{(E_p,\Delta')}$ is not related to the prime divisors T_j of Lemma 6.7 used in **Step 2** and **Step 3**. Let r be a positive integer such that $L := -r(K_{E_p} + \Delta')$ is a Cartier divisor. The $\mathbb{Q}$-divisors $D \ge 0$ with $D \sim_{\mathbb{Q}} -(K_{E_p} + \Delta')$ are the $\mathbb{Q}$-divisors $\frac{1}{rm}\operatorname{div}(s)$ with $m > 0$ and $s \in H^0(E_p, mL)\backslash\{0\}$. Thus by definition, the invariant T of [BJ20] for L and F equals $rT_{(E_p,\Delta')}(F)$. Let $\psi : Y \to E_p$ be a projective birational morphism from a normal projective

variety Y on which F is a prime divisor. Since $\mathrm{vol}(\psi^*L) = \mathrm{vol}(L)$, by [BJ20, Lemma 3.7(ii) and Corollary 3.9], the invariant S of [BJ20] for L and F equals $\mathrm{vol}(L)^{-1}\int_0^{+\infty}\mathrm{vol}(\psi^*L - tF)dt$, which is $rS_{(E_p,\Delta')}(F)$ after the substitution $t = rt'$, since vol is homogeneous of degree $N-5$, in accordance with [BJ20, Lemma 3.7(i) and Remark 3.8]. Since $\dim E_p = N-5$, by [BJ20, Lemma 2.6 and (3.1)], the invariant S is at least $\frac{1}{N-4}$ times the invariant T. Dividing by r, we obtain

$$S_{(E_p,\Delta')}(F) \geq \frac{1}{N-4}T_{(E_p,\Delta')}(F). \tag{7.19}$$

By (7.16) and (7.18), the $\mathbb{Q}$-divisor $D' := \frac{(N-4)\kappa_p}{M}D_g$ satisfies $D' \geq 0$ and $D' \sim_{\mathbb{Q}} -(K_{E_p} + \Delta')$. Thus

$$T_{(E_p,\Delta')}(F) \geq \mathrm{mult}_F D' = \frac{(N-4)\kappa_p}{M}\mathrm{mult}_F D_g.$$

By Lemma 6.7(3), E_p is $\mathbb{Q}$-factorial. Since $\Delta' - \Delta_p \geq 0$, $a(F,E_p,\Delta_p) \geq a(F,E_p,\Delta')$. By (7.17) and (7.19),

$$a(F,E_p,\Delta_p) \geq a(F,E_p,\Delta') \geq S_{(E_p,\Delta')}(F) \geq \frac{1}{N-4}T_{(E_p,\Delta')}(F) \geq \frac{\kappa_p}{M}\mathrm{mult}_F D_g.$$

Hence

$$a\Big(F,E_p,\Delta_p + \frac{\kappa_p}{M}D_g\Big) = a(F,E_p,\Delta_p) - \frac{\kappa_p}{M}\mathrm{mult}_F D_g \geq 0$$

for any prime divisor F over E_p. Thus $(E_p, \Delta_p + \frac{\kappa_p}{M}D_g)$ is lc. □

7.6. **Singular points with two or three non-zero coordinates.** In this subsection, we prove Proposition 7.13 and Theorem 7.1. At a singular point p of V_c at which exactly two of the coordinates u_9, u_{10}, u_{11} of degree 6 are non-zero, we have $\kappa_p = 6 < \frac{49}{6} = \frac{7}{6}e_*$, so Proposition 7.11 is not enough. In this case we restrict to the Cartier divisor $(r_i = 0)$ for one such index i, which is isomorphic to a fiber with $N = 10$ in Set-up 6.4 on which the non-zero coordinates of p are indexed by $I\backslash\{i\}$ and only one of them has degree 6, so that $e_* \leq 3$ and $\kappa_p \geq \frac{7}{6}e_*$ there (Table 2), and return to V_c by Theorem 2.5.

The first two conditions of the next lemma are the conditions under which Lemma 7.5, which needs $tM \leq \delta_p$, and Proposition 7.11, which needs $t \leq \frac{\kappa_p}{M}$, apply to the coefficient $t = \frac{7}{6J}$ and to the bound $M \leq e_*J$ of Lemma 7.8 on the lowest order M at p of a section of degree J. The third condition $6\delta_p > 7e_i'$ is the condition under which Proposition 7.6 handles the factor r_i of a section, which has degree d_i' and lowest order $42 = e_i'd_i'$ at p.

Lemma 7.12. *Notation and conditions as in Construction 6.5. Assume that*

$$\delta_p \geq \frac{7}{6}e_*, \qquad \kappa_p \geq \frac{7}{6}e_*, \qquad \text{and} \qquad 6\delta_p > 7e_i' \ \text{ for any } i \in I.$$

Then for any integer $J > 0$ and any non-zero element f of R'_J, the pair $(V', \frac{7}{6J}\mathrm{div}(f|_{V'}))$ is lc near p.

Proof. By Lemma 7.7, it suffices to show that $(V', \frac{7}{6J_1}\mathrm{div}(f_1|_{V'}))$ is lc near p for any integer $J_1 > 0$ and any non-zero element f_1 of R'_{J_1} which is not divisible by r_i for any $i \in I$. Since $k \geq 1$ (Lemma 6.8(1)), Proposition 7.6 for some $i \in I$, with the coefficient lowered to zero, shows that $(V', 0)$ is lc near p.

Since R' is a domain and $W \neq 0$ in R', $\varphi := f_1|_{V'}$ is a non-zero regular function on V'. Let $g \in A_p$ be the non-zero homogeneous component of lowest degree of the image of φ under the isomorphism $\widehat{\mathcal{O}}_{V',p} \cong \widehat{A_p}$ induced by ϖ_p, and let $M := \deg g$. If $M = 0$, then $\varphi(p) \neq 0$, so $\mathrm{div}(\varphi) = 0$ near p, and $(V', 0)$ is lc near p. Assume that $M > 0$. By Lemma 7.8, $M \leq e_*J_1$. Let $t := \frac{7}{6J_1}$. Then

$$tM \leq \frac{7}{6}e_* \leq \min\{\delta_p, \kappa_p\}.$$

In particular, $\delta_p > 0$. By Proposition 7.11, $(E_p, \Delta_p + \frac{\kappa_p}{M} D_g)$ is lc, where $D_g \geq 0$ is the $\mathbb{Q}$-divisor on E_p of Construction 6.5(7). By Lemma 6.7(3), E_p is $\mathbb{Q}$-factorial, so $K_{E_p} + \Delta_p + tD_g$ is $\mathbb{Q}$-Cartier. Since $D_g \geq 0$ and $t \leq \kappa_p/M$, $(E_p, \Delta_p + tD_g)$ is lc. Since $tM \leq \delta_p$, Lemma 7.5 implies that $(V', t \operatorname{div}(\varphi))$ is lc near p. □

Proposition 7.13. *Notation and conditions as in Set-up 6.1. Let $c \in \mathbb{C}^4 \backslash \{0\}$, let p be a singular point of V_c at which two or three of the coordinates $u_1, \dots, u_{11}$ are non-zero, let $J > 0$ be an integer, and let f be a non-zero element of $(R_c)_J$. Then $(V_c, \frac{7}{6J} D_f|_{V_c})$ is lc near p.*

Proof. Take $N = 11$, $\mathbf{w}'_l = \mathbf{w}_l$ for $1 \leq l \leq 11$, and $c' = c$ in Set-up 6.4. Then $R' = R_c$ and $V' = V_c$. Since W is invertible on $V_c = \overline{V}_c \backslash V_\infty$, we have $D_f|_{V_c} = \operatorname{div}(f|_{V_c})$, where $f|_{V_c}$ is as in Set-up 6.4(4). Let I, k, a_l $(l \in I)$, A_p, Z_p, E_p, Δ_p, δ_p, κ_p, e_*, and $r_l = U_l - a_l W^{d_l}$ $(l \in I)$ be the data of Construction 6.5 for V_c at p. By assumption, $k \in \{2, 3\}$. Since $\sum_{l=1}^{11} d_l = 151$, Construction 6.5(5) reads

$$\delta_p = \sum_{l \notin I} d_l - 42(4-k) = -17 + \sum_{l \in I} (42 - d_l), \qquad \kappa_p = \min\left(\left\{ \frac{\delta_p}{7} \right\} \cup \{ d_l \mid l \notin I \} \right). \tag{7.20}$$

By Lemma 6.8(3), $\delta_p \geq 25$. Thus $6\delta_p \geq 150 > 49 \geq 7e_l$ for any $l \in I$. Recall that $d_l = 6$ if and only if $l \in \{9, 10, 11\}$, if and only if $e_l = 7$.

Case 1: $\#(I \cap \{9, 10, 11\}) \leq 1$. Then $e_l = 7$ for at most one $l \in I$ and $e_l \leq 3$ for any other $l \in I$. Since $k \geq 2$, $e_* \leq 3$. Since $d_l \geq 6$ for any l and $\delta_p \geq 25$,

$$\kappa_p \geq \min\left\{ \frac{25}{7}, 6 \right\} = \frac{25}{7} > \frac{7}{2} \geq \frac{7}{6} e_*, \qquad \delta_p \geq 25 > \frac{7}{6} e_*.$$

By Lemma 7.12, $(V_c, \frac{7}{6J} \operatorname{div}(f|_{V_c}))$ is lc near p.

Case 2: $I = \{9, 10, 11\}$. By (7.20), $\delta_p = -17 + 3 \cdot 36 = 91$. Since $\{d_l \mid l \notin I\} = \{21, 14\}$, $\kappa_p = \min\{13, 14\} = 13$. Moreover, $e_* = 7$, and

$$\kappa_p = 13 > \frac{49}{6} = \frac{7}{6} e_*, \qquad \delta_p = 91 > \frac{49}{6}.$$

By Lemma 7.12, $(V_c, \frac{7}{6J} \operatorname{div}(f|_{V_c}))$ is lc near p.

Case 3: $I \cap \{9, 10, 11\} = \{i, j\}$ for two indices $i \neq j$. Then the third index $l \in \{9, 10, 11\}$ satisfies $l \notin I$ and $d_l = 6$, and $\delta_p \geq -17 + 2 \cdot 36 = 55 > 42$ by (7.20), so $\kappa_p = 6$. Since $e_* = 7$, we have $\kappa_p = 6 < \frac{49}{6} = \frac{7}{6} e_*$, and Lemma 7.12 cannot be used on V_c. We restrict to the divisor $(u_i = a_i)$.

Let $N = 10$, and index the coordinates of $\mathbb{A}^{10}$ by $l \in \{1, \dots, 11\} \backslash \{i\}$. In Set-up 6.4, take $d'_l := d_l$, $e'_l := e_l$, $\mathbf{w}'_l := \mathbf{w}_l$ for $l \neq i$, and $c' := c - a_i^7 \mathbf{w}_i$. Then $(d'_l)_{l \neq i} = (d_l)_{l \in \{1, \dots, 11\} \backslash \{i\}}$, which omits only the entry $d_i = 6$, and any four of the vectors $\mathbf{w}_l$, $l \neq i$, are linearly independent. Since $p \in V_c$ and $u_l(p) = 0$ for $l \notin I$, we have $c = \sum_{l \in I} a_l^{e_l} \mathbf{w}_l$, so

$$c' = \sum_{l \in I \backslash \{i\}} a_l^{e_l} \mathbf{w}_l \neq 0,$$

because $a_j \neq 0$ and the at most three vectors $\mathbf{w}_l$, $l \in I$, are linearly independent. Thus the conditions of Set-up 6.4 hold. Modulo r_i we have $U_i = a_i W^6$, so the defining relation of R_c becomes $\sum_{l \neq i} U_l^{e_l} \mathbf{w}_l = (c - a_i^7 \mathbf{w}_i) W^{42}$. Hence

$$R_c / (r_i) \cong R'$$

as graded rings, with $U_l \to U_l$ for $l \neq i$ and $W \to W$. Let

$$S := (r_i|_{V_c} = 0) = (u_i = a_i) \subset V_c.$$

The projection $\mathbb{A}^{11} \to \mathbb{A}^{10}$ forgetting u_i maps S isomorphically onto V', and we denote the image of p in V' again by p. For any homogeneous element f' of R_c with image $\bar{f}'$ in R', $(f'|_{V_c})|_S$ corresponds to $\bar{f}'|_{V'}$ under $S \cong V'$.

We show that p is a singular point of V'. Let ζ be a primitive 7-th root of unity. Since $e_i = 7$, the point $p_\zeta \in \mathbb{A}^{11}$ with $u_i(p_\zeta) = \zeta a_i$ and $u_l(p_\zeta) = u_l(p)$ for $l \neq i$ belongs to V_c, so $u_i - a_i$ is not identically zero on V_c. Since V_c is a normal variety with $\dim V_c = 7$ (Proposition 6.2(1)), any irreducible component of S is of dimension 6. Thus $V' \subset \mathbb{A}^{10}$ is a complete intersection of the four equations $\sum_{l\neq i} u_l^{e_l}\mathbf{w}_l = c'$ with $\dim V' = 6$. The Jacobian matrix of the four equations at p has columns $e_l u_l(p)^{e_l-1}\mathbf{w}_l$, $l \neq i$. Since $e_l \geq 2$, only the columns with $l \in I\backslash\{i\}$ are non-zero, so the rank of the Jacobian matrix at p is at most $\#(I\backslash\{i\}) \leq 2 < 4$. Hence p is a singular point of V'.

We denote by I^S, k^S, A_p^S, Z_p^S, ϖ_p^S, E_p^S, Δ_p^S, δ_p^S, κ_p^S, and e_*^S the data of Construction 6.5 for V' at p. Then $I^S = I\backslash\{i\}$ and $k^S = k-1$, and for $l \in I^S$, we have $d_l' = d_l$, so by Construction 6.5(1)(8), $a_l = u_l(p)$ and $r_l = U_l - a_l W^{d_l}$ both for V_c and for V'. Since $\sum_{l\neq i} d_l = 145$ and $N - 4 = 6$, Construction 6.5(5) reads

$$\delta_p^S = \sum_{l\notin I} d_l - 42(4-k^S) = -23 + \sum_{l\in I^S}(42-d_l), \qquad \kappa_p^S = \min\left(\left\{\frac{\delta_p^S}{6}\right\} \cup \{d_l \mid l \notin I\}\right).$$

Since $I \cap \{9,10,11\} = \{i,j\}$, either $I^S = \{j\}$, or $I^S = \{j,l\}$ with $d_l \in \{21,14\}$. The data in the three cases are listed in Table 2.

	I^S	$(d_l)_{l\notin I}$	δ_p^S	κ_p^S	e_*^S	$\frac{7}{6}e_*^S$
(a)	$\{j\}$	$(21^3, 14^5, 6)$	13	$\frac{13}{6}$	1	$\frac{7}{6}$
(b)	$\{j,l\}$, $d_l = 21$	$(21^2, 14^5, 6)$	34	$\frac{17}{3}$	2	$\frac{7}{3}$
(c)	$\{j,l\}$, $d_l = 14$	$(21^3, 14^4, 6)$	41	6	3	$\frac{7}{2}$

TABLE 2. The data of Construction 6.5 for V' at p when $I \cap \{9,10,11\} = \{i,j\}$.

In each case of Table 2, $\kappa_p^S \geq \frac{7}{6}e_*^S$ and $\delta_p^S \geq 13 > \frac{7}{6}e_*^S$. Moreover, $6\delta_p^S \geq 78 > 49 \geq 7e_l$ for any $l \in I^S$. By Lemma 7.12 for V' at p,

$$\left(V', \frac{7}{6J'}\operatorname{div}(\bar{g}|_{V'})\right) \text{ is lc near } p \text{ for any integer } J' > 0 \text{ and any non-zero } \bar{g} \in R'_{J'}. \tag{7.21}$$

Next we show that the different of (V_c, S) on S is zero near p. By Lemma 6.8(2) for V_c, r_i is a prime element of R_c, so $S = \operatorname{div}(u_i - a_i)$ is a prime Cartier divisor on V_c. By Lemma 6.8(1) for V', S is normal. By Proposition 6.2(2), V_c has only finitely many singular points. By Lemma 6.8(1)(2) for V', ϖ_p^S is étale at p, $\varpi_p^S(p) = 0$, and Z_p^S is smooth outside the origin, so V' is smooth on a punctured neighborhood of p. Possibly replacing V_c with an open neighborhood of p, we may assume that $V_c\backslash\{p\}$ and $S\backslash\{p\}$ are smooth. Since $\dim S = 6$, both V_c and S are smooth at any codimension one point of S, and the adjunction formula for a smooth divisor on a smooth variety shows that the different of (V_c, S) on S is zero, that is,

$$K_S = (K_{V_c} + S)|_S.$$

Now we treat f. Since $6\delta_p > 7e_l$ for any $l \in I$, Lemma 7.7 for V_c at p (that is, with $N = 11$, $\mathbf{w}_l' = \mathbf{w}_l$, and $c' = c$) shows that it suffices to prove that (V_c, B) is lc near p for any integer $J_1 > 0$ and any non-zero $f_1 \in (R_c)_{J_1}$ which is not divisible by r_l for any $l \in I$, where

$$B := \frac{7}{6J_1}\operatorname{div}(f_1|_{V_c}).$$

Since f_1 is not divisible by r_i, the image $\bar{f}_1$ of f_1 in R'_{J_1} is non-zero. Since R' is a domain and $W \neq 0$ in R', $\bar{f}_1|_{V'}$ is a non-zero function, so $B \geq 0$ has no common components with S, and $B|_S = \frac{7}{6J_1}\operatorname{div}(\bar{f}_1|_{V'})$ under $S \cong V'$. By (7.21) with $J' = J_1$ and $\bar{g} = \bar{f}_1$, $(S, B|_S)$ is lc near p. Possibly replacing V_c with a smaller open neighborhood of p, we may assume that $(S, B|_S)$ is lc.

Since $V_c \subset \mathbb{A}^{11}$ is defined by four equations and $\dim V_c = 7$, K_{V_c} is Cartier. Since S is Cartier and B is $\mathbb{Q}$-Cartier, $K_{V_c} + S + B$ is $\mathbb{Q}$-Cartier. Since S is normal and the different of (V_c, S) on S is zero, by [FST11, 11.1(v)], the different of $(V_c, S+B)$ on S is $B|_S$. By Theorem 2.5, $(V_c, S+B)$ is lc near S, in particular near p. Lowering the coefficient of S to zero, (V_c, B) is lc near p. □

Proof of Theorem 7.1. Since $c \neq 0$ and $h(0) = 0$, the origin of $\mathbb{A}^{11}$ does not belong to V_c, so any point of V_c has at least one non-zero coordinate. By Proposition 6.2(2), any singular point of V_c has at most three non-zero coordinates. Thus any point $p \in V_c$ is a smooth point of V_c, a singular point of V_c with exactly one non-zero coordinate, or a singular point of V_c with two or three non-zero coordinates. By Propositions 7.3, 7.10, and 7.13 respectively, $(V_c, \frac{7}{6J} D_f|_{V_c})$ is lc near p. Hence $(V_c, \frac{7}{6J} D_f|_{V_c})$ is lc.

By Proposition 6.9, $\overline{V}_c$ is klt and D_f is $\mathbb{Q}$-Cartier, so $(\overline{V}_c, \frac{1}{J} D_f)$ is a pair. Since $\frac{1}{J} < \frac{7}{6J}$, lowering the coefficient shows that $(\overline{V}_c, \frac{1}{J} D_f)$ is lc on V_c. By Proposition 7.4(2), $(\overline{V}_c, \frac{1}{J} D_f)$ is lc near V_∞. Since $\overline{V}_c = V_c \cup V_\infty$, $(\overline{V}_c, \frac{1}{J} D_f)$ is lc. □

8. Families of fibers of h

In this section, we prove two estimates for families of fibers of h, which Section 9 uses on the charts $y_j \neq 0$ of P as outlined in part (c) of the sketch of the proof in Section 1: Lemma 8.1 bounds the multiplicity of a section along the fibers, and Proposition 8.6 controls the degeneration of the family over the base.

8.1. **A saturation estimate.**

In this subsection, we bound the multiplicity of a section of a family of fibers of h along a prime divisor over the total space. Lemma 8.1 first removes the minimal multiplicity of the coefficients along the image of the prime divisor on the base. The remaining section does not vanish on the reduced fiber over that image, and inversion of adjunction bounds its multiplicity through the threshold on the fibers.

Lemma 8.1. *Notation and conditions as in Set-up 6.1. Let Y be a smooth variety, and let $\pi : T \to Y$ be either (a) the projection to Y of the closed subscheme $T \subset \mathbb{P}(1, 21^3, 14^5, 6^3) \times Y$ defined by $\sum_{i=1}^{11} U_i^{e_i} \mathbf{w}_i = c(y) W^{42}$, where $c : Y \to \mathbb{A}^4 \backslash \{0\}$ is a morphism, or (b) the projection $T := V_\infty \times Y \to Y$. Then T is a normal variety, K_T is $\mathbb{Q}$-Cartier, and for any prime divisor E over T the following hold.*

1. *If E is vertical over Y, with image F over Y and ramification index r, then $a(E,T) \geq r a(F,Y)$.*
2. *Let $J > 0$ be an integer, $q > 0$ a rational number, $\{m_\alpha\}$ the set of monomials of degree J in $W, U_1, \dots, U_{11}$ in which the exponents of U_1, U_2, U_3 are at most one and the exponent of U_4 is at most two, and which do not involve W in case (b), and φ_α regular functions on Y, not all zero. Let $D_s \geq 0$ be the divisor of zeros on T of $s := \sum_\alpha \varphi_\alpha m_\alpha$. Assume that $(\overline{V}_{c(y)}, \frac{1}{q} D_f)$ is lc for any closed point $y \in Y$ and any non-zero element f of $(R_{c(y)})_J$ in case (a), and that $(V_\infty, \frac{1}{q} D_f)$ is lc for any non-zero element f of $(R_\infty)_J$ in case (b). If E is horizontal over Y, then $\operatorname{mult}_E D_s \leq q a(E,T)$. If E is vertical over Y, with image F over Y and ramification index r, then*

$$\operatorname{mult}_E D_s \leq q\big(a(E,T) - r a(F,Y)\big) + r \min\{\operatorname{mult}_F \varphi_\alpha \mid \varphi_\alpha \neq 0\}.$$

The bounds on the exponents of $U_1, \dots, U_4$ make $\{m_\alpha\}$ a basis, independent of c, of the degree-J part of the graded ring of the family, also after base change (**Step 2** of the proof), so the coefficients φ_α and the minimum in (2) are determined by s. Case (a) requires c to take values in $\mathbb{A}^4 \backslash \{0\}$, since the klt property of the fibers (Proposition 6.9) holds only for $c \neq 0$.

Proof of Lemma 8.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that T is a normal variety, K_T is $\mathbb{Q}$-Cartier, and the locus of points of T at which π is not smooth intersects any closed fiber $T_y := \pi^{-1}(y)$ in a closed subset

of codimension at least two in T_y. **Steps 1–3** only use that Y is a smooth variety and, in case (a), that c is a morphism to $\mathbb{A}^4\backslash\{0\}$. We apply them below with other smooth varieties in place of Y and, in case (a), other morphisms to $\mathbb{A}^4\backslash\{0\}$ in place of c.

For any $b \in \mathbb{C}^4$ and $1 \le i \le 11$, the morphism

$$\begin{aligned}\big(\operatorname{Spec} R_b \cap (U_i = 1)\big) \times \mathbb{G}_m &\to \operatorname{Spec} R_b \cap (U_i \neq 0),\\ \big((W, U_1, \dots, U_{11}), \lambda\big) &\to (\lambda W, \lambda^{d_1} U_1, \dots, \lambda^{d_{11}} U_{11}),\end{aligned}$$

is a μ_{d_i}-torsor, hence étale. With W omitted, the corresponding morphism for R_∞ is also a μ_{d_i}-torsor, hence étale. Since R_b is a normal domain of dimension 8 (Proposition 6.2(1)) and $\operatorname{Spec} R_\infty$ is smooth outside the origin (Proposition 6.2(3)), $\operatorname{Spec} R_b \cap (U_i = 1)$ is normal of dimension 7, and $\operatorname{Spec} R_\infty \cap (U_i = 1)$ is smooth: since normality and smoothness ascend along étale morphisms, $(\operatorname{Spec} R_b \cap (U_i = 1)) \times \mathbb{G}_m$ is normal and $(\operatorname{Spec} R_\infty \cap (U_i = 1)) \times \mathbb{G}_m$ is smooth, and normality and smoothness descend along the faithfully flat projections to $\operatorname{Spec} R_b \cap (U_i = 1)$ and $\operatorname{Spec} R_\infty \cap (U_i = 1)$.

Assume (a). The open subset $T \cap (W \neq 0)$ is the closed subscheme $\{(u, y) \mid h(u) = c(y)\}$ of $\mathbb{A}^{11} \times Y$, with closed fibers $V_{c(y)}$. For $1 \le i \le 11$, $T \cap (U_i \neq 0)$ is the quotient of the closed subscheme

$$T_i := \left\{(W, U_1, \dots, U_{11}, y) \;\middle|\; U_i = 1,\ \sum_{l=1}^{11} U_l^{e_l} \mathbf{w}_l = c(y) W^{42}\right\} \subset \mathbb{A}^{11} \times Y$$

by μ_{d_i} acting on W and U_l with weights 1 and d_l, where $\mathbb{A}^{11}$ has the coordinates W and U_l, $l \neq i$. Let $\beta_i : T_i \to T \cap (U_i \neq 0)$ be the quotient morphism. The closed fiber of T_i over y is $\operatorname{Spec} R_{c(y)} \cap (U_i = 1)$. Thus each of $T \cap (W \neq 0), T_1, \dots, T_{11}$ is cut out by four equations in $\mathbb{A}^{11} \times Y$, and any closed fiber of it over Y is normal of dimension 7 (Proposition 6.2(1)). Hence it is a complete intersection of dimension $\dim Y + 7$. At a point at which the closed fiber is smooth, the Jacobian matrix of the four equations with respect to the coordinates of $\mathbb{A}^{11}$ has rank four, so the complete intersection is smooth over Y at that point. Since the singular locus of any closed fiber has codimension at least two in that fiber, the singular locus Σ of the complete intersection satisfies $\dim \Sigma \le \dim Y + 5$. Thus $T \cap (W \neq 0)$ and T_i are regular in codimension one, hence normal by [Har77, Proposition II.8.23]. Since a quotient of a normal quasi-projective scheme by a finite group is normal, $T \cap (U_i \neq 0)$ is normal. Since π is proper, Y is connected, and any closed fiber $T_y = \overline{V}_{c(y)}$ is a variety (Proposition 6.2(1)), T is connected. Thus T is a normal variety.

A non-identity element of μ_{d_i} multiplies W by a non-trivial root of unity. Thus the union Φ_i of the fixed loci of the non-identity elements of μ_{d_i} on T_i is contained in $T_i \cap (W = 0) = (\operatorname{Spec} R_\infty \cap (U_i = 1)) \times Y$. By Proposition 6.2(3), the fixed locus of any non-identity element of μ_{d_i} on $\operatorname{Spec} R_\infty \cap (U_i = 1)$ is of dimension at most 3. Hence $\dim \Phi_i \le \dim Y + 3$, $\dim(\Phi_i \cap \beta_i^{-1}(T_y)) \le 3$, and β_i is étale on $T_i \backslash \Phi_i$. In particular, β_i is étale in codimension one, so $K_{T_i} = \beta_i^*(K_T|_{T \cap (U_i \neq 0)})$. Complete intersections in smooth varieties are Gorenstein, so $K_{T \cap (W \neq 0)}$ and K_{T_i} are Cartier. By [KM98, Proposition 5.20(1)], applied to the restriction of β_i to any connected component of T_i, K_T is $\mathbb{Q}$-Cartier on $T \cap (U_i \neq 0)$. Thus K_T is $\mathbb{Q}$-Cartier. Moreover, π is smooth at any point of $T \cap (W \neq 0)$ and of $\beta_i(T_i \backslash \Phi_i)$ at which the closed fiber of π is smooth. Since $\dim(\beta_i(\Phi_i) \cap T_y) \le 3$ and the singular locus of T_y has codimension at least two in T_y, the claim on the locus where π is not smooth follows.

Assume (b). By the second paragraph of this step and Proposition 6.2(3), V_∞ is a normal variety, and $V_\infty \cap (U_i \neq 0)$ is the quotient of the smooth scheme $\operatorname{Spec} R_\infty \cap (U_i = 1)$ by μ_{d_i}. Thus $T = V_\infty \times Y$ is a normal variety with only quotient singularities, so T is $\mathbb{Q}$-factorial by [KM98, Proposition 5.15]. In particular, K_T is $\mathbb{Q}$-Cartier. The morphism π is smooth at (p, y) whenever V_∞ is smooth at p, and the singular locus of V_∞ has codimension at least two in V_∞ since V_∞ is normal.

Step 2. In this step, we show that $\{m_\alpha\}$ is a basis of the degree-J part of the graded ring of T over $\mathcal{O}_Y$, and we define D_s.

Let

$$\mathcal{R} := \mathcal{O}_Y[W, U_1, \dots, U_{11}] \Big/ \Big(\sum_{i=1}^{11} U_i^{e_i} \mathbf{w}_i - cW^{42} \Big)$$

in case (a), and

$$\mathcal{R} := \mathcal{O}_Y[U_1, \dots, U_{11}] \Big/ \Big(\sum_{i=1}^{11} U_i^{e_i} \mathbf{w}_i \Big)$$

in case (b), graded by $\deg W = 1$ and $\deg U_i = d_i$, so that $T = \operatorname{Proj}_Y \mathcal{R}$. By Set-up 6.1(1), the matrix $(\mathbf{w}_1\ \mathbf{w}_2\ \mathbf{w}_3\ \mathbf{w}_4)$ is invertible. Multiplying the vector equation by the inverse matrix, we obtain

$$U_l^{e_l} = g_l, \quad 1 \le l \le 4,$$

where $e_1 = e_2 = e_3 = 2$, $e_4 = 3$, and g_l is a homogeneous polynomial of degree 42 in $W, U_5, \dots, U_{11}$ with coefficients in $\mathcal{O}_Y$, which does not involve W in case (b). Thus $\mathcal{R}$ is a free module over $\mathcal{O}_Y[W, U_5, \dots, U_{11}]$ (over $\mathcal{O}_Y[U_5, \dots, U_{11}]$ in case (b)) with basis

$$U_1^{a_1} U_2^{a_2} U_3^{a_3} U_4^{a_4}, \quad a_1, a_2, a_3 \in \{0, 1\},\ a_4 \in \{0, 1, 2\}.$$

Hence the degree-J part $\mathcal{R}_J$ is a free $\mathcal{O}_Y$-module with basis $\{m_\alpha\}$. The presentation commutes with any base change $Y_1 \to Y$. In particular, $\{m_\alpha\}$ remains a basis after base change, and $\{m_\alpha\}$ is a basis of $(R_{c(y)})_J$ (resp. $(R_\infty)_J$) for any closed point $y \in Y$.

Let $y \in Y$ be a closed point at which not all φ_α vanish. Then $s(y) := \sum_\alpha \varphi_\alpha(y) m_\alpha$ is a non-zero element of $(R_{c(y)})_J$ (resp. $(R_\infty)_J$). Thus s/W^J is a non-zero regular function on $T \cap (W \neq 0)$, and s^{d_i}/U_i^J is a non-zero regular function on $T \cap (U_i \neq 0)$. Let

$$D_s := \operatorname{div}(s/W^J) \text{ on } T \cap (W \neq 0), \qquad D_s := \frac{1}{d_i} \operatorname{div}(s^{d_i}/U_i^J) \text{ on } T \cap (U_i \neq 0),$$

where only the second kind of chart occurs in case (b). On the intersection of two charts, the definitions coincide, since W^{d_i}/U_i and $U_k^{d_i}/U_i^{d_k}$ are units there. Thus $D_s \ge 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T. Since the local equations of D_s restrict to those of $D_{s(y)}$, we have $T_y \not\subset \operatorname{Supp} D_s$ and $D_s|_{T_y} = D_{s(y)}$. We use the notation s, $s(y)$, and D_s also when Y and c are replaced as in **Step 1**.

Step 3. In this step, we show the following. Let Δ be either 0 or $\frac{1}{q} D_s$, and in the second case assume that $s(y) \neq 0$ for any closed point $y \in Y$. Assume that $(T_y, \Delta|_{T_y})$ is lc for any closed point $y \in Y$. Then there exists a dense open subset $Y^\circ \subset Y$ such that $(\pi^{-1}(Y^\circ), \Delta|_{\pi^{-1}(Y^\circ)})$ is lc.

By **Steps 1 and 2**, (T, Δ) is a pair. Let $\eta : T' \to T$ be a log resolution of (T, Δ), let $G := \operatorname{Exc}(\eta) \cup \operatorname{Supp} \eta_*^{-1} \Delta$, and write $K_{T'} + \Delta' := \eta^*(K_T + \Delta)$, so that $\operatorname{Supp} \Delta' \subset G$. By generic smoothness [Har77, Corollary III.10.7], applied to $\pi \circ \eta$ and to the restriction of $\pi \circ \eta$ to any component of an intersection of components of G which dominates Y, there exists a dense open subset $Y^\circ \subset Y$ such that T' and any such component are smooth over Y°, and no component of an intersection of components of G which does not dominate Y intersects $(\pi \circ \eta)^{-1}(Y^\circ)$. Since $\dim \operatorname{Exc}(\eta) < \dim T'$, possibly shrinking Y°, we may assume that $\operatorname{Exc}(\eta)$ contains no component of $T'_y := (\pi \circ \eta)^{-1}(y)$ for any closed point $y \in Y^\circ$. Let $y \in Y^\circ$ be a closed point. Since η restricts to an isomorphism

$$T' \setminus \operatorname{Exc}(\eta) \to T \setminus \eta(\operatorname{Exc}(\eta)),$$

it maps $T'_y \setminus \operatorname{Exc}(\eta)$ isomorphically onto a non-empty open subset of T_y, which is irreducible since T_y is a variety (**Step 1**). Since T'_y is smooth, the components of T'_y are disjoint, and each of them intersects $T'_y \setminus \operatorname{Exc}(\eta)$. Hence T'_y is irreducible and the restriction $\eta_y : T'_y \to T_y$ of η is a proper birational morphism. Thus η_y is a log resolution of $(T_y, \Delta|_{T_y})$ (below we only use that η_y is a proper birational morphism from a smooth variety satisfying the crepant formula, and

$\mathrm{Exc}(\eta_y) = \mathrm{Exc}(\eta) \cap T'_y$ since T' is smooth over Y°, T is flat over Y as $\mathcal{R}$ is a free $\mathcal{O}_Y$-module (**Step 2**), and étaleness is checked fiberwise), and $G|_{T'_y}$ is a reduced snc divisor whose components are the components of $G_k \cap T'_y$ for the components G_k of G.

Moreover, $K_{T'}|_{T'_y} = K_{T'_y}$ since $\pi \circ \eta$ is smooth over Y°, and $K_T|_{T_y} = K_{T_y}$ since both sides are $\mathbb{Q}$-Cartier and they coincide outside a closed subset of codimension at least two in T_y by **Step 1**. By **Step 2**, $\Delta|_{T_y}$ is 0 or $\frac{1}{q}D_{s(y)}$. Hence

$$K_{T'_y} + \Delta'|_{T'_y} = \eta_y^*(K_{T_y} + \Delta|_{T_y}).$$

Let G_k be a component of G which intersects $(\pi \circ \eta)^{-1}(Y^\circ)$. Then G_k dominates Y. Let $y \in Y^\circ$ be a closed point and Γ a component of $G_k \cap T'_y$. Then Γ is a prime divisor on T'_y and $\mathrm{mult}_\Gamma(\Delta'|_{T'_y}) = \mathrm{mult}_{G_k} \Delta'$. Since $(T_y, \Delta|_{T_y})$ is lc,

$$\mathrm{mult}_{G_k} \Delta' = 1 - a(\Gamma, T_y, \Delta|_{T_y}) \leq 1.$$

Thus, over Y°, T' is smooth, G is snc, and all coefficients of Δ' are at most one. By [KM98, Corollary 2.31(3)], $a(E', T', \Delta') \geq 0$ for any prime divisor E' over $(\pi \circ \eta)^{-1}(Y^\circ)$. Therefore $(\pi^{-1}(Y^\circ), \Delta|_{\pi^{-1}(Y^\circ)})$ is lc by [KM98, Lemma 2.30].

Step 4. In this step, we assume that E is vertical over Y, with image F over Y and ramification index r, and we compute $a(E, T)$ on a base change of T to a smooth variety on which F is a divisor.

By Definition-Lemma 2.6, there exist a birational morphism $\psi : T'' \to T$ from a normal variety T'' on which E is a divisor, a birational morphism $\rho : \widetilde{Y} \to Y$ from a smooth variety $\widetilde{Y}$, and a morphism $\pi'' : T'' \to \widetilde{Y}$ with $\rho \circ \pi'' = \pi \circ \psi$, such that F is a prime divisor on $\widetilde{Y}$, $\pi''(E)$ is dense in F, and $r = \mathrm{mult}_E \pi''^* F$. Let $\widetilde{T} := T \times_Y \widetilde{Y}$, with projections $\widetilde{\rho} : \widetilde{T} \to T$ and $\widetilde{\pi} : \widetilde{T} \to \widetilde{Y}$. The morphism $\widetilde{\pi}$ is defined as in (a) (resp. (b)) with $\widetilde{Y}$ in place of Y and, in case (a), $c \circ \rho$ in place of c. By **Step 1**, $\widetilde{T}$ is a normal variety and $K_{\widetilde{T}}$ is $\mathbb{Q}$-Cartier. Since $\widetilde{\rho}$ is birational, so is the induced morphism $T'' \to \widetilde{T}$. Thus E is a prime divisor over $\widetilde{T}$, $\mathrm{center}_{\widetilde{T}} E$ is contained in $S := \widetilde{\pi}^* F$, and $\widetilde{\pi}$ maps the generic point of $\mathrm{center}_{\widetilde{T}} E$ to the generic point of F. Moreover, $\mathrm{mult}_E S = \mathrm{mult}_E \pi''^* F = r$.

We show that

$$K_{\widetilde{T}} = \widetilde{\rho}^* K_T + \widetilde{\pi}^*(K_{\widetilde{Y}} - \rho^* K_Y). \tag{8.1}$$

Let $T^{\mathrm{sm}} \subset T$ be the open subset where π is smooth. Let $\widetilde{T}^{\mathrm{sm}} \subset \widetilde{T}$ be the preimage of T^{sm} under $\widetilde{\rho}$. Then T^{sm} and $\widetilde{T}^{\mathrm{sm}}$ are smooth, $\widetilde{\pi}$ is smooth on $\widetilde{T}^{\mathrm{sm}}$, and the relative canonical divisor of a smooth morphism commutes with base change. Thus both sides of (8.1) coincide on $\widetilde{T}^{\mathrm{sm}}$. Since $\widetilde{\rho}$ maps any closed fiber of $\widetilde{\pi}$ isomorphically onto a closed fiber of π, **Step 1** implies that $\widetilde{T} \backslash \widetilde{T}^{\mathrm{sm}}$ intersects any closed fiber of $\widetilde{\pi}$ in a closed subset of codimension at least two in that fiber. Hence $\widetilde{T} \backslash \widetilde{T}^{\mathrm{sm}}$ has codimension at least two in $\widetilde{T}$. Since both sides of (8.1) are $\mathbb{Q}$-Cartier and $\widetilde{T}$ is normal, (8.1) holds.

Since $\widetilde{\pi}$ maps the generic point of $\mathrm{center}_{\widetilde{T}} E$ to the generic point of F, possibly shrinking $\widetilde{Y}$ to a neighborhood of the generic point of F, we may assume that $\widetilde{Y}$ is affine, F is smooth, $F = \mathrm{div}(\xi)$ for some $\xi \in \mathcal{O}(\widetilde{Y})$, and $K_{\widetilde{Y}} - \rho^* K_Y = (a(F, Y) - 1)F$. Then (8.1) becomes $\widetilde{\rho}^* K_T = K_{\widetilde{T}} + (1 - a(F, Y))S$. Since $\mathrm{mult}_E S = r$,

$$a(E, T) = a\big(E, \widetilde{T}, (1 - a(F, Y))S\big) = a(E, \widetilde{T}, S) + r a(F, Y). \tag{8.2}$$

Since $S = T \times_Y F$, the morphism $S \to F$ is defined as in (a) (resp. (b)) with the smooth variety F in place of Y and, in case (a), $c \circ \rho|_F$ in place of c. By **Step 1**, S is a normal variety. In particular, S is a normal prime divisor on $\widetilde{T}$. Since S is moreover Cartier, $\widetilde{T}$ is regular at the generic point of any prime divisor on S, so the different of $(\widetilde{T}, S)$ on S is zero [FST11, 11.1]. Also, $K_{\widetilde{T}} + S$ is $\mathbb{Q}$-Cartier.

Step 5. In this step, we prove (1). Let ρ, $\widetilde{T}$, $\widetilde{\pi}$, and S be as in **Step 4**.

For any closed point $z \in F$, the fiber of S over z is $T_{\rho(z)}$, which is klt: in case (a) by Proposition 6.9 since $c(\rho(z)) \neq 0$, and in case (b) since, by Proposition 6.2(3), $V_\infty \cap (U_i \neq 0)$ is the quotient of the smooth scheme $\operatorname{Spec} R_\infty \cap (U_i = 1)$ by μ_{d_i} and the quotient morphism is étale in codimension one, so [KM98, Proposition 5.20(4)], applied with $\Delta = 0$ to the restriction of the quotient morphism to any connected component, shows that V_∞ is klt. By **Step 3** applied to $S \to F$ with $\Delta = 0$, there exists a dense open subset $F^\circ \subset F$ such that S is lc over F°. Possibly replacing $\widetilde{Y}$ with $\widetilde{Y}\backslash(F\backslash F^\circ)$, we may assume that $F^\circ = F$. By **Step 4**, $\widetilde{T}$ is normal, S is a normal prime divisor, $K_{\widetilde{T}} + S$ is $\mathbb{Q}$-Cartier, and the different of $(\widetilde{T}, S)$ on S is zero. Since S is lc, Theorem 2.5 with $B = 0$ implies that $(\widetilde{T}, S)$ is lc near S. Since $\operatorname{center}_{\widetilde{T}} E \subset S$, we have $a(E, \widetilde{T}, S) \geq 0$, and (1) follows from (8.2).

Step 6. In this step, we prove (2).

Assume that E is horizontal over Y. Let $Y^1 := Y\backslash\bigcap_\alpha(\varphi_\alpha = 0)$, a dense open subset of Y. For any closed point $y \in Y^1$, $s(y)$ is a non-zero element of $(R_{c(y)})_J$ (resp. $(R_\infty)_J$) by **Step 2**, so $(T_y, \frac{1}{q}D_s|_{T_y}) = (T_y, \frac{1}{q}D_{s(y)})$ is lc by the assumption. By **Step 3** applied to $\pi^{-1}(Y^1) \to Y^1$ with $\Delta = \frac{1}{q}D_s$, there exists a dense open subset $Y^\circ \subset Y^1$ such that $(\pi^{-1}(Y^\circ), \frac{1}{q}D_s|_{\pi^{-1}(Y^\circ)})$ is lc. Since E is horizontal over Y, $\operatorname{center}_T E$ intersects $\pi^{-1}(Y^\circ)$. Thus

$$0 \leq a\big(E, T, \tfrac{1}{q}D_s\big) = a(E, T) - \tfrac{1}{q}\operatorname{mult}_E D_s,$$

that is, $\operatorname{mult}_E D_s \leq qa(E, T)$.

Assume that E is vertical over Y, with image F over Y and ramification index r, and let ρ, $\widetilde{T}$, $\widetilde{\rho}$, $\widetilde{\pi}$, S, and ξ be as in **Step 4**. Let

$$n := \min\{\operatorname{mult}_F \varphi_\alpha \mid \varphi_\alpha \neq 0\},$$

and fix α_0 with $\varphi_{\alpha_0} \neq 0$ and $\operatorname{mult}_F \varphi_{\alpha_0} = n$. Since $F = \operatorname{div}(\xi)$, $\varphi'_\alpha := (\varphi_\alpha \circ \rho)/\xi^n$ is a regular function on $\widetilde{Y}$ for any α, and $\varphi'_{\alpha_0}|_F \neq 0$. Possibly shrinking $\widetilde{Y}$ to a neighborhood of the generic point of F, we may assume that $\varphi'_{\alpha_0}|_F$ is nowhere zero on F. Let $s_0 := \sum_\alpha \varphi'_\alpha m_\alpha$ on $\widetilde{T}$. Since the pullback of s to $\widetilde{T}$ is $\xi^n s_0$, the local equations in **Step 2** imply $\widetilde{\rho}^* D_s = D_{s_0} + nS$. Since $\operatorname{mult}_E S = r$ (**Step 4**),

$$\operatorname{mult}_E D_s = \operatorname{mult}_E D_{s_0} + nr. \tag{8.3}$$

Since $\{m_\alpha\}$ remains a basis after base change to F (**Step 2**) and $\varphi'_{\alpha_0}|_F$ is nowhere zero, the section $s_0|_S := \sum_\alpha(\varphi'_\alpha|_F)m_\alpha$ of the family $S \to F$ satisfies $(s_0|_S)(z) \neq 0$ for any closed point $z \in F$. In particular, $S \not\subset \operatorname{Supp} D_{s_0}$ and $D_{s_0}|_S = D_{s_0|_S}$. For any closed point $z \in F$, the fiber of S over z is $T_{\rho(z)}$, and by **Step 2** the restriction of $D_{s_0|_S}$ to $T_{\rho(z)}$ is D_f for the non-zero element $f := \sum_\alpha \varphi'_\alpha(z)m_\alpha$ of $(R_{c(\rho(z))})_J$ (resp. $(R_\infty)_J$). By the assumption, $(T_{\rho(z)}, \frac{1}{q}D_f)$ is lc. By **Step 3** applied to $S \to F$ with $\Delta = \frac{1}{q}D_{s_0|_S}$, there exists a dense open subset $F^\circ \subset F$ such that $(S, \frac{1}{q}D_{s_0|_S})$ is lc over F°. Possibly replacing $\widetilde{Y}$ with $\widetilde{Y}\backslash(F\backslash F^\circ)$, we may assume that $F^\circ = F$.

Let $B := \frac{1}{q}D_{s_0}$. We check the hypotheses of Theorem 2.5 for $\widetilde{T}$, S, and B. By **Step 4**, $\widetilde{T}$ is normal and S is a normal prime divisor, in particular a reduced divisor. Since $S \not\subset \operatorname{Supp} D_{s_0}$, $B \geq 0$ has no common components with S. Since $K_{\widetilde{T}}$ is $\mathbb{Q}$-Cartier, S is Cartier, and D_{s_0} is $\mathbb{Q}$-Cartier (**Steps 1 and 2**), $K_{\widetilde{T}} + S + B$ is $\mathbb{Q}$-Cartier. Since the different of $(\widetilde{T}, S)$ on S is zero (**Step 4**), the different of $(\widetilde{T}, S + B)$ on S is $B|_S = \frac{1}{q}D_{s_0|_S}$ [FST11, 11.1(v)]. Since $(S, \frac{1}{q}D_{s_0|_S})$ is lc, Theorem 2.5 implies that $(\widetilde{T}, S + \frac{1}{q}D_{s_0})$ is lc near S. Since $\operatorname{center}_{\widetilde{T}} E \subset S$,

$$0 \leq a\big(E, \widetilde{T}, S + \tfrac{1}{q}D_{s_0}\big) = a(E, \widetilde{T}, S) - \tfrac{1}{q}\operatorname{mult}_E D_{s_0}.$$

By (8.2) and (8.3),

$$\operatorname{mult}_E D_s \leq qa(E, \widetilde{T}, S) + nr = q\big(a(E, T) - ra(F, Y)\big) + r\min\{\operatorname{mult}_F \varphi_\alpha \mid \varphi_\alpha \neq 0\}.$$

□

8.2. **A relative discrepancy estimate.** In this subsection, we bound the log discrepancies of divisors over a family of fibers of h whose base passes transversally through the zero locus of c.

Set-up 8.2. Notation and conditions as in Set-up 6.1. Let Y, c, Y_0, V, and π be as follows.

(1) Y is a smooth variety and $c = (c_1, c_2, c_3, c_4) : Y \to \mathbb{A}^4$ is a morphism.
(2) $Y_0 := c^{-1}(0)$ is either empty or a smooth closed subvariety of pure codimension four whose ideal sheaf $\mathcal{I}_{Y_0}$ is generated by c_1, c_2, c_3, c_4.
(3) $V := \{(u, y) \in \mathbb{A}^{11} \times Y \mid h(u) = c(y)\}$, and $\pi : V \to Y$ is the projection.

Condition (2) means that at any point of Y_0, the functions $c_1, \dots, c_4$ are part of a regular system of parameters of Y. Construction 8.4 uses condition (2) for the smoothness of Y_t (Lemma 8.5(1)) and, on the charts of the blow-up of Y_0, for the equality $\operatorname{mult}_F c_k = \operatorname{mult}_F \mathcal{I}_{Y_0}$ (Lemma 8.5(2)), and Section 9 uses condition (2) in Lemma 9.2 for $C = Y_0$ of codimension four. On the chart $y_j \neq 0$ of P, condition (2) follows from Lemma 3.2(2), since at any point of Y_0, at least four indices $l \neq j$ satisfy $y_l \neq 0$.

Lemma 8.3. *Notation and conditions as in Set-up 8.2. Then π is flat, V is a normal variety, and K_V is Cartier.*

Proof. The scheme V is the fiber product of $h : \mathbb{A}^{11} \to \mathbb{A}^4$ and $c : Y \to \mathbb{A}^4$. By Proposition 6.2(1), π is flat and the fiber of π over a closed point y is the normal variety $V_{c(y)}$ with $\dim V_{c(y)} = 7$. By Set-up 8.2(3), V is the closed subscheme of the smooth variety $\mathbb{A}^{11} \times Y$ defined by the four equations $h(u) = c(y)$. Since π is flat and its fibers over closed points are of dimension 7, any irreducible component of V is of dimension $\dim Y + 7$, so V is a local complete intersection of codimension four. In particular, V is Gorenstein [Mat89, Theorem 21.3], so K_V is Cartier. Since π is flat and Y is smooth, V is smooth at any smooth point of a fiber of π [Mat89, Theorem 23.7(ii)]. By Proposition 6.2(2), any fiber of π over a closed point has only finitely many singular points, so the singular locus of V is of dimension at most $\dim Y = \dim V - 7$. Thus V is regular in codimension one, and V is normal by [Har77, Proposition II.8.23(b)]. Since V is normal, the irreducible components of V are open and closed in V. Since π is flat, π is open, so any irreducible component of V dominates Y. If V had two irreducible components, then for a general closed point $y \in Y$, the fiber $V_{c(y)}$ would be the disjoint union of two non-empty open and closed subsets, which is not possible since $V_{c(y)}$ is irreducible (Proposition 6.2(1)). Thus V is a normal variety. □

Since c vanishes on Y_0 and case (a) of Lemma 8.1 requires c to take values in $\mathbb{A}^4 \setminus \{0\}$, Lemma 8.1 cannot be used near $\pi^{-1}(Y_0)$. The following construction blows up Y_0, takes a 42-nd root t of c_1, and substitutes $u_i = t^{d_i} U_i / W^{d_i}$, so that $h(u) - c(y)$ becomes $t^{42} W^{-42}$ times the equations of $\overline{\mathcal{V}}$, see (8.4). We use a projective family over the smooth base Y_t since Lemma 8.1 treats projective families with closed fibers $\overline{V}_{c'}$, $c' \neq 0$, and since, after permuting $c_1, \dots, c_4$, any prime divisor over V vertical over Y with center in Y_0 lifts to $\overline{\mathcal{V}}$ (Lemma 8.5(2)).

Construction 8.4. Notation and conditions as in Set-up 8.2, and assume that $Y_0 \neq \emptyset$.

(1) $\sigma : Y' \to Y$ is the blow-up of Y_0, $Y_1' \subset Y'$ is the open subset on which $\sigma^* c_1$ generates $\mathcal{I}_{Y_0} \cdot \mathcal{O}_{Y'}$, and $b_j := \sigma^* c_j / \sigma^* c_1$ for $j = 2, 3, 4$, which are regular functions on Y_1'.
(2) Y_t is the closed subscheme

$$Y_t := \{(y', t) \in Y_1' \times \mathbb{A}^1 \mid t^{42} = \sigma^* c_1(y')\},$$

and $\tau : Y_t \to Y$ is the natural morphism.
(3) $\overline{\mathcal{V}}$ is the closed subscheme of $\mathbb{P}(1, 21^3, 14^5, 6^3) \times Y_t$ defined by

$$\sum_{i=1}^{11} U_i^{e_i} \mathbf{w}_i = (1, b_2, b_3, b_4)^\top W^{42},$$

and $\mathcal{V}_\infty := (W = 0) \subset \overline{\mathcal{V}}$.

(4) $\phi : \overline{\mathcal{V}} \dashrightarrow V$ is the rational map given by $u_i = t^{d_i} U_i / W^{d_i}$ for $1 \le i \le 11$ and by τ on Y_t.

Lemma 8.5. *Notation and conditions as in Construction 8.4. Then the following hold.*

(1) Y_t is a smooth variety, $\overline{\mathcal{V}}$ is a normal variety, $K_{\overline{\mathcal{V}}}$ is $\mathbb{Q}$-Cartier, the projection $\overline{\mathcal{V}} \to Y_t$ is projective, any fiber of $\overline{\mathcal{V}} \to Y_t$ over a closed point is isomorphic to $\overline{V}_{c'}$ for some $c' \in \mathbb{C}^4 \backslash \{0\}$, and ϕ is dominant and generically finite.

(2) Let E be a prime divisor over V which is vertical over Y, with image F over Y and ramification index r. Assume that $\operatorname{center}_Y F \subset Y_0$ and $\operatorname{mult}_F c_1 = \operatorname{mult}_F \mathcal{I}_{Y_0}$. Then there exists a prime divisor E' over $\overline{\mathcal{V}}$ which ϕ maps onto E, that is, for some birational morphisms $\mathcal{V}' \to \overline{\mathcal{V}}$ and $V' \to V$ from normal varieties on which E' and E are divisors, respectively, ϕ induces a morphism $\phi' : \mathcal{V}' \to V'$ with $\phi'(E') = E$.

(3) Let E, F, and r be as in (2), let E' be a prime divisor over $\overline{\mathcal{V}}$ which ϕ maps onto E, and let r' be the multiplicity along E' of the pullback of E. Then E' is vertical over Y_t. Moreover, if F' is the image of E' over Y_t and r'' is the ramification index of E' over Y_t, then

$$r' a(E, V) = a(E', \overline{\mathcal{V}}) + 150 \operatorname{mult}_{E'} t + 16 \operatorname{mult}_{E'} \mathcal{V}_\infty,$$
$$r' r a(F, Y) = r'' a(F', Y_t) + 167 \operatorname{mult}_{E'} t, \quad \text{and} \quad 42 \operatorname{mult}_{E'} t = r' r \operatorname{mult}_F \mathcal{I}_{Y_0}.$$

Proof. The proof is long, so we divide it into several steps. By Lemma 8.3, V is a normal variety and K_V is Cartier. Let $\pi_t : \overline{\mathcal{V}} \to Y_t$ be the projection, $\tau_1 : Y_t \to Y_1'$ the projection, so that $\tau = \sigma \circ \tau_1$, and $\overline{\mathcal{V}}^\circ$ the open subset $\overline{\mathcal{V}} \backslash \mathcal{V}_\infty$. We also write $(t = 0)$ for the divisor of the function t on Y_t and for the divisor of $\pi_t^* t$ on $\overline{\mathcal{V}}$.

Step 1. In this step, we prove (1).

Since Y_0 is smooth of codimension four and $\mathcal{I}_{Y_0}$ is generated by $c_1, \dots, c_4$, the differentials $dc_1, \dots, dc_4$ are linearly independent at any point of Y_0. Thus, near any point of $\operatorname{Exc}(\sigma) \cap Y_1'$, the functions $\sigma^* c_1, b_2, b_3, b_4$ are part of a system of local coordinates of Y', and $\operatorname{Exc}(\sigma) \cap Y_1' = (\sigma^* c_1 = 0)$ as divisors on Y_1'. At a point of Y_t with $t \neq 0$, we have $\partial(t^{42} - \sigma^* c_1)/\partial t = 42 t^{41} \neq 0$, and at a point of Y_t with $t = 0$, we have $d(\sigma^* c_1) \neq 0$. Thus at any closed point y_t of Y_t, the image of $t^{42} - \sigma^* c_1$ in $\mathfrak{m}_{y_t} / \mathfrak{m}_{y_t}^2$, where $\mathfrak{m}_{y_t}$ is the maximal ideal of $Y_1' \times \mathbb{A}^1$ at y_t, is non-zero, so $\mathcal{O}_{Y_t, y_t}$ is a regular local ring [Mat89, Theorem 14.2], and Y_t is smooth. If moreover $t(y_t) = 0$, then $t, \sigma^* c_1, b_2, b_3, b_4$ are part of a regular system of parameters at y_t and $t^{42} - \sigma^* c_1 \equiv -\sigma^* c_1$ modulo $\mathfrak{m}_{y_t}^2$, so the images of t, b_2, b_3, b_4 in the cotangent space of Y_t at y_t are linearly independent, and t, b_2, b_3, b_4 are part of a system of local coordinates of Y_t near y_t [Mat89, Theorem 14.2]. Moreover, τ_1 is finite and flat of degree 42, hence open and closed. Since Y_t is smooth, the irreducible components of Y_t are open and closed in Y_t, so any irreducible component of Y_t maps onto Y_1'. As $\tau_1^{-1}(y')$ is one point for any $y' \in \operatorname{Exc}(\sigma) \cap Y_1' \neq \emptyset$, any irreducible component of Y_t contains $\tau_1^{-1}(\operatorname{Exc}(\sigma))$. Thus Y_t is irreducible.

The scheme $\overline{\mathcal{V}}$ is the scheme T in case (a) of Lemma 8.1 with $Y := Y_t$ and $c := (1, b_2, b_3, b_4)$, which maps Y_t to $\mathbb{A}^4 \backslash \{0\}$. By Lemma 8.1, $\overline{\mathcal{V}}$ is a normal variety and $K_{\overline{\mathcal{V}}}$ is $\mathbb{Q}$-Cartier. The projection π_t is projective, and the fiber of π_t over a closed point y_t is $\overline{V}_{c'}$ with $c' = (1, b_2(y_t), b_3(y_t), b_4(y_t)) \neq 0$.

On $\overline{\mathcal{V}}^\circ$, we use the coordinates $z_i := U_i / W^{d_i}$, so that

$$\overline{\mathcal{V}}^\circ = \{(z, y_t) \in \mathbb{A}^{11} \times Y_t \mid h(z) = (1, b_2(y_t), b_3(y_t), b_4(y_t))^\top\}.$$

Finally, consider the morphism

$$\Psi : \mathbb{A}^{11} \times Y_t \to \mathbb{A}^{11} \times Y, \quad (z, y_t) \to \big((t^{d_1} z_1, \dots, t^{d_{11}} z_{11}), \tau(y_t)\big).$$

Since $u_i^{e_i}$ is of weighted degree $d_i e_i = 42$, we have $h(t^{d_1} z_1, \dots, t^{d_{11}} z_{11}) = t^{42} h(z)$. Since $\tau^* c_1 = t^{42}$ and $\tau^* c_j = t^{42} b_j$ for $j = 2, 3, 4$, we have

$$\big(h(u) - c(y)\big) \circ \Psi = t^{42} \big(h(z) - (1, b_2, b_3, b_4)^\top\big). \tag{8.4}$$

Thus $\Psi(\overline{\mathcal{V}}^\circ) \subset V$, and $\phi = \Psi|_{\overline{\mathcal{V}}^\circ}$ is a morphism on $\overline{\mathcal{V}}^\circ$. Since V is a variety, π is surjective, as its fibers over closed points are the varieties $V_{c(y)}$, and c_1 is not identically zero on Y, as Y_0 is of codimension four, the open subset $V \cap \pi^{-1}(c_1 \neq 0)$ is dense in V. Let $(u, y) \in V$ be a closed point with $c_1(y) \neq 0$. Since σ is an isomorphism over $Y \backslash Y_0$, $y' = \sigma^{-1}(y) \in Y_1'$, and the points of $\overline{\mathcal{V}}^\circ$ mapping to (u, y) are the 42 points $(t^{-d_1}u_1, \dots, t^{-d_{11}}u_{11}, (y', t))$ with $t^{42} = c_1(y)$. Hence ϕ is dominant and generically finite. This proves (1).

Step 2. In this step, we compute K_{Y_t}.

By [Har77, Exercise II.8.5(b)], $K_{Y'} = \sigma^* K_Y + 3\operatorname{Exc}(\sigma)$. The morphism τ_1 is étale over $Y_1' \backslash \operatorname{Exc}(\sigma)$, and

$$\tau_1^*(\operatorname{Exc}(\sigma) \cap Y_1') = (\tau_1^* \sigma^* c_1 = 0) = (t^{42} = 0) = 42(t = 0).$$

By the Hurwitz formula, $K_{Y_t} = \tau_1^* K_{Y_1'} + 41(t = 0)$. Thus

$$K_{Y_t} = \tau^* K_Y + 167(t = 0), \qquad 167 = 3 \cdot 42 + 41. \tag{8.5}$$

Step 3. In this step, we compute a canonical divisor of $\overline{\mathcal{V}}$ on $\overline{\mathcal{V}}^\circ$.

Let X_0 be a smooth variety, $T_0 \subset X_0$ a normal closed subvariety of codimension four defined by regular functions $f = (f_1, \dots, f_4)$, and Φ a generator of ω_{X_0}. Let $\mathcal{I}$ be the ideal sheaf of T_0. Near a smooth point of T_0, the sequence

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{X_0} \otimes \mathcal{O}_{T_0} \to \Omega_{T_0} \to 0$$

is exact [Har77, Theorem II.8.17] and consists of locally free sheaves, and $\mathcal{I}/\mathcal{I}^2$ is generated by the images of $f_1, \dots, f_4$. Hence $df_1, \dots, df_4$ are linearly independent at the smooth points of T_0. We denote by $\operatorname{Res}_{T_0}(\Phi/f)$ the section of ω_{T_0} over the smooth locus of T_0 such that

$$\operatorname{Res}_{T_0}(\Phi/f) \wedge df_1 \wedge \cdots \wedge df_4 = \Phi|_{T_0}.$$

By the adjunction formula [Har77, Proposition II.8.20], $\operatorname{Res}_{T_0}(\Phi/f)$ generates ω_{T_0} at the smooth points of T_0. Since T_0 is normal, T_0 is smooth at the generic point of any prime divisor on T_0.

Near a closed point of Y, let η be a generator of ω_Y. Let $f := h(u) - c(y)$ on $\mathbb{A}^{11} \times Y$, $du := du_1 \wedge \cdots \wedge du_{11}$ for the coordinates u_i of $\mathbb{A}^{11}$, and $\eta_V := \operatorname{Res}_V(du \wedge \eta/f)$. Then η_V generates ω_V at the smooth points of V. Since V is a normal variety (Lemma 8.3), $K_V = \pi^* K_Y$. Let $s = (s_1, \dots, s_4) := h(z) - (1, b_2, b_3, b_4)^\top$ on $\mathbb{A}^{11} \times Y_t$ and $dz := dz_1 \wedge \cdots \wedge dz_{11}$. Since $du_i = t^{d_i} dz_i + d_i z_i t^{d_i - 1} dt$ on $\mathbb{A}^{11} \times Y_t$ and $dt \wedge \tau^* \eta = 0$, we have

$$\Psi^*(du \wedge \eta) = t^{151} dz \wedge \tau^* \eta, \qquad 151 = d_1 + \cdots + d_{11}.$$

By (8.4), $d(f_l \circ \Psi) = t^{42} ds_l$ along $\overline{\mathcal{V}}^\circ$ for $1 \leq l \leq 4$. Pulling back $\eta_V \wedge df_1 \wedge \cdots \wedge df_4 = du \wedge \eta|_V$ by Ψ, we obtain

$$\phi^* \eta_V \wedge t^{168} ds_1 \wedge \cdots \wedge ds_4 = t^{151} dz \wedge \tau^* \eta|_{\overline{\mathcal{V}}^\circ}.$$

The morphism τ_1 is étale over $Y_1' \backslash \operatorname{Exc}(\sigma)$, and σ is an isomorphism over $Y \backslash Y_0$, so $\tau^* \eta$ generates ω_{Y_t} on $(t \neq 0)$. Thus, on $\overline{\mathcal{V}}^\circ \cap (t \neq 0)$,

$$\phi^* \eta_V = t^{-17} \operatorname{Res}_{\overline{\mathcal{V}}^\circ}(dz \wedge \tau^* \eta / s), \qquad 17 = 4 \cdot 42 - 151. \tag{8.6}$$

By (8.5), $\tau^* \eta$ vanishes to order 167 along $(t = 0)$ as a section of ω_{Y_t}, that is, near any point of $(t = 0)$, we have $\tau^* \eta = t^{167} \eta_t$ for a generator η_t of ω_{Y_t}. Thus, near any point of $\overline{\mathcal{V}}^\circ \cap (t = 0)$,

$$\phi^* \eta_V = t^{150} \operatorname{Res}_{\overline{\mathcal{V}}^\circ}(dz \wedge \eta_t / s), \qquad 150 = 167 - 17,$$

and the residue on the right-hand side generates $\omega_{\overline{\mathcal{V}}^\circ}$ at the smooth points of $\overline{\mathcal{V}}^\circ$. Hence

$$K_{\overline{\mathcal{V}}}|_{\overline{\mathcal{V}}^\circ} = \pi_t^* K_{Y_t} - 17(t = 0) = \pi_t^* \tau^* K_Y + 150(t = 0). \tag{8.7}$$

Step 4. In this step, we compute the coefficient of $\mathcal{V}_\infty$ in $K_{\overline{\mathcal{V}}}$.

By Proposition 6.2(3), $\mathcal{V}_\infty = V_\infty \times Y_t$ is a prime divisor on $\overline{\mathcal{V}}$. Fix $1 \le k \le 11$. Let $T_k \subset \mathbb{A}^1 \times \mathbb{A}^{10} \times Y_t$, with coordinates W and U_i for $i \ne k$, be the closed subscheme defined by

$$\widehat{s} = (\widehat{s}_1, \dots, \widehat{s}_4) := \sum_{i \ne k} U_i^{e_i} \mathbf{w}_i + \mathbf{w}_k - (1, b_2, b_3, b_4)^\top W^{42} = 0,$$

and let $q_k : T_k \to \overline{\mathcal{V}} \cap (U_k \ne 0)$ be the quotient morphism by μ_{d_k}, which acts by $\zeta \cdot (W, U_i) = (\zeta W, \zeta^{d_i} U_i)$. Let $x \in T_k$ be a closed point with $W(x) = 0$. The coordinates $(U_i(x))_{1 \le i \le 11}$ with $U_k(x) = 1$ form a non-zero point of $\operatorname{Spec} R_\infty$, at which $\operatorname{Spec} R_\infty$ is smooth of dimension 7 (Proposition 6.2(3)). By [Har77, Theorem I.5.1], the 4×11 matrix $(\partial h / \partial U_i)_{1 \le i \le 11}$ is of rank 4 at x. By the weighted Euler identity,

$$\sum_{i=1}^{11} d_i U_i \frac{\partial h}{\partial U_i} = 42h = 0$$

at x, and $U_k(x) = 1$, so $\partial h / \partial U_k$ is a linear combination of the columns $\partial h / \partial U_i$ with $i \ne k$ at x. Thus $(\partial \widehat{s} / \partial U_i)_{i \ne k}$ is of rank 4 at x, so T_k is smooth at x and $(W = 0)$ is a smooth reduced divisor on T_k near x. Moreover, $T_k \cap (W = 0) = (\operatorname{Spec} R_\infty \cap (U_k = 1)) \times Y_t$, and μ_{d_k} acts trivially on Y_t. By Proposition 6.2(3), the action of μ_{d_k} on $\operatorname{Spec} R_\infty \cap (U_k = 1)$ is free at the generic point, so q_k is étale at the generic point of $(W = 0)$ and $q_k^* \mathcal{V}_\infty = (W = 0)$ near there.

On $T_k \cap (W \ne 0)$, the morphism q_k is given by $z_k = W^{-d_k}$ and $z_i = W^{-d_i} U_i$ for $i \ne k$. Let $dU' := \bigwedge_{i \ne k} dU_i$. Since $dz_k = -d_k W^{-d_k - 1} dW$ and $dz_i = W^{-d_i} dU_i - d_i U_i W^{-d_i - 1} dW$ for $i \ne k$, we have

$$q_k^*(dz \wedge \tau^* \eta) = \pm d_k W^{-152} dW \wedge dU' \wedge \tau^* \eta, \qquad 152 = 1 + d_1 + \cdots + d_{11}.$$

Moreover, $s \circ q_k = W^{-42} \widehat{s}$, so $d(s_l \circ q_k) = W^{-42} d\widehat{s}_l$ along $T_k \cap (W \ne 0)$. Since $\mathcal{V}_\infty$ dominates Y_t, t does not vanish at the generic point of $\mathcal{V}_\infty$, so $\tau^* \eta$ generates ω_{Y_t} and $dW \wedge dU' \wedge \tau^* \eta$ generates $\omega_{\mathbb{A}^1 \times \mathbb{A}^{10} \times Y_t}$ near the generic point of $(W = 0)$ on T_k. As in **Step 3**, we obtain

$$q_k^* \operatorname{Res}_{\overline{\mathcal{V}}^\circ}(dz \wedge \tau^* \eta / s) = \pm d_k W^{16} \operatorname{Res}_{T_k}(dW \wedge dU' \wedge \tau^* \eta / \widehat{s}), \qquad 16 = 4 \cdot 42 - 152,$$

and the residue on the right-hand side generates ω_{T_k} near the generic point of $(W = 0)$. By (8.6), $\operatorname{mult}_{\mathcal{V}_\infty} K_{\overline{\mathcal{V}}} = 16$. Together with (8.7), we obtain

$$K_{\overline{\mathcal{V}}} = \pi_t^* \tau^* K_Y + 150(t = 0) + 16 \mathcal{V}_\infty. \tag{8.8}$$

Since $K_{\overline{\mathcal{V}}}$ is $\mathbb{Q}$-Cartier and $\pi_t^* \tau^* K_Y$ and $(t = 0)$ are Cartier, $\mathcal{V}_\infty$ is $\mathbb{Q}$-Cartier. Moreover, $\pi \circ \phi = \tau \circ \pi_t$ on $\overline{\mathcal{V}}^\circ$, so $\pi_t^* \tau^* K_Y$ is the pullback of $K_V = \pi^* K_Y$.

Step 5. In this step, we prove (2).

Since V is a normal variety (Lemma 8.3), by Definition-Lemma 2.6, there exist a birational morphism $\psi : V' \to V$ from a normal variety V' on which E is a divisor, a birational morphism $g : Y'' \to Y$ from a smooth variety Y'', and a morphism $\pi' : V' \to Y''$ with $g \circ \pi' = \pi \circ \psi$, such that F is a prime divisor on Y'' and $\pi'(E)$ is dense in F. Let $\widetilde{Y} \to Y''$ be a birational morphism from a smooth variety which resolves the rational map $Y'' \dashrightarrow Y'$. Since Y'' is smooth, $\widetilde{Y} \to Y''$ is an isomorphism over the generic point of F, and by [Har77, Theorem II.4.7], the induced rational map $V' \dashrightarrow \widetilde{Y}$ is defined near the generic point of E. Possibly replacing Y'' with $\widetilde{Y}$ and V' with the normalization of the closure of the graph of $V' \dashrightarrow \widetilde{Y}$, we may assume that $g = \sigma \circ g''$ for a morphism $g'' : Y'' \to Y'$.

Since $\operatorname{center}_Y F \subset Y_0$, we have $g''(F) \subset \operatorname{Exc}(\sigma)$. Near the generic point of $g''(F)$, the ideal $\mathcal{I}_{Y_0} \cdot \mathcal{O}_{Y'}$ is generated by $\sigma^* c_k$ for some $1 \le k \le 4$, so $\operatorname{mult}_F c_k = \operatorname{mult}_F \mathcal{I}_{Y_0}$, and $\sigma^* c_1 / \sigma^* c_k$ is regular there with

$$\operatorname{mult}_F(\sigma^* c_1 / \sigma^* c_k) = \operatorname{mult}_F c_1 - \operatorname{mult}_F \mathcal{I}_{Y_0} = 0.$$

Thus $\sigma^* c_1 / \sigma^* c_k$ does not vanish at the generic point of $g''(F)$, so $\sigma^* c_1$ generates $\mathcal{I}_{Y_0} \cdot \mathcal{O}_{Y'}$ near it, that is, the generic point of $g''(F)$ is in Y_1'.

Let $Y_1'' := g''^{-1}(Y_1')$ and $V_1' := \pi'^{-1}(Y_1'')$, which are open neighborhoods of the generic points of F and E, respectively. By (8.4), Ψ restricts to an isomorphism from $\overline{\mathcal{V}}^\circ \cap (t \neq 0)$ onto $(V \times_Y Y_t) \cap (t \neq 0)$, with inverse $(u, y_t) \to ((t^{-d_i}u_i)_{1\leq i\leq 11}, y_t)$. Since ψ is an isomorphism over a dense open subset of V and $V \cap \pi^{-1}(c_1 \neq 0)$ is dense in V (**Step 1**), there exists a dense open subset of V_1' over which $V_1' \times_{Y_1'} Y_t$ is isomorphic via Ψ to an open subset of $\overline{\mathcal{V}}^\circ \cap (t \neq 0)$, which is irreducible as $\overline{\mathcal{V}}$ is a variety (**Step 1**). The generic point of any irreducible component of $V_1' \times_{Y_1'} Y_t$ dominating V_1' maps to the generic point of V_1', hence into that dense open subset, so $V_1' \times_{Y_1'} Y_t$ has a unique irreducible component dominating V_1'. Let $\widetilde{V}$ be the normalization of the irreducible component of $V_1' \times_{Y_1'} Y_t$ dominating V_1', and let $\widetilde{f} : \widetilde{V} \to V_1'$ and $\widetilde{\pi} : \widetilde{V} \to Y_t$ be the induced morphisms. Since τ_1 is finite, $\widetilde{f}$ is finite and surjective. Since ψ is birational, there exists a birational map $\chi : \widetilde{V} \dashrightarrow \overline{\mathcal{V}}$ with $\pi_t \circ \chi = \widetilde{\pi}$ and $\phi \circ \chi = \psi \circ \widetilde{f}$. Since $\widetilde{f}$ is finite and surjective, there exists a prime divisor $E_{\widetilde{V}}$ on $\widetilde{V}$ with $\widetilde{f}(E_{\widetilde{V}}) = E$.

The fiber product $\overline{\mathcal{V}} \times_{Y_t} \widetilde{V}$ is projective over $\widetilde{V}$ by (1), and χ induces a rational section of the projection to $\widetilde{V}$. Since $\widetilde{V}$ is normal, by [Har77, Theorem II.4.7], χ is defined near the generic point of $E_{\widetilde{V}}$. Let $\mathcal{V}'$ be the normalization of the closure of the graph of χ in $\widetilde{V} \times_{Y_t} \overline{\mathcal{V}}$, with projections $q : \mathcal{V}' \to \widetilde{V}$ and $p' : \mathcal{V}' \to \overline{\mathcal{V}}$. Then q is proper, birational, and an isomorphism near the generic point of $E_{\widetilde{V}}$. Let E' be the strict transform of $E_{\widetilde{V}}$ on $\mathcal{V}'$ and $\phi' := \widetilde{f} \circ q : \mathcal{V}' \to V'$. Then p' is birational, $\psi \circ \phi' = \phi \circ p'$, and $\phi'(E') = \widetilde{f}(E_{\widetilde{V}}) = E$. This proves (2).

Step 6. In this step, we prove that E' is vertical over Y_t and the first equality of (3).

Let E' be a prime divisor over $\overline{\mathcal{V}}$ which ϕ maps onto E, with $\mathcal{V}' \to \overline{\mathcal{V}}$, $V_E' \to V$, and $\phi_E' : \mathcal{V}' \to V_E'$ as in (2). In this step, $\mathcal{V}'$, q, p', ϕ', and $E_{\widetilde{V}}$ are attached to this E', not to the choices made in **Step 5**, while V', V_1', Y_1'', $\widetilde{V}$, $\widetilde{f}$, and $\widetilde{\pi}$ are as in **Step 5**. Since E is a divisor on both V_E' and V' of **Step 5**, the birational map $V_E' \dashrightarrow V'$ is an isomorphism between neighborhoods of the generic points of E. Log discrepancies and multiplicities along E' only depend on a neighborhood of the generic point of E'. Possibly replacing $\mathcal{V}'$ with such a neighborhood, we may assume that the composite of ϕ_E' and $V_E' \dashrightarrow V'$ is a morphism $\phi' : \mathcal{V}' \to V_1'$. The morphisms

$$\mathcal{V}' \to V_1' \to Y_1'' \to Y_1' \quad \text{and} \quad \mathcal{V}' \to \overline{\mathcal{V}} \to Y_t \to Y_1'$$

coincide on a dense open subset, hence coincide. Thus $\mathcal{V}'$ maps to $V_1' \times_{Y_1'} Y_t$, onto a dense subset of the component dominating V_1'. Since $\mathcal{V}'$ is normal, there exists a birational morphism $q : \mathcal{V}' \to \widetilde{V}$ with $\widetilde{f} \circ q = \phi'$. The closure of $q(E')$ is an irreducible closed subset of $\widetilde{V}$ whose image under the finite morphism $\widetilde{f}$ is E, so it is a prime divisor $E_{\widetilde{V}}$ on $\widetilde{V}$. Since $\widetilde{V}$ is normal, the local ring R of $\widetilde{V}$ at the generic point of $E_{\widetilde{V}}$ is a discrete valuation ring. As q is birational and maps the generic point of E' to the generic point of $E_{\widetilde{V}}$, the local ring R' of $\mathcal{V}'$ at the generic point of E' dominates R, so $R' = R$ [Har77, Theorem I.6.1A]. Thus $E' = E_{\widetilde{V}}$ as prime divisors over $\widetilde{V}$.

The image of E' in Y_1' is the image of E in Y_1', which is contained in $g''(F) \subset \mathrm{Exc}(\sigma)$. Hence the image of E' in Y_t is contained in $\tau_1^{-1}(\mathrm{Exc}(\sigma)) = (t = 0)$. In particular, E' is vertical over Y_t.

Let $B_{V'} := \psi^* K_V - K_{V_1'}$ and $B_{\widetilde{V}} := \widetilde{f}^* \psi^* K_V - K_{\widetilde{V}}$, so that

$$K_{\widetilde{V}} + B_{\widetilde{V}} = \widetilde{f}^*(K_{V_1'} + B_{V'}),$$

and $K_{V_1'} + B_{V'} = \psi^* K_V$ is Cartier (Lemma 8.3). Then $a(E, V) = a(E, V_1', B_{V'})$. Since V' is normal, E is Cartier near the generic point of E, and the ramification index of $\widetilde{f}$ along $E_{\widetilde{V}}$ is

$$\mathrm{mult}_{E_{\widetilde{V}}} \widetilde{f}^* E = \mathrm{mult}_{E'} \phi'^* E = r'.$$

By Lemma 2.3 applied to $\widetilde{f}$ and $B_{V'}$,

$$a(E', \widetilde{V}, B_{\widetilde{V}}) = a(E_{\widetilde{V}}, \widetilde{V}, B_{\widetilde{V}}) = r' a(E, V).$$

On the other hand, let $p' : \mathcal{V}' \to \overline{\mathcal{V}}$ be the birational morphism of (2) for this E'. Since $\pi \circ \psi \circ \widetilde{f} \circ q = \tau \circ \pi_t \circ p'$, by (8.8),

$$p'^*\big(K_{\overline{\mathcal{V}}} - 150(t=0) - 16\mathcal{V}_\infty\big) = p'^*\pi_t^*\tau^* K_Y = q^*\widetilde{f}^*\psi^* K_V = q^*(K_{\widetilde{V}} + B_{\widetilde{V}}).$$

As p' and q are birational, we obtain

$$a(E', \overline{\mathcal{V}}, -150(t=0) - 16\mathcal{V}_\infty) = a(E', \widetilde{V}, B_{\widetilde{V}}).$$

Hence

$$r' a(E, V) = a(E', \overline{\mathcal{V}}) + 150 \operatorname{mult}_{E'} t + 16 \operatorname{mult}_{E'} \mathcal{V}_\infty.$$

Step 7. In this step, we prove the second and the third equalities of (3).

Let $\widetilde{Y}_t$ be the normalization of the irreducible component of $Y_1'' \times_{Y_1'} Y_t$ dominating Y_t, and $\widetilde{g} : \widetilde{Y}_t \to Y_1''$ the induced finite morphism. The morphism $\widetilde{V} \to Y_1'' \times_{Y_1'} Y_t$ is dominant onto that component, and $\widetilde{V}$ is normal, so it factors through a morphism $\widetilde{V} \to \widetilde{Y}_t$. The closure of the image of $E_{\widetilde{V}}$ in $\widetilde{Y}_t$ is irreducible, maps onto a dense subset of F under the finite morphism $\widetilde{g}$, and is contained in the preimage of $(t=0)$. Thus it is a prime divisor F_0' on $\widetilde{Y}_t$. Since $\widetilde{Y}_t$ is normal, $\widetilde{Y}_t$ is smooth near the generic point of F_0'. Shrinking $\mathcal{V}'$ near the generic point of E' and applying Definition-Lemma 2.6 to the smooth locus of $\widetilde{Y}_t$, we see that F_0' is the image F' of E' over Y_t and $r'' = \operatorname{mult}_{E'} F'$, where F' is pulled back to $\mathcal{V}'$. Let $\rho := \operatorname{mult}_{F'} \widetilde{g}^* F$ be the ramification index of $\widetilde{g}$ along F'.

By [Har77, Exercise II.8.5(b)] and (8.5),

$$K_{Y_1'} - 3\operatorname{Exc}(\sigma) = \sigma^* K_Y \quad \text{and} \quad K_{Y_t} - 167(t=0) = \tau_1^*\big(K_{Y_1'} - 3\operatorname{Exc}(\sigma)\big).$$

Since σ is proper and birational, $K_{Y'} - 3\operatorname{Exc}(\sigma) = \sigma^* K_Y$, and $\sigma_* \operatorname{Exc}(\sigma) = 0$, we have $a(F, Y', -3\operatorname{Exc}(\sigma)) = a(F, Y)$ [KM98, Lemma 2.30]. By **Step 5**, the generic point of $g''(F)$, which is the generic point of the center of F on Y', is in Y_1', so $a(F, Y_1', -3\operatorname{Exc}(\sigma)|_{Y_1'}) = a(F, Y', -3\operatorname{Exc}(\sigma))$ [KM98, 2.24–2.25]. By Lemma 2.3 applied to τ_1, $B := -3\operatorname{Exc}(\sigma)|_{Y_1'}$, and the birational morphism $Y_1'' \to Y_1'$,

$$\rho a(F, Y) = a(F', Y_t, -167(t=0)) = a(F', Y_t) + 167 \operatorname{mult}_{F'} t.$$

Near the generic point of E', the pullback of F' is $r''E'$, so $\operatorname{mult}_{E'} t = r'' \operatorname{mult}_{F'} t$. We compute the multiplicity along E' of the pullback of F in two ways: through $\widetilde{Y}_t$, it is $r''\rho$, and through V', where the pullback of F is rE near the generic point of E, it is $r'r$. Thus $r''\rho = r'r$. Multiplying the equality $\rho a(F, Y) = a(F', Y_t) + 167 \operatorname{mult}_{F'} t$ by r'' and using $r''\rho = r'r$ and $\operatorname{mult}_{E'} t = r'' \operatorname{mult}_{F'} t$, we obtain

$$r' r a(F, Y) = r'' a(F', Y_t) + 167 \operatorname{mult}_{E'} t.$$

Finally, $\tau^* c_1 = t^{42}$ on Y_t, so $\operatorname{mult}_{E'} c_1 = 42 \operatorname{mult}_{E'} t$. Near the generic point of F, the divisor of $g^* c_1$ is $(\operatorname{mult}_F c_1)F$, so

$$\operatorname{mult}_{E'} c_1 = r' \operatorname{mult}_E c_1 = r' r \operatorname{mult}_F c_1 = r' r \operatorname{mult}_F \mathcal{I}_{Y_0}.$$

Thus $42 \operatorname{mult}_{E'} t = r' r \operatorname{mult}_F \mathcal{I}_{Y_0}$. This proves (3). □

In the following proposition, $\frac{17}{42} = \frac{4 \cdot 42 - 151}{42}$, where $4 \cdot 42$ is the sum of the degrees of the four equations of h and $151 = d_1 + \cdots + d_{11}$.

Proposition 8.6. *Notation and conditions as in Set-up 8.2. Then V is a normal variety with klt singularities, and V is smooth near $\pi^{-1}(Y_0)$. Moreover, for any prime divisor E over V which is vertical over Y, with image F over Y and ramification index r, we have*

$$a(E, V) \geq r\Big(a(F, Y) - \frac{17}{42} \operatorname{mult}_F \mathcal{I}_{Y_0}\Big),$$

where $\operatorname{mult}_F \mathcal{I}_{Y_0} := 0$ *if* $Y_0 = \emptyset$.

Proof. By Lemma 8.3, π is flat, V is a normal variety, and K_V is Cartier.

Since Y_0 is smooth of codimension four and $\mathcal{I}_{Y_0}$ is generated by $c_1, \dots, c_4$, there exists an open neighborhood Y° of Y_0 on which c is smooth. Hence $\pi^{-1}(Y^\circ) = \mathbb{A}^{11} \times_{\mathbb{A}^4} Y^\circ$ is smooth over $\mathbb{A}^{11}$, so V is smooth near $\pi^{-1}(Y_0)$.

Let $V^\circ := \pi^{-1}(Y \backslash Y_0)$. With $u_i = U_i/W^{d_i}$, V° is the open subset $T \cap (W \neq 0)$ of the variety T in case (a) of Lemma 8.1, applied to $Y \backslash Y_0$ and the restriction of c to $Y \backslash Y_0$, which takes values in $\mathbb{A}^4 \backslash \{0\}$.

We show that V is klt. Let E be a prime divisor over V. If the generic point of $\operatorname{center}_V E$ is contained in $\pi^{-1}(Y_0)$, then V is smooth at that point, so $a(E, V) \geq 1$ [KM98, Corollary 2.31(2)]. Otherwise, E is a prime divisor over V°. If E is vertical over Y, with image F over Y and ramification index r, then by Lemma 8.1(1), $a(E, V) \geq r a(F, Y) \geq r > 0$, as $a(F, Y) \geq 1$ for the smooth variety Y [KM98, Corollary 2.31(2)]. Suppose that E is horizontal over Y. Let $f : V^\natural \to V$ be a projective log resolution of V on which E is a divisor, and write

$$K_{V^\natural} = f^* K_V + \sum_j (a(E_j, V) - 1) E_j,$$

where the E_j are the prime divisors on $V^\natural$ exceptional over V. By generic smoothness [Har77, Corollary III.10.7], applied to $\pi \circ f$ and to the restrictions of $\pi \circ f$ to the strata of $\sum_j E_j$, there exists a dense open subset $Y^\natural \subset Y \backslash Y_0$ such that for any closed point $y \in Y^\natural$, the fiber $V_y^\natural$ is smooth, $\sum_j E_j|_{V_y^\natural}$ is a simple normal crossing divisor, and $V_y^\natural \to V_{c(y)}$ is a log resolution (birational for general y, since f is an isomorphism over a dense open subset of V, with $V_y^\natural$ connected by Zariski's main theorem [Har77, Corollary III.11.4]). Since $V_y^\natural$ and $V_{c(y)}$ are the fibers over the smooth point y of the flat morphisms $\pi \circ f$ and π, we have $K_{V_y^\natural} = K_{V^\natural}|_{V_y^\natural}$ and $K_{V_{c(y)}} = K_V|_{V_{c(y)}}$. As E dominates Y, $E|_{V_y^\natural} \neq 0$ for general y, and any prime divisor E_y on $V_y^\natural$ contained in $E|_{V_y^\natural}$ satisfies $a(E_y, V_{c(y)}) = a(E, V)$. Since $c(y) \neq 0$, $V_{c(y)} = \overline{V}_{c(y)} \backslash V_\infty$ is klt by Proposition 6.9. Thus $a(E, V) > 0$, and V is klt.

We prove the inequality. Let E be a prime divisor over V which is vertical over Y, with image F over Y and ramification index r. With the notation of Definition-Lemma 2.6, $\pi''(E)$ is dense in F, so the generic point of $\operatorname{center}_V E$ maps to the generic point of $\operatorname{center}_Y F$.

First, assume that $Y_0 = \emptyset$ or $\operatorname{center}_Y F \not\subset Y_0$. Then $\operatorname{mult}_F \mathcal{I}_{Y_0} = 0$, and E is a prime divisor over V°. By Lemma 8.1(1), $a(E, V) \geq r a(F, Y)$.

Next, assume that $\operatorname{center}_Y F \subset Y_0$. Since $\mathcal{I}_{Y_0}$ is generated by $c_1, \dots, c_4$, we have

$$\operatorname{mult}_F \mathcal{I}_{Y_0} = \min_k \operatorname{mult}_F c_k.$$

After replacing c by Mc and each $\mathbf{w}_i$ by $M\mathbf{w}_i$ for a 4×4 permutation matrix M, we may assume that $\operatorname{mult}_F c_1 = \operatorname{mult}_F \mathcal{I}_{Y_0}$. The replacement does not change V, Y_0, $\mathcal{I}_{Y_0}$, or the condition that any four of $\mathbf{w}_1, \dots, \mathbf{w}_{11}$ are linearly independent. Let $\overline{\mathcal{V}}$, Y_t, t, and $\mathcal{V}_\infty$ be as in Construction 8.4. By Lemma 8.5(2), there exists a prime divisor E' over $\overline{\mathcal{V}}$ which ϕ maps onto E. By Lemma 8.5(3), E' is vertical over Y_t. Let r' be the multiplicity along E' of the pullback of E, F' the image of E' over Y_t, r'' the ramification index of E' over Y_t, and $m := \operatorname{mult}_{E'} t$. The variety $\overline{\mathcal{V}}$ is the variety T in case (a) of Lemma 8.1 with $Y := Y_t$ and $c := (1, b_2, b_3, b_4)$. By Lemma 8.1(1),

$$a(E', \overline{\mathcal{V}}) \geq r'' a(F', Y_t).$$

Since $\mathcal{V}_\infty \geq 0$, $\operatorname{mult}_{E'} \mathcal{V}_\infty \geq 0$. By Lemma 8.5(3),

$$r' a(E, V) \geq r'' a(F', Y_t) + 150m = r' r a(F, Y) - 17m = r' r \Big(a(F, Y) - \frac{17}{42} \operatorname{mult}_F \mathcal{I}_{Y_0} \Big).$$

Dividing by $r' > 0$, we obtain the inequality. □

9. Log canonicity on P

In this section, we prove the following theorem. The proof of Theorem 9.1 is given at the end of this section.

Theorem 9.1. *Notation and conditions as in Set-up 3.3. Let $\ell > 0$ be an integer and f a non-zero element of $(R_P)_\ell$. Then $(P, \frac{42}{\ell} D_f)$ is lc.*

Section 10 uses Theorem 9.1 at the centers contained in P, as outlined in part (b) of the sketch of the proof in Section 1. We prove Theorem 9.1 on the open subsets $P \cap (y_j \neq 0)$ in Subsection 9.2 and near the locus in P where all the coordinates y_l vanish in Subsection 9.3.

9.1. **A polynomial estimate.** In this subsection, we prove Lemma 9.2, which we use on the charts $y_j \neq 0$ in Subsection 9.2, and part (1) of which we also use in Subsection 9.3. On the charts $y_j \neq 0$, the bound $\operatorname{mult}_F g \leq k a(F, \mathbb{A}^N)$ for a polynomial g of degree at most k is too weak: over Y_0, the family $V \to Y$ of Set-up 8.2 loses up to $\frac{17}{42} \operatorname{mult}_F \mathcal{I}_{Y_0}$ of log discrepancy, multiplied by the ramification index (Proposition 8.6). Lemma 9.2 differentiates along a constant direction transverse to the center, keeps the multiplicities of the other normal coordinates, and thus gains $3 \operatorname{mult}_F \mathcal{I}_{Y_0}$ for the subvariety Y_0 of codimension four.

Lemma 9.2. *Let $N \geq 1$ and $k \geq 1$ be integers, and g a non-zero polynomial on $\mathbb{A}^N$ of degree at most k. Then the following hold for any prime divisor F over $\mathbb{A}^N$.*

(1) $\operatorname{mult}_F g \leq k a(F, \mathbb{A}^N)$.

(2) If $C \subset \mathbb{A}^N$ is a smooth closed subvariety of pure codimension $r \geq 1$ with ideal sheaf $\mathcal{I}_C$, then

$$\operatorname{mult}_F g \leq k\big(a(F, \mathbb{A}^N) - (r-1) \operatorname{mult}_F \mathcal{I}_C\big).$$

Proof. Let $m := \deg g$, so that $0 \leq m \leq k$, and let g_m be the homogeneous part of degree m of g. For any prime divisor F_1, we set $\operatorname{mult}_{F_1} 0 := +\infty$.

We first prove the following inequality. Let S be a smooth variety of dimension N and $\varphi_1, \dots, \varphi_N$ regular functions on S such that $d\varphi_1 \wedge \cdots \wedge d\varphi_N$ vanishes nowhere on S, and let $\partial/\partial\varphi_1$ be the derivation of $\mathcal{O}_S$ dual to $d\varphi_1$ with respect to the basis $d\varphi_1, \dots, d\varphi_N$ of Ω_S. Then for any prime divisor F_1 over S and any regular function ψ on S,

$$\operatorname{mult}_{F_1} \frac{\partial \psi}{\partial \varphi_1} \geq \operatorname{mult}_{F_1} \psi - a(F_1, S) + \sum_{i=2}^{N} \operatorname{mult}_{F_1} \varphi_i. \tag{9.1}$$

To prove (9.1), take a higher model of S on which F_1 is a divisor, and local coordinates $\zeta_1, \dots, \zeta_N$ at a general point of F_1 with $F_1 = (\zeta_1 = 0)$. If a regular function χ on S has multiplicity b along F_1, then in the pullback of $d\chi$ the coefficient of $d\zeta_1$ has multiplicity at least $b-1$ along F_1, and the other coefficients have multiplicity at least b. Thus the coefficient of the pullback of $d\psi \wedge d\varphi_2 \wedge \cdots \wedge d\varphi_N$ with respect to $d\zeta_1 \wedge \cdots \wedge d\zeta_N$ has multiplicity at least $\operatorname{mult}_{F_1} \psi + \sum_{i \geq 2} \operatorname{mult}_{F_1} \varphi_i - 1$ along F_1. On the other hand,

$$d\psi \wedge d\varphi_2 \wedge \cdots \wedge d\varphi_N = \frac{\partial \psi}{\partial \varphi_1} d\varphi_1 \wedge \cdots \wedge d\varphi_N,$$

and $d\varphi_1 \wedge \cdots \wedge d\varphi_N$ generates ω_S, so the coefficient of the pullback of $d\varphi_1 \wedge \cdots \wedge d\varphi_N$ has multiplicity $a(F_1, S) - 1$ along F_1 by Definition 2.1. This implies (9.1).

We prove (1). Since $m \leq k$ and $a(F, \mathbb{A}^N) \geq 0$, it suffices to show that $\operatorname{mult}_F g \leq m a(F, \mathbb{A}^N)$. If $m = 0$, then $\operatorname{mult}_F g = 0$. If $m \geq 1$, then we choose $\mathbf{e} \in \mathbb{C}^N$ with $g_m(\mathbf{e}) \neq 0$ and linear coordinates $z_1, \dots, z_N$ on $\mathbb{A}^N$ with $z_1(\mathbf{e}) = 1$ and $z_i(\mathbf{e}) = 0$ for $i \geq 2$. Then $\partial/\partial z_1$ is the derivative in the direction $\mathbf{e}$, and $\partial^m g / \partial z_1^m = m! g_m(\mathbf{e})$ is a non-zero constant. Since $\operatorname{mult}_F z_i \geq 0$ for any i, applying (9.1) m times to $S = \mathbb{A}^N$ and $\varphi_i = z_i$ implies that

$$0 = \operatorname{mult}_F \frac{\partial^m g}{\partial z_1^m} \geq \operatorname{mult}_F g - m a(F, \mathbb{A}^N).$$

We prove (2). If $\operatorname{center}_{\mathbb{A}^N} F \not\subset C$, then $\operatorname{mult}_F \mathcal{I}_C = 0$, and (2) is (1). Assume that $\operatorname{center}_{\mathbb{A}^N} F \subset C$. Let o be a closed point of $\operatorname{center}_{\mathbb{A}^N} F$ and $T_oC \subset \mathbb{C}^N$ the tangent space of C at o, so that $\dim T_oC = N - r < N$. The vectors $\mathbf{e} \in \mathbb{C}^N \backslash T_oC$ form a non-empty open subset of $\mathbb{C}^N$, and if $m \geq 1$, then so do the vectors $\mathbf{e}$ with $g_m(\mathbf{e}) \neq 0$. We choose $\mathbf{e} \in \mathbb{C}^N \backslash T_oC$, with $g_m(\mathbf{e}) \neq 0$ if $m \geq 1$, and an r-dimensional linear subspace $L \subset \mathbb{C}^N$ such that $\mathbf{e} \in L$ and $L \cap T_oC = 0$. Let $\eta : \mathbb{A}^N \to \mathbb{A}^{N-r}$ be a linear projection with kernel L and components $\eta_1, \dots, \eta_{N-r}$, and let $z_1, \dots, z_r$ be linear functions on $\mathbb{A}^N$ such that $(z_1, \dots, z_r, \eta_1, \dots, \eta_{N-r})$ are linear coordinates on $\mathbb{A}^N$, $z_1(\mathbf{e}) = 1$, and $z_i(\mathbf{e}) = 0$ for $i \geq 2$. Then the derivation $\partial/\partial z_1$ with respect to $(z_1, \dots, z_r, \eta_1, \dots, \eta_{N-r})$ is the derivative in the direction $\mathbf{e}$, and hence $\partial^m g/\partial z_1^m = m! g_m(\mathbf{e})$ is a non-zero constant.

Since $L \cap T_oC = 0$, the restriction $\eta|_C$ is étale at o. Let C° be an irreducible open neighborhood of o in C on which $\eta|_{C^\circ}$ is étale, let

$$Y^\circ := \mathbb{A}^N \times_{\mathbb{A}^{N-r}} C^\circ,$$

and let $\lambda : Y^\circ \to \mathbb{A}^N$ be the first projection, which is étale. Via $(y, c) \to (c, z_1(y), \dots, z_r(y))$, we have $Y^\circ \cong C^\circ \times \mathbb{A}^r$. For $1 \leq i \leq r$, let θ_i be the pullback to Y° of the function $z_i|_{C^\circ}$ under the second projection, and let $\xi_i := z_i \circ \lambda - \theta_i$. We also write η_l for $\eta_l \circ \lambda$. The diagonal $C' := \{(c, c) \mid c \in C^\circ\}$ is the closed subvariety $(\xi_1 = \dots = \xi_r = 0)$ of Y°, and $C' \cong C^\circ$. Since λ is étale, $\lambda^{-1}(C)$ is smooth of pure dimension $N - r$, so C' is a connected component of $\lambda^{-1}(C)$. Possibly replacing Y° with the complement of the other components of $\lambda^{-1}(C)$, we may assume that $\lambda^{-1}(C) = C'$, and hence $\mathcal{I}_C \cdot \mathcal{O}_{Y^\circ} = (\xi_1, \dots, \xi_r)$.

Since $\eta|_{C^\circ}$ is étale, $d\eta_1, \dots, d\eta_{N-r}$ form a basis of Ω_{C°. Thus each $d\theta_i$ is a combination of $d\eta_1, \dots, d\eta_{N-r}$, and

$$d\xi_1 \wedge \dots \wedge d\xi_r \wedge d\eta_1 \wedge \dots \wedge d\eta_{N-r} = \lambda^*(dz_1 \wedge \dots \wedge dz_r \wedge d\eta_1 \wedge \dots \wedge d\eta_{N-r}),$$

which vanishes nowhere on Y°. Moreover, with respect to the coordinates $\xi_1, \dots, \xi_r, \eta_1, \dots, \eta_{N-r}$, the derivation $\partial/\partial\xi_1$ kills any function pulled back from C°, as the differential of such a function is a combination of $d\eta_1, \dots, d\eta_{N-r}$. Since $z_i \circ \lambda = \xi_i + \theta_i$, we obtain

$$\frac{\partial(\psi \circ \lambda)}{\partial \xi_1} = \frac{\partial \psi}{\partial z_1} \circ \lambda$$

for any regular function ψ on $\mathbb{A}^N$. In particular,

$$\frac{\partial^m (g \circ \lambda)}{\partial \xi_1^m} = m! g_m(\mathbf{e}) \neq 0.$$

We lift F to Y°. Let $X' \to \mathbb{A}^N$ be a birational morphism from a normal variety on which F is a divisor, and let $X'' := X' \times_{\mathbb{A}^N} Y^\circ$. Then $X'' \to X'$ is étale, so X'' is normal, and any component of X'' maps onto a dense open subset of X'. Let $A \subset \mathbb{A}^N$ be the open subset over which $X' \to \mathbb{A}^N$ is an isomorphism. Then $X'' \to Y^\circ$ is an isomorphism over the irreducible open subset $\lambda^{-1}(A)$ of Y°, and any component of X'' intersects the preimage of $\lambda^{-1}(A)$. Thus X'' is irreducible, and $X'' \to Y^\circ$ is birational. Let X'_λ be the preimage in X' of the open subset $\lambda(Y^\circ)$ of $\mathbb{A}^N$ (λ is étale, hence flat and open [Har77, Exercise III.9.1]). Since $X'' = X'_\lambda \times_{\lambda(Y^\circ)} Y^\circ$ and $\lambda : Y^\circ \to \lambda(Y^\circ)$ is étale and surjective, the relative canonical divisor of X'' over Y° is the pullback of the relative canonical divisor $K_{X'/\mathbb{A}^N}|_{X'_\lambda}$ of X'_λ over $\lambda(Y^\circ)$ [Kol13, 2.14]. Since $\lambda(Y^\circ)$ is an open neighborhood of $o \in \operatorname{center}_{\mathbb{A}^N} F$, the preimage of F on X'' is non-empty, and we let F° be a component of the preimage of F on X''. Since $X'' \to X'$ is étale, a local parameter of X' at the generic point of F pulls back to a local parameter of X'' at the generic point of F°. Thus

$$a(F^\circ, Y^\circ) = a(F, \mathbb{A}^N), \quad \operatorname{mult}_{F^\circ}(\psi \circ \lambda) = \operatorname{mult}_F \psi, \quad \text{and} \quad \operatorname{mult}_{F^\circ}(\mathcal{I}_C \cdot \mathcal{O}_{Y^\circ}) = \operatorname{mult}_F \mathcal{I}_C$$

for any regular function ψ on $\mathbb{A}^N$. Moreover, $\operatorname{center}_{Y^\circ} F^\circ \subset \lambda^{-1}(C) = C'$, so $\operatorname{mult}_{F^\circ} \xi_i \geq \operatorname{mult}_F \mathcal{I}_C$ for any i, and $\operatorname{mult}_{F^\circ} \eta_l \geq 0$ for any l. Applying (9.1) to Y°, the functions

$\xi_1, \dots, \xi_r, \eta_1, \dots, \eta_{N-r}$, and F°, we obtain

$$\operatorname{mult}_{F^\circ} \frac{\partial \psi}{\partial \xi_1} \geq \operatorname{mult}_{F^\circ} \psi - a(F, \mathbb{A}^N) + (r-1) \operatorname{mult}_F \mathcal{I}_C \tag{9.2}$$

for any regular function ψ on Y°. Applying (9.2) to $\psi = \partial^i(g \circ \lambda)/\partial \xi_1^i$ for $0 \leq i \leq m-1$ and adding up, we obtain

$$0 = \operatorname{mult}_{F^\circ} \frac{\partial^m (g \circ \lambda)}{\partial \xi_1^m} \geq \operatorname{mult}_F g - m\big(a(F, \mathbb{A}^N) - (r-1) \operatorname{mult}_F \mathcal{I}_C\big).$$

Finally, $\sum_{i=1}^r (\xi_i = 0)$ is an snc divisor on Y° with coefficients one, so $(Y^\circ, \sum_{i=1}^r (\xi_i = 0))$ is lc [KM98, Corollary 2.31(3)]. Thus

$$a(F, \mathbb{A}^N) = a(F^\circ, Y^\circ) \geq \sum_{i=1}^r \operatorname{mult}_{F^\circ} \xi_i \geq r \operatorname{mult}_F \mathcal{I}_C,$$

so

$$a(F, \mathbb{A}^N) - (r-1) \operatorname{mult}_F \mathcal{I}_C \geq \operatorname{mult}_F \mathcal{I}_C \geq 0.$$

Since $m \leq k$, (2) follows. □

9.2. **The charts $y_j \neq 0$.** In this subsection, we prove Theorem 9.1 on the open subsets $P \cap (y_j \neq 0)$ of P.

The normal form below writes f as a sum of products of a monomial in the u_i and a polynomial in the y_l. In the proof of Proposition 9.4, Theorem 7.1 bounds the monomials along the fibers of h, Lemma 9.2 bounds the polynomials on the base $\mathbb{A}^{17}$, and the degree identity of Definition-Lemma 9.3 combines the bounds into the coefficient $\frac{42}{\ell}$. The exponents of $u_1, \dots, u_4$ are restricted so that the monomials do not depend on the fiber and are the monomials of Lemma 8.1(2).

Definition-Lemma 9.3. Notation and conditions as in Set-up 3.3. Let $\ell > 0$ be an integer and f a non-zero element of $(R_P)_\ell$. Then there exist distinct monomials $m_1, \dots, m_s$ in $u_1, \dots, u_{11}$, in which the exponents of u_1, u_2, u_3 are at most one and the exponent of u_4 is at most two, and non-zero homogeneous polynomials $p_1, \dots, p_s \in \mathbb{C}[y_1, \dots, y_{18}]$, such that $f = \sum_{\alpha=1}^s p_\alpha m_\alpha$ in R_P, and such an expression of f is unique up to the order of the terms. Moreover, if $43j_\alpha$ is the degree of m_α and k_α is the degree of p_α as a polynomial, then $43j_\alpha + 42k_\alpha = \ell$ for any α. In particular, $\ell = 43J + 42k$ for $J := \max_\alpha j_\alpha$ and $k := \min_\alpha k_\alpha$.

Proof. Since $P = (x = 0)$,

$$R_P = \mathbb{C}[u_1, \dots, u_{11}, y_1, \dots, y_{18}] \Big/ \Big(\sum_{i=1}^{11} \boldsymbol{v}_i u_i^{e_i} + \sum_{l=1}^{18} \boldsymbol{v}_{11+l} y_l^{43} \Big).$$

By Proposition 3.4(1), any four of the vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{29}$ are linearly independent. We have $e_1 = e_2 = e_3 = 2$ and $e_4 = 3$. By Lemma 3.2(1), applied to the 29 variables $u_1, \dots, u_{11}, y_1, \dots, y_{18}$ and the indices of u_1, u_2, u_3, u_4, the ring R_P is a free module over $\mathbb{C}[u_5, \dots, u_{11}, y_1, \dots, y_{18}]$ with basis the monomials $u_1^{b_1} u_2^{b_2} u_3^{b_3} u_4^{b_4}$, where $b_1, b_2, b_3 \in \{0, 1\}$ and $b_4 \in \{0, 1, 2\}$. Thus R_P is a free module over $\mathbb{C}[y_1, \dots, y_{18}]$ with basis the set of monomials in $u_1, \dots, u_{11}$ in which the exponents of u_1, u_2, u_3 are at most one and the exponent of u_4 is at most two. This proves the existence and the uniqueness of the expression $f = \sum_{\alpha=1}^s p_\alpha m_\alpha$ with distinct monomials m_α and non-zero polynomials p_α.

The monomials m_α and the coordinates y_l are homogeneous elements of R_P. Taking the homogeneous part of degree ℓ of each $p_\alpha m_\alpha$, we obtain an expression $f = \sum_{\alpha=1}^s p'_\alpha m_\alpha$ with $p'_\alpha \in \mathbb{C}[y_1, \dots, y_{18}]$. Since $m_1, \dots, m_s$ are distinct elements of a basis of the free $\mathbb{C}[y_1, \dots, y_{18}]$-module R_P, we have $p'_\alpha = p_\alpha$, and hence $p_\alpha m_\alpha$ is homogeneous of degree ℓ for any α. Since $\deg y_l = 42$ and $\deg u_i = 43 d_i$, the polynomial p_α is homogeneous, the degree of m_α is $43j_\alpha$, where j_α is the degree of m_α with respect to the weights $d_1, \dots, d_{11}$, and $43j_\alpha + 42k_\alpha = \ell$. In particular, an index α maximizing j_α minimizes k_α, and $\ell = 43J + 42k$. □

Proposition 9.4. *Notation and conditions as in Set-up 3.3. Let $\ell > 0$ be an integer and f a non-zero element of $(R_P)_\ell$. Then for any $1 \le j \le 18$, the pair $(P, \frac{42}{\ell} D_f)$ is lc on $P \cap (y_j \neq 0)$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce Proposition 9.4 to an estimate for the multiplicities of a regular function on a cover of $P \cap (y_j \neq 0)$, and we introduce the quantities used in the estimate.

Let $P_j := P \cap (y_j \neq 0)$. Let $V := \widetilde{P}_{y_j}$ and let $q_j : V \to P_j$ be the quotient morphism by μ_{42} (Lemma 3.7). Then V is smooth and q_j is étale in codimension one (Lemma 3.7(1)(2)). Let $Y := \mathbb{A}^{17}$ with coordinates y_l, $l \neq j$, and

$$c(y) := -\boldsymbol{v}_{11+j} - \sum_{l \neq j} \boldsymbol{v}_{11+l} y_l^{43}.$$

We use the notation of Set-up 6.1 with $\mathbf{w}_i := \boldsymbol{v}_i$ for $1 \le i \le 11$, any four of which are linearly independent by Proposition 3.4(1). Then $V = \{(u, y) \in \mathbb{A}^{11} \times Y \mid h(u) = c(y)\}$, so V and the projection $\pi : V \to Y$ are as in Set-up 8.2(3), once we check Set-up 8.2(2).

The subvariety $Y_0 = c^{-1}(0)$ is the intersection of the cone $\sum_{l=1}^{18} \boldsymbol{v}_{11+l} y_l^{43} = 0$ in $\mathbb{A}^{18}$ with $(y_j = 1)$. By Lemma 3.2(2), applied to the vectors $\boldsymbol{v}_{12}, \dots, \boldsymbol{v}_{29} \in \mathbb{C}^4$ and the exponents 43, any closed point of Y_0 has at least five non-zero coordinates in $\mathbb{A}^{18}$, and hence at least four non-zero coordinates $y_{l_1}, \dots, y_{l_4}$ with $l_1, \dots, l_4 \neq j$. Since $\partial c / \partial y_l = -43 y_l^{42} \boldsymbol{v}_{11+l}$ and $\boldsymbol{v}_{11+l_1}, \dots, \boldsymbol{v}_{11+l_4}$ are linearly independent, the Jacobian matrix of (c_1, c_2, c_3, c_4) has rank four at any point of Y_0. Thus at any closed point y of Y_0, the images of c_1, c_2, c_3, c_4 in $\mathfrak{m}_y / \mathfrak{m}_y^2$ are linearly independent, so $\mathcal{O}_{Y,y}/(c_1, c_2, c_3, c_4)$ is a regular local ring of dimension $\dim Y - 4$ [Mat89, Theorem 14.2], and hence an integral domain [Mat89, Theorem 14.3]. Since the support of the nilradical of $\mathcal{O}_{c^{-1}(0)}$, if non-empty, contains a closed point, the closed subscheme $c^{-1}(0)$ of Y is reduced. Hence $c^{-1}(0)$ is smooth of pure codimension four, and $\mathcal{I}_{Y_0}$ is generated by c_1, c_2, c_3, c_4. Thus Set-up 8.2(2) holds. Moreover, $Y_0 \neq \emptyset$: for four indices $l_1, \dots, l_4$ different from j, the vectors $\boldsymbol{v}_{11+l_1}, \dots, \boldsymbol{v}_{11+l_4}$ form a basis of $\mathbb{C}^4$, so there exists $y \in Y$ with $c(y) = 0$ and $y_l = 0$ for $l \notin \{j, l_1, \dots, l_4\}$. We use $Y_0 \neq \emptyset$ for $C = Y_0$ in Lemma 9.2(2) and for Construction 8.4.

Let f_V be the regular function on V given by $f_V(u, y) := f(u, y_1, \dots, y_{j-1}, 1, y_{j+1}, \dots, y_{18})$. The rational function f^{42}/y_j^ℓ is regular on P_j with divisor $42 D_f|_{P_j}$, and the pullback of f^{42}/y_j^ℓ to V is f_V^{42}. Thus $q_j^* D_f = \operatorname{div}(f_V)$. Since q_j is étale in codimension one, $K_V = q_j^* K_{P_j}$ [Kol13, 2.39 and (2.41.3)], and P_j is $\mathbb{Q}$-factorial [KM98, Lemma 5.16]. By Lemma 2.3, applied to q_j and $\frac{42}{\ell} D_f|_{P_j}$, it suffices to show that $(V, \frac{42}{\ell} \operatorname{div}(f_V))$ is lc. Since V is smooth, it suffices to show that for any prime divisor E over V,

$$\operatorname{mult}_E f_V \le \frac{\ell}{42} a(E, V). \tag{9.3}$$

Let $f = \sum_\alpha p_\alpha m_\alpha$ be the expression of Definition-Lemma 9.3, and let $\bar{p}_\alpha$ be the polynomial on Y defined by

$$\bar{p}_\alpha(y) := p_\alpha(y_1, \dots, y_{j-1}, 1, y_{j+1}, \dots, y_{18}).$$

Since p_α is a non-zero homogeneous polynomial of degree k_α, the polynomial $\bar{p}_\alpha$ is non-zero of degree at most k_α. Thus

$$f_V = \sum_\alpha \bar{p}_\alpha(y) m_\alpha(u).$$

We fix an index α_0 with $j_{\alpha_0} = J$, so that $k_{\alpha_0} = k$.

Let E_1 be a prime divisor over a normal variety which maps onto a prime divisor E_2 over a normal variety, and let e be the multiplicity along E_1 of the pullback of E_2. Near the generic point of E_2, a regular function ψ is a unit times a power of a local parameter of E_2, so

$$\operatorname{mult}_{E_1} \psi = e \operatorname{mult}_{E_2} \psi, \tag{9.4}$$

where on the left we take the pullback of ψ. Let E be a prime divisor over V which is vertical over Y, with image F over Y and ramification index r. By Definition-Lemma 2.6 and (9.4) with $(E_1, E_2, e) = (E, F, r)$, for any regular function ψ on Y,

$$\operatorname{mult}_E(\psi \circ \pi) = r \operatorname{mult}_F \psi. \tag{9.5}$$

Let

$$\Lambda_1 := a(E,V) - r\Big(a(F,Y) - \frac{17}{42}\operatorname{mult}_F \mathcal{I}_{Y_0}\Big),$$
$$\Lambda_2 := r\big(a(F,Y) - 3\operatorname{mult}_F \mathcal{I}_{Y_0}\big), \quad \Lambda_3 := r \operatorname{mult}_F \mathcal{I}_{Y_0}.$$

Then $\Lambda_1 \geq 0$ by Proposition 8.6, and $\Lambda_3 \geq 0$. If $\operatorname{center}_Y F \subset Y_0$, then near the generic point of $\operatorname{center}_Y F$, the functions $c_1, \ldots, c_4$ are part of a regular system of parameters of Y, so $(Y, \sum_{i=1}^4 (c_i = 0))$ is lc there [KM98, Corollary 2.31(3)], and

$$a(F,Y) \geq \sum_{i=1}^4 \operatorname{mult}_F c_i \geq 4 \operatorname{mult}_F \mathcal{I}_{Y_0}.$$

If $\operatorname{center}_Y F \not\subset Y_0$, then $\operatorname{mult}_F \mathcal{I}_{Y_0} = 0$. In both cases, $a(F,Y) \geq 4 \operatorname{mult}_F \mathcal{I}_{Y_0}$, so $\Lambda_2 \geq \Lambda_3 \geq 0$. Since $\frac{17}{42} - 3 + \frac{109}{42} = 0$,

$$a(E,V) = \Lambda_1 + \Lambda_2 + \frac{109}{42}\Lambda_3. \tag{9.6}$$

If $k \geq 1$, then by Lemma 9.2(2), applied to the codimension-four subvariety $C = Y_0$ of Y and the polynomial $\bar{p}_{\alpha_0}$ of degree at most k, we have

$$r \operatorname{mult}_F \bar{p}_{\alpha_0} \leq kr\big(a(F,Y) - 3\operatorname{mult}_F \mathcal{I}_{Y_0}\big) = k\Lambda_2. \tag{9.7}$$

If $k = 0$, then $\bar{p}_{\alpha_0}$ is a non-zero constant, and (9.7) also holds.

Since $\ell = 43J + 42k$ and $J, k \geq 0$, we have $\max(J,k) \leq \frac{\ell}{42}$. For a prime divisor E over V which is vertical over Y, with $\Lambda_1, \Lambda_2, \Lambda_3$ defined for E before (9.6), consider the inequality

$$\operatorname{mult}_E f_V \leq J\Lambda_1 + k\Lambda_2 + \frac{J}{42}\Lambda_3. \tag{9.8}$$

If (9.8) holds, then since $\Lambda_1, \Lambda_2, \Lambda_3 \geq 0$, (9.6) implies that

$$\operatorname{mult}_E f_V \leq J\Lambda_1 + k\Lambda_2 + \frac{J}{42}\Lambda_3 \leq \max(J,k)\Big(\Lambda_1 + \Lambda_2 + \frac{109}{42}\Lambda_3\Big)$$
$$= \max(J,k) a(E,V) \leq \frac{\ell}{42} a(E,V).$$

Hence it suffices to show that $\operatorname{mult}_E f_V \leq J a(E,V)$ for any prime divisor E over V which is horizontal over Y, and that (9.8) holds for any prime divisor E over V which is vertical over Y.

Step 2. In this step we treat the case $J = 0$.

If $J = 0$, then $m_\alpha = 1$ for any α, so $f_V = \bar{p}_{\alpha_0} \circ \pi$ and $k = \frac{\ell}{42} \geq 1$. Let E be a prime divisor over V. If E is horizontal over Y, then $\bar{p}_{\alpha_0} \circ \pi$ does not vanish identically on E, so $\operatorname{mult}_E f_V = 0$. If E is vertical over Y, then by (9.5) and (9.7), $\operatorname{mult}_E f_V = r \operatorname{mult}_F \bar{p}_{\alpha_0} \leq k\Lambda_2$, which is (9.8).

Step 3. In this step we treat the case $J > 0$ when E is horizontal over Y, or E is vertical over Y with image F such that $\operatorname{mult}_F \mathcal{I}_{Y_0} = 0$.

If E is horizontal over Y, then the image of $\operatorname{center}_V E$ in Y is dense. If E is vertical over Y with image F and $\operatorname{mult}_F \mathcal{I}_{Y_0} = 0$, then $\operatorname{center}_Y F \not\subset Y_0$, and the image of $\operatorname{center}_V E$ in Y is dense in $\operatorname{center}_Y F$. In both cases, $\operatorname{center}_V E$ intersects $\pi^{-1}(Y \backslash Y_0)$. On $Y \backslash Y_0$, the morphism c takes values in $\mathbb{A}^4 \backslash \{0\}$. Let $T \to Y \backslash Y_0$ be the family of Lemma 8.1, case (a), for the restriction of c to $Y \backslash Y_0$. With $u_i = U_i / W^{d_i}$, $\pi^{-1}(Y \backslash Y_0)$ is the open subset $T \cap (W \neq 0)$ of T. Replacing Y with $Y \backslash Y_0$ and V with $\pi^{-1}(Y \backslash Y_0)$, we may regard E as a prime divisor over T, and $a(E,T) = a(E,V)$. The image F, the ramification index r, $a(F,Y)$, and $\operatorname{mult}_F \bar{p}_{\alpha_0}$ do not change under the replacement.

Let

$$f_T := \sum_\alpha \bar{p}_\alpha(y) m_\alpha(U) W^{J-j_\alpha}$$

and D_{f_T} the divisor of zeros of f_T on T. The monomials $m_\alpha(U)W^{J-j_\alpha}$ are distinct monomials of degree J in $W, U_1, \dots, U_{11}$, in which the exponents of U_1, U_2, U_3 are at most one and the exponent of U_4 is at most two. On $T \cap (W \neq 0)$, we have $f_T/W^J = f_V$, so $\operatorname{mult}_E D_{f_T} = \operatorname{mult}_E f_V$. By Theorem 7.1, $(\overline{V}_{c(y)}, \frac{1}{J} D_{f'})$ is lc for any closed point $y \in Y$ and any non-zero element f' of $(R_{c(y)})_J$. Thus the assumption of Lemma 8.1(2) holds for $q = J$. If E is horizontal over Y, then by Lemma 8.1(2), applied to $s = f_T$ and $q = J$,

$$\operatorname{mult}_E f_V \le Ja(E,T) = Ja(E,V).$$

If E is vertical over Y, then $\Lambda_3 = 0$ and $\Lambda_1 = a(E,V) - ra(F,Y)$, so by Lemma 8.1(2) and (9.7),

$$\operatorname{mult}_E f_V \le J\big(a(E,V) - ra(F,Y)\big) + r \operatorname{mult}_F \bar{p}_{\alpha_0} \le J\Lambda_1 + k\Lambda_2,$$

which is (9.8).

Step 4. In this step we treat the case $J > 0$ when E is vertical over Y with image F such that $\operatorname{mult}_F \mathcal{I}_{Y_0} > 0$.

Then $\operatorname{center}_Y F \subset Y_0$. Since $\mathcal{I}_{Y_0}$ is generated by $c_1, \dots, c_4$, we have $\operatorname{mult}_F \mathcal{I}_{Y_0} = \min_i \operatorname{mult}_F c_i$. After replacing c by Mc and each $\mathbf{w}_i$ by $M\mathbf{w}_i$ for a 4×4 permutation matrix M, we may assume that $\operatorname{mult}_F c_1 = \operatorname{mult}_F \mathcal{I}_{Y_0}$. The replacement does not change V, Y_0, $\mathcal{I}_{Y_0}$, f_V, or the condition that any four of $\mathbf{w}_1, \dots, \mathbf{w}_{11}$ are linearly independent.

Let σ, Y_1', b_2, b_3, b_4, Y_t, t, τ, $\overline{\mathcal{V}}$, $\mathcal{V}_\infty$, and ϕ be as in Construction 8.4. By Lemma 8.5(2), there exists a prime divisor E' over $\overline{\mathcal{V}}$ which ϕ maps onto E. Let r' be the multiplicity along E' of the pullback of E. By Lemma 8.5(3), E' is vertical over Y_t, and if F' is the image of E' over Y_t and r'' is the ramification index of E' over Y_t, then

$$\begin{aligned} r'a(E,V) &= a(E', \overline{\mathcal{V}}) + 150 \operatorname{mult}_{E'} t + 16 \operatorname{mult}_{E'} \mathcal{V}_\infty, \\ r'ra(F,Y) &= r''a(F', Y_t) + 167 \operatorname{mult}_{E'} t, \quad \text{and} \quad 42 \operatorname{mult}_{E'} t = r'r \operatorname{mult}_F \mathcal{I}_{Y_0} = r'\Lambda_3. \end{aligned} \tag{9.9}$$

By Lemma 8.5(1), Y_t is a smooth variety, and $\overline{\mathcal{V}} \to Y_t$ is the family T of Lemma 8.1, case (a), for the morphism $(1, b_2, b_3, b_4) : Y_t \to \mathbb{A}^4 \backslash \{0\}$. Let

$$H_f := \sum_\alpha t^{j_\alpha} (\bar{p}_\alpha \circ \tau) m_\alpha(U) W^{J-j_\alpha},$$

and D_{H_f} the divisor of zeros of H_f on $\overline{\mathcal{V}}$. Since $u_i = t^{d_i} U_i / W^{d_i}$ under ϕ, we have $m_\alpha(u) = t^{j_\alpha} m_\alpha(U)/W^{j_\alpha}$, and hence the pullback of f_V under ϕ is H_f/W^J. Since ϕ is dominant (Lemma 8.5(1)) and is given by τ on Y_t, the morphism τ is dominant. Moreover, $t^{42} = \sigma^* c_1 \neq 0$. Thus the coefficient $t^J(\bar{p}_{\alpha_0} \circ \tau)$ of H_f is non-zero.

By Proposition 6.2(3), $\mathcal{V}_\infty = V_\infty \times Y_t$ is a prime divisor on $\overline{\mathcal{V}}$. Fix $1 \le i \le 11$, and let

$$\widehat{\mathcal{V}}_i := \Big\{ (W, U_1, \dots, U_{11}, y) \in \mathbb{A}^{12} \times Y_t \;\Big|\; U_i = 1,\ \sum_{l=1}^{11} U_l^{e_l} \mathbf{w}_l = (1, b_2, b_3, b_4)^\top W^{42} \Big\}.$$

Then $\overline{\mathcal{V}} \cap (U_i \neq 0)$ is the quotient of $\widehat{\mathcal{V}}_i$ by μ_{d_i}, and the quotient morphism is étale at the generic point of $\widehat{\mathcal{V}}_i \cap (W = 0)$, since μ_{d_i} acts trivially on Y_t and freely at the generic point of $\operatorname{Spec} R_\infty \cap (U_i = 1)$ (Proposition 6.2(3)). Moreover, $\widehat{\mathcal{V}}_i \cap (W = 0)$ is reduced at its generic point (as $\widehat{\mathcal{V}}_i \cap (W = 0)$ is the product of $\operatorname{Spec} R_\infty \cap (U_i = 1)$ and Y_t, and $\operatorname{Spec} R_\infty \cap (U_i = 1)$ is étale in codimension one over the normal variety $V_\infty \cap (U_i \neq 0)$ by Proposition 6.2(3)). Thus W vanishes to order one along $\mathcal{V}_\infty$, and $\operatorname{div}(H_f/W^J) = D_{H_f} - J\mathcal{V}_\infty$. By (9.4), applied to (E', E, r'), (E, F, r), and (E', F', r''), we obtain

$$\begin{aligned} r' \operatorname{mult}_E f_V &= \operatorname{mult}_{E'} D_{H_f} - J \operatorname{mult}_{E'} \mathcal{V}_\infty, \quad \text{and} \\ r'' \operatorname{mult}_{F'} \big(t^J (\bar{p}_{\alpha_0} \circ \tau)\big) &= J \operatorname{mult}_{E'} t + r'r \operatorname{mult}_F \bar{p}_{\alpha_0}. \end{aligned} \tag{9.10}$$

By Theorem 7.1, $(\overline{V}_{c'}, \frac{1}{J}D_{f'})$ is lc for any $c' \in \mathbb{C}^4\backslash\{0\}$ and any non-zero element f' of $(R_{c'})_J$. Thus Lemma 8.1(2), applied to $\overline{\mathcal{V}} \to Y_t$, $s = H_f$, $q = J$, and E', implies that

$$\operatorname{mult}_{E'} D_{H_f} \le J\big(a(E', \overline{\mathcal{V}}) - r'' a(F', Y_t)\big) + r'' \operatorname{mult}_{F'} \big(t^J(\bar{p}_{\alpha_0} \circ \tau)\big). \tag{9.11}$$

By (9.9),

$$a(E', \overline{\mathcal{V}}) - r'' a(F', Y_t) = r'\big(a(E, V) - r a(F, Y)\big) + 17 \operatorname{mult}_{E'} t - 16 \operatorname{mult}_{E'} \mathcal{V}_\infty. \tag{9.12}$$

Combining (9.10), (9.11), and (9.12), we obtain

$$r' \operatorname{mult}_E f_V \le Jr'\big(a(E, V) - r a(F, Y)\big) + 18J \operatorname{mult}_{E'} t + r'r \operatorname{mult}_F \bar{p}_{\alpha_0} - 17J \operatorname{mult}_{E'} \mathcal{V}_\infty.$$

By (9.9), $18J \operatorname{mult}_{E'} t = \frac{18}{42} Jr'\Lambda_3$, and $a(E, V) - r a(F, Y) = \Lambda_1 - \frac{17}{42}\Lambda_3$. Since $\mathcal{V}_\infty \ge 0$, we have $\operatorname{mult}_{E'} \mathcal{V}_\infty \ge 0$. Dividing by r' and using (9.7), we obtain

$$\operatorname{mult}_E f_V \le J\Lambda_1 + \frac{J}{42}\Lambda_3 + r \operatorname{mult}_F \bar{p}_{\alpha_0} \le J\Lambda_1 + k\Lambda_2 + \frac{J}{42}\Lambda_3,$$

which is (9.8).

By **Steps 2–4**, $\operatorname{mult}_E f_V \le Ja(E, V)$ for any prime divisor E over V which is horizontal over Y, and (9.8) holds for any prime divisor E over V which is vertical over Y. Thus $(P, \frac{42}{\ell} D_f)$ is lc on $P \cap (y_j \ne 0)$ by **Step 1**. □

9.3. **The locus $y_1 = \cdots = y_{18} = 0$.** In this subsection, we prove Proposition 9.5, and then Theorem 9.1. The locus $P \cap (y_1 = \cdots = y_{18} = 0)$ is disjoint from the charts $y_j \ne 0$, where Proposition 9.4 uses the morphism to the y-coordinates. Instead, we blow up the smooth punctured cone over P along the preimage of $P \cap (y_1 = \cdots = y_{18} = 0)$, and reduce the problem by inversion of adjunction (Theorem 2.5) to the exceptional divisor.

Proposition 9.5. *Notation and conditions as in Set-up 3.3. Let $\ell > 0$ be an integer and f a non-zero element of $(R_P)_\ell$. Then $(P, \frac{42}{\ell} D_f)$ is lc near $P \cap (y_1 = \cdots = y_{18} = 0)$.*

Proof. The proof is long, so we divide it into several steps. Let $\widehat{P} := \operatorname{Spec} R_P\backslash\{0\}$, $q_P : \widehat{P} \to P$ the natural morphism, and $Z := \widehat{P} \cap (y_1 = \cdots = y_{18} = 0)$. For a regular function $g \ne 0$ on a smooth variety, $\operatorname{div} g$ denotes the divisor of zeros of g. We use Set-up 6.1 with $\mathbf{w}_i := \boldsymbol{v}_i$ for $1 \le i \le 11$, any four of which are linearly independent by Proposition 3.4(1). Let m_α, p_α, j_α, k_α, J, and k be as in Definition-Lemma 9.3 for f, so that $f = \sum_\alpha p_\alpha m_\alpha$ and $\ell = 43J + 42k$.

Step 1. In this step we reduce Proposition 9.5 to a statement on $\widehat{P}$.

By Proposition 3.4(1), R_P is a graded normal domain with $(R_P)_0 = \mathbb{C}$, generated by $u_1, \dots, u_{11}, y_1, \dots, y_{18}$. Since P is quasismooth (Proposition 3.4(5)), $\operatorname{Spec} R_P\backslash\{0\}$ is smooth. Let z be one of $u_1, \dots, u_{11}, y_1, \dots, y_{18}$ and w the weight of z. By Lemma 3.7(2), the quotient morphism $\widehat{P} \cap (z = 1) \to P \cap (z \ne 0)$ by μ_w is étale in codimension one, so the pullback of D_f to $\widehat{P} \cap (z = 1)$ is $\operatorname{div}(f|_{\widehat{P} \cap (z=1)})$. Let $\mathbb{G}_m$ act on $\widehat{P}$ by $t \cdot (u_i, y_j) = (t^{43d_i} u_i, t^{42} y_j)$. The morphism

$$(\widehat{P} \cap (z = 1)) \times \mathbb{G}_m \to \widehat{P} \cap (z \ne 0), \quad (\xi, t) \to t \cdot \xi,$$

is finite étale and pulls f back to $t^\ell f(\xi)$. Thus $q_P^* D_f = \operatorname{div} f$. Since

$$q_P^{-1}\big(P \cap (y_1 = \cdots = y_{18} = 0)\big) = Z,$$

Lemma 2.10, applied with R_P in place of R, a point in place of Y, and $P \cap (y_1 = \cdots = y_{18} = 0)$ in place of Σ, shows that it suffices to prove that

$$\Big(\widehat{P}, \frac{42}{\ell} \operatorname{div} f\Big) \text{ is lc near } Z.$$

Step 2. In this step we describe Z and the blow-up of $\widehat{P}$ along Z.

We have

$$R_P/(y_1,\dots,y_{18}) = \mathbb{C}[u_1,\dots,u_{11}]\Big/\Big(\sum_{i=1}^{11} \boldsymbol{v}_i u_i^{e_i}\Big),$$

and $u_i \to U_i$ defines an isomorphism of rings $R_P/(y_1,\dots,y_{18}) \cong R_\infty$ which multiplies degrees by 43. By Proposition 6.2(1)(3), R_∞ is a normal domain, $\dim \operatorname{Spec} R_\infty = \dim V_\infty + 1 = 7$, and $\operatorname{Spec} R_\infty\backslash\{0\}$ is smooth. Since $\dim \widehat{P} = \dim P + 1 = 25$, the isomorphism $Z \cong \operatorname{Spec} R_\infty\backslash\{0\}$, $u_i \to U_i$, shows that Z is a smooth irreducible closed subvariety of $\widehat{P}$ of codimension 18, whose ideal sheaf I_Z is generated by $y_1,\dots,y_{18}$. Hence at any point of Z, the functions $y_1,\dots,y_{18}$ are part of a regular system of parameters of $\widehat{P}$, and I_Z/I_Z^2 is a free $\mathcal{O}_Z$-module with basis $\eta_1,\dots,\eta_{18}$, where η_j is the image of y_j.

Let $\rho : \widehat{P}' \to \widehat{P}$ be the blow-up along Z and G the exceptional divisor. Then $\widehat{P}'$ is smooth, $G \cong Z \times \mathbb{P}^{17}$, where $\eta_1,\dots,\eta_{18}$ are the homogeneous coordinates of $\mathbb{P}^{17}$, and

$$K_{\widehat{P}'} = \rho^* K_{\widehat{P}} + 17G \tag{9.13}$$

[Har77, Theorem II.8.24 and Exercise II.8.5(b)]. For $1 \le j_0 \le 18$, let $\widehat{P}'_{j_0} \subset \widehat{P}'$ be the open subset on which $I_Z\mathcal{O}_{\widehat{P}'}$ is generated by y_{j_0}. On $\widehat{P}'_{j_0}$, the functions $y'_j := y_j/y_{j_0}$ are regular, $G \cap \widehat{P}'_{j_0} = (y_{j_0} = 0)$, and $G_{j_0} := G \cap \widehat{P}'_{j_0}$ is the open subset $Z \times (\eta_{j_0} \ne 0) \cong Z \times \mathbb{A}^{17}$ of G, on which $y'_j|_{G_{j_0}} = \eta_j/\eta_{j_0}$.

Step 3. In this step we show that $\rho^* \operatorname{div} f = D' + kG$ for a divisor $D' \ge 0$ on $\widehat{P}'$ which does not contain G, and we compute $D'|_G$.

For any α, $k_\alpha = (\ell - 43j_\alpha)/42 \ge k$, with equality if and only if $j_\alpha = J$. Let

$$\bar{f} := \sum_{j_\alpha = J} p_\alpha(\eta_1,\dots,\eta_{18}) m_\alpha|_Z \tag{9.14}$$

be the initial form of f along Z, a homogeneous polynomial of degree k in $\eta_1,\dots,\eta_{18}$ with coefficients regular functions on Z, and let $\operatorname{div} \bar{f}$ be the divisor of zeros of $\bar{f}$ on G, that is, the divisor of $\bar{f}/\eta_{j_0}^k$ on G_{j_0} for any j_0. On $\widehat{P}'_{j_0}$, we have

$$p_\alpha(y_1,\dots,y_{18}) = y_{j_0}^{k_\alpha} p_\alpha(y'_1,\dots,y'_{18})$$

with $y'_{j_0} := 1$. Thus $f \circ \rho = y_{j_0}^k f'_{j_0}$, where

$$f'_{j_0} := \sum_\alpha y_{j_0}^{k_\alpha - k} p_\alpha(y'_1,\dots,y'_{18})(m_\alpha \circ \rho), \qquad f'_{j_0}|_{G_{j_0}} = \frac{\bar{f}}{\eta_{j_0}^k}.$$

We show that $\bar{f} \ne 0$. First, the restrictions $m_\alpha|_Z$ with $j_\alpha = J$ are linearly independent over $\mathbb{C}$. Since Z is a non-empty open subset of the integral scheme $\operatorname{Spec} R_P/(y_1,\dots,y_{18}) \cong \operatorname{Spec} R_\infty$ (**Step 2**), restriction to Z is injective on $R_P/(y_1,\dots,y_{18})$, so it suffices to show that if $c_\alpha \in \mathbb{C}$ $(j_\alpha = J)$ satisfy $\sum_{j_\alpha=J} c_\alpha m_\alpha \in (y_1,\dots,y_{18})R_P$, then all c_α are zero. Suppose not, and let $c_{\alpha_0} \ne 0$. Write $\sum_{j_\alpha=J} c_\alpha m_\alpha = \sum_{j=1}^{18} y_j h_j$ with $h_j \in (R_P)_{43J-42}$, where $h_j = 0$ if $43J < 42$. Replacing each non-zero h_j by the expression of Definition-Lemma 9.3 and grouping the terms by their monomial factor, we obtain $\sum_j y_j h_j = \sum_\beta q_\beta n_\beta$, where the n_β are distinct monomials as in Definition-Lemma 9.3 and the q_β are non-zero homogeneous polynomials in $y_1,\dots,y_{18}$ of positive degree. Since $\deg n_\beta = 43J - 42\deg q_\beta < 43J$, no n_β is equal to an m_α with $j_\alpha = J$. Thus

$$c_{\alpha_0} m_{\alpha_0} = -\sum_{\alpha \ne \alpha_0, j_\alpha = J} c_\alpha m_\alpha + \sum_\beta q_\beta n_\beta,$$

and after deleting the terms with $c_\alpha = 0$, the right-hand side is an expression as in Definition-Lemma 9.3 which does not involve m_{α_0}. Since R_P is a domain and $u_i \ne 0$ in R_P (the equations of P have degree $1806 > 43d_i$), $c_{\alpha_0} m_{\alpha_0} \ne 0$. This contradicts the uniqueness in Definition-Lemma 9.3. Now choose α_1 with $j_{\alpha_1} = J$ and a monomial η^γ of degree k whose coefficient in

$p_{\alpha_1}(\eta)$ is non-zero. The coefficient of η^γ in $\bar{f}$ is a linear combination of the $m_\alpha|_Z$ with $j_\alpha = J$ in which $m_{\alpha_1}|_Z$ has a non-zero coefficient, so it is non-zero. Hence $\bar{f} \neq 0$.

Since G is irreducible and $f'_{j_0}|_{G_{j_0}} \neq 0$ for any j_0, we get $\operatorname{mult}_G \rho^* \operatorname{div} f = k$. Thus $D' := \rho^* \operatorname{div} f - kG \geq 0$ is a divisor which does not contain G, and $D'|_G = \operatorname{div} \bar{f}$.

Step 4. In this step we show that $(G, \frac{42}{\ell} \operatorname{div} \bar{f})$ is lc when $J > 0$.

Fix $1 \leq j_0 \leq 18$, and let $\mathbb{A}^{17}$ be the affine space with coordinates η_j/η_{j_0}, $j \neq j_0$, so that $G_{j_0} \cong Z \times \mathbb{A}^{17}$. For $j_\alpha = J$, let

$$\varphi_\alpha := p_\alpha\Big(\frac{\eta_1}{\eta_{j_0}}, \dots, \frac{\eta_{18}}{\eta_{j_0}}\Big),$$

a polynomial on $\mathbb{A}^{17}$. Then $\varphi_\alpha \neq 0$, $\deg \varphi_\alpha \leq k$, and

$$\frac{\bar{f}}{\eta_{j_0}^k} = \sum_{j_\alpha = J} \varphi_\alpha m_\alpha|_Z.$$

Let $m_\alpha(U_1, \dots, U_{11}) \in R_\infty$ be the image of m_α under the isomorphism $R_P/(y_1, \dots, y_{18}) \cong R_\infty$, $u_i \to U_i$, of **Step 2**. Since $\deg m_\alpha = 43J$ and $\deg u_i = 43d_i$, $m_\alpha(U_1, \dots, U_{11})$ is a monomial of degree J in R_∞, in which the exponents of U_1, U_2, U_3 are at most one and the exponent of U_4 is at most two. Let $T := V_\infty \times \mathbb{A}^{17}$,

$$s := \sum_{j_\alpha = J} \varphi_\alpha m_\alpha(U_1, \dots, U_{11}),$$

and $D_s \geq 0$ the divisor of zeros of s on T.

In case (b) of Lemma 8.1(2), take $\mathbb{A}^{17}$ in place of Y, $\frac{6J}{7}$ in place of q, the coefficients φ_α for $j_\alpha = J$, and zero coefficients for the other monomials. The assumption of Lemma 8.1(2) then holds by Proposition 7.4(1). For any prime divisor F over $\mathbb{A}^{17}$ and any α with $j_\alpha = J$, we have

$$\operatorname{mult}_F \varphi_\alpha \leq k a(F, \mathbb{A}^{17}).$$

If $k \geq 1$, then this is Lemma 9.2(1). If $k = 0$, then φ_α is a non-zero constant, and $\operatorname{mult}_F \varphi_\alpha = 0$.

Let E be a prime divisor over T. If E is horizontal over $\mathbb{A}^{17}$, then $\operatorname{mult}_E D_s \leq \frac{6J}{7} a(E, T)$, so $a(E, T) \geq 0$. If E is vertical over $\mathbb{A}^{17}$, with image F over $\mathbb{A}^{17}$ and ramification index r (Definition-Lemma 2.6), then Lemma 8.1(1)(2) implies that $a(E, T) - r a(F, \mathbb{A}^{17}) \geq 0$ and

$$\operatorname{mult}_E D_s \leq \frac{6J}{7}\big(a(E, T) - r a(F, \mathbb{A}^{17})\big) + rk a(F, \mathbb{A}^{17}),$$

where $a(F, \mathbb{A}^{17}) > 0$ since $\mathbb{A}^{17}$ is smooth. In both cases,

$$\operatorname{mult}_E D_s \leq \max\Big\{\frac{6J}{7}, k\Big\} a(E, T) \quad \text{and} \quad a(E, T) \geq 0. \tag{9.15}$$

Since $\ell = 43J + 42k$, $\frac{42}{\ell} \max\{\frac{6J}{7}, k\} \leq 1$. By Proposition 6.2(3), T has quotient singularities, so D_s is $\mathbb{Q}$-Cartier [KM98, Proposition 5.15]. Thus (9.15) implies that

$$a\Big(E, T, \frac{42}{\ell} D_s\Big) = a(E, T) - \frac{42}{\ell} \operatorname{mult}_E D_s \geq 0,$$

that is, $(T, \frac{42}{\ell} D_s)$ is lc.

Let $q_\infty : \operatorname{Spec} R_\infty \backslash \{0\} \to V_\infty$ be the natural morphism. By Proposition 6.2(3), R_∞, generated by $U_1, \dots, U_{11}$ of degrees $d_1, \dots, d_{11}$, satisfies the assumptions of Lemma 2.10. For $1 \leq i \leq 11$, the quotient morphism

$$\operatorname{Spec} R_\infty \cap (U_i = 1) \to V_\infty \cap (U_i \neq 0)$$

by μ_{d_i} is étale in codimension one, and the finite étale morphism

$$(\operatorname{Spec} R_\infty \cap (U_i = 1)) \times \mathbb{G}_m \to \operatorname{Spec} R_\infty \cap (U_i \neq 0), \quad (\xi, t) \to t \cdot \xi,$$

pulls s back to $t^J s$. Thus

$$(q_\infty \times \mathrm{id}_{\mathbb{A}^{17}})^* D_s = \operatorname{div} s.$$

Lemma 2.10, applied with R_∞ in place of R, $\mathbb{A}^{17}$ in place of Y, and T in place of Σ, implies that

$$\Big((\operatorname{Spec} R_\infty\backslash\{0\})\times\mathbb{A}^{17},\frac{42}{\ell}\operatorname{div} s\Big)\text{ is lc.}$$

Under the isomorphism $Z\cong\operatorname{Spec} R_\infty\backslash\{0\}$ of **Step 2**, $m_\alpha|_Z$ corresponds to $m_\alpha(U_1,\dots,U_{11})$, so s corresponds to $\bar{f}/\eta_{j_0}^k$ on G_{j_0}. Hence $(G_{j_0},\frac{42}{\ell}\operatorname{div}\bar{f})$ is lc. Since $G_1,\dots,G_{18}$ cover G, $(G,\frac{42}{\ell}\operatorname{div}\bar{f})$ is lc.

Step 5. In this step we show that $(G,\frac{42}{\ell}\operatorname{div}\bar{f})$ is lc when $J=0$.

If $J=0$, then f has a single term p_1m_1 with $m_1=1$, $f=p_1$ is a homogeneous polynomial of degree $k=\frac{\ell}{42}\geq 1$ in $y_1,\dots,y_{18}$, so $\frac{42}{\ell}=\frac{1}{k}$, and $\bar{f}=p_1(\eta_1,\dots,\eta_{18})$. Fix $1\leq j_0\leq 18$, and let $\mathbb{A}^{17}$ and φ_1 be defined as in **Step 4**, so that $\varphi_1\neq 0$, $\deg\varphi_1\leq k$, and $\operatorname{div}\bar{f}|_{G_{j_0}}$ is the pullback of $\operatorname{div}\varphi_1$ under the projection $Z\times\mathbb{A}^{17}\to\mathbb{A}^{17}$. By Lemma 9.2(1), $\operatorname{mult}_F\varphi_1\leq ka(F,\mathbb{A}^{17})$ for any prime divisor F over $\mathbb{A}^{17}$. Thus $(\mathbb{A}^{17},\frac{1}{k}\operatorname{div}\varphi_1)$ is lc. Let $\psi:A'\to\mathbb{A}^{17}$ be a log resolution of $(\mathbb{A}^{17},\operatorname{div}\varphi_1)$ such that the strict transform of $\operatorname{div}\varphi_1$ is smooth. Then $\mathrm{id}_Z\times\psi$ is a log resolution of $(G_{j_0},\operatorname{div}\bar{f}|_{G_{j_0}})$ such that the strict transform of $\operatorname{div}\bar{f}|_{G_{j_0}}$ is smooth. The exceptional prime divisors of $\mathrm{id}_Z\times\psi$ are $Z\times E'$, where E' runs over the exceptional prime divisors of ψ. Since

$$K_{Z\times A'/Z\times\mathbb{A}^{17}}=\operatorname{pr}_2^*K_{A'/\mathbb{A}^{17}}\quad\text{and}\quad\operatorname{div}\bar{f}|_{G_{j_0}}=\operatorname{pr}_2^*\operatorname{div}\varphi_1,$$

where pr_2 denotes the second projections, we have

$$a\Big(Z\times E',G_{j_0},\frac{1}{k}\operatorname{div}\bar{f}|_{G_{j_0}}\Big)=a\Big(E',\mathbb{A}^{17},\frac{1}{k}\operatorname{div}\varphi_1\Big)\geq 0.$$

Moreover, the coefficients of $\frac{1}{k}\operatorname{div}\bar{f}|_{G_{j_0}}$ are those of $\frac{1}{k}\operatorname{div}\varphi_1$, hence at most one. By [KM98, Corollary 2.32(2)], $(G_{j_0},\frac{42}{\ell}\operatorname{div}\bar{f})$ is lc. Since $G_1,\dots,G_{18}$ cover G, $(G,\frac{42}{\ell}\operatorname{div}\bar{f})$ is lc.

Step 6. In this step we finish the proof.

By **Steps 4 and 5**, $(G,\frac{42}{\ell}\operatorname{div}\bar{f})$ is lc. Since $\widehat{P}'$ and G are smooth, $(K_{\widehat{P}'}+G)|_G=K_G$. Since D' is a Cartier divisor which does not contain G, the different of $(\widehat{P}',G+\frac{42}{\ell}D')$ on G is $\frac{42}{\ell}D'|_G$ [FST11, 11.1(v)], which equals $\frac{42}{\ell}\operatorname{div}\bar{f}$ by **Step 3**. By Theorem 2.5, $(\widehat{P}',G+\frac{42}{\ell}D')$ is lc near G. By (9.13) and **Step 3**,

$$K_{\widehat{P}'}+\frac{42}{\ell}D'+\Big(\frac{42k}{\ell}-17\Big)G=\rho^*\Big(K_{\widehat{P}}+\frac{42}{\ell}\operatorname{div}f\Big),$$

and $18-\frac{42k}{\ell}\geq 17>0$ since $42k\leq\ell$. Since ρ is proper and $\rho^{-1}(Z)=G$, there exists an open neighborhood $\widehat{P}^\circ$ of Z such that $(\widehat{P}',G+\frac{42}{\ell}D')$ is lc on $\rho^{-1}(\widehat{P}^\circ)$. Let E be a prime divisor over $\widehat{P}$ whose center on $\widehat{P}$ intersects $\widehat{P}^\circ$. Then the center of E on $\widehat{P}'$ intersects $\rho^{-1}(\widehat{P}^\circ)$, and by [KM98, Lemma 2.30],

$$a\Big(E,\widehat{P},\frac{42}{\ell}\operatorname{div}f\Big)=a\Big(E,\widehat{P}',G+\frac{42}{\ell}D'\Big)+\Big(18-\frac{42k}{\ell}\Big)\operatorname{mult}_EG\geq 0.$$

Thus $(\widehat{P},\frac{42}{\ell}\operatorname{div}f)$ is lc on $\widehat{P}^\circ$, and Proposition 9.5 follows from **Step 1**. □

Proof of Theorem 9.1. By Proposition 9.5, the locus $P\cap(y_1=\cdots=y_{18}=0)$ has an open neighborhood P° such that $(P,\frac{42}{\ell}D_f)$ is lc on P°. By Proposition 9.4, $(P,\frac{42}{\ell}D_f)$ is lc on $P\cap(y_j\neq 0)$ for any $1\leq j\leq 18$. Since P° and $P\cap(y_j\neq 0)$, $1\leq j\leq 18$, cover P, the center on P of any prime divisor over P intersects one of them. Thus $(P,\frac{42}{\ell}D_f)$ is lc. □

10. Canonicity of movable linear systems

In this section, we prove Theorem 10.1, which bounds the multiplicities of movable linear systems on X along exceptional prime divisors. Centers intersecting U are treated in Section 5.

Theorem 10.1. *Notation and conditions as in Set-up 3.3. Let $\ell > 0$ be an integer, $\mathcal{M} \subset |\ell H|$ a movable linear system, and E an exceptional prime divisor over X. Then*

$$\frac{26}{\ell} \operatorname{mult}_E \mathcal{M} \leq a(E, X) - 1,$$

and the inequality is strict if $\operatorname{center}_X E \subset P$.

Part (b) of the sketch of the proof in Section 1 explains why centers contained in P need a margin that terminality of X does not provide. The following lemma shows that the multiplicities of the semi-invariant coordinates along E have fractional parts governed by one common congruence.

Lemma 10.2. *Let $\widetilde{Y}$ be a smooth variety with a faithful action of μ_d, $Y := \widetilde{Y}/\mu_d$, and let $z_1, \ldots, z_s$ be non-zero regular functions on $\widetilde{Y}$ with $\zeta^* z_i = \zeta^{\chi_i} z_i$ for any $\zeta \in \mu_d$, where $\chi_1, \ldots, \chi_s$ are integers. Let $Y' \to Y$ be a birational morphism from a normal variety, $\widetilde{Y}'$ the normalization of the component of $Y' \times_Y \widetilde{Y}$ dominating $\widetilde{Y}$, $\widetilde{E}$ a prime divisor on $\widetilde{Y}'$, and r the ramification index of $\widetilde{Y}' \to Y'$ along $\widetilde{E}$. Then there exists an integer $0 \leq b < d$ such that*

$$\frac{1}{r} \operatorname{mult}_{\widetilde{E}} z_i - \frac{b\chi_i}{d} \in \mathbb{Z}$$

for any $1 \leq i \leq s$.

Proof. The group μ_d acts on $Y' \times_Y \widetilde{Y}$ through the second factor and preserves the unique component dominating $\widetilde{Y}$. Hence μ_d acts on $\widetilde{Y}'$, the morphism $\widetilde{Y}' \to Y'$ is μ_d-invariant, and $\zeta^* z_i = \zeta^{\chi_i} z_i$ on $\widetilde{Y}'$ for any $\zeta \in \mu_d$ and any $1 \leq i \leq s$. Let $Y'_0 \subset Y'$ be the open subset over which $Y' \to Y$ is an isomorphism. Over Y'_0, the morphism $\widetilde{Y}' \to Y'$ coincides with the quotient morphism $\widetilde{Y} \to Y$ over the image of Y'_0. Thus μ_d acts faithfully on $\widetilde{Y}'$, and the induced morphism $\widetilde{Y}'/\mu_d \to Y'$ is finite and birational. Since Y' is normal, $\widetilde{Y}'/\mu_d \to Y'$ is an isomorphism, that is, $\widetilde{Y}' \to Y'$ is the quotient by μ_d.

Let $E \subset Y'$ be the image of $\widetilde{E}$ and $\mathcal{O}_E$ the local ring of Y' at the generic point of E. Let $\mathcal{O}$ be the local ring of $\widetilde{Y}'$ at the generic point of $\widetilde{E}$, $\mathfrak{m}$ the maximal ideal of $\mathcal{O}$, π a generator of $\mathfrak{m}$, and $\kappa := \mathcal{O}/\mathfrak{m}$. Since Y' and $\widetilde{Y}'$ are normal, $\mathcal{O}_E$ and $\mathcal{O}$ are discrete valuation rings. The semilocal ring of $\widetilde{Y}'$ along the preimage of E is the integral closure of $\mathcal{O}_E$ in the fraction field of $\mathcal{O}$, and μ_d acts on it faithfully with invariant ring $\mathcal{O}_E$.

Let $I \subset \mu_d$ be the subgroup of the elements ζ with $\zeta(\widetilde{E}) = \widetilde{E}$ that act trivially on $\widetilde{E}$. Any $\zeta \in I$ acts on $\mathcal{O}$ and acts trivially on κ. Since κ has characteristic zero, we have $|I| = r$ by [Ser79, Chapter I, Section 7, Proposition 21 and Corollary]. By [Ser79, Chapter II, Section 3, Theorem 1 and Corollary 4], the extension $\widehat{\mathcal{O}}_E \subset \widehat{\mathcal{O}}$ of the completions of $\mathcal{O}_E$ and $\mathcal{O}$ is Galois with group the stabilizer of $\widetilde{E}$ in μ_d, its ramification index is r, and its residue field is κ. Since

$$\widehat{\mathcal{O}}/\widehat{\mathfrak{m}}^2 = \mathcal{O}/\mathfrak{m}^2,$$

the space $\mathfrak{m}/\mathfrak{m}^2$ does not change, and neither does I, the subgroup of the stabilizer acting trivially on κ. Since κ has characteristic zero, by [Ser79, Chapter IV, Section 2, Proposition 7 and Corollaries 1 and 2] applied to $\widehat{\mathcal{O}}_E \subset \widehat{\mathcal{O}}$, the map

$$\psi : I \to \kappa^*, \quad \zeta \to \text{the class of } \frac{\zeta^* \pi}{\pi} \text{ in } \kappa$$

is an injective group homomorphism whose image consists of roots of unity. Since $\mathbb{C} \subset \kappa$ is algebraically closed, any root of unity in κ belongs to $\mathbb{C}$. Since μ_d is cyclic and $|I| = r$, I is the unique subgroup μ_r of μ_d of order r, which may be a proper subgroup of μ_d. Thus ψ is an injective homomorphism $\mu_r \to \mu_r \subset \mathbb{C}^*$, hence an automorphism of μ_r. Therefore there exists an integer n coprime to r such that $\psi(\zeta) = \zeta^n$ for any $\zeta \in \mu_r$, that is,

$$\zeta^* \pi - \zeta^n \pi \in \mathfrak{m}^2 \quad \text{for any } \zeta \in \mu_r.$$

Fix $1 \le i \le s$ and let $m_i := \operatorname{mult}_{\widetilde{E}} z_i$. Since z_i is non-zero and $\widetilde{Y}' \to \widetilde{Y}$ is birational, we may write $z_i = v_i \pi^{m_i}$, where v_i is a unit of $\mathcal{O}$. Let $\zeta \in \mu_r$. Since ζ acts trivially on κ, we have $\zeta^* v_i \in v_i(1+\mathfrak{m})$, and $\zeta^*\pi \in \zeta^n \pi(1+\mathfrak{m})$. Thus

$$\zeta^{\chi_i} v_i \pi^{m_i} = \zeta^* z_i = (\zeta^* v_i)(\zeta^*\pi)^{m_i} \in \zeta^{nm_i} v_i \pi^{m_i}(1+\mathfrak{m}).$$

Dividing by $v_i\pi^{m_i}$, we obtain

$$\zeta^{\chi_i} - \zeta^{nm_i} \in \mathfrak{m} \cap \mathbb{C} = \{0\}.$$

Since $\zeta \in \mu_r$ is arbitrary,

$$\chi_i \equiv nm_i \pmod{r}.$$

Let n' be an integer such that $nn' \equiv 1 \pmod{r}$. Then $m_i \equiv n'\chi_i \pmod{r}$. Since $r = |I|$ divides d, $n'd/r$ is an integer. Let b be the remainder of $n'd/r$ modulo d, so that $0 \le b < d$. Then

$$\frac{m_i}{r} - \frac{b\chi_i}{d} = \frac{m_i - n'\chi_i}{r} + \frac{(n'd/r - b)\chi_i}{d} \in \mathbb{Z}.$$

Since b does not depend on i, Lemma 10.2 follows. □

Proof of Theorem 10.1. If $\operatorname{center}_X E \cap U \ne \emptyset$, then the inequality is Theorem 5.1. Since $U = X \setminus P$, we may assume that $\operatorname{center}_X E \subset P$, and we prove the strict inequality. The proof is long, so we divide it into several steps.

Step 1. In this step we use inversion of adjunction to reduce the theorem to the inequality

$$16a(E,X) + 26 \operatorname{mult}_E P > 42. \tag{10.1}$$

By Proposition 3.4(4), any member of $|\ell H|$ is the divisor of zeros D_s of a non-zero element $s \in R_\ell = H^0(X, \mathcal{O}_X(\ell H))$. Since P is a prime divisor and $\mathcal{M}$ has no fixed components, there exists $D_s \in \mathcal{M}$ with $P \not\subset \operatorname{Supp} D_s$. If $s \in xR_{\ell-1}$, then $D_s \ge P$. Thus the image f of s in $(R_P)_\ell$ is non-zero.

Since X is $\mathbb{Q}$-factorial (Proposition 3.4(3)), D_s is $\mathbb{Q}$-Cartier. Here $D_s|_P$ is the pullback of D_s to P [FST11, 11.1(v)]. We show that $D_s|_P = D_f$. Let z be one of the coordinates $u_1, \dots, u_{11}, y_1, \dots, y_{18}$, w the weight of z, and $g : \widetilde{X}_z \to X \cap (z \ne 0)$ the quotient by μ_w (Lemma 3.7(2)). By Lemma 3.7(1), $\widetilde{P}_z = (x = 0) \subset \widetilde{X}_z$ is smooth. The coordinate rings of $\widetilde{X}_z$ and $\widetilde{P}_z$ are $R/(z-1)R$ and $R_P/(z-1)R_P$. Since R and R_P are graded domains (Proposition 3.4(1)) and z has positive weight, the product of $z-1$ with a non-zero element of R or of R_P has non-zero homogeneous components in two different degrees. Hence any non-zero homogeneous element of R (resp. R_P) is a non-zero regular function on $\widetilde{X}_z$ (resp. $\widetilde{P}_z$). In particular, f is a non-zero regular function on $\widetilde{P}_z$.

On $X \cap (z \ne 0)$, the rational functions x^w/z and s^w/z^ℓ have divisors wP and wD_s, since z has no zeros there. Their pullbacks to $\widetilde{X}_z$, where $z = 1$, are x^w and s^w. Since the pullback under the finite morphism g of the divisor of a rational function ϕ is $\operatorname{div}(g^*\phi)$, we obtain $g^*P = \operatorname{div}(x)$ and $g^*D_s = \operatorname{div}(s)$. Let $g_P : \widetilde{P}_z \to P \cap (z \ne 0)$ be the quotient by μ_w. Since the rational function f^w/z^ℓ on $P \cap (z \ne 0)$ has divisor wD_f and pulls back to f^w, we have $g_P^* D_f = \operatorname{div}(f)$. Thus

$$g_P^*(D_s|_P) = (g^* D_s)|_{\widetilde{P}_z} = \operatorname{div}(f) = g_P^* D_f.$$

The first equality holds since $g \circ \iota_{\widetilde{P}_z} = \iota_P \circ g_P$, where $\iota_{\widetilde{P}_z}$ and ι_P are the inclusions of $\widetilde{P}_z$ into $\widetilde{X}_z$ and of $P \cap (z \ne 0)$ into $X \cap (z \ne 0)$, the second holds since $g^* D_s = \operatorname{div}(s)$ and f is non-zero on $\widetilde{P}_z$, and the third is the computation above. Since pullback under the finite surjective morphism g_P is injective on $\mathbb{R}$-divisors, $D_s|_P = D_f$ over $P \cap (z \ne 0)$. Any point of P has a non-zero coordinate among $u_1, \dots, u_{11}, y_1, \dots, y_{18}$, so the open subsets $P \cap (z \ne 0)$ cover P, and $D_s|_P = D_f$.

By Lemma 3.7(3), P is normal and $(K_X + P)|_P = K_P$. Since D_s and P have no common components, [FST11, 11.1(v)] implies that the different of $(X, P + \frac{42}{\ell} D_s)$ on P is $\frac{42}{\ell} D_s|_P = \frac{42}{\ell} D_f$. By Theorem 9.1, $(P, \frac{42}{\ell} D_f)$ is lc. The variety X is normal (Proposition 3.4(1)), $\frac{42}{\ell} D_s \ge 0$ has no

common components with P, and $K_X + P + \frac{42}{\ell} D_s$ is $\mathbb{R}$-Cartier. By Theorem 2.5, $(X, P + \frac{42}{\ell} D_s)$ is lc near P. Since $\operatorname{center}_X E \subset P$,

$$a(E,X) - \operatorname{mult}_E P - \frac{42}{\ell} \operatorname{mult}_E D_s = a\Big(E, X, P + \frac{42}{\ell} D_s\Big) \geq 0.$$

Since $\operatorname{mult}_E \mathcal{M} \leq \operatorname{mult}_E D_s$ (Definition 2.2(3)), we obtain

$$a(E,X) - 1 - \frac{26}{\ell} \operatorname{mult}_E \mathcal{M} \geq a(E,X) - 1 - \frac{26}{42}\big(a(E,X) - \operatorname{mult}_E P\big) = \frac{16a(E,X) + 26 \operatorname{mult}_E P}{42} - 1.$$

Thus it suffices to prove (10.1).

Step 2. In this step we choose a quotient chart near the generic point of $\operatorname{center}_X E$ and derive the congruence (10.3) from Lemma 10.2.

Since the open subsets $X \cap (z \neq 0)$, where z runs over $u_1, \dots, u_{11}, y_1, \dots, y_{18}$, cover $P \supset \operatorname{center}_X E$, we may choose $z \in \{u_1, \dots, u_{11}, y_1, \dots, y_{18}\}$ such that z does not vanish identically on $\operatorname{center}_X E$. Let w be the weight of z and $g : \widetilde{X}_z \to X \cap (z \neq 0)$ the quotient by μ_w as in **Step 1**. Any $\zeta \in \mu_w$ acts on $\widetilde{X}_z$ by $\zeta^* c = \zeta^q c$ for any coordinate c of weight q. Since x has weight 1, the action of μ_w is faithful. By **Step 1**, each of the 29 coordinates other than z is a non-zero regular function on $\widetilde{X}_z$.

Let $Y' \to X \cap (z \neq 0)$ be a birational morphism from a normal variety on which E is a prime divisor, $\widetilde{Y}'$ the normalization of the component of $Y' \times_{X \cap (z \neq 0)} \widetilde{X}_z$ dominating $\widetilde{X}_z$, $\widetilde{E}$ a prime divisor on $\widetilde{Y}'$ whose image on Y' is E, and r the ramification index of $\widetilde{Y}' \to Y'$ along $\widetilde{E}$, which is the ramification index in Lemma 2.3 applied to g. Since g is étale in codimension one, $K_{\widetilde{X}_z} = g^* K_X$ by the Hurwitz formula. By Lemma 2.3 with $B = 0$, and since $g^* P = \operatorname{div}(x)$ by **Step 1**,

$$a(\widetilde{E}, \widetilde{X}_z) = r a(E,X) \quad \text{and} \quad \operatorname{mult}_{\widetilde{E}} x = \operatorname{mult}_{\widetilde{E}} g^* P = r \operatorname{mult}_E P. \tag{10.2}$$

By Lemma 10.2, applied to $\widetilde{X}_z$ with the action of μ_w ($\widetilde{X}_z$ is smooth by Lemma 3.7(1)), to the 29 coordinates other than z, and to their weights, there exists an integer $0 \leq b < w$ such that, with $\theta := b/w \in [0,1)$,

$$\frac{1}{r} \operatorname{mult}_{\widetilde{E}} c - q\theta \in \mathbb{Z} \tag{10.3}$$

for any coordinate $c \neq z$, where q is the weight of c. For $c = x$, (10.2) and (10.3) imply that $\operatorname{mult}_E P - \theta \in \mathbb{Z}$.

Step 3. In this step we bound $a(E,X)$ from below by the multiplicities of the coordinates along $\widetilde{E}$.

Let $\widetilde{V} := \operatorname{center}_{\widetilde{X}_z}(\widetilde{E})$ and let $\mathcal{S}$ be the set of coordinates other than z that vanish on $\widetilde{V}$. A coordinate $c \neq z$ belongs to $\mathcal{S}$ if and only if $\operatorname{mult}_{\widetilde{E}} c > 0$. Since $g(\widetilde{V}) \subset \operatorname{center}_X E \subset P$ and $g^* P = \operatorname{div}(x)$, we have $x \in \mathcal{S}$.

Let $p \in \widetilde{V}$ be a general point. Then a coordinate vanishes at p if and only if it vanishes on $\widetilde{V}$. By Lemma 3.7(1), there exist four coordinates $c_1, \dots, c_4$ other than z that do not vanish at p, and the 25 coordinates other than $z, c_1, \dots, c_4$, minus their values at p, form a regular system of parameters of $\widetilde{X}_z$ at p. The elements of $\mathcal{S}$ vanish at p, so they are among the 25 coordinates other than $z, c_1, \dots, c_4$. Thus $\sum_{c \in \mathcal{S}} \operatorname{div}(c)$ is an snc divisor near p, and $(\widetilde{X}_z, \sum_{c \in \mathcal{S}} \operatorname{div}(c))$ is lc near p by [KM98, Corollary 2.31(3)]. Since $p \in \widetilde{V}$, we have

$$a(\widetilde{E}, \widetilde{X}_z) - \sum_{c \in \mathcal{S}} \operatorname{mult}_{\widetilde{E}} c = a\Big(\widetilde{E}, \widetilde{X}_z, \sum_{c \in \mathcal{S}} \operatorname{div}(c)\Big) \geq 0.$$

Dividing by r and using (10.2), we obtain

$$a(E,X) \geq \operatorname{mult}_E P + \frac{1}{r} \sum_{c \in \mathcal{S} \setminus \{x\}} \operatorname{mult}_{\widetilde{E}} c. \tag{10.4}$$

Step 4. In this step we prove (10.1). For a real number t, we write $\{t\} := t - \lfloor t \rfloor$.

Case 1. $b = 0$. Then $\theta = 0$, and $\operatorname{mult}_E P \in \mathbb{Z}$ by **Step 2**. Since $x \in \mathcal{S}$, $\operatorname{mult}_{\widetilde{E}} x > 0$, so $\operatorname{mult}_E P \geq 1$ by (10.2). Since X is terminal (Proposition 3.4(2)) and E is exceptional over X, $a(E, X) > 1$. Hence

$$16a(E, X) + 26 \operatorname{mult}_E P > 16 + 26 = 42.$$

Case 2. $0 < b < w$. Then $0 < \theta < 1$. Since $\operatorname{mult}_E P - \theta \in \mathbb{Z}$ and $\operatorname{mult}_E P > 0$, we have $\operatorname{mult}_E P \geq \theta$.

Let q be one of the weights 903, 602, 258, 42 such that $\{q\theta\} > 0$. Since $w\theta = b \in \mathbb{Z}$, we have $q \neq w$, so z is not of weight q. Let c be any coordinate of weight q. By (10.3), $\frac{1}{r} \operatorname{mult}_{\widetilde{E}} c - \{q\theta\} \in \mathbb{Z}$, so $\operatorname{mult}_{\widetilde{E}} c$ is not a multiple of r. In particular, $\operatorname{mult}_{\widetilde{E}} c > 0$, so $c \in \mathcal{S}$ and $\frac{1}{r} \operatorname{mult}_{\widetilde{E}} c \geq \{q\theta\}$. The numbers of coordinates of weight 903, 602, 258, and 42 are 3, 5, 3, and 18, respectively, and each of them is at least 3. By (10.4),

$$a(E, X) \geq \operatorname{mult}_E P + 3\big(\{903\theta\} + \{602\theta\} + \{258\theta\} + \{42\theta\}\big).$$

Since $1 + 903 + 602 + 258 + 42 = 1806$ and the weight w of z, which is one of 903, 602, 258, 42, divides 1806,

$$\theta + \{903\theta\} + \{602\theta\} + \{258\theta\} + \{42\theta\} \equiv 1806\theta = \frac{1806}{w} b \equiv 0 \pmod{\mathbb{Z}}.$$

Since $\theta > 0$, the left-hand side is a positive integer, so

$$\{903\theta\} + \{602\theta\} + \{258\theta\} + \{42\theta\} \geq 1 - \theta.$$

Therefore

$$16a(E, X) + 26 \operatorname{mult}_E P \geq 42 \operatorname{mult}_E P + 48(1 - \theta) \geq 42\theta + 48(1 - \theta) = 48 - 6\theta > 42.$$

Thus (10.1) holds in both cases, and **Step 1** implies the strict inequality. □

11. Proof of the main theorem

Proof of Theorem 1.4. Since

$$(e_1, \dots, e_{11}) = (2, 2, 2, 3, 3, 3, 3, 3, 7, 7, 7) = (42/d_1, \dots, 42/d_{11})$$

and $(43d_1, \dots, 43d_{11}) = (903^3, 602^5, 258^3)$, the variety X in Theorem 1.4 is the variety X in Set-up 3.3, and H is the Weil divisor H in Set-up 3.3.

(1) By Proposition 3.4(1)(2)(3), X is a $\mathbb{Q}$-factorial Fano variety of dimension 25 with terminal singularities, $\operatorname{Cl}(X) = \mathbb{Z}[H]$, and $-K_X \sim 26H$.

(2) Following [SZ19, Section 2.1], a *movable boundary* on X is an expression $a\mathcal{M}$, where $a \in \mathbb{Q}$ and $\mathcal{M}$ is a movable linear system on X, and $(X, a\mathcal{M})$ is *canonical* if for any integer $k \gg 0$ and general members $D_1, \dots, D_k$ of $\mathcal{M}$, the pair $(X, \frac{a}{k} \sum_{i=1}^{k} D_i)$ is canonical in the sense of [KM98, Definition 2.34]. By (1), X is a $\mathbb{Q}$-factorial Fano variety of Picard number one with terminal singularities. Thus, by the Noether–Fano criterion [SZ19, Theorem 2.3], X is birationally superrigid in the sense of [SZ19, Section 2.3] provided that $(X, a\mathcal{M})$ is canonical for any movable boundary $a\mathcal{M} \sim_{\mathbb{Q}} -K_X$.

Let $a\mathcal{M} \sim_{\mathbb{Q}} -K_X$ be a movable boundary on X. Since $\operatorname{Cl}(X) = \mathbb{Z}[H]$, there exists an integer ℓ such that $\mathcal{M} \subset |\ell H|$. Since $-K_X \sim 26H$, we have $a\ell[H] = 26[H]$ in $\operatorname{Cl}(X) \otimes \mathbb{Q} = \mathbb{Q}[H]$, so $a\ell = 26$. By Proposition 3.4(4), $|\ell H| = \emptyset$ for $\ell < 0$. Thus $\ell > 0$ and

$$a = \frac{26}{\ell}.$$

Let $h : X' \to X$ be a log resolution of X such that the linear system $h_*^{-1}\mathcal{M}$ is base point free. Such an h exists by [KM98, Theorem 0.2] applied to the base ideal of the strict transform of $\mathcal{M}$ on a resolution of X. Let $k \geq 2a$ be an integer. For general members $D_1, \dots, D_k$ of $\mathcal{M}$, we have $\operatorname{mult}_E D_i = \operatorname{mult}_E \mathcal{M}$ for any h-exceptional prime divisor E and any i, where $\operatorname{mult}_E \mathcal{M}$ is as in Definition 2.2(3), since, for each of the finitely many h-exceptional E, the members D

with $\operatorname{mult}_E D > \operatorname{mult}_E \mathcal{M}$ form a proper closed subset of $\mathcal{M}$. Moreover, $\sum_{i=1}^k h_*^{-1}D_i + \operatorname{Exc}(h)$ is an snc divisor by Bertini's theorem. By Definition 2.2(1) and $\operatorname{mult}_E D_i = \operatorname{mult}_E \mathcal{M}$, we have $h^*D_i = h_*^{-1}D_i + \sum_E (\operatorname{mult}_E \mathcal{M})E$ for any i, where E runs over the h-exceptional prime divisors. Since $K_{X'} = h^*K_X + \sum_E (a(E,X)-1)E$, we obtain

$$K_{X'} + \frac{a}{k}\sum_{i=1}^k h_*^{-1}D_i + \sum_E \big(1 - a(E,X) + a\operatorname{mult}_E \mathcal{M}\big)E = h^*\Big(K_X + \frac{a}{k}\sum_{i=1}^k D_i\Big). \tag{11.1}$$

By Theorem 10.1,

$$a\operatorname{mult}_E \mathcal{M} = \frac{26}{\ell}\operatorname{mult}_E \mathcal{M} \le a(E,X) - 1$$

for any h-exceptional prime divisor E. Thus

$$a\Big(E, X, \frac{a}{k}\sum_{i=1}^k D_i\Big) = a(E,X) - a\operatorname{mult}_E \mathcal{M} \ge 1$$

for any h-exceptional prime divisor E. Moreover, on the left-hand side of (11.1), the coefficient of $h_*^{-1}D_i$ is $\frac{a}{k} \le \frac{1}{2}$ for any i, and the coefficient of E is $1 - a(E,X) + a\operatorname{mult}_E \mathcal{M} \le 0$ for any h-exceptional prime divisor E. Hence, by [KM98, Corollary 2.31(3)] and [KM98, Lemma 2.30], $a(F, X, \frac{a}{k}\sum_{i=1}^k D_i) \ge 1$ for any prime divisor F over X which is exceptional over X'. Therefore $(X, \frac{a}{k}\sum_{i=1}^k D_i)$ is canonical, and $(X, a\mathcal{M})$ is canonical. Thus X is birationally superrigid in the sense of [SZ19, Section 2.3], that is, any birational map from X to a Mori fiber space is an isomorphism.

We show that X is birationally superrigid in the sense of Definition 2.11. Let $f : Y \to T$ be a Mori fiber space such that Y is birational to X. Then the birational map $X \dashrightarrow Y$ is an isomorphism. In particular, $\rho(Y) = 1 = \rho(Y/T)$, so $\rho(T) = 0$ and T is a point, since for $\dim T > 0$ the pullback of an ample divisor on T would be a non-zero nef class on Y that is not ample, contradicting $\rho(Y) = 1$. Thus X is birationally rigid. Moreover, $X \to \operatorname{Spec}\mathbb{C}$ is a Mori fiber space by (1), so any birational map $X \dashrightarrow X$ is an isomorphism, that is, $\operatorname{Bir}(X) = \operatorname{Aut}(X)$. Therefore X is birationally superrigid.

(3) follows from Theorem 4.1. □

Proof of Corollary 1.5. By Theorem 1.4(1)(2), X is a $\mathbb{Q}$-factorial Fano variety of Picard number one with terminal singularities, and X is birationally superrigid. In particular, X is birationally rigid by Definition 2.11. Thus X satisfies the hypotheses of Conjectures 1.1 and 1.3.

Since $\operatorname{Cl}(X) = \mathbb{Z}[H]$, we have $\operatorname{rk}\operatorname{Cl}(X) = 1$. Following [CSZ23, Section 1], a Fano variety with terminal singularities and class group of rank one is birationally rigid if it is not birational to any Mori fiber space other than itself. Since X is birationally rigid in the sense of Definition 2.11, for any Mori fiber space $f : Y \to T$ such that Y is birational to X, T is a point and $Y \cong X$. Thus X is birationally rigid in the sense of [CSZ23, Section 1], and X satisfies the hypotheses of Conjecture 1.2.

However, X is not K-stable by Theorem 1.4(3). Thus none of Conjectures 1.1, 1.2, and 1.3 holds. □

References

[AK00] D. Abramovich and K. Karu, *Weak semistable reduction in characteristic* 0, Invent. Math. **139** (2000), no. 2, 241–273.

[Ben70] B. M. Bennett, *On the characteristic functions of a local ring*, Ann. of Math. (2) **91** (1970), 25–87.

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[BJ20] H. Blum and M. Jonsson, *Thresholds, valuations, and K-stability*, Adv. Math. **365** (2020), 107062.

[BH98] W. Bruns and J. Herzog, *Cohen–Macaulay rings*, revised edition, Cambridge Studies in Advanced Mathematics **39**, Cambridge University Press, Cambridge (1998).

[CO24] L. Campo and T. Okada, *K-stability of Fano threefold hypersurfaces of index 1*, arXiv:2409.09492.

[Che01] I. Cheltsov, *Log canonical thresholds on hypersurfaces*, Sb. Math. **192** (2001), no. 8, 1241–1257.

[Che05] I. Cheltsov, *Birationally rigid Fano varieties*, Uspekhi Mat. Nauk **60** (2005), no. 5, 71–160; translation in Russian Math. Surveys **60** (2005), no. 5, 875–965.

[Che09a] I. Cheltsov, *Extremal metrics on two Fano varieties*, Mat. Sb. **200** (2009), no. 1, 97–136; translation in Sb. Math. **200** (2009), no. 1, 95–132.

[Che09b] I. Cheltsov, *Log canonical thresholds of three-dimensional Fano hypersurfaces*, Izv. Ross. Akad. Nauk Ser. Mat. **73** (2009), no. 4, 77–152; translation in Izv. Math. **73** (2009), no. 4, 727–795.

[CP17] I. Cheltsov and J. Park, *Birationally rigid Fano threefold hypersurfaces*, Mem. Amer. Math. Soc. **246** (2017), no. 1167.

[CPW14] I. Cheltsov, J. Park, and J. Won, *Log canonical thresholds of certain Fano hypersurfaces*, Math. Z. **276** (2014), no. 1–2, 51–79.

[CSZ23] I. Cheltsov, A. Sarikyan, and Z. Zhuang, *Birational rigidity and alpha invariants of Fano varieties*, arXiv:2304.11333.

[CPR00] A. Corti, A. V. Pukhlikov, and M. Reid, *Fano 3-fold hypersurfaces*, in: *Explicit birational geometry of 3-folds*, London Math. Soc. Lecture Note Ser. **281**, Cambridge Univ. Press, Cambridge (2000), 175–258.

[dF13] T. de Fernex, *Birationally rigid hypersurfaces*, Invent. Math. **192** (2013), no. 3, 533–566; Erratum, Invent. Math. **203** (2016), 675–680.

[dF23] T. de Fernex, *Birational rigidity and K-stability of Fano hypersurfaces with ordinary double points*, Rend. Circ. Mat. Palermo (2) **72** (2023), no. 2, 1049–1065.

[Dol82] I. Dolgachev, *Weighted projective varieties*, in: *Group actions and vector fields (Vancouver, B.C., 1981)*, Lecture Notes in Math. **956**, Springer, Berlin (1982), 34–71.

[Ess25] L. Esser, *Rational weighted projective hypersurfaces*, Int. Math. Res. Not. IMRN **2025** (2025), no. 1, Paper No. rnae261.

[FST11] O. Fujino, K. Schwede, and S. Takagi, *Supplements to non-lc ideal sheaves*, RIMS Kôkyûroku Bessatsu **B24** (2011), 1–46.

[Fuj17] K. Fujita, *K-stability of log Fano hyperplane arrangements*, arXiv:1709.08213.

[Fuj19a] K. Fujita, *A valuative criterion for uniform K-stability of $\mathbb{Q}$-Fano varieties*, J. Reine Angew. Math. **751** (2019), 309–338.

[Fuj19b] K. Fujita, *K-stability of Fano manifolds with not small alpha invariants*, J. Inst. Math. Jussieu **18** (2019), no. 3, 519–530.

[Ful98] W. Fulton, *Intersection theory*, second edition, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **2**, Springer-Verlag, Berlin (1998).

[Gro05] A. Grothendieck, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux (SGA 2)*, augmenté d'un exposé de M. Raynaud, Documents Mathématiques (Paris) **4**, Société Mathématique de France, Paris (2005).

[DGO25] T. Duarte Guerreiro and T. Okada, *A survey on birational rigidity of threefold weighted complete intersections*, arXiv:2508.13025.

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics **52**, Springer-Verlag, New York–Heidelberg (1977).

[IF00] A. R. Iano-Fletcher, *Working with weighted complete intersections*, in: *Explicit birational geometry of 3-folds*, London Math. Soc. Lecture Note Ser. **281**, Cambridge Univ. Press, Cambridge (2000), 101–173.

[KOW18] I.-K. Kim, T. Okada, and J. Won, *Alpha invariants of birationally rigid Fano three-folds*, Int. Math. Res. Not. IMRN **2018** (2018), no. 9, 2745–2800.

[KOW23] I.-K. Kim, T. Okada, and J. Won, *K-stability of birationally superrigid Fano 3-fold weighted hypersurfaces*, Forum Math. Sigma **11** (2023), Paper No. e93.

[Kol13] J. Kollár, *Singularities of the minimal model program*, with a collaboration of S. Kovács, Cambridge Tracts in Math. **200**, Cambridge University Press, Cambridge (2013).

[Kol19] J. Kollár, *The rigidity theorem of Fano–Segre–Iskovskikh–Manin–Pukhlikov–Corti–Cheltsov–de Fernex–Ein–Mustaţă–Zhuang*, in: *Birational geometry of hypersurfaces*, Lect. Notes Unione Mat. Ital. **26**, Springer, Cham (2019), 129–164.

[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).

[Lan02] S. Lang, *Algebra*, revised third edition, Graduate Texts in Mathematics **211**, Springer-Verlag, New York (2002).

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry II*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **49**, Springer-Verlag, Berlin (2004).

[Li17] C. Li, *K-semistability is equivariant volume minimization*, Duke Math. J. **166** (2017), no. 16, 3147–3218.

[LZ21] Y. Liu and Z. Zhuang, *Birational superrigidity and K-stability of singular Fano complete intersections*, Int. Math. Res. Not. IMRN **2021** (2021), no. 1, 382–401.

[Mat89] H. Matsumura, *Commutative ring theory*, second edition, translated from the Japanese by M. Reid, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge (1989).

[Mum74] D. Mumford, *Abelian varieties*, 2nd ed., Tata Institute of Fundamental Research Studies in Mathematics **5**, Oxford University Press, London (1974).

[OO13] Y. Odaka and T. Okada, *Birational superrigidity and slope stability of Fano manifolds*, Math. Z. **275** (2013), no. 3–4, 1109–1119.

[OS12] Y. Odaka and Y. Sano, *Alpha invariant and K-stability of $\mathbb{Q}$-Fano varieties*, Adv. Math. **229** (2012), no. 5, 2818–2834.

[Oka26] T. Okada, *K-stability and birational rigidity of Fano 3-fold weighted complete intersections*, arXiv:2609.01389.

[PST17] M. Pizzato, T. Sano, and L. Tasin, *Effective nonvanishing for Fano weighted complete intersections*, Algebra Number Theory **11** (2017), no. 10, 2369–2395.

[Puk17] A. V. Pukhlikov, *The $4n^2$-inequality for complete intersection singularities*, Arnold Math. J. **3** (2017), no. 2, 187–196.

[Ser79] J.-P. Serre, *Local fields*, Grad. Texts in Math. **67**, Springer-Verlag, New York (1979).

[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.

[SZ19] C. Stibitz and Z. Zhuang, *K-stability of birationally superrigid Fano varieties*, Compos. Math. **155** (2019), no. 9, 1845–1852.

[Suz17] F. Suzuki, *Birational rigidity of complete intersections*, Math. Z. **285** (2017), no. 1–2, 479–492.

[Tia87] G. Tian, *On Kähler–Einstein metrics on certain Kähler manifolds with $C_1(M) > 0$*, Invent. Math. **89** (1987), no. 2, 225–246.

[Zhu20] Z. Zhuang, *Birational superrigidity and K-stability of Fano complete intersections of index 1*, Duke Math. J. **169** (2020), no. 12, 2205–2229.

[Zhu21] Z. Zhuang, *Optimal destabilizing centers and equivariant K-stability*, Invent. Math. **226** (2021), no. 1, 195–223.

School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Email address: liujihao@math.pku.edu.cn